TOMATOES

2nd Edition

Edited by Ep Heuvelink

CROP PRODUCTION SCIENCE IN HORTICULTURE

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Tomatoes, 2nd Edition

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Preface

The first edition of the book *Tomatoes* in the CABI Publishing Series *Crop Production Science in Horticulture* was published in 2005. Since then, our knowledge on tomato has greatly extended. The tomato genome has been sequenced, making tomato even more than before a model fruit-bearing crop. Great progress has been made in open field and greenhouse tomato production, and in our understanding of tomato crop physiology, fruit quality and postharvest physiology. Since 2004, almost 24,000 scientific papers have been published with 'tomato' or 'tomatoes' in their title.

As with the first edition, this book is not a monograph but each chapter is written by experts in their particular subject areas. These subject areas are almost the same as in the previous edition; however, from the 19 current authors only six also contributed to the first edition. Furthermore, a chapter on organic tomato production has been added. Authors come from different regions of the world, giving the book a more international nature.

Tomato is food, it is a crop of great economic and scientific importance, and can also be fun – as shown in the photograph overleaf.

I thank the authors of the different chapters sincerely for their time spent, as well as for their patience. I am grateful to CABI Publishing and in particular to Rachael Russell for giving me the opportunity to be the editor of the tomato volume in their series.

I hope readers will find in this second edition the information they were looking for and I look forward to comments and suggestions for improvement.

Ep Heuvelink
Wageningen, June 2018

Every year the tiny village of Buñol in Valencia (Spain) hosts the largest tomato war in the world: 'La Tomatina'. In this festival, at the peak of the tomato season, for 2 hours, participants pelt each other with ripe, red fruit and the streets turn into rivers of tomato juice.

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The Global Tomato Industry

J. Miguel Costa and Ep Heuvelink

CLASSIFICATION AND TAXONOMY

Tomato is one of the world's major fresh and processed fruit and is the second most important vegetable crop after the potato worldwide. Tomato belongs to the *Solanaceae* (nightshade family), genus *Solanum*, section *Lycopersicon*. The *Solanaceae* family includes other important (vegetable) crops like chilli and bell peppers (*Capsicum* spp.), potato *(Solanum tuberosum*), aubergine (*Solanum melongena*), tomatillo (*Physalis ixocarpa*), tamarillo or tree tomato (*Solanum betaceum*) and tobacco (*Nicotiana tabacum*). Information on the genetic variation within *Solanum*, section *Lycopersicon* (13 species: tomato and its wild relatives), is provided in Chapter 2.

In 1753, Linnaeus named tomato *Solanum lycopersicum*. Fifteen years later, Philip Miller moved it to its own genus, naming it *Lycopersicon esculentum*. This name came into wide use but was in breach of the plant naming rules. Although the name *Lycopersicon lycopersicum* may be found, it is not used because it violates the International Code of Nomenclature barring the use of tautonyms in botanical nomenclature. Genetic evidence has now shown that Linnaeus was correct to put the tomato in the genus *Solanum*, making *Solanum lycopersicum* the correct name. Both names, however, will probably be found in the literature for some time.

USES AND AVAILABLE VARIETIES AND TYPES

The popularity of tomatoes relates to the fact that it can be eaten in multiple forms, either fresh or processed. Among the processed products are: (i) tomato preserves (e.g. whole peeled tomatoes, tomato pulp and juice, tomato puree, pickled tomato and tomato paste; (ii) dried tomatoes (tomato powder, tomato flakes, dried tomato fruits); and (iii) tomato-based foods (e.g. tomato soup, tomato sauces, chilli sauce and ketchup). Tomato has become a model species to study plant and fruit physiology, biochemistry and genetics (Vitale *et al.*, 2014), and the tomato genome has been sequenced (Solanke and Kumar, 2013). Tomato has several relevant traits, such as fleshy fruit, a sympodial shoot and compound leaves, that cannot be studied in other model plants (e.g. rice and *Arabidopsis*).

Tomato is grown as an annual crop mainly in temperate climates, but plants and fruit suffer physiological injury under low non-freezing temperatures (e.g. below 12°C; see Chapters 4 and 10). Varieties for processing or for the fresh market have different growth habits. Processing tomatoes have determinate growth, dwarf habit, uniform and concentrated fruit set and ripening, tough skins and high soluble-solids content. Cultivation is made in the open field, either by direct seeding or by using transplants. In the past decade direct seeding has been largely replaced by transplanting, resulting in a better stand establishment, reduced weed competition and higher success rate and yield (Barrett, 2015).

Due to their growth habit, processing tomatoes do not require trellising or staking and single harvest is done mechanically (Fig. 1.1F). Tomatoes for fresh consumption are grown in open field systems or under protection (greenhouses), have determinate (open field) or indeterminate growth (greenhouses) and require trellising (Fig. 1.1A–C). Fruit quality is promoted by staking, as it avoids fruit contact with the soil, whereas pruning favours fruit size (see Chapter 4). Harvesting is done by hand. The number of varieties for the fresh market is large and it ranges from the popular classic round tomato to niche/regional types (e.g. 'Coeur de Boeuf') (Table 1.1; Fig. 1.1H).

THE GLOBAL TOMATO INDUSTRY: PRODUCTION AND MARKET

General introduction

The tomato industry is a global, diversified and innovative industry. The main production areas are located in temperate zones and characterized by long summer periods and winter precipitation. However, tomatoes are also produced in (sub)tropical climates. Tomato production occurs either in open field or under protection (greenhouses). Harvest can be manual (mostly for fresh consumption) or mechanical (mostly for the processing industry) (Fig. 1.1F). A large variation exists in the cultivated varieties and landraces, cultivation practices, organizational structure of the supply chain, certification and marketing (Bellec-Gauche *et al.*, 2015).

Global tomato production (fresh and processed) has strongly increased in the past five decades. In 1961, production was 27.6 million tonnes, in 2002 this was 116.5 million and in 2014 it was estimated at 171 million tonnes (Table 1.2). China, the European Union (EU), India, the USA and Turkey accounted for almost 70% of global production in 2014. Asia leads with about

Fig. 1.1. (A) A modern Dutch tomato greenhouse: cultivation on stone wool placed on hanging gutters (see Chapter 9). (B) Plastic tunnel in South Europe (courtesy Revista APH). (C) Tomato production in a Chinese solar greenhouse. (D) Field-grown tomato for processing, using a biodegradable plastic film for soil covering (courtesy FILMAGREGA project, ISA, University of Lisbon). (E) View from a typical processing tomato field in Ribatejo, Portugal. (F) Mechanical harvest of processing tomato (courtesy M. Reis, Universidade do Algarve). (G) Processing tomato variety. (H) Fresh tomato cultivar 'Coeur de Boeuf' (courtesy Mon Petit Marche, Switzerland).

Table 1.1. Major groups/types of tomato landraces and varieties.

Type	Characteristics and uses
Classic round	Round shape, 2–3 locules, average fruit weight 70–100 g and diameter 4.7–6.7 cm. It is the most popular variety, used in salads, grilling, baking or frying, soups and sauces.
Plum and baby plum	Oval shape. Baby plum tomato is smaller. The flesh is firm and less juicy in the centre. Used for barbecue, and processed for pizzas and pasta dishes.
Beefsteak	Flattened shape, with 5 or more locules, fruit weight 180–250 g. Large variability in shape, colour (red, pink), texture and flavour. Used for stuffing and baking whole, salads and sandwiches.
Cherry and cocktail	Smaller than classic tomatoes, weight 10–20 g, diameter 1.6–2.5 cm. Cherry tomatoes are smaller than the cocktail ones. Both are very sweet. Cherry tomatoes are generally red, but golden, orange and yellow varieties do exist. Almost all cocktail varieties are sold attached to the stem ('on-the-vine') (not in USA). They are eaten whole and raw or cooked. Cocktail tomatoes are used in salads or skewered whole for grilling.
Vine or truss (cluster)	Marketed when still attached to the fruiting stem; the stem gives the distinctive tomato aroma. Fruit ripening is uniform within the cluster and keeps a fresh green calyx and vine after harvest.
Regional varieties, landraces and non-hybrids	Coeur de Boeuf (France), Raf (Almeria, Spain), Tomàtiga de Ramellet (Balleares, Spain), Marmande (France), Mezzo tempo (Abruzzo, Italy), Spagnoletta (Latium, Italy).

Sources: Baldina *et al.* (2016); BritishTomatoes (2017); Galmés *et al.* (2011); Rodríguez-Burruezo *et al.* (2016).

60% of the world's production, and the Americas and Europe account for about 15% and 13%, respectively (FAOSTAT, 2016). The cultivated area in 2014 reached 4.3 million hectares and the five leading countries were China, India, the USA, Turkey and Egypt (Table 1.2). China and India together account for about 60% of global cultivated area (Table 1.2). The ratio between fresh versus processing tomato production varies strongly among countries. For example, in India, processing tomatoes are < 1% of the total production, whereas this is about 96% in the USA or Italy (Table 1.2). Worldwide processing tomato production was about 41 million tonnes in 2015 (WPTC, 2016) and the largest producers were the USA, China, Italy, Spain and Turkey (Table 1.2). These five countries together represent about 85% of global production of processing tomato.

Tomatoes for fresh consumption are mainly grown in open field systems; however, greenhouse production has expanded in recent decades, especially in the Mediterranean basin (e.g. Spain, Turkey, Portugal, Morocco) and in Middle

Table 1.2. The top ten tomato-producing countries. Annual data 2014–2016.

Country	Total production[a] (× 10^6 t)	Tomatoes for processing[b,c,d] (× 10^6 t)	Tomatoes for processing[b,c,d] (% total)	Total harvested area[a] (× 10^3 ha)
1. China	52.72	5.60	11	1002
2. India	18.74	0.13[g]	< 1	882
3. USA	14.51	13.40	95[e]	163
4. Turkey	11.85	2.70–3.9	25–30[d]	319
5. Egypt	8.29	0.25	3	214
6. Iran	5.97	1.35	23	159
7. Italy	5.62	5.40	96	104[h]
8. Spain	4.89	3.03	62	63[h]
9. Brazil	4.30	1.30	33[f]	64
10. Mexico	3.54	0.04	1	95
EU[h]	17.90[h]	60	40	257[h]
World total	**171**	**41**	**24**	**5023**

[a]FAOSTAT (2016); [b]WPTC (2016); [c]AMITOM (2015, 2016); [d]http://www.turkstat.gov.tr/; [e]USDA (2016e); [f]ABH (2016); [g]Subramanian (2016); [h]Eurostat (2016).

and Latin America (Brazil and Mexico), North America (USA, Canada) as well as in Australia and Russia. China is the largest consumer of tomato worldwide, followed by the EU, Mediterranean Africa and North American Free Trade Agreement (NAFTA) countries (Canada, Mexico, USA) (Branthôme, 2010). Consumption is stable or slightly decreasing in mature markets (e.g. EU, North America) (Freshfel, 2015) but keeps increasing in countries of Africa, Asia, South America and the Middle East (Agrotypos, 2016).

Trading in fresh tomatoes and processed products is a major global business. In 2015, world exports of fresh tomatoes reached 7.1 billion euro (Workman, 2018). Trade of fresh produce occurs mainly between neighbouring countries, due to freight and tariff advantages and because fresh tomato is a perishable commodity.

In 2015, the world leader in tomato export volume was Mexico followed by The Netherlands (USDA, 2016a,b). However, Dutch exports have a higher economic value than those of Mexico.

Europe (North, Eastern and Mediterranean) and Turkey

The EU is among the top five tomato producers worldwide. The EU represents around 10% of global tomato production and 40% of the global production for processing (European Commission, 2016; Eurofresh Distribution, 2016a). The crop is grown on 257,000 ha with a production output of 17.9 million tonnes. Italy, Spain and Portugal are the leading producers and represent about 70% of EU production. Open field cultivation is typical of southern Mediterranean EU countries, whereas protected cultivation is more common in the Northern Atlantic and continental EU members. Northern producing systems are highly technological and capital intensive (Fig. 1.1A) and focus on fresh tomato production, whereas Mediterranean countries focus on processing tomato in open field systems. However, fresh tomato production in greenhouses has expanded in countries like Spain, Italy, France, Greece and Portugal (Fig. 1.1B).

Italy

Italy's tomato production in 2015 was 5.6 million tonnes (96% for processing) (Table 1.2). Italy is the world's seventh largest producer and leads tomato production in the EU with a 36% share (Eurostat News, 2016). The Italian tomato processing industry accounts for 14% of world production and about 50% of EU production, with an estimated value of €3.2 billion, half of which is exported (ANICAV, 2016). Tomato processing companies are concentrated in the regions of Campania (north) and in Puglia (south) (AMITOM, 2016). In the north, farms are larger, favouring mechanization and professional management. Growers use hybrid seeds and transplants resulting in higher yields (75–100 t/ha) as compared with the south (about 70 t/ha) (AMITOM, 2016). Planting begins in early May, harvest occurs in mid-July and the season ends by the middle–end of September. In southern Italy, companies have a family-type structure and an average size of 4–10 ha. In 2015, about 26,000 ha were dedicated to fresh tomato production, 50% of which was protected cultivation (Dall'Olio, 2016). Sicily leads protected cultivation of tomato with 6500 ha and an output of 380,000 t (Testa *et al*., 2014; Dall'Olio, 2016). Yields are in the range of 20–25 kg/m² (Battistel, 2011). In the north, in more modern greenhouses, with heating and carbon dioxide (CO_2) enrichment and a long cycle (January–December), yields up to 55 kg/m² are realized (Battistel, 2011). Italy has a large number of regional varieties of salad tomatoes and it focuses on selection and breeding of these varieties for both organic and low-input farming (Campanelli *et al*., 2015).

Spain

Spain is the second largest tomato producer in the EU (Eurostat, 2016) and the eighth largest worldwide (Table 1.2) with a cultivated area of about 55,000 ha (Table 1.2). Spain leads the EU fresh tomato market, with a production of 950,000 t (about 27% of EU production) (Table 1.2) and an area of 27,000 ha (MAGRAMA, 2015). Tomatoes are the most important exported fresh vegetable (679,000 t representing €678.6 million in 2016) and the province of Almeria alone accounts for about 60% of Spanish exports. Germany, France and the UK are the main markets, whereas imports are mainly from Portugal, The Netherlands and Morocco (FEPEX, 2016; Mili, 2016).

Production of tomato for fresh consumption is mainly done under protected cultivation on an estimated area of 24,000 ha. The provinces of Almeria and Murcia lead greenhouse production with an area of 10,300 ha and 2700 ha, respectively (MAGRAMA, 2015). Production of round tomato is dominant but the segment of 'specialties' is increasing. Average tomato production in Almeria is 16.8 kg/m^2 (Martínez *et al.*, 2016). Growers have used plastic structures called 'parral' for several decades (Costa and Heuvelink, 2000) but the area of modern plastic multi-tunnels and biological production has been increasing in recent years (Martínez *et al.*, 2016). Greenhouse cultivation is still done in *enarenado* (artificial soil produced by the 'sanding' technique), with a 30 cm layer of soil put on top of the natural soil, then 2–3 cm compost and a top-layer of sand, but cultivation on inert substrates has increased. Production peak occurs in December/January. Cooperatives have controlled the supply chain for a long time and they are involved in mergers to become more competitive (Costa and Heuvelink, 2000; Martínez *et al.*, 2016).

Regarding processing tomato, the cultivated area in 2015 was about 27,700 ha (AMITOM, 2016). Extremadura represents about 80% of the Spanish production area, distributed over Andalucía, Navarra and Aragón, Castilla-La Mancha. Production in 2016 was estimated to be 2.95 million tonnes, with an average yield of 80 t/ha (WPTC, 2016).

Portugal

In 2015 Portugal became the third largest producer within the EU, just behind Spain and Italy, with a cultivated area of 19,300 ha and a production of 1.4 million tonnes (Eurostat, 2016), mostly for processing (Fig. 1.1E–G). Since the end of last century the sector has experienced increased concentration and more professional organization and receives foreign investment (American and Japanese). Drip irrigation and mechanization have become common practice, which has increased yield and production efficiency. However, high energy costs are a limitation for both the processing tomato industry and greenhouse production. As the internal market is small, Portugal is focused on exports. Tomatoes for processing are mainly produced in Ribatejo Oeste and Alentejo. There are nine large processing plants and 24 producer organizations (POs) (GPP, 2013; WPTC, 2016). In 2015, the average yield was about 90 t/ha (INE, 2016). Ribatejo and Baixo Alentejo (southern Portugal) are the largest and most productive regions, due to novel cultivation procedures. Tomato is the most important greenhouse crop in Portugal with about 1400 ha and a production of about 96,000 t, mainly located in Ribatejo Oeste and Algarve and using plastic tunnels (Fig. 1.1B). Cherry tomato is produced in glasshouses for exports (Baptista *et al.*, 2014).

Greece

Greece had a production area of 17,000 ha in 2015, with a production of 1.08 million tonnes (Eurostat, 2016), mostly for fresh consumption. Tomato is the most important greenhouse vegetable, with a production of about 600,000 t in 2014 (RVO, 2015a), mainly located in northern regions. Recent investments in modern glasshouses and geothermal greenhouses have been made in

Crete, Peloponnese and Western Greece. Production of tomato for processing is decreasing and reached 0.5 million tonnes in 2015 (WPTC, 2016).

The Netherlands

Tomato is the most commonly grown vegetable in Dutch greenhouses and cultivation has steadily increased in recent years. In 2000 the area amounted to 1133 ha; 10 years later it had increased by 50% and thereafter stabilized, whereas the production volume almost doubled (520,000 t in 2000 and 900,000 t in 2016). The average size of a tomato farm is 6.2 ha of glasshouses. The Netherlands is also a major exporter of fresh tomatoes, being second worldwide, in volume just behind Mexico (USDA, 2016b). Dutch tomato production takes place in about 1750 ha of glasshouses (of which 1260 ha are tomatoes on the vine) and about 40% of this area is equipped with supplementary light. Cultivation is on substrates, mainly stone wool, and only 32 ha of organic production is done in soil. The market is well organized; about 85% of the production is sold via growers' associations (USDA, 2016b) and about 90% is exported, mostly to Germany and the UK (USDA, 2016b). The Netherlands continues to lead intensification with 60–70 kg/m^2 for year-round production. Dutch tomatoes are appreciated for their constant quality, very low residues (due to biological pest control) and, very importantly, reliable delivery. Cultivation without supplementary light starts in December and harvest extends from March to November. The use of modern technology is typical of the Dutch production system (Fig. 1.1A). Growers have good levels of education and organization. Innovation is promoted by the academies and by private companies. A major recent development is Agriport A7, a large-scale greenhouse area in North Holland, just 30 min from Amsterdam, which accommodates modern greenhouse farms with areas of up to 100 ha (AgriportA7, 2017).

Belgium

Fresh tomato is produced in greenhouses with similar technological levels to those of Belgium's Dutch neighbours. In both countries production is hydroponic, mainly on stone wool. The production area is about 500 ha and total production volume in 2015 was 246,000 t (loose, vine, plum tomatoes) plus 22,000 t of specialties (De Blasier, 2016). Ninety per cent is marketed via producer organizations (POs) located in Flandres and 70% is exported (De Blasier, 2016). The tendency is for a decrease in beefsteak tomato and an increase in specialities.

France

The country produced 603,000 t of fresh tomato in 2016 on 2230 ha, of which 1900 ha are in greenhouses (AGREST, 2016). Fresh tomatoes are mainly produced under protected cultivation and this tends to increase, mainly in Brittany (36% of total production), the South-East, the South-West, Loire Valley and Languedoc-Roussillon (Bellec-Gauche *et al.*, 2015). France imports about 400,000 t per year, mainly from Morocco and Spain (Bellec-Gauche *et al.*, 2015). Imports occur between October and March via Saint Charles (Perpignan). French POs control 70% of fresh tomato production, which is collected, packed and sold through these organizations (Bellec-Gauche *et al.*, 2015). In 2015, 194,000 t of processing tomatoes were produced in about 2550 ha, with an average production of 75–80 t/ha, but production can reach 100 t/ha in the *Sud-Est* Mediterranean region (AMITOM, 2016).

Turkey

Turkey is the fourth largest producer of tomato worldwide with about 12 million tonnes in 2014 and 2015 of which 25% for processing (Table 1.2). Tomato production represents about 25% of the country's total fresh fruit and vegetable production (Aksoy and Kaymak, 2016), and it is mainly produced in about 21,700 ha of greenhouses. Greenhouse tomato expanded fast between 2005 and 2014 and became the country's second most important crop species in value and the most exported vegetable (Engİndenİz and Ucar, 2016). Tomatoes are grown throughout Turkey but those for fresh consumption are mainly produced in Mediterranean climate conditions in Antalya and Mersin provinces (USDA, 2009). Izmir, Canakkale and Mersin are other major producing regions. Cultivation in greenhouses is done in winter and early spring. Production of processing tomatoes is mainly located in Aegean and Marmara regions (USDA, 2009; Aksoy and Kaymak, 2016). Turkey is self-sufficient in tomato and a major global exporter with about 60% exported fresh and frozen to Russia (Aksoy and Kaymak, 2016). However, the 2014 Russian embargo on food imports from EU countries (FAO, 2014) closed borders to imports of vegetables from Turkey in 2015 and diverted Turkish tomatoes to EU members (e.g. Romania, Bulgaria, Austria, Hungary, Poland, Greece), due to their geographical proximity and lower transportation cost (Mili, 2016).

Russia

In 2014, Russia produced 2.8 million tonnes of tomato on 118,000 ha (FAOSTAT, 2016). This is insufficient to supply the internal market and Russia is the second largest importer of tomato worldwide, with Turkey as its main supplier (USDA, 2016c). In 2014, Russia imported about 846,000 t (USDA, 2015). However, due to the Russian embargo, both the EU and Turkish exports to Russia abruptly decreased between 2013 and 2015 (FAO, 2014). This novel situation changed the Russian tomato sector. Market prices have increased (Uzun, 2016) and greenhouse production expanded in southern Russia (Mulderij, 2016). To make Russia less dependent on imports, the Russian government supported investments in the greenhouse production sector in order to increase greenhouse area from 1800 ha (2013) up to 4700 ha by 2020 (Flandres Investment Trade, 2014). Most greenhouses in Russia were built in the 1970s and 1980s, and so far have not been modernized or rebuilt, which has resulted in higher production costs and the unprofitability of many businesses. Production of tomato for processing reached 90,000 t in 2015 and is mainly located in Astrakhan, Krasnodar and Volgograd (WPTC, 2015).

South Asia (China, India and Japan)

China

With a total of 52.7 million tonnes in 2015 and a cultivated area of about 1 million hectares (Table 1.2), China is the world's leader in tomato production. Chinese production represents more than one-third of global production and is mostly for domestic use. China is the world's largest exporter of processed tomato products, such as tomato sauce and paste. Xinjiang is the major processing region in China, with local processed volume totalling 5.011 million tonnes and accounting for about 74% of national production in 2011 (WPTC, 2012). Xinjiang's cultivation calendar is quite similar to the one for California: in March, fields are prepared for planting; April to July is the growth period; and harvest and processing occur between August and October. Other important production regions for processing tomatoes are Inner Mongolia, Gansu province and Ningxia region (WPTC, 2012). Chinese competitiveness lies in low production costs, which negatively affects global market prices. China does not have preferential access to EU markets for processed tomatoes but nevertheless it manages to increase exports to the EU (European Commission, 2012).

Tomatoes for fresh consumption are produced throughout China, but four provinces (Shandong, Xinjiang, Hebei and Henan) represent about 50% of total Chinese production. Shandong (north-east China) has a cultivated area of 79,000 ha (RVO, 2015b) and produces 60% of China's fresh tomatoes in winter, in 35,000 ha of solar greenhouses (RVO, 2015b). The peri-urban areas of Shanghai and Beijing are major producers of fresh tomato (Costa *et al.*, 2004). In China, greenhouse production is largely based on small farms (600–2000 m^2) using solar-type greenhouses (Fig. 1.1C) (Costa *et al.*, 2004; Zhang *et al.*, 2010). In the past decade foreign investments (e.g. from Taiwan and Japan) have been made in new greenhouses, and novel varieties have been introduced via foreign and national breeders. Production of cherry tomatoes is a recent trend and more investments are expected for the southern provinces of Guangdong and Guangxi and on Hainan Island (South China) (Mulderij, 2016).

India

India is the fourth largest tomato producer worldwide with a total production of 18.7 million tonnes and a harvested area of 882,000 ha (Table 1.2). Area and production have increased since 2010, as a result of increased market demand, mainly for processed tomato products, which showed an annual growth rate of about 30% in the past 3 years (Subramanian, 2016). India's global tomato exports are minor in comparison with the global trade in fresh tomatoes. The most important states producing tomatoes are Andhra Pradesh, Karnataka, Madhya Pradesh and Telangana (Subramanian, 2016). Fresh tomatoes are available year-round and less than 1% of the tomato production is used for processing (Table 1.2). Several multinationals are present in India but there are limitations such as the lack of suitable varieties for processing, outdated processing technology and high postharvest losses (Subramanian, 2016). Yields per unit of area are the lowest amongst the world's ten largest tomato producers, which is related to the fact that production is largely based on smallholders, using extensive crop systems and agronomic practices in which tomato is grown in rotation with other vegetables. Production of tomatoes in large landholdings and using intensive cultivation practices has still to be adopted in India (Subramanian, 2016).

Japan

Japanese growers are focused on tomato for fresh consumption. In 2014, the production area was 12,400 ha and a production output of about 740,000 t (FAOSTAT, 2016). In global terms, Japan's processing industry is small with only 35,000 t being processed in 2015 (AMITOM, 2016). Fresh tomatoes are produced in glasshouses, high plastic tunnels or under rain shelters. Field cultivation occurs only during summer time (Otsuka, 2011). Japan imports mainly from South Korea (964,000 t in 2010) and from the USA, Mexico, New Zealand, Canada and Australia (Otsuka, 2011). The Japanese market continues to increase and tends to be more diversified, with large opportunities for beefsteak tomatoes (Mulderij, 2016). The Japanese vegetable market is increasingly demanding ready-made meals and convenience food and it is expected that 'minimal-processed vegetables' will be on demand as part of the trend towards 'convenience' foods (USDA, 2016d). This applies to tomato products. Food safety and premium products are other priorities for Japanese consumers.

South Korea

South Korea produces mainly greenhouse tomato, with an area of 5200 ha in 2005. Growers specialize in Japanese varieties (e.g. 'Momotaro') for exporting to Japan. Tomato is among the top ten crops, with a production of 480,000 t in 2014, which has increased rapidly in recent years (325,000 t in 2010) (AAFC, 2016a).

North America

USA

The USA is the world's third largest tomato producer (Table 1.2) and the world's leader in production and export of processed tomatoes. It is also one of the world's largest tomato importers, as internal production only accounts for about 40% of the country's needs. The US import share grew from 22% in 1980 to 52% in 2010. Mexico is the main supplier and to a much lesser extent Canada. Consumption of fresh tomatoes is stable and accounts for about 25% of total tomato consumption, most being as processed products (sauces, juice, tomato paste) (Eurofresh Distribution, 2016a). California's tomato processors anticipated contracting a total of 11.5 million tonnes in 2016, 11% less than in 2015 (Wells and Bond, 2016). This decline partly derives from severe drought events in California, higher disease/pest pressures and large stocks resulting from record high production in 2014 and 2015 and the strong US dollar. California accounts for about 96% of the country's production in processing tomato and 30% of tomato for fresh consumption (USDA, 2016e). California combines long, warm and dry growing seasons with the use of novel hybrids and technologies (field production and processing plants), and strong investment in research and development for a long time (Mitchell *et al.*, 2001; Barrett, 2015). Processing tomato in California is grown in rotation with crops such as cotton, garlic, onions, melons and wheat in San Joaquin Valley, or with wheat and edible dry beans in Sacramento Valley (Mitchell *et al.*, 2001). Among all the vegetable crops produced in the USA, the total value of fresh tomato production is the highest (Asci *et al.*, 2014). California and Florida lead production and represent two-thirds of the total production area. Production of fresh tomato has decreased in recent decades due to imports from Europe, Canada and mainly Mexico. Mexican tomato competes directly with the winter and early spring market for field-grown tomato from Florida and greenhouse tomatoes (Asci *et al.*, 2014). The trade conflict between US and Mexican fresh tomato producers has continued for a few decades, and the revised agreement (the 'Suspension Agreement') between the two countries became effective on March 2013 to regulate imports from Mexico and includes anti-dumping measures (Asci *et al.*, 2016). Greenhouse tomatoes are increasing their fresh-tomato market share (USDA, 2016e) and receive higher prices than field-grown ones, due to a positive perception on quality (Asci *et al.*, 2014). The USA exports fresh tomato to Canada, with a small volume going to Japan (USDA, 2016e).

Canada

Tomato is the most important field vegetable produced in Canada, with a total production of 402,000 t in 2015 (AAFC, 2016b). Tomatoes are also the most important greenhouse commodity (266,000 t) produced on about 550 ha (AAFC, 2016b). Canadian growers are technologically advanced and production is highly comparable to northern European countries in technology, investment and crop management. Ontario leads greenhouse tomato production. Canadian tomato exports reached 125,000 t in 2015 and were almost exclusively to the USA (124,000 t) (AAFC, 2016b).

Central and South America

Mexico

Mexico is the world leader in tomato exports. The production area in 2015 and 2016 was around 55,000 ha (Flores, 2015), though for unknown reasons this figure differs considerably from the 95,000 ha mentioned by FAO for 2014 (Table 1.2). The greenhouse area of tomato in Mexico was estimated at about 15,000 ha in 2015/16 (USDA, 2016a). The sector has the technological support of different foreign partners including Dutch, Spanish, Canadian and Israeli. Greenhouse/shade-house production is concentrated in the states of Sinaloa, Baja California and Jalisco (USDA, 2016a). Greenhouse production uses cultivars of indeterminate habit, cutting off lateral buds, in long crop cycles (up to 11 months), with 15–25 clusters per plant and using 2–3 plants/m^2 (Del Castillo *et al.*, 2012). Average yields have increased from 23 t/ha in 1990 to 50 t/ha in 2015/16. Greenhouse/shade-house yields generally range from 150 t/ha to 200 t/ha, depending on the technology used. For example, Sinaloa can grow Roma tomatoes (saladette) in open field with yields of about 32 t/ha, while it can grow them under protected systems with yields in the range of 87–128 t/ha (Flores, 2015). Mexican greenhouse technology and growing conditions are heterogeneous, with a large range of structures (Thornsbury and Jerardo, 2012). There are few high-tech projects, and yields remain lower than those of North American countries. Foreign investment from the USA and Canada promoted protected cultivation, and high-quality tomatoes are available for export and to supply the growing domestic market (Thornsbury and Jerardo, 2012).

Brazil

Brazil emerged as the ninth largest tomato producer worldwide in 2014 (Table 1.2) with a production area of 64,000 ha and a production of 4.3 million tonnes which results in an average yield of 67 t/ha. Mechanization of transplanting and harvesting and the use of modern hybrids and drip irrigation have increased yield (Santos *et al.*, 2011). Statistics for 2015 reported a smaller cultivated area (57,000 ha, of which 19,000 ha were for processing), which could be due to the ongoing crises in the sector (ABH, 2016). The states of Goiás, São Paulo and Minas Gerais are the most important producing regions. Goiás leads production for processing, resulting from its warm and dry climate conditions and the availability of large irrigated areas by drip or sprinkler (centre pivot) irrigation (Rocco and Morabito, 2016). Production of processing tomato is estimated to be around 1.3 million tonnes, which makes Brazil one of the largest global players in the processing tomato industry (Table 1.2). Fresh tomatoes are produced all over the country, in particular near the major consumption centres. Companies are small to medium size and use staking and furrow or drip irrigation. São Paulo, Minas Gerais, Goiás and Bahía are the most important producing regions for fresh tomato (ABH, 2016). Marketing of fresh tomato occurs via wholesale markets, *entrepostos* (warehouses) or 'Centrais de Abastecimento' (CEASA). There are 63 CEASAs all over Brazil (ABH, 2016). Small growers sell their tomatoes directly to wholesalers, retailers and supermarkets, without intermediates, which represents a major change in a sector that was previously dominated by a network of intermediates (SEAPA, 2016).

Oceania

Australia

In Australia the cultivated area of tomato was 10,000 ha in 2001 and it decreased to 5430 ha in 2016 (ABS, 2016). Production of processing tomato is mainly concentrated in the state of Victoria (about 90% of production). Despite this decrease in area, the industry shows a dynamic growth: 234,000 t in 2014–2015 and about 285,000 t in 2015–2016 (ABS, 2016). This follows recent expansion of processing and canning plants in Victoria (ABS, 2016). About 2700 ha were planted in 2014–2015 and the average yields are quite high at about 100 t/ha (PHA, 2016). Australia consumes around 550,000 t of processed tomato, and the majority comes from Italy and China (PHA, 2016). The main varieties grown in Australia are Heinz cultivars, and 99% of the production area is irrigated using sub-surface drip lines (PHA, 2016). Regarding fresh production, several large greenhouse investments have been reported in recent years, creating new opportunities for the Australian tomato industry so that it becomes less dependent on imports (HIN, 2016).

New Zealand

In New Zealand, fresh tomato is grown in open field systems as well as in greenhouses. In 2015, there was a total area of 765 ha, with 600 ha for processing tomato, 45 ha for fresh production in open field and 120 ha for greenhouse production, with a total production of 108,200 t (FreshFacts, 2016). Production is mainly for the internal market, though the country exports to Japan and Australia. Australia is easy to access by sea, but Australian domestic production is increasing and there are campaigns against imported produce (TomatoesNZ Inc, 2016).

Africa and the Middle East

Egypt

Egypt leads tomato production on the African continent with 8.3 million tonnes produced in 2014 (Table 1.2). Production is mostly for fresh consumption, mainly located in the Nile delta and 98% of total acreage in open field. Due to the increasing demand from Western Europe, larger farmers are investing in protected cultivation of cherry tomato, which is increasing fast (Nijs, 2014). Egyptian tomatoes are mainly exported to Saudi Arabia and Europe (e.g. The Netherlands), which for logistical reasons and price are more attractive than Spain (Mili, 2016).

Tunisia

Tunisia's tomato production was estimated at 1.25 million tonnes in 2014 (FAOSTAT, 2016) and the planted area was about 19,000 ha with an average yield of 66 t/ha. About 50% of total production is processed, which takes place in Cap-Bon in the north-east (AMITOM, 2016). Dry tomatoes are also produced and exported. The EU and Tunisia have been negotiating a trade agreement to facilitate exchange of agricultural products, which may favour future exports of Tunisian tomato to the EU (Almeria Verde, 2016).

Israel

The total tomato production area in Israel in 2014 was about 5500 ha with an output of 426,000 t (FAOSTAT, 2016). The country produced about 220,000 t of processing tomato in 2015 (WPTC, 2016). Processing tomatoes are grown in collective farms (kibbutz and moshav), with intensive mechanization and the use of drip irrigation. Yields can reach 120 t/ha but production costs are high, which makes the processing industry less competitive than in other Mediterranean countries (WPTC, 2016). In recent years, the number of growers has decreased due to low domestic market prices. The country reached its peak in production in 1999 with around 600,000 t, but volume has declined steadily since then, down to 450,000 t. The impact of this decline was mainly seen in exports, as domestic demand has increased. Exports dropped by 65% in 5 years, down to 15,000 t in 2015 (Mulderij, 2016).

Morocco

In 2014, the total tomato production area in Morocco was about 16,000 ha with a total production of about 1.2 million tonnes (FAOSTAT, 2016). The production of tomatoes is mainly concentrated in the regions of Souss-Massa-Drâa and Doukkala-Abda. Other important sites are the eastern regions of Oued-Eddahab-Lagouira, Greater Casablanca and Rabat-Zemmour-Zaer (DEPF, 2014). Greenhouse tomato is the most important early-season vegetable. About 5000 ha of greenhouses for tomato production are located mainly along the Atlantic coast and in the Souss-Massa region (south-west), with average yields of about 150 t/ha, which can reach 220–250 t/ha under optimal conditions (SIRRIMED, 2014). In the Souss-Massa region tomato is planted in greenhouses from July to September for winter harvest. Morocco leads in exports to the EU. About one-third of production is exported. Indeed, the country doubled its export volume of tomato to the EU in the past decade, reaching almost 486,000 t in 2014, and making the country the fourth largest tomato exporter worldwide. However, a major handicap is still the expensive logistics, which can account for about 30% of the final price of the exported product (DEPF, 2014). Tomato for processing is mainly cultivated in the regions of Gharb and Tangiers/Tetouan.

South Africa

South Africa is the major regional tomato producer in sub-Saharan Africa. Tomatoes are produced in all South African provinces. Limpopo province is the major production area with 3590 ha (2700 ha in Northern Lowveld and 890 ha in far northern areas of Limpopo). Other main producing areas are Mpumalanga province (770 ha) and the Border area of Eastern Cape province (450 ha) (DAFF, 2015). Production is limited in winter months and tomatoes can only be produced in frost-free areas during winter or under protection such as tunnels (DAFF, 2012). South Africa exports to the EU (Belgium, Germany, Italy, The Netherlands, Sweden, UK) with a preferential tariff of 0%.

FUTURE PERSPECTIVES FOR THE TOMATO SECTOR

Production and global market

Tomato production is a global and highly diversified industry. Global tomato production is currently around 170 million tonnes, with 75% for the fresh market and 25% for processing. Production and quality of fresh and processing tomato are expected to continue to increase via the use of novel cultivars, more precise crop management, improved technologies for production and processing, and more skilled farmers and managers. However, the trend in per capita consumption is less positive. Over the period 2004–2013, average annual per capita tomato consumption in the EU declined by 20% (Eurofresh Distribution, 2016b).

Field tomato production will take advantage of increasingly accessible monitoring technologies for small–medium farmers (e.g. lower-cost smart sensors, remote sensing) to optimize water and nitrogen use efficiency, predict yield and quality and optimize pest and disease control (Mahlein, 2016). Modern greenhouse production will benefit from a higher degree of computerization and automation, to cope with labour scarcity and to reduce costs in countries such as The Netherlands or Japan. Spain, France and Israel will lead innovation in production under plastic structures (greenhouses, high and low tunnels).

Spain and The Netherlands will keep leading the supply of fresh tomato to the EU, but other countries such as Portugal, Italy and North African countries (e.g. Morocco, Egypt, Tunisia) will increase their share. More bilateral agreements between the EU and North African countries are expected in the coming years. In addition, Turkey can emerge as a major player, but political issues may negatively affect exports to the EU and Russia. Also the Balkan countries (e.g. Albania, Serbia and Macedonia) are emerging as potential suppliers to the EU (Eurofresh Distribution, 2016b). On the American continent, the move to greenhouse production is expected to continue in Mexico and also in the USA, while in South American countries such as Brazil the production of fresh and processed tomato tends to expand.

China and India are expected to increase their production. Greenhouse production is expected to support growth in China. However, both China and India have limitations due to water scarcity and pollution, deficient logistics, a lack of advanced technologies in cultivation and processing, as well as a fragmented supply chain (Costa *et al.*, 2004; Subramaian, 2016; Kang *et al.*, 2017).

Mexico will lead global exports of tomato, but the Dutch are expected to remain as leaders in global logistics of fresh tomato with excellent transport and tracking systems diminishing storage period, improving quality and lowering costs (Rabobank, 2016). Optimizing logistics is still a major issue in regions such as India and African countries (e.g. Morocco), addressing large postharvest losses and the need to increase exports (Arah *et al.*, 2015).

Environment and sustainability

The tomato industry has a substantial environmental impact resulting from the intensive use of water, nutrients, biocides and energy (Karakaya and Özilgen, 2011; Brodt *et al.*, 2013). More sustainable field and greenhouse production is envisaged, especially in more advanced countries. Here, growers face challenges in fully integrating information about plant and climate into crop management decisions to optimize production efficiency and quality with minimal costs of energy, labour and other inputs (see Chapter 9). The concept of the ‘closed greenhouse’ is a reality in countries like The Netherlands as a means to save energy as well as increase yields. However, low-cost solutions are needed for less developed economies and smaller growers/companies.

The EU has launched several strategic initiatives related to environmental issues, including the Resource Efficiency Roadmap, the 2020 Biodiversity Strategy and the Low Carbon Economy Roadmap. This clearly demonstrates the increasing relevance of environmental legislation for EU's agribusiness as well as for EU's trade partners. Circular economy practices and minimal waste production must be implemented by the modern tomato industry, to accommodate stricter environmental legislation (e.g. EU) and more demanding consumers. Innovative recycling solutions must be developed and implemented. Energy production from tomato waste (Amón *et al.*, 2015) and the use of biodegradable plastics are already being tested (Fig. 1.1D). Minimizing wastewater production in processing plants (Caputo *et al.*, 2015) needs further testing and implementation.

Food supply chains are increasingly associated with environmental and socio-economic impacts (Sala *et al.*, 2017) and organic tomato production has been increasing in the EU, but it is still a small market and yields per unit of area in general are lower than conventional production. Consumer preferences for produce that is free from pesticides and GMO, along with environmental concerns, drive the expansion of the organic sector, especially for fresh greenhouse production. New biological control agents, biopesticides and cultural practices to increase plant resilience to biotic and abiotic stresses and precise nutrient supply are needed for more competitive organic tomato production. Furthermore, research on the nutritive value of organic tomato needs attention (see Chapter 11).

Easier and robust environmental impact assessment tools and sustainability programmes are required. For example, the use of water and carbon footprint (Lovarelli *et al.*, 2016) and life cycle analysis (LCA) (Dias *et al.*, 2017) can help to improve sustainability, but methodological adjustments are needed for a wider use by the industry (Sala *et al.*, 2017). Life cycle inventories should cope with limitations on data availability and quality and data representativeness (Sala *et al.*, 2017). Combining LCA with other indicators (e.g. water footprint, ecological footprint and other indexes) is envisaged for more robust environmental impact assessment. Social and economic sustainability is also highly relevant for the tomato industry. Increased sustainability involves improved working conditions and education of employees (from pickers and greenhouse workers to engineers and managers), and improved safety rules for production and handling produce. This especially applies to countries that depend on low-skill migrant temporary workers.

The sector also faces a major challenge related to climate change and increasingly adverse climate conditions, and problems with water scarcity and low quality (e.g. Mediterranean Basin countries, North China and India). Protected cultivation could help to partially mitigate those effects, but more solutions at plant and technical level are needed. Breeding can play an important role here: hundreds of (re)sequenced genomes are available for tomato (The 100 Tomato Genome Sequencing Consortium, 2014) as well as novel tools such as bio-informatics to support breeders (see Chapter 2).

Market trends and politics

Tomato growers must cope with increasing pressure on prices by large retailers. In more mature and developed markets, the focus on local and more ‘tasty’ regional varieties is increasing (Rodríguez-Burruezo *et al.*, 2016). Despite a growing consumer segment that values excellent taste in tomatoes, the industry remains focused on delivering large volumes at low prices (Kouwenhoven and Nalla, 2016).

New living styles based on vegetarianism and veganism can be a positive development for the tomato sector. In addition, lifestyle attributes such as variation, convenience, price, health, taste and food become more important in developed economies. Niche markets for fresh produce ‘specialities’ are growing in several countries, especially in more mature markets. The industry must be aware of all these quality parameters perceived by consumers (Frez-Muñoz *et al.*, 2016). Improving yield traits but also the overall fruit quality is thus a priority. Improved knowledge on the genetic and molecular controls of quality traits, the development of high-throughput phenotyping methods for plant and fruit, and the development of integrative approaches to unravel genotype × environment × management (G×E×M) interactions will help to attain the industry’s objectives, though finding viable compromises between yield and quality remains a challenge (see Chapter 5).

Another important market trend among European and North American consumers is the demand for locally produced food (Van der Velden, 2016). Indeed, while global chains tend to perform better in the economic sense, local chains score best on socio-environmental and ethical issues (Bellec-Gauche *et al.*, 2015).

Regional and global geopolitics, currency issues and environmental legislation will keep influencing the tomato industry. Changes in commercial rules and relationships between countries (e.g. Russian embargo on EU and Turkish produce, the UK’s Brexit or US policies in relation to Mexico) will contribute to changes in production location centres and future investments. EU markets more open to imports from North African countries via bilateral agreements will influence the tomato sectors and will also increase competition between African countries (e.g. Tunisia, Morocco, Egypt) (DEPF, 2014). Future developments of the industry must also involve international cooperation at research and education levels to ultimately increase the sustainability of the tomato industry at regional and global levels.

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2

Genetics and Breeding

Sjaak van Heusden and Pim Lindhout

INTRODUCTION

Tomato breeding is a success story, as illustrated by the many cultivars available now that are grown all over the world in a variety of conditions and that harbour a wide range of traits such as vigorous growth, resistances to pests and diseases, high fruit quality, etc. Tomato breeding follows the classical approach of identifying superior genotypes, making of crossings and selection in the progenies. Nowadays, breeding is facilitated by molecular marker techniques and the known high-quality tomato sequence. Genetic transformation is possible, but genetic modification is not used in commercial tomato breeding. The future trend is genomics-based breeding.

GENETIC VARIATION WITHIN *SOLANUM* L. SECT. *LYCOPERSICON*

The cultivated tomato reached its present form and place in the human diet/lifestyle after many centuries of domestication. Initial development was probably reached by selecting preferred genotypes in existing germplasm. In a predominantly inbreeding species genetic variation tends to decrease, even without selection. Tomato has suffered from severe genetic bottlenecks in the past 600 years as the crop was carried from the New World to Europe, then back to North America where breeding programmes were initiated with the materials available in these regions. As a consequence, genetic variation in the cultivated tomato species has always been considered as extremely limited. It is likely that no crosses with related wild species were deliberately made until the 20th century.

The lack of diversity is illustrated by resequencing various tomato cultivars. Recently, methods have been developed that make it possible to obtain a lot of sequence information (next generation sequencing (NGS)). Despite the tens of thousands of single nucleotide polymorphisms (SNPs) that have been identified by comparing sequences of cultivars (Tang *et al.*, 2006; Sim *et al.* 2012a,b; Viquez-Zamora *et al.*, 2013), the level of variation, detectable at the DNA level, is low compared with other crops such as corn and potato. However, these SNPs may explain the wealth of morphological variation observed among tomato (*Solanum lycopersicum*) accessions. Until recently, tomato has been maintained as self-pollinated varieties. Spontaneous, or induced, mutants creating unusual phenotypes are readily detected in self-pollinated species. Such mutants in tomato have been carefully documented, preserved and propagated. Recessive point mutations become visible after self pollination. These mutants often are associated with obvious deviating phenotypes. A large population of deviating phenotypes as observed in gene banks or in the genetic stock of a breeder shows the abundant variation (Rick, 1991). A number of these mutants have been used for crop improvement. Linkage relationships of many of the mutants were studied from the 1950s to the 1980s, resulting in the generation of the tomato morphological map (http://solgenomics.net/). This map, the more recent molecular maps and the published tomato sequence (Tomato Genome Consortium, 2012) can be accessed on the Solanaceae Genomics Network database (http://solgenomics.net/).

Wild species of tomato

At this moment, 13 species are known that belong in the *Solanum* section *Lycopersicon*. *Solanum pimpinellifolium*, *S. cheesmaniae* and *S. galapagense* are in the same group as *S. lycopersicum*. They are all crossable, and the level of DNA differences between *S. galapagense* and *S. cheesmaniae* is extremely low (Viquez-Zamora *et al.*, 2013). Many more sequence differences are found between tomato and *S. chmielewskii*, *S. habrochaites*, *S. neorickii* and *S. pennellii* but they are still crossable with tomato. The remaining five species (*S. arcanum*, *S. chilense*, *S. corneliomulleri*, *S. huaylasense* and *S. peruvianum*) are more difficult to cross with tomato but often techniques such as embryo rescue will make crosses possible.

Diversity among and within wild species of the *Solanum* L. sect. *Lycopersicon*

The lack of genetic diversity within cultivated tomato is not a barrier to accomplish progress in tomato breeding as there is much variation readily available in tomato's wild relatives. The variation among the species within the genus *Solanum* L. sect. *Lycopersicon* is tremendous (Fig. 2.1). Numerous investigations have studied specific traits present in the related wild species and are exploiting this variation for tomato improvement. Nearly all of the wild species have contributed valuable traits for crop improvement, from the closely related red-fruited self-pollinated *S. pimpinellifolium, S. cheesmaniae* and *S. galapagense* (Fig. 2.2) to the more distantly related green-fruited self-incompatible species *S. pennellii* and *S. arcanum*. Crosses (Fig. 2.3) are sometimes easy but sometimes also more challenging, such as crosses between *S. lycopersicum* and *S. chilense* or *S. arcanum*. These crosses often require embryo rescue to obtain F_1 progeny. Also, in later generations, additional barriers that are caused by genetic interactions may impede further progress, for instance in the development of recombinant inbred lines (RILs).

Because of their self-incompatibility and mode of reproduction, outcrossing species harbour a higher level of genetic variation than largely self-pollinating species. Miller and Tanksley (1990) assessed this variation using molecular markers. It is striking that more genetic variation was observed within a single accession of a self-incompatible species than in all accessions of any of the self-compatible species.

In order to use related wild species of tomato to introduce new favourable traits, genetic studies and the identification of closely linked molecular markers to gene(s) of interest are needed. This requires segregating populations that are derived from crosses between two parents with a difference in

Fig. 2.1. Some variation in fruit colour, shape and size of the *Solanum* section *Lycopersicon*.

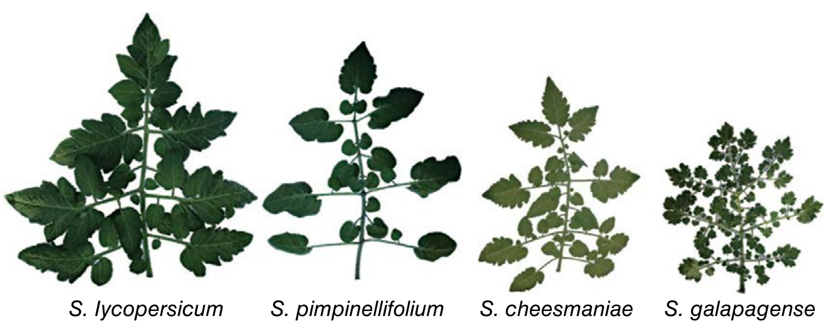

Fig. 2.2. Top leaves of tomato (*Solanum lycopersicum*) and its closest related *Solanum* species.

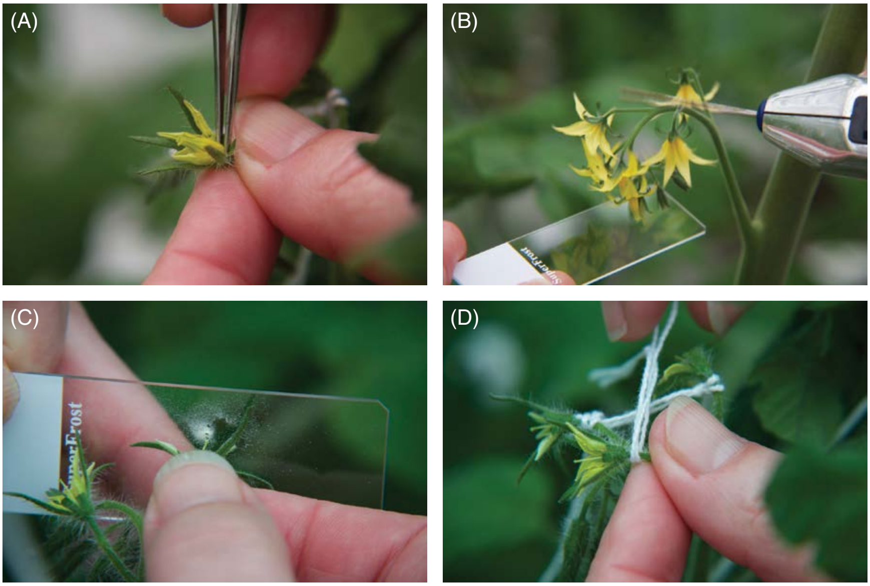

Fig. 2.3. Crossing tomatoes. (A) Emasculation of female parent. (B) Pollen collection. (C) Crossing. (D) Labelling.

the trait of interest. The most common populations for quantitative trait locus (QTL) analysis are F_2 (population obtained after selfing a hybrid) or BC_1 (population obtained after crossing a hybrid with one of its parents). RILs as a permanent population can be studied for one or more interesting traits in multiple environments with many replicates. However, the efficiency of detecting a particular QTL in a segregating population is still low, partly because other QTLs are segregating and major QTLs are masking the minor ones. For this reason, Eshed and Zamir (1994) proposed the use of introgression line (IL) libraries. An IL population consists of a series of lines harbouring a unique segment from a wild progenitor introgressed into a uniform, cultivated genetic background (Fig. 2.4). Ideally, all lines together present the complete genome of the wild progenitor. Until now, several IL populations have been reported and they are respectively derived from *S. pennellii* LA716 (Eshed and Zamir, 1994), *S. habrochaites* LA1777 (Monforte and Tanksley, 2000), *S. habrochaites* LA407 (Francis *et al.*, 2001), *S. habrochaites* LYC4 (Finkers *et al.*, 2007) and *S. chmielewskii* LA1840 (Mathieu *et al.*, 2009). In addition to the species within *Solanum* L. sect. *Lycopersicon*, introgressions have been made with *Solanum lycopersicoides* and *S. sitiens* (Canady *et al.*, 2005; Ji *et al.*, 2004).

In conclusion, there is limited genetic variation in cultivated tomato species, but considerable genetic variation exists among and within the wild crossable species that can be utilized, with varying degrees of difficulty, for crop improvement. Breeders would naturally use the sources closest to cultivated tomato, for ease,

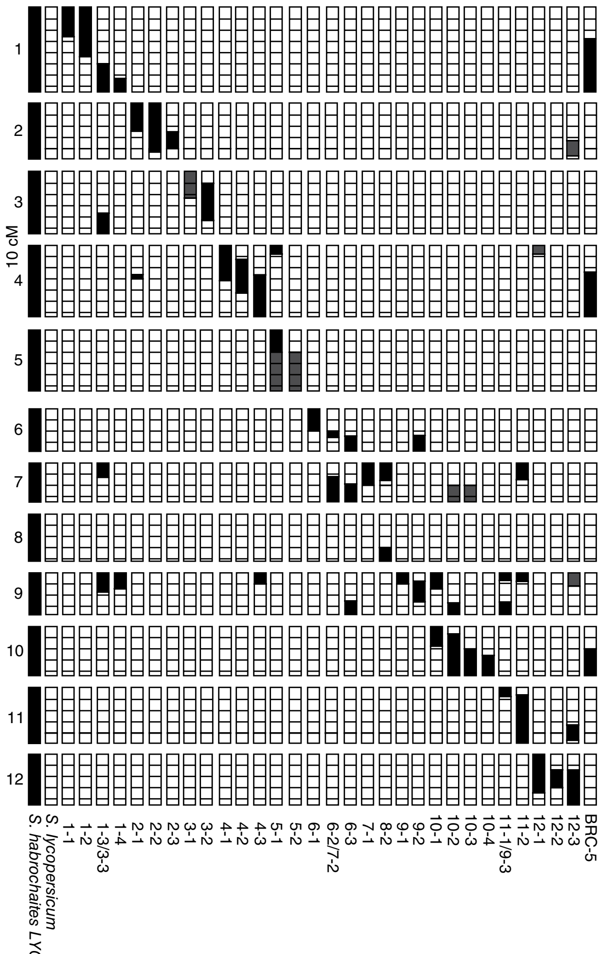

Fig. 2.4. An introgression line population of *S. habrochaites* (black) with as background *S. lycopersicum* (white). In total the population represents 80–85% of the *S. habrochaites* genome. Reprinted by permission from Springer Nature: Theoretical and Applied Genetics, The construction of a Solanum habrochaites LYC4 introgression line population and the identification of QTLs for resistance to *Botrytis cinerea*, Finkers *et al*. (2007).

efficiency and rapidity of the transfer. However, none of the species is out of reach for breeding when sufficient time, labour and space are available.

Collection and preservation of genetic resources

The collection, description, propagation and distribution of genetic materials are of utmost importance for tomato breeding. In the second half of the 20th century, numerous expeditions were made to the Andes region to collect accessions of various wild tomato species. It is noteworthy to mention Dr Charles Rick from the University of California, who understood the importance of germplasm collection (Rick, 1991). He collected and described numerous accessions of wild species from the *Solanum* L. sect. *Lycopersicon* which are still maintained at the Genetic Stock Centre in Davis, California. These materials are available upon request for the scientific community as well as for the commercial breeding industry. Tomato germplasm is maintained in several other world gene banks. The Rio de Janeiro Convention on biological diversity has strictly regulated the collection of new germplasm in order to get a fair and equitable sharing of benefits arising from genetic resources.

Within a gene bank, the accessions should be fully and accurately described. In addition to information on location site, full passport data and taxonomic classification, this includes data about the native habitat and geographical and climatic characteristics. Morphology, such as leaf shape, plant growth habit, fruit size and shape are also important descriptors (Table 2.1). The curators are responsible for maintaining the variability within accessions. More plants are needed for propagation and maintenance of out-crossing self-incompatible accessions compared with self-compatible plants. It is difficult to guarantee that no special traits get lost in the multiplication due to genetic drift. Abiotic and biotic stress characteristics such as tolerance to salt, fungal pathogens and insect pests may also be included. Evaluating these traits is expensive and gene banks are generally underfunded. Therefore these data are lacking and researchers are asked to share the results of their evaluations. Regrettably researchers do not always feel morally obliged to do so. In considering the use of wild germplasm for crop improvement, it should be noted that useful genes of the wild species in the *Solanum* L. sect. *Lycopersicon* might be masked. For example, the small-fruited wild species *S. pimpinellifolium* has genes that potentially can increase fruit size when these genes are present in a cultivated tomato background (Monforte and Tanksley, 2000).

Cytogenetics

Tomato has a diploid genome with 12 chromosome pairs. At the pachytene stage, the chromosomes of tomato are very distinct and are distinguishable by their

Table 2.1. A descriptors list for some fruit traits evaluated at the Centre for Genetic Resources, Wageningen, The Netherlands.

Trait	Description
Immature fruit colour	(+ = present, 0 = pale, 1–9 = intensity of green collar/ greenback: 1 = very light, 9 = very dark)
Mature fruit colour	(A = red, B = orange, C = pink, D = yellow, E = yellow green, F = red with yellow spots, G = light green)
Fruit setting ability	(0 = no flowers set, 5 = 50% of flowers set, 9 = all flowers set)
Fruit ribbing	(0 = absent, 1 = very little, …, 9 = very high)
Fruit top hardness	No description available
Fruit cracking tendency	(0 = no expression, 1 = very low, …, 9 = very high expression)
Fruit firmness	No description available
Fruit shape	Sonatine = 5 (1 = very flat, …, 9 = very long)
Fruit section shape	(1 = very round, 3 = round, 5 = angular, 7 = irregular, 9 = very irregular)
Fruit green edge	(+ = present, 0 = absent, 1–9 = intensity of green edge: 1 = very light, 9 = very dark)
Fruit locule number	Count and estimate the average number, using a few plants
Fruit yellow top	(0 = absent, 1 = very few, 3 = few, 5 = half of the fruit, 7 = most fruits, 9 = all fruits)
Fruit fleshiness	Sonatine = 3 (1 = very low, …, 9 = very high)
Fruit outerwall thickness	Measurement (9 = 9 mm or more)
Pericarp thickness	Sonatine = 3 (1 = very little, 3 = little, 5 = intermediate, 7 = flesh, 9 = very fleshy)
Fruit puffiness	(0 = no expression, 1 = very low, …, 9 = very high expression)
Pedicel jointlessness	(0 = absent, + = present)
Blotchy ripening	(0 = absent, + = present)
Harvest time begins	(2,3,4,5 = 4,3,2,1 weeks before Sonatine, 6 = Sonatine, 7,8,9 = 1,2,3 weeks after Sonatine)
Blossom-end rot	(0 = absent, + = present, ++ = severe)

length and banding patterns. The total length of the tomato chromosomes at the metaphase stage is about 50 μm. Haploid production in tomato is not available so far, though some reports indicate that there might be possibilities (patent: Tomato anther culture, US 4835339; Zagorska *et al.*, 1998; Chambonnet, 2012). Haploids are useful for a fast screening of the role of recessive alleles influencing a certain trait. However, the success rate of making haploids in tomato is extremely low and hence haploidization is not used in scientific research or in practical breeding.

Crossing a diploid with a tetraploid-induced genotype can produce triploids. Triploids and tetraploids, as well as aneuploids, are produced from the progenies of triploids. Tetraploids are characterized by thicker and darker leaves, reduced internode length, increased stem thickness and reduced fertility as well as reduced fruit set and fruit quality. Consequently tetraploid tomatoes are not useful commercially. This is also true for all other non-diploid tomatoes.

Nowadays the location of unique DNA sequences can be visualized by fluorescent *in situ* hybridization (FISH). FISH can help in solving problems in obtaining sequences in the right order (Szinay *et al.*, 2008) and chromosomal rearrangements can be visualized among different *Solanum* species. FISH has been frequently used to assign larger DNA stretches to specific tomato chromosomes. This has been instrumental in assembling the complete tomato genome sequence (Tomato Genome Consortium, 2012).

Rearrangements like inversions can be the reason for cold spots of recombination and consequently for linkage drag that cannot be minimized by further crosses.

Molecular linkage maps

Originally, the tomato morphological map was generated using visually distinguishable morphological traits. It was therefore constructed from data from numerous linkage analyses, most of which were based on segregation of three to six markers. Later, different forms of isozymes could be identified and added to the morphological map (Tanksley and Mutschler, 1990). These isozymes are considered to be the first generation of molecular markers. Initially, isozymes greatly contributed to mapping studies; however, the limitation of isozymes as molecular markers is that they represent only a limited number of loci and that the protein separation techniques are expensive and not high-throughput. This is why isozymes are no longer used.

With the advent of DNA-based molecular markers, high-density genetic maps could be generated using data from only one segregating population. The first high-density map was published in 1991 (Tanksley *et al.*, 1992). This map contained 1030 DNA markers (restriction fragment length polymorphisms (RFLPs)). The majority of these markers are potentially useful in any segregating population within the *Solanum* L. sect. *Lycopersicon*. This is due to the fact that the order of these markers is conserved over all species in the *Solanum* L. sect. *Lycopersicon*. The genetic relationship of tomato with other more distantly related species such as potato, pepper and aubergine has been studied by comparing genetic linkage maps. The tomato map is very similar to the potato map, but is different from the pepper map (Wu *et al.*, 2009). The similarities between genomes of related Solanaceae allow a comparison of the map position of similar genes. Indeed, the map positions of genes and the gene repertoire appear to be conserved as well. All these data point to an ancient common ancestor of these related Solanaceae species.

The landscape of molecular marker technology has changed at an unprecedented speed. The ability to get a lot of sequence information at relatively low costs made it possible to compare sequences and to find sequence differences (Tang *et al*., 2006; see also http://www.bioinformatics.nl/tools/snpweb). For genotyping populations, other high-throughput methods have been developed that only require minor quantities of DNA. This makes the SNP the marker of choice at the moment and SNPs facilitate the generation of a high-density maps in any region of interest. International corporations are also focusing on identifying protein and metabolic fingerprints of tomato. Together with the DNA sequence information, this research field is described as genomics. It is expected that these developments will result in an important step forward in our knowledge and understanding of the structure of the tomato genome. The challenge will be how to effectively interpret and use all the information resources. Bioinformatics will play an important role in this.

Applications of molecular markers

SNPs and high-density tomato genetic linkage maps enabled a whole series of new applications. The advantage of SNP markers is the possibility of analysing thousands of markers in any population simultaneously, allowing the mapping of any gene by co-segregation. Generally, four mapping strategies can be followed: (i) linkage disequilibrium mapping; (ii) segregation analysis of interspecific populations; (iii) QTL mapping; and (iv) marker-assisted selection (MAS).

Linkage disequilibrium mapping

Linkage disequilibrium (LD) (or genome-wide association mapping) is a method to detect linkage without generating dedicated segregating populations but instead using a large number of cultivars. The aim is to look for associations of markers and traits. Van Berloo *et al*. (2008) calculated on the basis of markers that the average window around a gene of interest in tomato, where not too many recombinations have occurred, is 15 centimorgans (cM). The size of this window is also determined by the choice of material. When all cultivated tomatoes, including vintage and contemporary ones, were considered, Robbins *et al*. (2010) identified a linkage decay of 6–8 cM. Within contemporary processing varieties, LD decayed over 6–14 cM, compared with 3–16 cM within fresh market varieties.

Segregation analysis of interspecific populations

Interspecific segregating populations – which are obtained when, for example, susceptible lines are crossed with resistant wild relatives – enable genetic studies and help to get a real estimate about the number of loci involved. Linkage analysis within such populations often reveals co-segregation of the resistant phenotype with the presence of specific markers. These markers are now SNPs with a known position on the tomato sequence. With a fully annotated sequence (where for all gene sequences the most likely function has been added) (Tomato Genome Consortium, 2012), it is possible to find all genes with their putative functions in the region delimited by the linked markers. The size of the fragment under investigation depends on the relation of the genetic linkage map distances and the physical map distances. Recombination hot and cold spots can result in large differences between genetic and physical maps (Fig. 2.5) (Viquez-Zamora *et al*., 2013). If necessary, more markers in the region of interest and more progeny plants are screened to fine map a favourite gene. An example of such an approach is given by Firdaus *et al*. (2012) in which a major gene for whitefly resistance originating from *S. galapagense* is tagged.

QTL mapping

A somewhat different approach is needed when interest lies with 'field' or quantitative resistance. A segregating population is needed in which marker

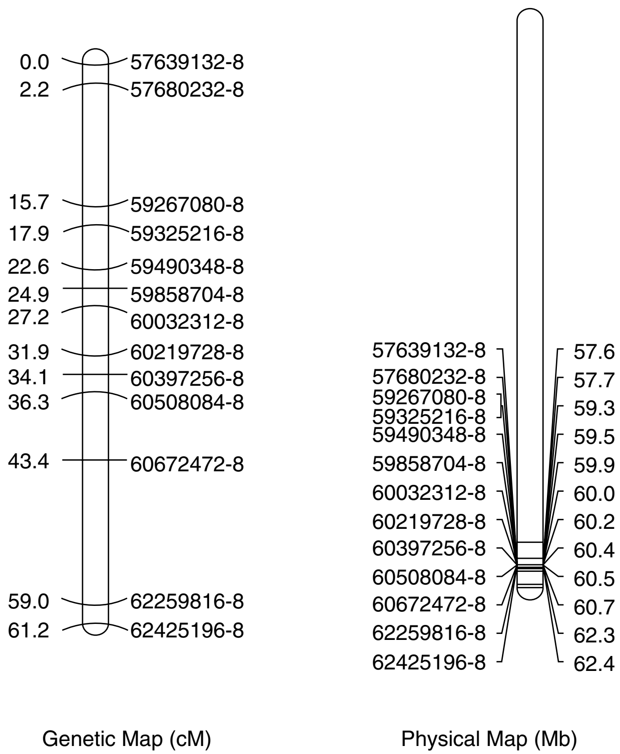

Fig. 2.5. Comparison of genetic or linkage map (cM, centimorgans) with a physical map (Mb, megabase pairs) of chromosome 8 for an F_2 population between two cherry-type tomatoes (57639132-8 stands for position 57.6 million bp on chromosome 8).

polymorphisms have been determined and where all individual plants are phenotyped (quantitative characters). Correlations between marker genotypes and phenotypes can be calculated along the linkage map. In this way, for each position on the genome the probability that a QTL is present is determined. When significant correlations are found, confidence intervals are assigned for QTLs that influence the trait of interest. Reliable and reproducible phenotyping is the most important factor in QTL mapping. To be able to do such a reliable test, often more replicates with genetically identical plants are necessary. For F_2 and BC_1 populations, this means that the plants have to be propagated either by cuttings or by *in vitro* propagation. RILs and ILs are homozygous and the numerous seeds collected on these plants are all genetically identical to the parental plant. This allows repeated screenings in time and space, leading to a more accurate QTL mapping and to knowledge about genotype × environment (G×E) interactions.

Knowledge on the position of these QTLs can be applied in a marker-assisted selection (MAS) approach and can serve as a starting point for genome walking and gene cloning procedures. Numerous QTLs in tomato have now been identified and mapped for traits such as cold and salt tolerance, soluble solid content and other traits involved in fruit quality as well as for resistance genes to pests and pathogens (e.g. Fridman *et al.*, 2004; Firdaus *et al.*, 2012; Van Heusden *et al.*,1999).

Marker-assisted selection (MAS)

After the identification of markers linked to traits of interest and for QTL explaining enough of the variation, the traits can be introgressed into tomato using these markers, instead of selecting for the trait itself. This can enhance the efficiency of breeding, especially if difficult-to-score traits are involved. MAS can already be done in the seedling stage, which reduces the greenhouse space needed in a breeding programme. Breeding by design is a method to combine many favourable traits in a single genotype (Peleman and Rouppe van der Voort, 2003).

Map-based cloning

A high-density linkage map also enables isolation of individual genes through chromosome walking or chromosome landing. A nice example is the isolation of the gene conferring positional sterility in tomato (Gorguet *et al.*, 2009). After identifying roughly the location of the locus in an interspecific cross of *S. lycopersicum* and *S. pimpinellifolium*, a fine mapping experiment pointed to only one gene, namely Tomato Dehiscence Polygalacturonase. Only one nucleotide difference was responsible for non-opening anthers in the mutant. This approach is becoming more and more feasible because of next-generation sequencing and the availability of total annotated genome sequences. By now many genes have been cloned that are involved in quantitative characters such as fruit size, taste and yield.

HISTORY OF TOMATO BREEDING

The tomato was brought to the Old World from Central and South America in the 16th century but was initially considered as a curiosity and not as an edible plant species. For two centuries, magic powers were assigned to tomato fruits and they were designated golden apple or love apple (*pommo d'oro* and *pomme d'amour*). In the 18th century tomato was gradually accepted as edible food and its production area and usage spread from southern Europe to the north. Immigrants took the tomato to North America. Professional seed sellers not only sold seeds but also tried to improve tomato cultivars by mass selection as a primitive form of breeding. So, the first breeding companies combined seed merchandizing with seed propagation and selection. As the production area increased, pathogens and pests appeared more frequently, causing an increasing demand for resistant varieties. In the first half of the 20th century the first crosses were made with wild species to introgress disease-resistance genes. One of the first examples was *Cladosporium fulvum* resistance from *S. pimpinellifolium* in 1934. This research was done in the public sector because the new cultivars did not give much return on the breeder's investment as growers could still propagate seeds on their own nursery, frustrating breeders in increasing the seed price.

This caused an increasing awareness that innovations should be protected. To protect their intellectual properties, breeders needed to obtain intellectual property (IP) rights for their cultivars. In 1948 the Plant Breeders Law was established. This gave the breeders exclusive ownership over their varieties but with a breeder's exemption, allowing other breeders to use commercial varieties as a source in their own breeding germplasm. This gave a boost to breeders' activities, as competition was not allowed to commercialize these protected varieties. In the second half of the 20th century, national governments founded public research institutes to support breeding activities to secure good seeds for growers. In addition, the Consultative Group on International Agricultural Research (CGIAR), a global partnership formed in France in 1971, established more than a dozen international organizations to increase food security in regions in the world where national administration did not have the capacity to establish national breeding programmes. In Taiwan the Asian Vegetables Research and Development Centre (AVRDC) was established, but this has never achieved full international recognition because of the political status of the hosting country.

In 1930 the first hybrid cultivars were developed in maize. This boosted the maize yield, which has increased at least fivefold since then. It took longer before this breeding system was introduced in tomato. In 1946 the first hybrid tomato cultivar, 'Single Cross', was released. Hybrids combine good characters from both parents that might give heterosis – the phenomenon that F_1 progenies perform better than each of the parents. Finally, upon seed propagation by selfing, the progenies segregate for the traits that vary between the parents such as disease resistances, discouraging seed propagation by growers. The advantages of hybrid varieties over true-breeding varieties were so great that the growers accepted paying higher prices for hybrid seeds. This covered not only the higher costs for hybrid seed production but also the investments in breeding (and research) programmes. Within a few decades all new tomato cultivars for the higher end of the (fresh) market were hybrids. It took longer for the processing tomato industry, as the seed prices are much lower for this bulk product. Breeding companies could invest more in professional breeding programmes, and new varieties were continually developed that combined several resistance genes with other good agronomic characters such as high yield, resistance to extreme conditions and high fruit quality. The success of hybrids is reflected in the high seed price of tomatoes, which might reach more than €1 per individual seed for some rare niche markets. As the added value of the product from one tomato plant, raised from one expensive seed, may easily exceed €30, the contribution of the seed price to the overall production costs is still below 5%. It is this dynamic cycle of innovative product introduction at higher prices that has developed the tomato breeding industry to one of the strongest research-and-development-driven industries, with often 15–25% of the total turnover invested in R&D.

TOMATO HYBRID BREEDING

In hybrid breeding two main steps are designated: (i) inbred line improvement; and (ii) testing experimental hybrids generated by crossing the inbred lines.

Inbred line improvement

Usually, inbred lines originate from existing advanced lines that are crossed to combine good characters, following by inbreeding, or by inbreeding existing cultivars from competition. The latter is allowed as the Plant Breeders Law allows the usage of existing IP-protected cultivars to enhance the breeder's germplasm. In the USA, cultivars can be patent protected and are then not available for further breeding by others.

The aim is to generate completely homozygous lines by inbreeding. In each generation of selfing, the percentage of heterozygotes in a diploid crop is reduced by 50%. After five of six generations of selfing the lines are essentially homozygous (> 95%). In tomato, alternative strategies such as haploidization by gynogenesis or androgenesis are not yet successful.

As hybrids are generated by crossing parent lines, much care is taken in developing lines with unique characters that are combined into a hybrid cultivar. Often, the parent lines are grouped into heterotic groups based on the occurrence of dominant resistance genes. In this way new varieties that combine resistance genes can easily be generated. Such a cultivar is then heterozygotic for these traits and the selfed progeny derived from such a hybrid cultivar will segregate for these resistance genes, hence discouraging growers from generating their own seeds by selfing a hybrid cultivar.

In a typical breeding programme, for each market segment, two parent groups or heterotic groups are maintained and continuously improved. When market segments show overlap, sometimes parent groups can be combined, like round cluster tomato and round loose tomato. However, often market constraints are so severe that unique parent pools are needed. The total number of parent lines per parent pool varies in the range of 10–50, depending on the level of segregation among the parents.

Hybrid production and testing

Inbreeding results in a decrease of heterozygosity and an increase of homozygosity. After several generations of inbreeding most, if not all, loci are homozygous and all seeds of the next generation are genetically identical and also identical to the previous generation.

In a typical hybrid breeding programme, after four to six generations of inbreeding, crosses are made to generate the first generation of hybrids. These crosses are randomly made between individual lines of the parent pools. The hybrids are tested in the next season under conditions that are as close as possible to the growers' practices. Initially, the testing starts with only a few plants of many different hybrids. After selection, the best ones are retested for another two or three seasons to get a good impression of their performance. Later, selected hybrids are tested at growers' sites and their performance is compared with leading market varieties. Finally, the best hybrid is selected and introduced to the market after granting breeder's rights. To show the added value of the new hybrid, demonstration trials are set up whereby many growers are invited to evaluate the new hybrids. Often, growers get some free seeds of new varieties to test them.

An additional advantage of the intensive testing of hybrids in the first one or two cycles is the knowledge about general combining ability of the parent lines. With this knowledge the breeder can proceed with further testing and improvement. A typical breeding programme may test 10–50 hybrids per market segment per year.

BREEDING GOALS

The goals of public and private tomato breeding programmes vary widely, depending on technical feasibilities, market needs and resources. To be successful, growers must produce high yields of high-quality fruit at minimal production costs. Therefore, many of the breeding goals focus on characteristics that reduce production costs or ensure reliable production of high yields and high fruit quality. These can be grouped in the following trait groups: (i) yield; (ii) resistance to biotic stress; (iii) resistance to abiotic stress; and (iv) fruit quality.

Breeding for yield

While for most field crops yield is by far the most important agricultural trait, yield is less relevant for breeding horticultural crops like tomato. Still, yield will always contribute to the grower's revenues and hence growers will prefer high-yielding varieties. In a typical tomato breeding programme, parent lines are selected for good plant performance, but no extensive yield data are collected. After the experimental hybrids are generated, they are mainly evaluated for fruit characters as these are crucial for any market segment. The parents are selected with the best general combining abilities (GCA). In the second year of hybrid testing, progenies from two parents with good GCA are tested and preliminary yield data may be recorded but large-scale yield data are not generated. This is not done until the latest stage of testing hybrids as these determinations are time, labour and space consuming. Even so, under growers' conditions considerable yield differences can occur when the same variety is used. In tomato, heterosis is quite small, but occasionally combinations between two parents with good GCA outperform the rest. This cannot be predicted and hence is designated specific combining ability (SCA).

Breeding for resistance to biotic stresses

Tomato hosts more than 200 species of a wide variety of pests and pathogens that can cause severe losses. Often, these pests and pathogens have to be controlled by using chemical compounds such as fungicides or pesticides. These methods may not be fully effective; they may also raise production costs and require compliance with chemical use laws. In addition they cause concern regarding potential risk for the growers, the consumers and for the environment. Therefore, one of the most important focus traits in tomato breeding is breeding for resistance to the most destructive pests and pathogens.

Nature has provided a great wealth of resistances in the genus Solanum, section *Lycopersicon* (e.g. Lindhout, 2002; Van Berloo and Lindhout, 2001). Many of the resistances are simply inherited and tomato breeders have had remarkable success in transferring disease resistance genes into cultivated tomato. As a consequence, modern cultivars can carry more than a dozen resistance genes. These resistances can be grouped according to the pathogen.

Virus resistance

The *Tm2* gene for control of the tomato strain (ToMV) of *Tobacco mosaic virus* (TMV) was obtained from *S. arcanum* in the 1950s. It is used in nearly all greenhouse tomatoes and some field tomatoes and is still effective after more than 50 years.

A devastating virus in tropical and subtropical areas is *Tomato yellow leaf curl virus* (TYLCV). This virus is transmitted by whitefly (*Bemisia tabaci*) and can only partially be controlled by controlling whitefly. In *S. arcanum* and in *S. chilense* some semi-dominant resistance genes to TYLCV have been identified that reside close to the centromere of chromosome 6. The gene order in this region is not constant, as some inversions have been identified. This has greatly hampered detailed mapping studies (Verlaan *et al.*, 2011).

Another important virus is *Tomato spotted wilt virus* (TSWV), which has a very large host range and is transmitted by thrips. The TSWV resistance gene *Sw-5* has been identified in *S. arcanum* and is now present in many modern varieties. *Cucumber mosaic virus* (CMV) is mainly present in subtropical and tropical tomato production fields and is transmitted by aphids or mechanically. No good level of natural resistance is available yet but transgenic plants carrying a viral gene may have high levels of resistance (see section 'Biotechnology and transgenic or genetically modified organisms', p. 52, this volume).

Pepino mosaic virus (PepMV) was first recorded in Europe in 1999. PepMV has rapidly spread and is now one of the most common viruses for greenhouse production. All cultivars are very susceptible to this mechanically transmitted virus and no resistance has been reported in wild species. Often no obvious symptoms are visible, while the virus may still cause reduced growth and development of tomato plants and can hence also reduce yield, especially under suboptimal conditions. Though the damage by PepMV is limited, breeders are very concerned as each seed lot has to be free of viruses. PepMV occurs so frequently that it is extremely hard to organize completely virus-free breeding programmes. Therefore, often, breeders produce their seeds in regions where the chances of contamination by PepMV are much less, such as tropical regions at higher altitudes.

Bacterium resistance

Bacterial wilt is one of the most devastating bacterial diseases that threatens the tomato crop in tropical areas. The causal agent is the pathogen *Ralstonia solanacearum* (= *Pseudomonas solanacearum*). No good levels of resistance are available yet but some genes have been identified that cause a partial resistance to this pathogen. Similarly, partial resistance to *Clavibacter michiganensis* has been identified in *S. arcanum*, though this resistance does not completely block the growth of bacteria in infected plants (Sen *et al.*, 2013). The *Pto* gene confers resistance to bacterial blight caused by *Pseudomonas syringae*. The tomato–*Pseudomonas* interaction is a very well studied model system for plant defence responses (Pedley and Martin, 2003).

Fungus resistance

The dominating group of pathogens that infect tomato are fungi. For most pathogens, resistance genes have been identified. Breeders have preferentially used complete or qualitative resistances that are monogenic and race specific. However, often these resistance genes are easily rendered ineffective by adaptation of the pathogen, forcing breeders to introduce a new gene. A good example is *Cladosporium fulvum* where breeders have used at least four resistance genes (*Cf-2*, *Cf-4*, *Cf-5* and *Cf-9*) and none of them is yet completely effective (Lindhout *et al.*, 1989) (Table 2.2). However, numerous additional genes are available and can be used for breeding (Haanstra *et al.*, 2000). A pathogen that has been more successfully controlled by breeding is *Verticillium albo-atrum* or *Verticillium dahliae*. For more than 50 years the *Ve* gene has effectively conferred resistance to *Verticillium* race 1. *Fusarium* wilt is controlled by the genes *I*, *I-2* and *I-3*, which all confer resistance to specific races. As *Fusarium* is a soil-born fungus, the spread of new virulent races is quite slow and consequently resistance *I*-genes may remain effective for decades.

Alternaria solanii is a widespread pathogen that causes early blight disease. Resistance that protects the stems, and to a lesser extent the leaves, has been transferred to fresh market tomato lines and is present in some hybrid varieties. There are two powdery mildew species that attack the tomato crop: *Leveillula taurica* only occurs in subtropical and tropical areas and can be controlled by the *Lt* gene; whereas *Oidium neolycopersici* occurs worldwide in mainly greenhouse tomatoes. This fungus has, after its first occurrence in 1986, readily spread over the world but can be controlled by several *Ol*-genes that are available in modern hybrid cultivars. As in potato, *Phytophthora infestans* is widespread and can cause the disease late blight, mainly in field tomatoes. Strong resistance has been found in *S. pimpinellifolium* and *S. habrochaites* and resistent cultivars have been generated. A very common fungus is *Botrytis cinerea*, which is not a major disease but mainly infects plants that are growing weakly, for instance at the end of the season. Some QTLs for resistance have been described and are available for breeding (Finkers *et al.*, 2007).

Insect resistance

Insect pests can be controlled by chemical protection agents, which have for a long time been the most important method of insect control. Unlike resistance genes to fungi or viruses, no simple monogenic resistances to insects have been identified in tomato. As a consequence, all modern tomato cultivars are susceptible to insects. Very high levels of resistance in *S. galapagense* were mainly caused by a single QTL/gene (Firdaus *et al.*, 2013) but no cultivars are on the market yet. The high resistance to insects of the wild species may be caused by toxic compounds such as acyl-sugars or 2,3-tridecanone or sesquiterpenes (Rodriguez *et al.*, 1993; Maliepaard *et al.*, 1995; Firdaus *et al.*, 2013; Bleeker *et al.*, 2012).

Table 2.2. Virulence spectrum of some races of *Cladosporium fulvum* from The Netherlands, France and Poland on tomato genotypes (Lindhout *et al*., 1989).

		Races of *Cladosporium fulvum*[a]											
		Netherlands								France	Poland		
Tomato genotypes	Resistance genes	2	4	5	2.4	2.4.11	2.4.5	2.4.5.11	2.4.5.9.11	2.5.9	4.11	2.4.11	2.4.9.11
Moneymaker	*Cf-0*	S	S	S	S	S	S	S	S	S	S	S	S
Vetomold	*Cf-2*	S	R	R	S	nd	S	nd	S	S	R	S	S
Purdue 135	*Cf-4*	R	S	R	S	nd	S	nd	S	R	S	S	S
Vagabond	*Cf-2, Cf-4*	R	R	R	S	S	S	S	S	R	R	R	R
Ontario 7717	*Cf-5*	R	R	S	R	R	S	S	S	S	R	R	R
Ontario 7818	*Cf-6*	R	R	R	R	R	R	R	R	R	R	R	R
Ontario 7522	*Cf-8*	R	S	R	S	nd	S	nd	S	R	S	S	S
Ontario 7719	*Cf-9*	R	R	R	R	R	R	R	S	S	R	R	S
Ontario 7716	*Cf-11*	R	R	R	R	S	R	S	S	R	S	S	S

S, susceptible; R, resistant; nd, not determined.
[a]The race names refer to the absence of an *Avr* gene that interacts with the corresponding *Cf-* gene (Race 2 is irulent on tomato lines carrying only *Cf-2*).
Note: The genes *Cf-2*, *Cf-4*, *Cf-5* and *Cf-9* have been exploited in tomato breeding, but these resistances are ineffective as *Cladosporium* races have occurred that lack *Avr2*, *Avr4*, *Avr5* and/or *Avr9*.

In many crops, resistance to insect pests has been achieved by transgenic plants carrying a *Bacillus thuringiensis* construct. In tomato, there is no such product on the market as it is still legally not allowed in many countries.

Breeding for resistance to abiotic factors

The optimal conditions for tomato cultivation are in subtropical regions, similar to the higher altitudes in the Andes, where tomato was originally domesticated. As a consequence, tomato production is suboptimal in large parts of the tomato-growing areas. The most important stress factors are high or low temperatures, excessive water or drought, and soil salinity. In the tomato germplasm much genetic variation for tolerance to these stresses exists, making breeding for these traits technically feasible and commercially attractive (Kazmi *et al.*, 2012).

Cold tolerance

Cold tolerance is important for field production in regions where night temperatures are low, or for winter cultivation as in the Mediterranean area. For heated greenhouse cultivations, there is a constant need for energy saving as this is an important aspect of production costs. The aim is to optimize the whole system. Depending on the conditions and market needs, a heated-greenhouse grower should decide what the most optimal temperature regime is. This is also related to plant growth, humidity, fungal attack, market prices, etc. Therefore greenhouse growers, as well as open field growers, need varieties that are tolerant of changes in temperature.

Most genotypes are adversely affected at temperatures below 13°C. Prolonged exposure to temperatures below 6°C can even result in plant death. There is a huge variation for low-temperature tolerance in tomato and its crossable wild relatives, and this variation has been extensively investigated. High-altitude accessions of *S. habrochaites* and *S. arcanum* that grow at altitudes of over 3000 m show a better adaptation to growth under low-temperature conditions than the cultivated tomato (Venema *et al.*, 2000).

Low-temperature tolerance can be specified into several characteristics, e.g. seed germination, growth rate and fruit set. The last characteristic is mainly determined by poor anther dehiscence and pollination. Cold tolerance is developmentally regulated and there is no clear correlation in cold tolerance at different developmental stages. Although many sources of cold tolerance have been reported and even used in breeding programmes, not much is known about the genetic factors underlying this trait, except that the inheritance is complex. Fruit set seems to be controlled by recessive factors (Kemp, 1965; Kalloo and Banerjee, 1990). Also, genetic loci have been identified that affect seed germination at low temperatures. It is expected that more traits involved in cold tolerance will be determined and mapped using molecular marker techniques. For greenhouse cultivation, the best approach is probably to breed for high yields under non-stress conditions. Most likely, higher yields will then also be obtained under suboptimal conditions (Goodstal *et al.*, 2005).

Heat resistance

In most parts of the world, high temperatures during growth of tomato in summer can affect production negatively. One of the most sensitive stages is fruit set. Most cultivars lose their blossom after 4 h at 40°C or a day/night temperature regime of 34°C/20°C, though this may vary between genotypes and depends on humidity and soil moisture. Factors involved in heat resistance have been well characterized and heat-resistant germplasm is available (Paupière *et al.*, 2014). Factors that are involved in heat resistance are photosynthesis, translocation, flower production, flower morphology (stigma exertion), anther dehiscence, pollen viability and pollen germination. Recessive genes that control a greater flower number have been identified, as well as an additive gene for fruit set and partially dominant genes for stigma exertion with additive components (Chen and Tanksley, 2004). Considering the large amount of physiological characteristics involved in heat resistance, each controlled by one or several genes, marker-assisted selection (MAS) might attribute to the introgression of all these genes in commercially interesting cultivars (Grilli *et al.*, 2007).

Drought resistance

Drought resistance is not considered to be one of the major issues in breeding. Nevertheless, research has been performed in order to extend the tomato-growing areas. One approach is to breed for drought escape, by breeding for very early or very late varieties. Generally, it is considered that cultivars with a very short life cycle are a good option, so that the drought periods can be avoided. Variation has been observed for this trait between different tomato cultivars and some accessions of the wild relatives *S. pennellii*, *S. cheesmaniae* and *S. pimpinellifolium* (Easlon and Richards, 2009).

Resistance to flooding

Flooding of tomato causes anaerobiosis, which in turn stimulates the production of ethylene, which causes epinasty. Sensitivity to flooding was found to be higher at higher soil temperatures. One indication of sensitivity to flooding is the accumulation of proline. Flooding tolerance has been found in some tomato cultivars (Ezin *et al.*, 2010).

Salt resistance

Irrigation may increase the amount of salts present in the soil. Tomato is moderately sensitive to salinity and alkalinity. At high levels of salt, a rapid decrease of yield occurs, due to decrease in fruit size (see Chapter 4). Several accessions of *S. cheesmaniae*, *S. arcanum* and *S. pennellii* have been found to be salt tolerant. Several developmental stages have been tested, and seed germination was found to be the most sensitive stage. In most cases, salt-tolerant genotypes absorbed more sodium (Na^+) (Cuartero *et al.*, 2006).

Physiological disorders

The tomato fruit is subject to several physiological disorders, such as blossom-end rot, catface, fruit cracking and blotchy ripening (see Chapter 5), which affect fruit quality negatively. The inheritance of different forms of cracking is well understood and modern cultivars can be stored for some weeks while maintaining a good fruit quality (high shelf-life, see below). Also, adjustment of nutrition may prevent these physiological disorders (see Chapter 6) .

Breeding for specific production systems

Since different production systems demand specific characters or traits, breeding has to meet these different characters. The main distinction that can be made in production systems is based upon the difference between processing and fresh market tomatoes. The processing tomatoes for ketchup, paste, etc. are grown in the open field and mechanically harvested, whereas fresh market tomatoes are picked manually during the whole season.

Several characters are necessary for machine harvesting of tomatoes. Determinate growth habit, uniform ripening and jointlessness are important traits, but most important of all is dry matter content or Brix yield.

Most fresh market tomatoes are indeterminate (see Chapter 3). In heated greenhouses in the northern hemisphere tomato plants are grown year-round. They produce marketable fruits from March to November, or year-round when artificial light is applied in winter. In recent decades the fruit type has shifted from a uniform size and shape to a wide spectrum of different types such as cherry, truss, yellow, beefsteak, pink, Roma, orange, pear-shaped, long-shelf-life tomatoes, etc. Worldwide, some 100 different product market combinations may exist. This puts a great constraint on those who are breeding tomato cultivars for all these markets. However, in the tomato germplasm a large variation is present and the majority of the alleles in the wild species have not been disclosed yet. Tomato breeding will continue to develop new products with new innovative traits that are required in all these markets.

Breeding for quality

There is a growing consumer demand for high quality. Quality is a combination of visual scorable characters (e.g. size, shape and colour) and sensory factors (e.g. sugar, acidity and taste). However, consumer perception of quality is also heavily influenced by-product appearance and labels such as eco-, sun-ripened, on the vine, heirloom, organic, transgenic, etc. To meet consumer demands, bigger wholesale retailers and supermarkets are putting more emphasis on the control of the whole production chain, such as cultivar choice, production conditions, ecological foot print, local-for-local and handling. Only a few tomato products are now presented with a brand name (e.g. Tasty Tom) that should offer a reliable quality but may comprise different cultivars. However, these products are still niche-market, as the majority of the consumers are not willing to pay for quality.

Flavour

Flavour comprises mainly sugars, acids and volatile compounds, the latter being of particular importance for fresh fruit (Thakur *et al.*, 1996). Some predictions about flavour can be made by measuring the acidity and degrees Brix (°Bx), which is equivalent to the soluble solids content. However, taste panels are necessary for a final evaluation of the taste of potential new cultivars. Tomato taste is not controlled by a single gene, but many genetic loci may play a role and each locus has a partial or quantitative contribution to the trait. These are designated quantitative trait loci (QTLs). Many QTLs that are associated with quality traits have been identified from *S. pimpinellifolium*, *S. pennellii*, *S. arcanum* and *S. habrochaites* (Causse *et al.*, 2003; Lecomte *et al.*, 2004; Yates *et al.*, 2004; Mathieu *et al.*, 2009; Tikunov *et al.*, 2013). QTLs for soluble solids contents (°Bx), firmness, stem retention (the ease by which the tomato can be harvested), pH, viscosity, puffiness (air in locules), colour and fruit shape have been mapped on the tomato genome. Mostly, wild species harbour QTLs that have positive effects and other QTLs that have negative alleles for a quality trait. For example, QTLs have been observed from the green-fruited *S. habrochaites* that give a less intense colour, while another QTL has been observed that is involved in more intense red coloration. This has also been observed for traits involved in yield. For example, wild species with smaller fruits might carry QTLs that give rise to increased fruit size when crossed to tomato, as has been observed for *S. pimpinellifolium*, *S. arcanum* and *S. pennellii*. QTLs increasing soluble solid contents (°Bx) have been found in all of the above-mentioned wild species, as well as QTLs increasing total yield. However, QTLs increasing the total amount of soluble solids, represented by °Bx × yield, were less frequent and observed only in *S. pennellii*, *S. pimpinellifolium* and *S. habrochaites*.

The traits that partially determine flavour are °Bx and acidity. Breeding and selection have been used to improve these traits. In addition, volatiles are involved in flavour. Breeding for volatiles has not been carried out yet, as much less is known about the relationship between flavour and aroma and these volatiles. Moreover, expensive measurements are necessary to determine the concentration of these compounds in plants, hampering the screening of larger populations. In ripening tomato fruit, approximately 400 different volatile compounds have been observed. Only a small number of these compounds have been identified as important components of flavour and aroma. These compounds are mainly aldehydes and alcohols. The enzyme alcoholdehydrogenase (Adh) converts aldehydes into alcohols. In tomato, two Adh enzymes have been identified, of which only Adh2 has been observed in ripening fruit. Transgenic tomatoes have been obtained with elevated levels of Adh2 expression, resulting in increased levels of hexanol and Z-3-hexenol. Plants with higher Adh2 expression were found to have a more intense ripe fruit flavour.

Colour

The redness of tomato is determined by the colour of the skin and of the flesh. The skin colour varies from yellow to colourless, while the flesh colour varies between green and red. The flesh colour depends on the amount and type of carotenoid pigments, which are C_{40} isoprenoid derivatives. There are two types of carotenoids in tomato: carotenes and xanthophylls. There is not much variation in xanthophyll levels between green and red fruits, whereas the level of carotenes increases tremendously (Liu *et al.*, 2003). One of the most apparent differences between green and red fruits is the level of lycopene. During ripening, there is a 500-fold increase in the level of lycopene in the tomato fruit. Lycopene comprises 90–95% of the total pigmentation in tomato. More attention has been drawn to lycopene since it was found that this antioxidant is associated with resistance to certain forms of human cancer, in particular cancer of the prostate. The antioxidant activity of lycopene also protects degradation of β-carotene, which is a source of vitamin A. QTLs for elevated levels of lycopene have been identified in a cross between tomato and *S. pimpinellifolium*. Several mutants have been identified that are involved in fruit colour. Some of these are not only affected in the carotene biosynthesis but also in other fruit ripening processes. The *ogc* mutant has elevated levels of carotene, which is at the expense of the level of β-carotene. Two *hp* mutants are recessive mutations in two different genes that show elevated levels of lycopene as well as β-carotene, which accompany a general elevated level of chlorophyll. The *hp* mutants also showed higher levels of flavonoids and sucrose, the latter at the expense of the levels of glucose and fructose (Van Tuinen *et al.*, 1997).

Ripening/shelf-life

Ripening of tomato is a well studied process (Giovannoni *et al.*, 1999). It is of interest to tomato breeders since it affects several quality traits, some of which have been mentioned above, such as colour, flavour and soluble solids content. Another factor that is especially important for fresh market tomatoes is shelf-life. During ripening, several processes occur that affect the storage of the fruit negatively. Some genes involved in ripening (e.g. polygalacturonase and ethylene synthase) have been cloned and transgenic tomatoes blocking the expressing of these genes have an extended shelf-life.

In the period after World War II most focus was on the increase in the production of food, including tomato. In the 1980s the European tomato market was saturated. This coincided with the increase in consumers' living standards to a level that they could afford to be more critical about the quality of the product. Consumers realized that greenhouse tomato production on artificial substrate and with a well controlled environment was so artificial that they started complaining about the quality. Growers and retailers started investigations to improve quality. It was already known that tomatoes could be picked better at riper stages but this reduced the shelf-life to a great extent. In Israel, Professor Nachum Kedar had a long-term research programme to improve the shelf-life of tomato by using the mutants 'non-ripening' (*nor*) and 'ripening inhibitor' (*rin*). It took more than 25 years to release a variety, designated 'Daniella', that ripened much more slowly than the usual tomatoes and had a harder skin and texture. The shelf-life was about 2 months, compared with the shelf-life of the other tomatoes of about 14 days. This had a great impact on tomato production in Europe. Growers in more subtropical regions, like Spain, Morocco, Tunisia and Italy, turned to 'Daniella', which could be shipped to the more northern countries. Growers in the latter areas suddenly had new competitors producing large quantities of tomatoes, with good shelf-life and at low prices. The taste of 'Daniella' was much worse than that of the existing cultivars but, remarkably, consumers did not complain. Their concern about the other tomatoes was more the way of production than the quality itself. Growers in the more northern countries, such as Belgium, the UK and The Netherlands, responded by gradually bringing a wide range of tomato types with clearly different appearances to the market: from a bulk product to a high-quality product. Retailers are gradually controlling the complete production chain, including the choice of cultivars, production and transport systems and selling to the consumers. These are merchandised with brand names like 'Gartenaroma' or 'Tasty Tom' (see www.tastytom.nl) and the system became more profitable for growers. This is a good example showing that breeders may deliver better varieties, but only when this is combined with clear agreements with all partners in the complete product chain might this lead to a successful branded product with added value.

Parthenocarpy

Normally, tomato produces seeded fruits upon fertilization. This happens under normal weather conditions, but in extreme conditions (cold, heat or lack of light) there is no fruit set, due to lack of viable pollen, which prevents pollination and fertilization. This occurs regularly in many cultivation areas. Hence, breeders aim at the development of commercially viable parthenocarpic tomato lines for these regions. Three sources of facultative parthenocarpy, *pat-1, pat-2, pat-3/pat-4*, have long been available for tomato breeding (Gorguet *et al*., 2005), while recently three to four new loci have been identified, designated *pat-6/pat-7* and *pat-8/pat-9*. The *pat-6* and *pat-8* genes are in the same locus and might be the same (Gorguet *et al*., 2008). The development of parthenocarpic fruit under control of these genes seems to be triggered by a deregulation of the hormonal balance in some specific tissues. Auxins and gibberellins are considered as the key elements in parthenocarpic fruit development of those lines, mainly by means of an increased level of those hormones in the ovary, which may substitute for fertilization to trigger fruit development. The presence of the *pat* genes often results in pleiotropic effects such as reduced taste, growth and yield. This has hampered the release of commercial cultivars.

Alternatively, obligate parthenocarpic tomatoes have been obtained by expressing the *rolB* gene from *Agrobacterium rhizogenes*, which enhances auxin sensitivity. This gene was expressed using the TPRP-F1 promoter, which is highly specific for ovaries, young fruits and developing embryos. Though these genetically modified tomatoes have given promising results in terms of quality and quantity of seedless fruit production, concerns from consumers still inhibit the release of transgenic cultivars.

Seed production

Tomato is predominantly an inbreeding species. In the field, some out-crossing takes place, partly by the wind, but also by larger insects such as bees and butterflies. In greenhouses without insects, pollination is strongly reduced. Pollination is stimulated by mechanical shaking of the plants or by (bumble) bees that are released in the greenhouse.

Until 1946 all tomato cultivars were true breeding lines that were essentially homozygous and produced progeny that were identical to the parents. Cultivars could easily be maintained and propagated by the growers by collecting seeds from their own crop.

The technology of tomato seed production is quite simple and efficient. Large quantities of seeds can be easily obtained from a single plant. Each fruit can carry more than 100 seeds, each cluster about ten fruits and each plant five to 15 clusters.

Just like in the production of tomato fruits, seed production is influenced by environmental factors, such as light intensities, soil condition and nutrients. Stress conditions may have an adverse effects on the germination of the seed and seedling vigour. Harvesting and extracting seeds before full fruit maturity is detrimental to seed viability and further germination.

As tomato seeds have a high value, growers want guarantees about germination as well as about absence of pathogens in the seeds. Therefore, phytosanitation protocols have been issued to minimize the chance of contaminations. Nowadays, tomato seeds are often produced in tropical regions at higher altitudes, where labour is cheap and the occurrence of tomato pathogens is low.

Fruits can be harvested by hand or mechanically. Seed extraction from the fruits is usually completely mechanized. The extraction results in a mixture of gelatinous seed, which is separated into seeds and debris. After extraction, seeds can be cleaned either by fermentation or by application of 0.7% HCl.

Seeds are then filtered, washed with water and air dried for 2–4 days at temperatures of 20–30°C.

To control the seed-borne *Tobacco mosaic virus* (ToMV), seeds can be treated with sodium carbonate. Drying should be uniform and slow, because fast drying shrinks the seed coat around the embryo and reduces seed quality (Stevens and Rick, 1986). Tomato seeds can be easily stored for some years at room temperature and for decades at –20°C and at low humidity of < 20%.

The costs for producing F_1 hybrid seeds are high. Since tomato is an inbreeding crop, female parental lines are emasculated and pollinated with pollen from the male parent line. This procedure is labour and time consuming. Alternatively, male sterile parents are preferred as these do not require emasculation. Many male sterility (ms) genes have been identified but are hardly used yet due to maintenance problems of the ms lines as well as to the occurrence of some self-pollination (Gorguet *et al.*, 2009). Some progress has been made using cytoplasmic male sterility (CMS), which has been observed in a BC_{10} of *S. arcanum* with *S. pennellii* (Petrova *et al.*, 1998). This CMS has been backcrossed with an F_1 between *S. lycopersicum* and *S. pennellii* in order to overcome the incompatibility between *S. lycopersicum* and *S. pennellii*. However, this CMS has not led to a commercial success.

BIOTECHNOLOGY AND TRANSGENIC OR GENETICALLY MODIFIED ORGANISMS (GMO)

Since the 1970s, technologies have been developed in tomato to regenerate plants from tissue or single cells, to introduce genes by transformation, or to break the borders between incongruent species by somatic hybridization. The potentials for breeding are evident. With biotechnology, the exploitable genetic variation for tomato breeding increased to almost unlimited levels. In fact, tomato was the first food crop for which transgenic fruits were commercially available. The first transgenic cultivar 'Flavr-Savr'™ was released in 1994, soon followed by a transgenic product with longer shelf-life (Murray *et al.*, 1995). After the first enthusiastic reactions, 'Flavr-Savr' eventually did not become a commercial success. This was due to the other unfavourable characters in the transgenic cultivar. At that time, there were no hurdles such as public non-acceptance of transgenic food products. However, this attitude changed at the end of the last century and now no transgenic tomato products are available on the market.

As stated above, tomato breeding companies have invested in biotechnology but do not release transgenic cultivars on the market. The reasons are the complicated and expensive patent filing that is associated with transgenic crops, lack of public acceptance, and the wealth of the available genetic variation to accomplish the same goals without transgenic crops.

In 2016, the global area of transgenic crops reached 185 million hectares and the accumulated area (planted since 1996) reached 2.1 billion hectares

(ISAAA, 2016). Since this has not caused any direct casualty, one might expect that public acceptance will gradually change.

EPILOGUE

In this chapter, a summary has been presented about the achievements of tomato breeding in the past century and the developments in fundamental and applied tomato research. It has been shown that a large variation is present and exploitable from wild crossable species but the vast majority remains still hidden in the tomato germplasm. To explore and exploit this variation, introgression lines that carry small introgressed chromosome fragments from related wild species in a cultivated tomato background can be useful. It is remarkable that until now any wild accession under study has been shown to harbour at least several new genes of agronomic interest. It is virtually impossible to exploit all individual wild accessions by generating genetic libraries of introgression lines. Considering the thousands of *Lycopersicon* accessions in gene banks and probably even more that are still growing untouched in the Andes, such an approach is too laborious. We are now in the genomics era and, for tomato, hundreds of (re)sequenced genomes are available (Aflitos *et al.*, 2014). Gene annotations are helpful to identify the candidate genes that might be underlying important traits; after confirmation these genes can be exploited (Fig. 2.6). The generation

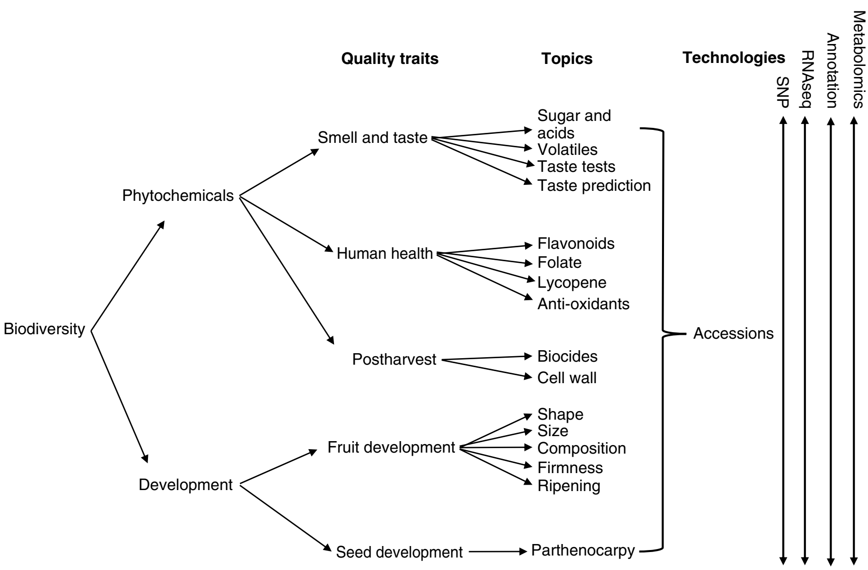

Fig. 2.6. Relationship between various traits, components and technologies applicable in tomato fruit quality genomics (from Arnaud Bovy, Wageningen University & Research, Wageningen, The Netherlands).

of sequence data is no longer rate limiting. Bio-informatics is at the centre of linking databases and proposing structure function relationships. Still, for each single gene the function has to be determined on a case-by-case basis. All the technical and scientific developments will contribute to a better understanding of the function of tomato genes. This allows us to search for allelic variation of these genes in the *Lycopersicon* germplasm ('allele mining'). Techniques like (eco)tilling and genome editing will greatly facilitate the identification of useful genes in the wild germplasm (Comai *et al*., 2004).

In the end, breeders have their genetic stock in the seed store and a large database about the seeds. Breeders' capital will shift from the field to the computer. They will select the best combinations of genotypes and design a programme to combine the traits in their cultivar. This is the era of genomics breeding.

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3

Developmental Processes

Ep Heuvelink and Robert C.O. Okello

INTRODUCTION

Growth and development in plants are strongly related, but the interpretation of plant responses to environmental factors is facilitated by making a distinction between these two processes. Plant development is defined as a series of identifiable events resulting in a qualitative (germination, flowering, etc.) or quantitative (number of leaves, number of flowers, etc.) change in plant structure. Growth is defined as an irreversible increase in plant or organ dimensions over time, e.g. length, width, diameter, area, volume and mass (Dambreville *et al.*, 2015).

Plant development starts with fertilization. Fertilization, the fusion of pollen and egg nuclei, produces a diploid zygote, which differentiates into the embryo, the vital next generation of the plant. Embryo development, or embryogenesis, is accompanied by production of storage tissues, such as endosperm or megagametophyte, and maternal integument layers, which in mature seed become the testa (seed coat). Hormones (e.g. auxin) play important roles in seed development and maturation. Precocious germination (vivipary) is controlled by abscisic acid (ABA) during late stages of seed development. However, Wang *et al.* (2016) reported vivipary occurring in one accession of tomato *rin* mutant fruit not directly associated with ABA, and concluded that hypo-osmolality in *rin* fruit may be an important factor in permitting limited viviparous germination. Single-gene mutations leading to defects in synthesis or sensitivity to ABA in tomato result in formation of non-dormant seeds (Bewley *et al.*, 2013).

Developmental processes such as germination, root and leaf development, inflorescence formation, flower and fruit development all have a strong impact on tomato yield and are discussed in detail in this chapter. Photoreceptors such as phytochromes play an important role in the control of plant developmental processes. *Solanum lycopersicum* contains five phytochromes: A, B1, B2, E and F (Eckstein *et al.*, 2016).

SEED AND SEED GERMINATION

The seed of a commercial tomato cultivar is a flattened ovoid, up to 5 mm long, 4 mm wide and 2 mm deep, consisting principally of the embryo, endosperm and testa or seed coat. The testa is covered with large soft hairs, which tend to bind with other seeds. Thousand-seed weight varies between 2 g (cherry tomatoes) and 4 g. One hybrid tomato seed may cost over €0.40, which makes it three times more expensive than gold (on a weight basis). Tomato seeds store easily and retain their viability for long periods between 5°C and 25°C and a fairly wide range of relative humidity (RH), and so no particular special storage measures need be taken. James *et al.* (1964) reported 90% and 59% germination after 15 and 30 years of storage, respectively. Mathematical models have been developed to predict tomato seed longevity as a function of storage conditions (e.g. Sinício *et al.*, 2009).

Germination performance of the seed is influenced by its characteristics, for example seed size (more rapid germination is associated with smaller seeds, possibly due to a reduced endosperm thickness). Harvesting and extraction of tomato seed before full maturity is detrimental to seed viability and germination, due to the presence of large numbers of immature seeds, whereas in late harvests viviparous germination can be a problem (Demir and Samit, 2001). Dias *et al.* (2006) evaluated tomato seed quality obtained from fruits harvested from trusses on the plant at 70 days after anthesis. Fruit weight decreased from the first to the sixth truss; however, seeds extracted from fruits of the fourth to sixth trusses had, overall, better quality than those from the first to third truss. On the same truss, distal fruit tended to produce seeds of higher physiological quality than proximal fruit.

Seed germination is a central process in the generative life cycle of higher plants. This complex process is regulated by a large number of environmental and endogenous factors. Germination begins with water uptake by the seed (imbibition or rehydration) and ends with emergence of the embryonic axis, usually the radicle, through the structures surrounding it. Cell elongation is necessary, and is generally accepted to be sufficient, for the completion of radicle protrusion; cell division is not essential. Radicle protrusion is sometimes referred to as 'visible germination' (Bewley *et al.*, 2013). The criterion for germination used by physiologists is radicle emergence; seed analysts extend this interpretation by classification of seedlings as either normal or abnormal. Tomato seed is considered a model system for germination research (Nonogaki, 2006).

Quality of seed lots is usually determined by a standard germination test conducted under ideal conditions in a laboratory. Both total germination percentage and uniformity in germination are important characteristics of a seed lot. High germination percentages do not necessarily mean high emergence in practice, especially under suboptimal, fluctuating field conditions. For direct-sown tomato crops seed lots with a high vigour, indicated by more rapid germination or greater tolerance of stressful conditions (Bewley *et al.*, 2013), are essential. Seed vigour differs from germination in that vigour emphasizes the germination rate (rapid and uniform) and its application in forecasting field emergence rather than laboratory performance. Seed vigour tests can be divided into categories according to ageing, cold, conductivity, seedling performance and tetrazolium tests (AOSA, 2009). 'Accelerated ageing' or 'controlled deterioration' tests utilize the extreme reduction in seed longevity by the combination of high temperature and high humidity (e.g. 40–50°C and 75–100% RH) to rank seed lots (Bewley *et al.*, 2013). For greenhouse production, seeds are often germinated in controlled-environment chambers, which guarantee optimal conditions. When seeds are sown into multiplates or peat pots, a germination percentage of about 100% normal seedlings is required, as opposed to sowing in seed trays and transplanting manually, which allows for some selection. Germination rate can be improved by pre-sowing seed treatments with growth regulators. Stimulatory effects have been demonstrated for gibberellins (GA) and various synthetic auxins. Venkatasubramanian and Umarani (2007) evaluated several seed priming methods to improve tomato seed performance. Priming allows the seed water content to be adjusted to a level that permits the seeds – provided they are freely aerated – to go through all the essential preparatory process of germination but prevents cell elongation and in consequence radicle emergence. During this waiting period, the less developed or slower seeds tend to catch up with the faster ones so that subsequent germination is faster and more uniform.

In cultivated tomato, seeds are considered non-dormant, whereas some studies with wild types suggest that ABA plays a role in a slight dormancy. Seed germination of several GA biosynthesis deficient mutants absolutely depends on the addition of GA to the medium during imbibition. Physiological, biochemical and genetic evidence suggests a role for GA in weakening the structures covering the embryo during germination. GA biosynthesis in developing seeds is involved in embryo growth and the prevention of seed abortion (Kucera *et al.*, 2005). These authors present an integrated view of the molecular genetics, physiology and biochemistry used to unravel how hormones control seed dormancy release and germination. ABA is a positive regulator of dormancy induction and most likely also maintenance, while it is a negative regulator of germination. GA releases dormancy, promotes germination and counteracts ABA effects. Ethylene and brassinosteroids (BR) promote seed germination and also counteract ABA effects.

Environmental factors strongly influence seed germination. Tomato seeds characteristically germinate best in the dark; and light will inhibit germination in some cultivars. These responses are dependent on phytochrome action. Far-red light has been reported to inhibit germination, whereas red light (at 37°C) promoted germination, suggesting that the presence of P_{fr} (the far-red absorbing form of phytochrome) in the seed is a prerequisite for germination.

Phytochromes A and B2 were shown to play specific roles, acting antagonistically in far-red light (Eckstein *et al.*, 2016). The ability of some light-sensitive cultivars to germinate in the dark has been attributed to high levels of P_{fr} present in dehydrated seed. Effects of water, salinity and temperature on tomato seed germination are discussed below.

Water and salinity

Without water, no germination will occur. Water absorption by seeds is characterized by three phases. In the first phase (imbibition), rapid uptake of water will occur, whether the seed is viable or not. In the following phase, hydration of the cotyledons may take place but the moisture content of the seed remains apparently constant. A resurgence in water uptake marks the growth phase associated with radical protrusion and subsequent growth.

The rate and degree of water uptake for germination are affected by temperature, water content and salinity. In general, tomato seeds germinate at soil water potentials ranging from just above the wilting point to field capacity, though optimum germination is usually obtained at 50–75% field capacity. Reduced germination at high soil water content is probably caused by reduced oxygen availability.

Water stress is often associated with increased soil salinity, which may affect germination in two ways: firstly by creating an osmotic potential that impedes water uptake; and secondly by entry of deleterious ions. High salt levels tend to decrease tomato seed germination, though tomatoes seem to be more tolerant of saline conditions during germination than during subsequent growth.

Temperature

Mathematical models have been developed to describe germination in response to temperature on the basis of thermal time or heat units (i.e. degree-days). The basis of thermal time is that the temperature in excess of the minimum temperature for germination, multiplied by time to a given germination percentage, is a constant. Constant heat units imply a linear relationship between germination rate and temperature. This seems unlikely for germination of tomato seeds, since optimum temperatures between 20°C and 25°C have been reported. However, over the suboptimal temperature range (13–25°C) this linearity has been observed by Bierhuizen and Wagenvoort (1974). These authors reported a minimum temperature of 8.7°C for germination and a heat sum requirement of 88 degree-days to achieve 50% germination, though in a further paper they considered 13°C as the ‘practical’ minimum temperature for germination (Wagenvoort and Bierhuizen, 1977). Membrane lipid changes appear to be involved in cultivar differences in the ability to germinate at low temperatures, whereas failure to germinate at high temperatures may be due to thermodormancy, a condition thought to be related to an interaction between temperature and phytochrome action.

LEAF APPEARANCE AND LEAF GROWTH

Leaf appearance or unfolding rate (LUR) is not necessarily equal to the rate of initiation of leaf primordia at the apex, but, in the long term, equality is expected. LUR is predominantly influenced by temperature (De Koning, 1994). Since three leaves are initiated between two successive trusses for 'indeterminate' types, LUR rate is simply three times the truss appearance rate. LUR shows an optimum response to increasing temperature. This relationship is close to linear at 17–23°C and as a rule of thumb three leaves unfold each week at 20°C. Meristem temperature determines leaf initiation rate and Savvides *et al.* (2013) determined that meristem temperature can deviate substantially (–2.6°C to 3.8°C for tomato) from air temperature under moderate environments.

De Koning (1994) observed no effects of fruit load, leaf removal or planting density on LUR. Heuvelink and Marcelis (1996) also reported little influence of assimilate supply on LUR (Table 3.1). Higher light interception per plant due to a lower planting density hardly affected LUR, whereas assimilate supply was strongly improved. However, Savvides *et al.* (2014) reported a reduction in leaf initiation rate in young tomato plants when light level was reduced below 6.5 mol/m^2/day (Fig. 3.1). A higher light level could result in a higher meristem temperature and hence reflect an indirect effect of light on leaf initiation rate, but Savvides *et al.* (2014) reported a constant meristem temperature among the light treatments, and hence reduced leaf initiation rate at low light level is likely to be determined by the reduction in local photosynthate availability. This contrast with Heuvelink and Marcelis (1996) may be explained by assimilate supply limiting LUR in seedlings, because of the low leaf area and thus low light interception.

The size of a compound tomato leaf is immensely variable. Leaves of greenhouse cultivars are typically 0.5 m long, a little less in breadth, with a large terminal leaflet and up to eight large lateral leaflets, which may themselves be compound. Many smaller leaflets (folioles) may be interspersed with the large leaflets. Stomata are present on the underside (abaxial side) of the leaflet and are often present in smaller numbers on the top (adaxial size) (Gay and Hurd, 1975).

Leaf size, both area and mass, is influenced by growth conditions (Table 3.1), with an increased assimilate supply resulting in larger leaves of higher mass. However, an individual leaf grown under high irradiance is usually smaller in area but larger in mass. The same is observed under

Table 3.1. Effect of treatment on the number of leaves visible, total dry weight, vegetative dry weight, average area and dry weight of individual leaves and specific leaf area (SLA) at the end of the experiments with tomato. For the latter three parameters, only the leaves still on the plant were taken into account. Means within an experiment, followed by the same letter, were not significantly different ($P = 0.05$) (Heuvelink and Marcelis, 1996).

Experiment	Treatment	No. of leaves visible	Total dry weight (g/plant)	Vegetative dry weight (g/plant)	Leaf area (cm^2/leaf)	Leaf dry weight (g/leaf)	SLA (cm^2/g)
1	1.6 plants/m^2	59.5^{a}	636^{a}	256^{a}	467^{a}	3.17^{a}	147^{a}
	2.1 plants/m^2	60.3^{a}	512^{b}	196^{b}	396^{b}	2.36^{b}	169^{a}
	3.1 plants/m^2	57.8^{a}	364^{c}	141^{c}	310^{c}	1.61^{c}	193^{b}
2	1 fruit per truss	68.6^{a}	385^{a}	294^{a}	322^{a}	2.67^{a}	123^{a}
	3 fruits per truss	71.7^{a}	536^{b}	262^{a}	352^{a}	2.53^{a}	140^{a}
	7 fruits per truss	69.4^{a}	519^{b}	176^{b}	329^{a}	1.58^{b}	209^{b}
3	1 fruit per truss	64.8^{a}	414^{a}	341^{a}	494^{a}	3.37^{a}	147^{a}
	7 fruits per truss	59.3^{b}	444^{a}	210^{b}	503^{a}	2.21^{b}	232^{b}
4	50% truss removal	62.7^{a}	471^{a}	252^{a}	419^{a}	2.71^{a}	155^{a}
	Control	62.3^{a}	488^{a}	182^{b}	320^{a}	1.91^{b}	166^{a}
5	50% leaf removal[1]	37.3^{a}	221^{a}	110^{a}	331^{a}	3.22^{a}	104^{a}
	Control	37.0^{a}	266^{b}	136^{a}	299^{a}	2.54^{b}	118^{a}

[1]Removal of every other leaf (at a length of 5 cm) above the first truss.

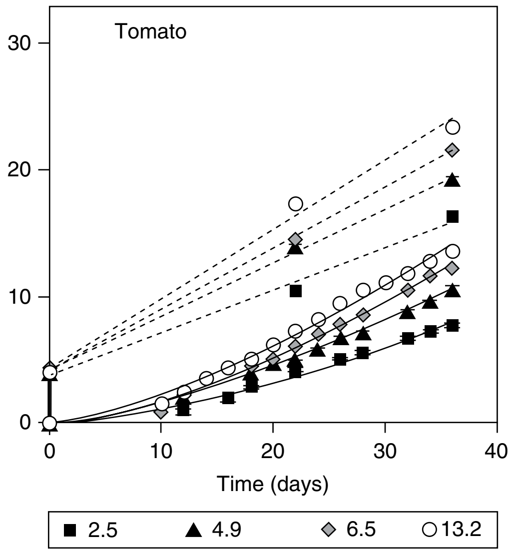

Fig. 3.1. Time course of the number of leaves initiated (dashed lines) and unfolded (length 10 cm; solid lines) in tomato plants grown under four diel photosynthetic photon flux densities (PPFD) (2.5, 4.9, 6.5 and 13.2 mol/m²/day). (Reproduced from Savvides *et al.*, 2014, with permission from CSIRO Publishing.)

enriched carbon dioxide (CO_2) conditions. Except in low light conditions, a higher temperature usually increases the rate of expansion of individual leaves, but expansion period is decreased relatively more strongly, resulting in a reduced area per leaf. On the other hand, plant leaf area usually increases with temperature, since LUR increases. Leaf expansion is improved by a high turgor pressure, which is favoured by a suppressed transpiration rate. Indeed, tomato leaf area increases with humidity but the effect is small. In the long run, high humidity may cause calcium deficiencies, which may lead to up to 50% leaf area reduction (Holder and Cockshull, 1990).

Leaf thickness increases with assimilate supply, resulting in a lower specific leaf area (SLA) (leaf area per unit of leaf dry mass). This is shown in Table 3.1 for wider plant spacing and reduced fruit load, but the same has been reported for increased irradiance (Fan *et al.*, 2013) or elevated CO_2 concentrations (Qian *et al.*, 2012). A higher temperature usually decreases leaf thickness and increases SLA.

Prolonged periods of surplus assimilates may lead to the so-called short-leaf syndrome (SLS), i.e. crisp, grey-green to purple-coloured leaves, sometimes with necrotic spots and leaf tips (Nederhoff *et al.*, 1992). These authors have suggested that SLS results from calcium deficiency in the apex, an indirect consequence of a low sink–source ratio resulting in fast-growing organs. Short leaves can be avoided by increasing the plant density, or, in an early-planted year-round crop, by maintaining an extra shoot on part of the plants in spring, thus increasing stem density. In both cases the sink–source ratio is increased.

STEM DEVELOPMENT

At the tip of the main stem is the shoot apical meristem (SAM), a region of active cell division where new leaves and flower parts are initiated. It is dome-shaped and protected by newly formed leaves. Leaves are arranged alternately with a 2/5 phyllotaxy. Shoot development in tomato can be separated into two different phases. During the initial vegetative phase, the SAM forms metamers consisting of an elongated internode, a leaf and a bud (Fig. 3.2). After the formation of between 7 and 11 metamers, the primary shoot SAM is transformed into an inflorescence meristem (IM). While the IM develops into an inflorescence, a second phase of shoot development is initiated by the outgrowth of the bud in the axil of the youngest leaf primordium. This sympodial shoot grows vigorously, displaces the developing inflorescence to a lateral position and transfers its subtending leaf to an elevated position above the inflorescence. After the formation of three leaves, the SAM of the sympodial shoot is also transformed into an IM and develops into an inflorescence. The main axis is again continued by the sympodial shoot in the axil of the youngest leaf primordium. In 'indeterminate' cultivars the process is repeated indefinitely with inflorescences every three leaves. These types are favoured in greenhouses as they produce high yields over an extended period. In 'determinate' types each axis produces a limited number of inflorescences and strong axillary buds develop at the base of the stem, producing a bushy habit, which is ideal for growing unsupported in the open. Microscopic studies reveal that axillary buds in tomato are formed early in development in all axils of leaf primordia. Due to the weak correlative inhibition by the shoot apex, these buds develop into side shoots without a resting phase, repeating at least part of the development of the primary shoot. As all nodes of a side shoot can again form lateral shoots, this pattern can be repeated indefinitely, resulting in a very bushy growth habit (Schmitz and Theres, 1999). These side shoots are removed when 'indeterminate' cultivars are grown. The side shoot from the leaf axil immediately below an inflorescence is the strongest and can seriously compete with the main apex.

Cao *et al.* (2016) subjected tomato plants to night break (NB) treatments with different frequencies ranging from every 1, 2, 3 and 4 h using red light (RL) with an intensity of 20 μmol/m^2/s. These authors showed that with the increase of RL NB frequency, plant height decreased, stem diameter increased and flower initiation was delayed (both in days and number of leaves below the first inflorescence).

The rate of stem elongation generally increases with temperature, with stem length primarily determined by day temperature (optimum > 28°C) (Langton and Cockshull, 1997). Reduction in length growth resulting from a so-called negative DIF (difference between day and night temperature) has been reported by many authors and for a large variety of crops, including tomato (Fig. 3.3). Bours (2014) concluded that negative DIF affects growth by directly affecting the phase and amplitude of circadian clock genes, which in

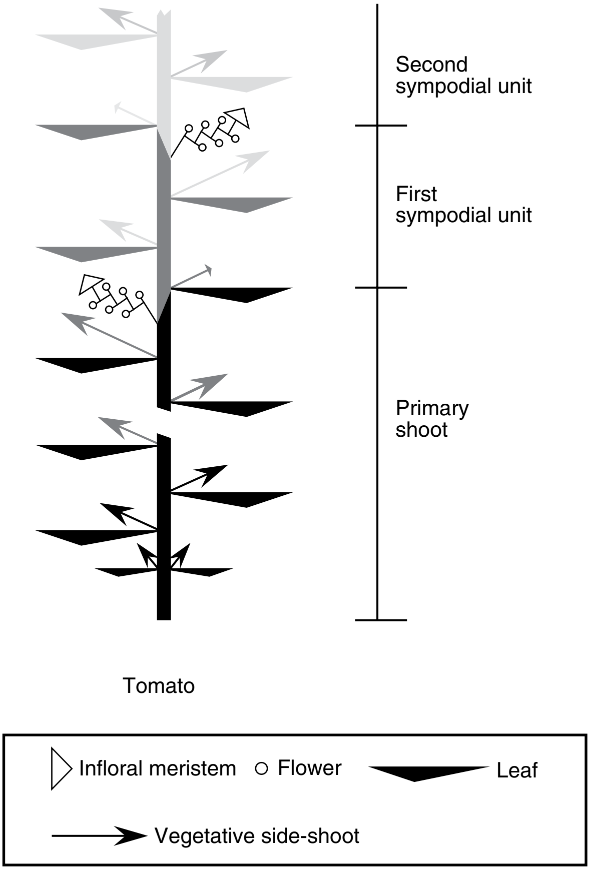

Fig. 3.2. Schematic representation of shoot development in tomato. The primary shoot is illustrated in black, lateral axes of successive order in grey. The primary shoot meristem forms the first segment of the sympodial stem terminating in the first inflorescence. In all leaf axils, vegetative side shoots are formed. The side shoot in the axil of the youngest leaf primordium grows vigorously, pushes the inflorescence to a lateral position and transfers its subtending leaf to an elevated position above the inflorescence. After the formation of three nodes the sympodial shoot also terminates in an inflorescence. Sympodial shoots of progressively higher order continue the elaboration of the main axis indefinitely. Inflorescences consist of a series of terminal flowers originating from sympodial inflorescence axes of progressively higher order. (Reproduced from Schmitz and Theres, 1999, with permission from Elsevier.)

turn control downstream processes such as starch metabolism and hormone signalling pathways. Auxin and ethylene signalling pathways affected by negative DIF show significant crosstalk and interconnect with the circadian clock at several positions. Bours (2014) stressed the unique position of the photoreceptor

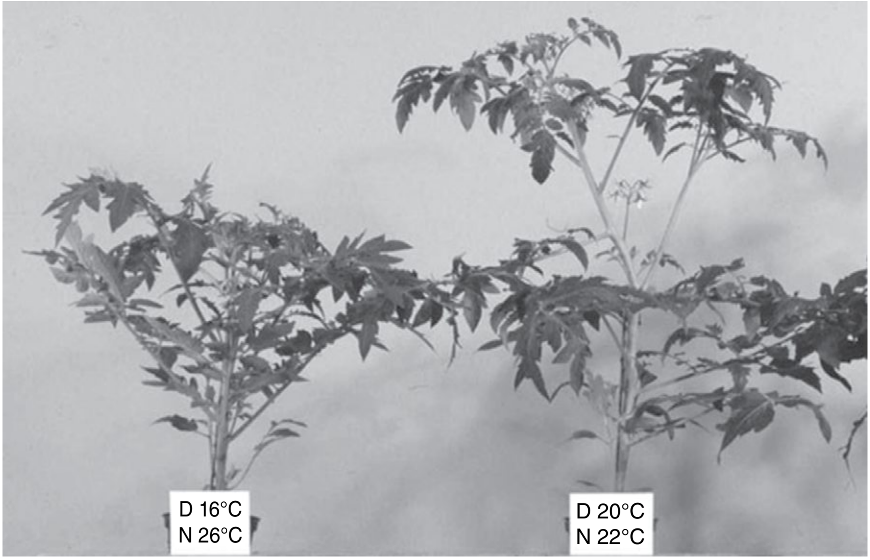

Fig. 3.3. Tomato plants (8 weeks after sowing, 40 days after start of temperature treatments) grown in climate chambers at a light intensity of 30 W/m^2 at three temperature regimes (D = day temperature; N = night temperature; day length was 12 h).

phytochrome B in this regulation. Gibberellin metabolism is almost certainly involved in the regulation of stem extension by temperature (Langton and Cockshull, 1997). The effect of increased root temperature on stem elongation is relatively small.

Under daily total irradiances above about 2 MJ/m^2/day the heights of plants of the same dry mass are constant. Reducing daily irradiance below this level generally increases the rate of stem elongation. The result is a weaker, thinner but taller stem. CO_2 enrichment generally produces taller plants but this is because of faster growth. Plants of similar dry or fresh mass are slightly shorter if grown with additional CO_2. High humidity improves stem elongation, though effects are only small. Holder and Cockshull (1990) did not observe any effect of humidity treatments (0.1, 0.2, 0.4 and 0.8 kPa vapour pressure deficit, maintained continuously for 28 days) on plant height. Salt stress (high electrical conductivity (EC) in the root environment) reduces stem elongation. Shaking both stresses and toughens plants, resulting in a reduced stem length, a phenomenon known as thigmomorphogenesis (Picken *et al.*, 1986).

Axillary shoot development is promoted by low temperature, negative DIF or short days. A period of 5 min of far-red light at the end of a light period has been shown to suppress side-shoot growth (Tucker, 1976), which may be of interest for greenhouse crop production, where manual removal of side shoots demands a lot of labour. In tomato, several mutants are defective in axillary meristem initiation. The lateral suppressor (ls) mutant is characterized by the absence of side shoots, except for the sympodial shoot and the lateral shoot immediately below (subfloral side shoots) (Schmitz and Theres, 1999).

ROOT DEVELOPMENT

Tomato seedlings develop a tap root, which can grow longer than 0.5 m but it is often damaged in culture. About 60% of the root system lies in the top 0.3 m of soil (Rendon-Poblete, 1980). Adventitious roots, similar in structure to the laterals, develop under favourable conditions from the stem, particularly near the base. They are also initiated on the underside of horizontal portions of stem, enabling the plant to re-root in nature.

Nowadays, many greenhouse tomato cultivations are on artificial substrates, e.g. stone wool, perlite or sometimes in nutrient solution only (nutrient film technique (NFT)). NFT enables studies of root growth and development, and both NFT and substrate cultures allow for a much more controlled root environment. In research focused on roots, sometimes plants are grown in aerated containers with nutrient solution; however, these roots have a different morphology from those grown in soil.

Water supply and nutrition have profound effects on plant growth (see Chapter 6). Roots grown at about 14°C were thicker, whiter and less branched than at higher soil temperatures, but nutrition influences this response (Gosselin and Trudel, 1982). Chapter 4 presents effects of root temperature on dry matter production and partitioning. High humidity may encourage aerial adventitious root growth.

FLOWERING

Samach and Lotan (2007) summarized progress in understanding the environmental cues that affect the initial transition to flowering in tomato, and the genes that are involved in this transition and additional transitions occurring on the sympodial shoot. Environmental cues discussed were day length, light intensity and growth temperature. Most of the genes isolated so far seem to play similar roles in *Arabidopsis* flowering. This section is largely based on Dieleman and Heuvelink (1992) and an earlier review on flowering in tomato (Picken *et al.*, 1985) has been consulted occasionally. In general the tomato plant initiates at least six to eight leaves preceding the first inflorescence (number of leaves preceding inflorescence (NLPI)). Inheritance of this characteristic was found to be simple; one single gene pair determines the NLPI. Complete dominance is suggested for smaller node numbers over large node numbers to the first inflorescence. The first inflorescence terminates the main axis of growth and subsequent extension growth is made by the lateral shoot in the axil of the last initiated leaf (see 'Stem development', above).

There are two flowering types, which are incorrectly termed 'determinate' and 'indeterminate'. Botanically, both types are determinate since each inflorescence terminates the main axis. Tomato does not respond to seed vernalization, i.e. NLPI is not reduced by low temperatures applied to imbibed seeds. The sensitive period, determining the position on the stem at which the first inflorescence will develop, lasts approximately 10 days, starting at cotyledon expansion, though this period can be longer depending on the cultivar. No single environmental factor can be regarded as critical for the control of flowering in tomato. Environmental factors such as light, temperature, CO_2, nutrition, moisture and growth regulators directly or indirectly influence flower initiation.

The NLPI can be seen as the result of two processes: the rate of leaf initiation and the time to initiation of the first inflorescence, which determines the end of the vegetative phase. The slopes of the lines in Fig. 3.1 indicate the rate of leaf initiation. The same rate of leaf initiation can result in a different NLPI, if the number of days to initiation of the first inflorescence is different. Also, a higher rate of leaf initiation combined with a lower number of days to initiation of the first inflorescence may result in the same NLPI.

Factors affecting the number of leaves preceding the first inflorescence

A lower temperature, either by day or by night, during the sensitive period of tomato seedlings causes a smaller NLPI. This effect of mean diurnal temperature on NLPI is mainly determined by the effect of temperature on the rate of leaf production. Temperature does not affect the time to flower initiation if light intensity is high. At low light intensities a higher temperature increases the time to flower initiation compared with a lower temperature. Root temperatures appeared to have little or no effect on the NLPI.

The rate of leaf production in a tomato seedling increases with increasing light intensity (Fig. 3.1). On the other hand, the number of days to flower initiation decreases with increasing light intensity, this effect being stronger than the effect on the rate of leaf production. This results in a lower NLPI at higher light intensities. Effects of light and temperature on both the rate of leaf initiation and the time to flower initiation, and hence on NLPI, interact. Flowering in tomato is not usually affected by photoperiod but some cultivars are regarded as quantitative short-day plants. Several authors have reported a smaller NLPI with shorter photoperiods.

CO_2 enrichment increases the rate of leaf initiation slightly and decreases the time to flower initiation a little, thus resulting in no clear effect on the NLPI. Removal of parts of the cotyledons immediately after they had unfolded increases NLPI, the increase being greater the larger the area removed. Removal of the first two foliage leaves from 6-day-old seedlings reduces the NLPI, whereas NLPI is not significantly altered by the removal of further foliage leaves. Generally, the higher the fertilizer level (nitrogen, phosphorus, potassium (NPK)), the lower is the NLPI. However, flowering is delayed when salt or water stress is applied in early stages of development.

The effects that growth regulators have on the NLPI vary with the method of application, the concentration, timing, environment and variety. Application of auxins as well as application of gibberellin as a foliar spray cause an increase in NLPI. However, application of gibberellin to imbibed seeds does not influence NLPI. Application of auxin or gibberellin transport inhibitors decreases NLPI; for example, an NLPI as low as 2.4 has been observed after application of 10^{-3} M TIBA (2,3,5-triiodobenzoic acid) foliar spray, which inhibits auxin transport.

Most of the observations mentioned above can be explained by nutrient diversion hypothesis for flowering (Sachs and Hackett, 1969). According to this hypothesis, the amount of assimilate available to the apex during the sensitive phase has to reach a certain minimum before flower initiation can take place. This level of assimilates is influenced by the assimilate production rate and its distribution. Factors that increase the total amount of assimilates in the plant as well as those increasing the competitive potential of the apex decrease the NLPI.

Appearance rate of new trusses

Most tomato cultivars develop ‘units’, comprising three leaves and internodes, terminated by an inflorescence, which build up the sympodial stem (Fig. 3.2). Hence, appearance rate of new inflorescences (trusses), following the first inflorescence on the sympodial stem, often called the flowering rate, is one-third of leaf appearance rate (see ‘Leaf appearance and leaf growth’, pp. 63–65, this volume). This means that the description of environmental effects on leaf appearance rate also holds for truss appearance rate. Thus, truss appearance rate after the first truss is predominantly determined by temperature. De Koning (1994) observed an almost linear response between temperature (range 17–23°C) and truss appearance rate (0.11–0.15 trusses per day). Truss appearance rate is equally related to day and night temperature and therefore responds to the 24 h mean temperature. No effects of fruit load, leaf removal or plant density on truss appearance rate have been observed (De Koning, 1994). Hence, sink–source ratio has no significant influence on truss appearance rate in tomato. Also, root temperature, EC in the root environment, air CO_2 concentration and air humidity hardly affect truss appearance rate.

FLOWER DEVELOPMENT, POLLINATION AND FRUIT SET

Once a tomato plant has terminated its vegetative phase, several processes following the initiation of an inflorescence determine whether fruits will start to grow. Flower buds have to develop into flowers, pollination and fertilization have to be successful and fruit set should occur (Fig. 3.4). All of these processes are considered in the present section, mainly based on Chapters 4 and 5 of the tomato book by Atherton and Rudich (1986) and a review on pollination and fruit set by Picken (1984).

Inflorescence size and flower bud abortion

The number of flowers initiated in an inflorescence depends on cultivar and environmental conditions. The inflorescence is a monochasial cyme in which the vegetative axis terminates in the king flower (Picken *et al*., 1985). Increased irradiance or decreased plant density and decreased temperatures positively influence the number of flowers formed in an inflorescence, with the temperature effect being larger at high irradiance compared with low irradiance. Low air temperatures (< 10°C) during inflorescence initiation promote inflorescence branching, usually resulting in more flowers per inflorescence. It is the mean diurnal temperature, rather than day or night temperature alone, that

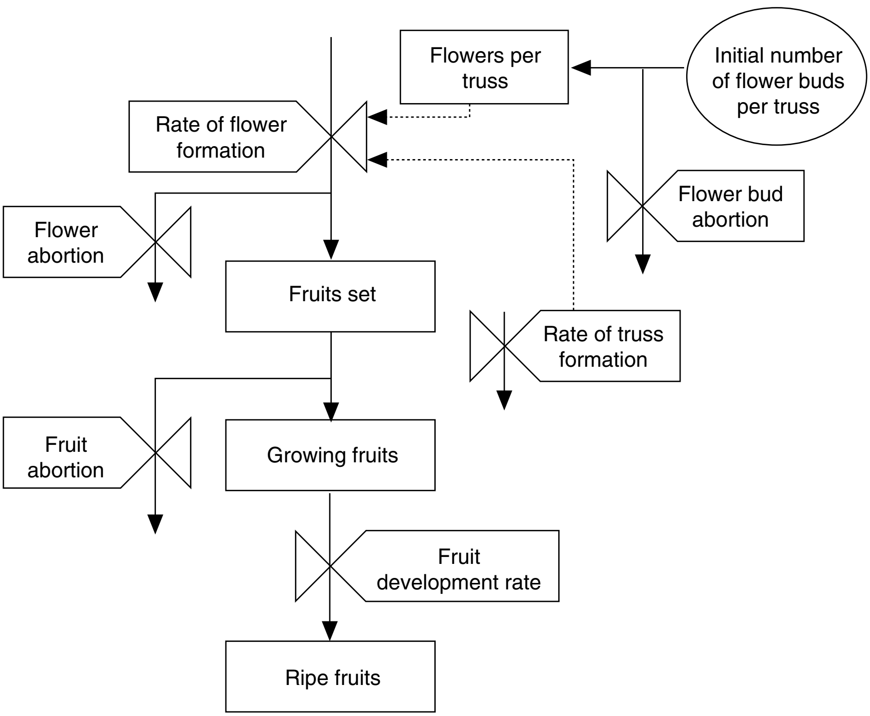

Fig. 3.4. A relational diagram of the number of growing fruits on an indeterminate tomato plant (De Koning, 1994).

controls branching and flower number (Hurd and Cooper, 1967). Similar to air temperature, a reduction in root temperature promotes the number of flowers in the first inflorescence. Inflorescence size is reduced when plants are grown under restricted water supply or in small pots containing a small volume of growing medium. Photoperiod or CO_2 level show no influence on the number of flowers formed in an inflorescence.

Flower development after initiation is primarily influenced by temperature: higher temperatures result in faster flower development. Increased day temperatures are reported to be more effective in promoting flower development than corresponding increases in night temperatures.

Especially under conditions where photosynthesis is low (low light intensities, low CO_2 and/or short days), increased temperatures will stimulate flower bud abortion. High temperature effects on flower abortion may be a consequence of failure of fruit set rather than a direct effect of temperature (see below). Overall it appears that low assimilate availability during flower development stimulates abortion of flower buds. Furthermore, when plants are grown at the same total daily irradiance, flower bud abortion is more evident in long days than in short days.

Pollination

Fruit set, defined here as the proportion of open flowers that subsequently set fruit of a marketable size, may fail for many reasons. Flower/fruit abortion is defined as the proportion of flowers that fail to yield fruit of marketable size and hence fruit set is one minus abortion. In tomato, flower abortion occurs frequently, whereas fruit abortion is uncommon, though sometimes distal fruits stop growing at a small size and never ripen. Failure of pollen production (number and viability of pollen) or pollination, pollen germination, pollen tube growth, ovule production, fertilization or fruit swelling all may result in poor fruit set. However, poor fruit set in low light conditions is most frequently caused by failure of pollen production or pollination.

The potential number of pollen grains is genetically determined. For example, some modern TMV-resistant cultivars, containing the *Tm-2* gene, produce less pollen of a slightly lower viability. Pollen development is adversely affected by low light. At severe carbohydrate deficiency, meiosis is abnormal in some pollen mother cells. With less severe carbohydrate stress, pollen development is variable. When microspore development ceases at an early stage, pollen grains are shrunken and irregular, whereas when microspore development proceeds normally until after mitotic division, the pollen grains are morphologically perfect. Abnormal flowers with rudimentary petals, stamens and pistils produced under low light produce sterile pollen. High temperature (40°C) damages pollen, the most critical stage for pollen development being meiosis, which occurs about 9 days before anthesis in plants grown at 20°C.

In contrast, meiosis appears unaffected by low temperature. A day/night temperature of 32°C/26°C compared with 28°C/22°C did not cause a significant change in the number of pollen grains produced; however, it significantly decreased the number of fruit set, pollen viability and the number of pollen grains released (Sato *et al.*, 2006). Failure of tomato fruit set under a moderately increased temperature above optimal was due to the disruption of sugar metabolism and proline translocation during the narrow window of male reproductive development. Also failures in anther dehiscence at high temperatures have been mentioned (Arrizumi *et al.*, 2013). Literature reports conflict where possible decrease in pollen quality at low temperatures is concerned, possibly because of interactions with light and root temperature.

Pollination, the successful transfer of pollen grains to the stigmatic surface, depends on pollen release, flower characteristics and pollen grain adherence to the stigma. Although pollen is mature and ready for transfer at anthesis, the stigma becomes receptive about 2 days previously and remains so for up to 4 days or more. All modern cultivars are self-pollinated. Flowers are not frequently visited by insects, since they do not produce nectar. Air movement in the field is enough to stimulate self-pollination but wind speed in greenhouses is too low. In greenhouses, pollination is promoted by vibrating each truss mechanically with an electric 'bee' two or three times a week, or, more recently, by the introduction of bumblebees. It is most effective in winter to vibrate flowers at around midday, maybe as a consequence of the higher temperature and irradiance at this time. However, tomato flowers close at night and open during the day, which may also contribute to the increased effectiveness of vibration at midday.

At high humidities, pollen tends to remain inside the anthers, whereas at low humidities it may not adhere to the stigma. However, the effects of relative humidity between 50% and 90% are small. Pollination can be adversely affected by abnormalities in flower structure. For self-pollination, the stigma must lie within the tip of the anther cone. Stigma exsertion beyond the cone enables the pollen to escape before reaching the stigma, and it may also lead to desiccation of the stigmatic surface. The length of the style is both genetically determined and affected by growing conditions. Low light, high nitrogen levels and high temperature have been shown to increase stigma exsertion. Other structural defects that reduce fruit set include, for example, poor development of the endothecum at higher temperature, which is essential for dehiscence, and splitting of the staminal cone and/or fasciation of the style at low light. Pollen adherence to the stigmatic surface depends not only on relative humidity (see above) but also on temperature, 17–24°C being optimal.

Germination of pollen grains and fertilization

Pollen germination depends on pollen viability and environmental factors. The time taken for pollen to germinate decreases with increasing temperature.

The germination percentage is greatly reduced at temperatures outside the range 5–37°C. The growth rate of the pollen tube increases with temperature between 10°C and 35°C but is reduced outside this range. Sato *et al.* (2000) concluded from their work that cultivar differences in pollen release and germination under heat stress are the most important factors determining their ability to set fruit. Humidity and light only marginally affect pollen tube growth.

Low light may reduce the size of flowers and ovaries, and ovule development may cease under such conditions before or soon after embryo sac formation. Ovule viability may be adversely affected by high temperature (40°C) 5–9 days before anthesis at about the time of meiosis in the ovule mother cells.

Fertilization takes place once the nuclei from the pollen tubes penetrate viable ovules. Because of an increased tube growth rate, higher temperatures reduce the time to fertilization. Fertilization is not greatly affected by growth conditions, though exposure to high temperature (40°C) can have adverse effects on fertilization and on processes that immediately precede or follow it. For example, the endosperm can deteriorate at high temperature 1–4 days after pollination.

Fruit initiation

Fruit set is positively correlated with assimilate availability, which may be expressed as the source–sink ratio in the plant (Fig. 3.5). Tomato fruit set depends on successful pollination and fertilization, which trigger the fruit developmental programme through activation of auxin and gibberellin signalling pathways. However, the role of each of these two hormones is still poorly understood. De Jong *et al.* (2009) reviewed the role of auxin and gibberellin in tomato fruit set and presented a model integrating the role of both hormones. The level of cytokinines is up-regulated 5 days after anthesis, suggesting a positive correlation between cytokinines and cell division. Application of synthetic cytokinin to pre-anthesis ovaries resulted in parthenocarpic fruit formation by activating cell division (Matsuo *et al.*, 2012). The precise role of seeds in the initiation of fruit growth in tomato has not been determined, though it has been suggested that they may be sources of auxin, which stimulates fruit swelling. Gibberellins are also involved in fruit growth initiation, for high levels of endogenous gibberellins in the ovaries have been observed in cultivars that exhibit parthenocarpy. The genetic background of parthenocarpy in tomato is discussed in Chapter 2.

FRUIT DEVELOPMENT

Like most developmental processes, the rate at which an initiated fruit progresses towards ripeness is primarily dependent on temperature. The relation

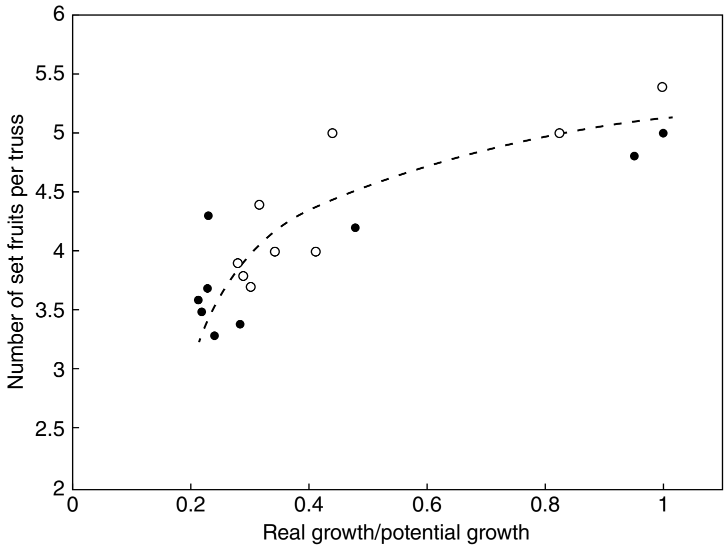

Fig. 3.5. Number of set fruits on the first nine inflorescences of beefsteak tomato 'Capello', grown in CO_2-enriched (○) or non-enriched (•) polycarbonate greenhouses, as a function of the ratio between real and potential fruit growth calculated at fruit set of each inflorescence, pruned to seven flowers. (Reprinted, with permission, from Bertin, 1995, *Annals of Botany*, 75.)

between fruit developmental stage and fruit growth rate is also presented here and the genetic control of fruit development is discussed. First, fruit morphology and anatomy are considered. Main references for this section are Chapter 5 of Atherton and Rudich (1986) and the dissertations of De Koning (1994) and Okello (2015).

Fruit morphology, anatomy and development rate

Botanically, a tomato fruit is a berry consisting of seeds within a fleshy pericarp developed from an ovary. Fruits of the cultivated species (*Solanum lycopersicum*) have two to several carpels and final masses from a few to several hundred grams. Based on the number of carpels, cultivars are grouped as: round tomatoes (two to three carpels), beefsteak tomatoes (more than five carpels) and the nowadays popular intermediate types (three to five carpels). Tomato fruits are composed of flesh (pericarp walls and skin) and pulp (placenta and locular tissue including seeds) (Fig. 3.6). In general, the pulp accounts for less than one-third of the fruit fresh mass. Covering the epidermis is a thin cuticle.

The period between anthesis and fruit maturity decreases with increasing averaged temperature from 14°C to 26°C, with a small effect of DIF (De Koning,

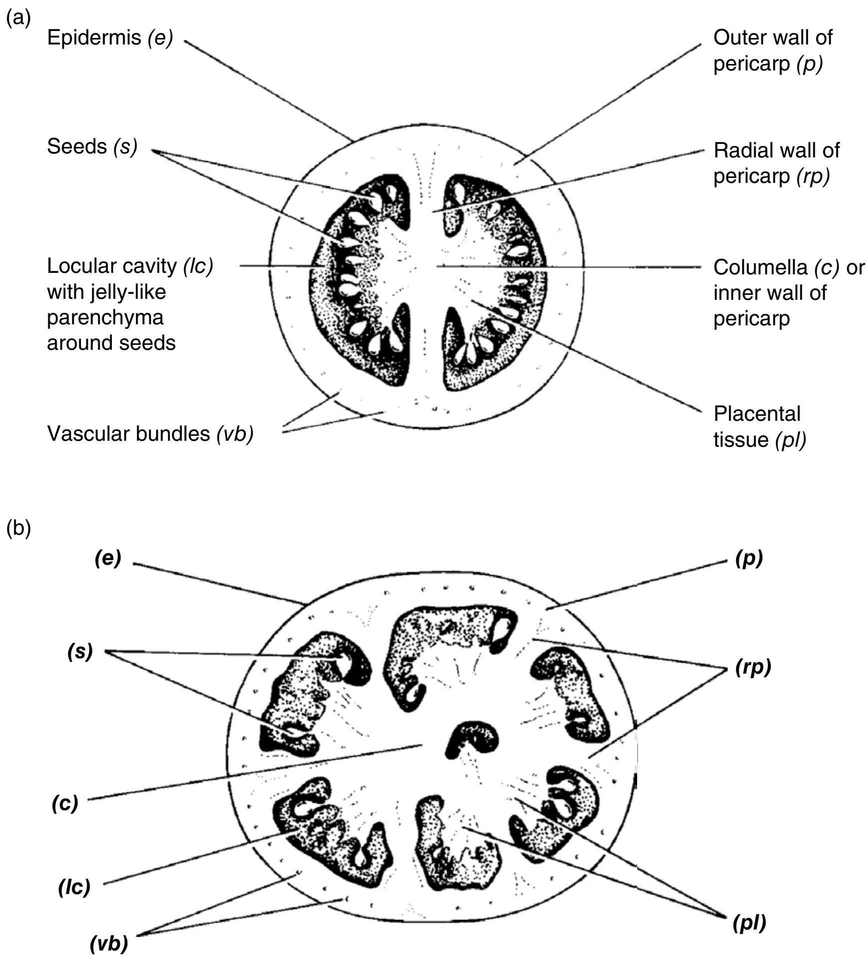

Fig. 3.6. Anatomy of tomato fruits with (a) bilocular or (b) multilocular structure shown as transverse sections (copyright Glasshouse Crops Research Institute).

1994). The fruit growth period can be well described by linearly relating its reciprocal (i.e. fruit development rate) to temperature (Fig. 3.7). This means for harvest-ripe fruits a certain temperature sum has to be reached. This concept of heat units has been explained in the section on germination (pp. 60–63, this volume). For cv. 'Counter', accurate predictions of fruit growth period may be made assuming a temperature sum of 940 degree-days and a base temperature of 4°C. However, the temperature sensitivity of the fruit development rate varies during fruit development, being high during the first weeks after anthesis when cell division and elongation take place, and then decreasing when only cell elongation occurs. When the fruit is close to maturity, increasing temperature enhances

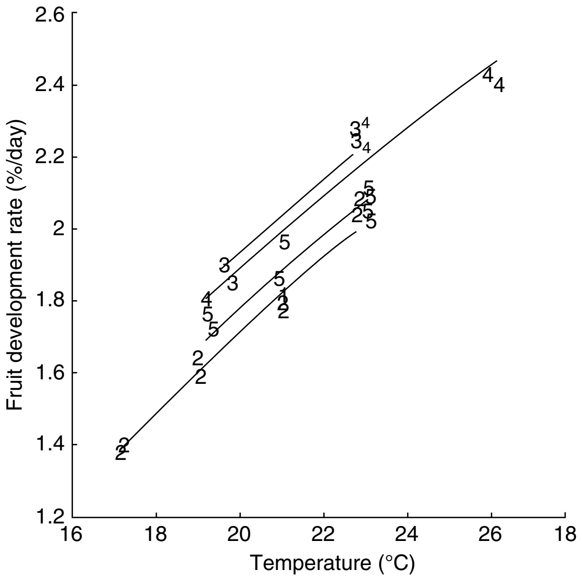

Fig. 3.7. Relationship between temperature and fruit development rate of tomato for five experiments: (1) cv. 'Counter' in spring; (2) cv. 'Calypso' in spring; (3) cv. 'Counter' in summer; (4) cv. 'Calypso' in summer; (5) cv. 'Liberto' in spring (De Koning, 1994).

fruit ripening (De Koning, 1994). Other factors, such as plant density, light intensity, CO_2, air humidity, fruit load, plant age or salinity in the root environment, have no or only a small effect on fruit growth period. Severe water stress shortens the duration of fruit growth period and fruit affected by blossom-end rot will ripen 1–2 weeks earlier. Fruit ripening is considered in detail in Chapter 5.

Fruits grown under source limitation reach the potential growth rate, i.e. growth rate under non-limiting assimilate supply, about 2 weeks after the source limitation has been lifted by severe fruit pruning (Fig. 3.8). This is an interesting observation, as apparently it is the fruit developmental stage that determines its potential growth rate and not the fruit size, which reflects the history of assimilate availability experienced by the fruit.

Genetic control of fruit development

Tomato fruit development is a four-phase process (Fig. 3.9) that begins with initiation of the floral meristem, carpel formation and ovary growth. The second phase involves pollination, fertilization, fruit set and resumption of cell division. During the third phase, cells expand and undergo endoreduplication. Initiation of ripening is the fourth phase and it marks the beginning of fruit senescence. Reviews by Arrizumi *et al.* (2013), Karlova *et al.* (2014) and Azzi *et al.* (2015) on fruit development have been consulted in writing up this section.

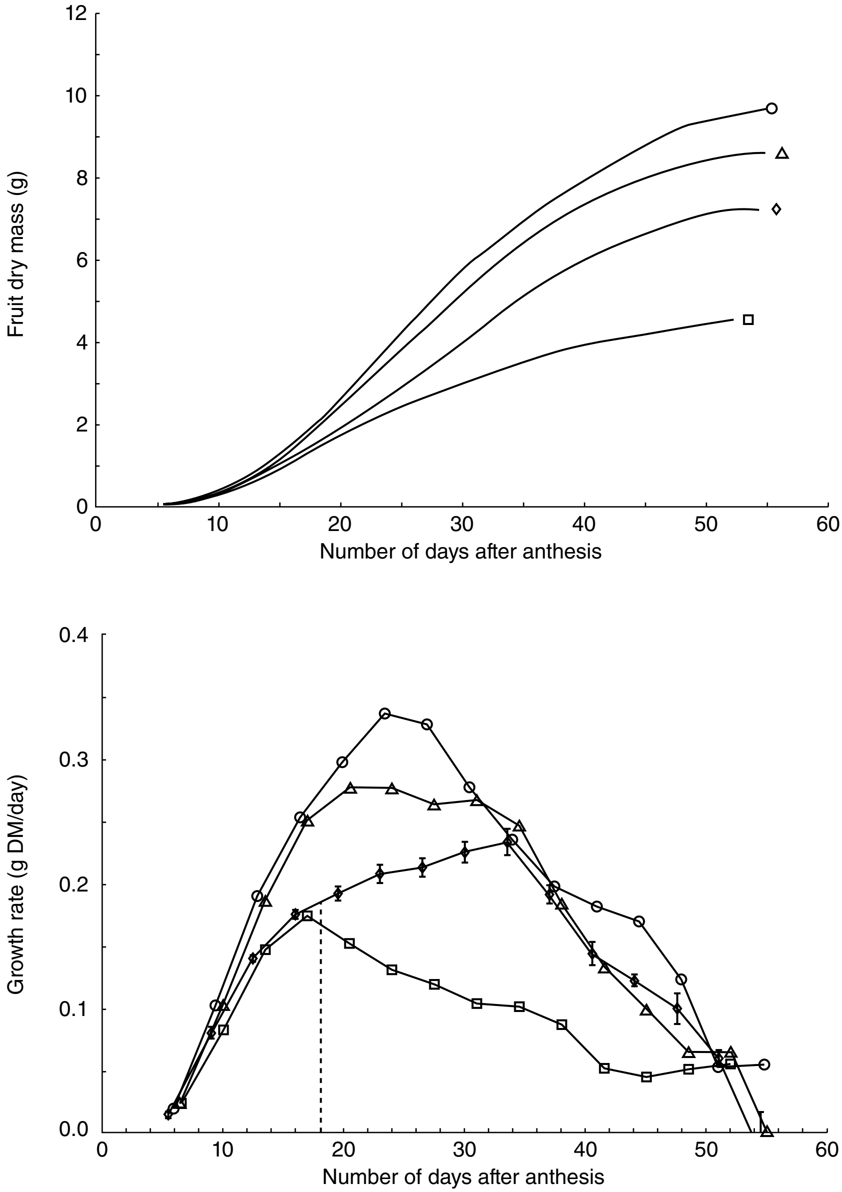

Fig. 3.8. Tomato (cv. 'Counter') fruit growth rate as a function of days after anthesis. Fruits were grown with one (○), two (△) or eight (□) fruits per truss, or seven fruits were removed from each truss from plants with eight fruits per truss at 18 days after anthesis (◊). Growth was calculated from diameter measurements on the second and third fruit on the fourth truss (Heuvelink and Wubben, unpublished data, 1999).

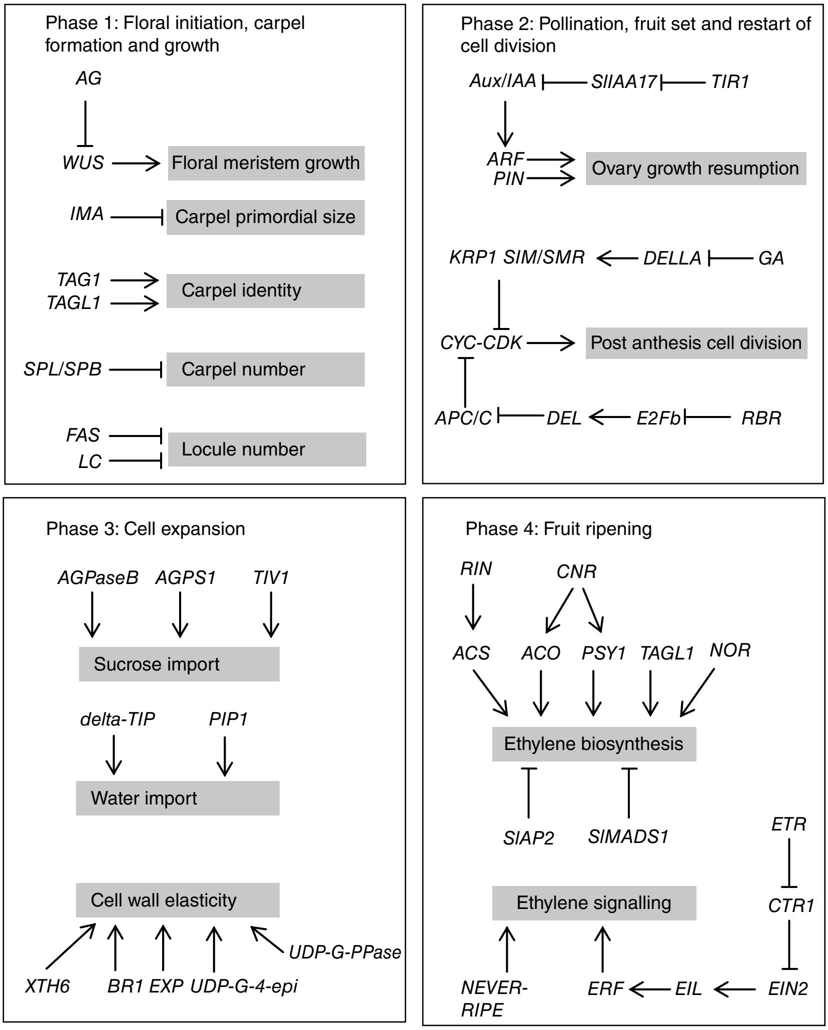

Fig. 3.9. Schematic representation of transcriptional regulation of the four phases of tomato fruit development: (1) floral initiation, carpel formation and growth; (2) pollination, fruit set and restart of cell division; (3) cell expansion; (4) fruit ripening. Lines ending with an arrow represent a positive feedback; blocked lines represent negative feedback. Abbreviations are explained in the text.

Phase 1: Floral initiation, carpel formation and ovary growth

Tomato flower induction occurs when the shoot apical meristem (SAM) is converted into an inflorescence meristem that produces floral meristems (FM). Unlike SAM, the FM exhibits determinate growth through repression of *WUSCHEL* (*WUS*) by *AGAMOUS* (*AG*) gene encoded transcription factors.

WUS promotes cell division and prevents premature differentiation. The FM produces four floral whorl primordia, sepals, petals, stamens and carpels. The fourth innermost whorl, the carpel primordium, fuses to form the ovary with locules that contain ovules. Carpel primordial size is regulated by *INHIBITOR OF MERISTEM ACTIVITY* (*IMA*). Overexpression of *IMA* causes formation of small carpel primordia and ultimately small flowers and fruits. Carpel identity is controlled by *TOMATO AGAMOUS 1* (*TAG1*) and *TOMATO AGAMOUS-LIKE 1* (*TAGL1*). Knock-down of *TAG1* causes defects in stamens, loss of determinacy, and production of flowers-in-flowers or fruit-in fruit-phenotype, while knock-down of *TAGL1* causes style trichome loss and production of a thin pericarp. Another group of carpel number regulators consists of *SQUAMOSA* promoter binding protein-like (SPL/SPB) transcription factors. Knock-down mutants of these transcription factors produce extra carpels and fruit-like structures at the fruit stylar end. The tomato fruit is a multi-carpelar berry and its size is dependent on locule number. The largest variation in locule number has so far been attributed to two transcription factor encoding genes, *FASCIATED* (*FAS*) and *LOCULE NUMBER* (*LC*). The mutant forms (*fas* and *lc*) present in most large-fruited domestic tomato are responsible for high locule number.

Phase 2: Flower pollination, fertilization of ovules, fruit set and re-start of cell division

Before anthesis, cell division in the ovary ceases temporarily. Following successful pollination and fertilization, cell division resumes (fruit set) due to auxin and gibberellic acid (GA) biosynthesis. Auxin transporters (PIN-FORMED (PIN)) and receptors (TRANSPORT INHIBITOR RESPONSE 1 (TIR1)) are important during fruit set, development and growth. Mounet *et al.* (2012) have shown that mutant tomato plants with silenced *SlPIN4* gene produce small parthenocarpic fruits. TIR1 controls degradation of AUXIN/INDOLE-3-ACETIC ACID (Aux/IAA) transcriptional repressors in the presence of auxin, leading to release of auxin response factors (ARFs). Parthenocarpic fruits are formed when *TIR1*, *Aux/IAA* and *ARF* genes are mis-expressed. Larger fruits form when transcriptional repressors of auxin transporters or receptors are silenced. Su *et al.* (2014) have shown that when the repressor (SlIAA17) of *Aux/IAA* was silenced in tomato, larger fruits with thick pericarp were formed. Development of unpollinated ovaries is usually inhibited through reduced expression of genes encoding GA biosynthesis enzymes. Transcript levels of these genes (e.g. those encoding GA 20-oxidases) increase after successful pollination, leading to synthesis of active GA_1 and GA_4. GA interacts with its receptor (GA INSENSITIVE DWARF1 (GID1)) to target DELLA proteins for proteolytic degradation, hence release of GA-responsive gene repression. Silencing of *SlDELLA1* and repression of *SlGA20ox1* causes production of pollen with reduced viability and small elongated facultative parthenocarpic fruits.

Following fruit set, cell division proceeds for about 2 weeks but can be longer (up to 25 days after anthesis) in some cultivars. Cell division is a four-phase process consisting of DNA synthesis (S) and mitosis (M) and two gap phases. During M phase, a cell divides into two. Newly formed cells expand during gap phase 1 (G1) in preparation for S phase, when cellular DNA content doubles. S phase is followed by a second gap phase (G2) in which cell size doubles in preparation for M phase. Phase transition is regulated by catalytic cyclin-dependent kinases (CDK) that are dependent on cyclins (CYC) for activation. Cyclins determine CYC–CDK complex stability, localization and substrate specificity. Seven CDKs and eight CYCs have been reported in tomato (Czerednik, 2012). Activity of the CYC–CDK complex can be modified by (i) proteolytic destruction of the CYC subunit (e.g. by the anaphase promoting complex/cyclosome (APC/C)), and (ii) inactivation by CDK inhibitors (e.g. SIM/SMR, WEE1 and Kip-related protein 1 (KRP1)). Cell cycle regulation can be upstream of CYCs and CDKs through transcription factors. E2F transcription factors are upstream cell cycle regulators that target genes (DEL) involved in DNA repair and chromatin dynamics at the transition between G1 and S. The RETINOBLASTOMA-RELATED (RBR) protein binds the E2Fb transcription factor to prevent cell cycle progression. G1 to S phase transition occurs when CYC–CDK dimers phosphorylate RBR and release E2F transcription factors. Some proteins (e.g. FW2.2) inhibit cell division but their mode of action is still elusive. FW2.2 accounts for up to 30% of the variation in fruit fresh weight of domesticated and wild tomato. *FW2.2* gene expression decreases in tomato fruits when they are grown at high temperature (Okello *et al.*, 2015).

Phase 3: Cell expansion and endoreduplication

Cell expansion is a process that starts after fruit set and continues until the end of fruit growth. By the end of fruit growth, mesocarp cells can attain a size that is 30,000 times that of cells at anthesis. This increase in cell size is caused by import of water and sucrose, and accumulation of macromolecules and organelles. Imported sucrose is broken down into glucose and fructose by vacuolar acid invertase. Glucose molecules are used in the formation of ADP glucose, the building blocks of starch. ADP glucose formation is catalysed by ADP glucose pyrophosphorylase (ADPGPP) while starch synthase catalyses starch formation from ADP glucose. ADPGPP consists of a large and small subunit encoded by *AGPS1* and *AGPaseB* respectively. Towards the end of fruit growth, starch is broken down into fructose and glucose in a reaction catalysed by starch phosphorylase. The expression of genes encoding these enzymes markedly influences fruit growth. Silencing of the gene (*TIV1*) encoding vacuolar invertase biosynthesis, for example, causes reduced fruit size, high sucrose but low fructose and glucose content. Guan and Janes (1991) demonstrated remarkable differences in fruit size of *in vitro* grown fruits exposed to light or darkness. They attributed the large fruit phenotype observed in light to higher rates of starch accumulation due to activation of ADPGPP by light.

Water uptake during cell expansion is driven by an osmotic potential gradient caused by carbohydrate accumulation. Water channels (aquaporins) facilitate water entry into cells. The expression of two aquaporin encoding genes (delta-tonoplast integral protein (delta-TIP); plasma membrane intrinsic protein (PIP1)) increases between 28 and 42 days after anthesis (Prudent *et al*., 2010). Cell expansion following water and carbohydrate uptake depends on cell wall elasticity. Therefore factors influencing cell wall elasticity such as new cell wall deposition, composition, bonding between wall components and enzymatic modification of the wall significantly affect cell expansion. Expansin (*EXP*) genes encode proteins that act as a molecular grease between wall components, thereby increasing cell wall elasticity. Prudent *et al*. (2010) studied the expression of two cell wall synthesis associated genes (UDP-glucose-4-epimerase (*UDP-G-4-epi*); UDP-glucose-pyrophosphorylase (*UDP-G-PPase*)) and three cell wall degradation genes (polygalacturonase (*PG*) and two xyloglucan endotransglycosylases (*XTH6* and *BR1*)) in tomato. Their findings showed that under non-source-limiting conditions, *UDP-G-4-epi* expression increases with fruit maturity but declines towards ripening. The expression of *PG* is similar to that of *UDP-G-4-epi* except that no decline in expression is observed towards ripening. Expression of *UDP-G-PPase*, on the other hand, does not change while that of *XTH6* and *BR1* drops initially and levels off during fruit development.

Cell expansion in tomato fruit is positively correlated with endoreduplication, a modified cell cycle in which cells undergo a repeated gap and DNA synthesis phase leading to ploidy levels ranging between 4C and 512C (where C is the haploid DNA content in picograms). Inhibitors of the CYC–CDK complex such as WEE1, KRP1, SIAMESE (SIM)/SIAMESE-RELATED (SMR) proteins promote endoreduplication. Promotion of endoreduplication also occurs through inhibitory competition between E2F transcription factors. For example, E2Fc transcription factors inhibit activity of the cell cycle promoting *DEL1* when they occupy the DNA binding site occupied by E2Fb during the normal cell cycle. Endoreduplication causes an increase in nuclear size. A positive effect of endoreduplication on transcription has also been shown in some studies. It is thought that the positive correlation between ploidy level and cell size stems from a causal relationship between nuclear and cytoplasmic growth as proposed in the karyoplasmic ratio theory.

Phase 4: Ripening

Ripening marks the beginning of senescence. It involves a change in colour from green to red or yellow. Colour change is associated with accumulation of carotenoids which results from breakdown of the thylakoid membrane in chloroplasts. Dark-grown fruit are white due to lack of chloroplasts but they contain organelles (etioplasts and amyloplasts) that are capable of carotenoid accumulation and red colour formation during ripening. Expression of carotenoid biosynthesis genes (e.g. *PHYTOENE SYNTHASE* (*PSY1*)) increases during ripening. PSY1 catalyses the first step in the carotenoid biosynthetic pathway and is a precursor for the red pigment, lycopene. Flavonoids and anthocyanin also accumulate in the fruit peel during ripening and significantly affect fruit colour.

Tomato fruit are climacteric, hence they exhibit marked increase in respiration and ethylene production at the onset of ripening. Ethylene production begins with conversion of *S*-adenosymethionine (SAMe) to 1-aminocyclopropane-1-carboxylic acid (ACC) by ACC synthase (ACS). ACC is then converted into ethylene in a reaction catalysed by ACC oxidase (ACO). Regulation of ripening via ethylene signalling occurs through ethylene biosynthesis and perception by receptors. Important regulators in the ethylene biosynthetic pathway include MADS-domain protein RIPENING INHIBITOR (RIN), COLORLESS NON-RIPENING (CNR), and NAC domain transcription factor encoded by the gene underlying the non-ripening (*nor*) mutation. Mutants (*rin* and *cnr*) in which RIN and CNR expression are suppressed produce fruits that are unable to ripen. Fruits of these mutants neither produce elevated levels of ethylene nor respond to exogenous ethylene application.

RIN regulates autocatalytic ethylene production genes and plays a role in aroma formation by regulating *LIPOXYGENASE* (*LOX*) genes. The *rin* mutants exhibit suppressed expression of two genes, *SlACS2* and *SlACS4*, involved in autocatalytic ethylene production. RIN also interacts with other ripening-promoting transcription factors, for example TOMATO AGAMOUS-LIKE 1 (TAGL1) and FRUITFUL 1 and 2 (FUL1/TDR4 and FUL2/MBP7). TAGL1 induces autocatalytic ethylene production while effects of FUL1 and FUL2 are ethylene independent (Bemer *et al.*, 2012). Surprisingly, RIN appears to play a role in negative regulation of ethylene biosynthesis through promotion of the expression of two genes (*SlAP2a* and *SlMADS1*) that negatively influence fruit ripening. Plants in which *SlAP2a* is repressed produce ripe fruits that are orange in colour, faster senescence and high levels of ethylene production. Expression of *SlMADS1* is high in mature green fruits but decreases as fruits ripen. Plants with silenced *SlMADS1* produce fruits that ripen earlier and produce more ethylene compared with the wild type. CNR, on the other hand, up-regulates the expression of ripening-related genes like *PSY1*, *LOX* and *ACO1*. Mutants (*cnr*) are incapable of carotenoid biosynthesis because they lack phytoene and other carotenoid precursors. However, CNR also promotes expression of a negative regulator of ethylene biosynthesis, *SlAP2*. The final step in ethylene biosynthesis requires activity of ACO. Silencing of its transcription factor, LeHB-1, reduces expression of *LeACO1*, leading to inhibition of ripening. Another recently discovered gene (*SlNAC4*) appears to be a positive regulator of ripening (Zhu *et al.*, 2014). Suppression of its expression results in fruits with delayed ripening, decreased ethylene biosynthesis, suppressed chlorophyll degradation and reduced carotenoid phenotype.

Ethylene receptors are encoded by *ETHYLENE RESPONSE* (*ETR*) genes. *ETR* genes play a role in ethylene signalling by removing the block on ETHYLENE INSENSITIVE 2 (EIN2) exerted by CONSTITUTIVE TRIPLE RESPONSE 1 (CTR1). The released EIN2 promotes *EIN3/EIN3-like* (*EIL*) transcription factor gene activity which activates transcription of ETHYLENE RESPONSE FACTORS (ERFs). In general, ERFs positively mediate ethylene signalling. For example, *LeERF1* positively mediates ethylene signalling while *Sl-ERF.B3* is active in feedback mechanisms that regulate ethylene production and responses. Other genes that regulate ripening via ethylene signal transduction include ethylene receptor genes, *NEVER-RIPE*, *ETR6* and *GREEN-RIPE*.

SUMMARY

- Germination rate depends linearly on temperature in the suboptimal temperature range, with a minimum temperature of 8.7°C and a heat sum requirement of 88 degree-days; 13°C is considered a 'practical' minimum temperature for germination.
- Leaf and truss appearance rate and fruit development rate depend linearly on temperature (range 17–23°C) and show hardly any influence of assimilate supply (e.g. light intensity, CO_2 concentration, fruit load, plant density or removal of just-unfolded leaves).
- Number of leaves below the first inflorescence correlates negatively with assimilate availability to the apex during a sensitive phase (approximately the first 10 days from cotyledon expansion).
- Fruit set is positively correlated with assimilate availability, and both auxin and gibberellins play a role. Poor fruit set under high or low temperatures results from poor pollen quality.
- The identity of tomato carpels is regulated by *TOMATO AGAMOUS 1* (*TAG1*) and *TOMATO AGAMOUS-LIKE 1* (*TAGL1*) genes.
- Locule number is a major determinant of fruit size. Variation in fruit size is attributed to two transcription factors FASCIATED (FAS) and LOCULE NUMBER (LC).
- Fruit set is dependent on genes encoding auxin transporters (PIN) and receptors (TIR1).
- Cyclins and cyclin-dependent kinases orchestrate cell division during early tomato fruit growth.
- Genes that regulate carbohydrate metabolism, endoreduplication and aquaporins determine the extent of cell expansion.
- Tomato fruit ripening is regulated by genes encoding cell wall associated proteins, ethylene biosynthesis and perception, and synthesis of carotenoids and anthocyanin.

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4

Crop Growth and Yield

Ep Heuvelink, Tao Li and Martine Dorais

INTRODUCTION

The fresh fruit yield of a tomato crop can be calculated from total crop biomass production, biomass partitioning (harvest index) and fruit dry matter content (Fig. 4.1). Besides affecting yield, these attributes also influence product quality (e.g. fruit size and taste; see Chapter 5).

Yields of field crops usually range between 40 and 100 t/ha, whereas yields from year-round cultivation in greenhouses in north-west Europe or North America easily exceed 500 t/ha, and yields as high as 700–900 t/ha are obtained in high-tech greenhouses without supplementary light (SL). With the use of SL, over 1000 t/ha have been produced, according to Verheul *et al.* (2012), who estimated that for Norway a yield potential of tomato production of 125–140 kg/m^2 using artificial light is realistic. The main reasons for a much higher yield for greenhouse crops compared with field-grown tomato crops are: (i) the length of the cultivation period (11–12 months; over 35 trusses harvested per stem) and thus higher cumulative light interception and biomass production; (ii) the control and optimization of environmental factors (carbon dioxide (CO_2), temperature, humidity and light); and (iii) intensive cultural practices (hydroponics, fertigation, biological and/or integrated pest and disease control, plant density, additional stems, leaf and fruit pruning, SL). Annual tomato production per unit greenhouse area in The Netherlands increased by 113% between 1983 and 2010 (De Gelder *et al.*, 2012). This increase was due to several factors, including improved climate control, improved crop and pest management and better control of the rooting medium. Also, genetic improvement played an important role, as yield of modern greenhouse tomato cultivars is about 40% higher than that of old cultivars (Higashide and Heuvelink, 2009). This increase is not related to a higher harvest index but rather it is due to an increase in light use efficiency (LUE) following morphological and physiological changes through breeding, which results in an increase in total dry matter production. For fresh market field tomato in North Carolina, genetic gain observed from 1975 to 2009

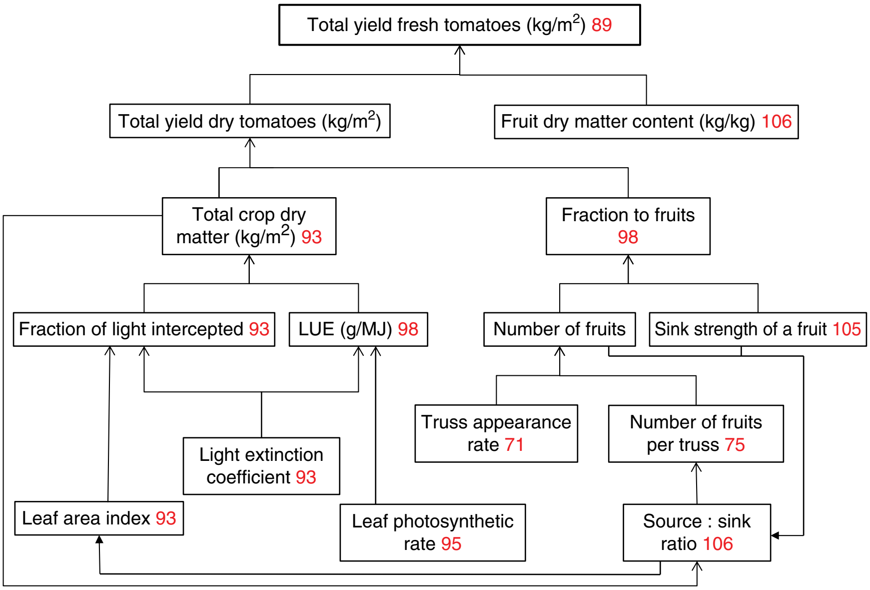

Fig. 4.1. Tomato fruit yield and its underlying components (numbers refer to page numbers in this book). Tomato fruit fresh yield can be higher, because the total fruit dry yield is higher and/or because of more dilution with water (lower fruit dry matter content). (Adapted from Highashide and Heuvelink (2009.)

(~227 kg/ha per year) results mainly from early marketable yield instead of total yield. Total yield reached its maximum level by 2000 and then levelled off, but marketable yield is still being improved. Fruit yield of processing tomato was genetically improved by 1.54% per year in California over a 20-year period (Panthee and Gardner, 2011).

Biomass production is primarily determined by crop photosynthesis, while photosynthesis to a large extent depends on light interception, which furthermore varies with leaf area. Moreover, a high biomass production does not necessarily result in a high yield, since only the tomato fruit is of economic interest. Partitioning to the fruit may be low because of poor fruit set, and consequently low fruit load, or by the low capacity of individual fruits to import photo-assimilate (fruit sink strength). Finally, fruit dry matter content is of major importance, since this parameter determines which fresh fruit mass results from the dry mass partitioned into the fruit. For processing tomato, dry matter content expressed as soluble solids or sucrose accumulation is of considerable economic importance for the food industry as a major determinant of palatability and processing characteristics (see Chapter 5).

Two important interactions (feedback mechanisms) between dry matter production and dry matter distribution in tomato can be distinguished (Fig. 4.2): (i) flower and/or fruit abortion at low source–sink ratio, resulting

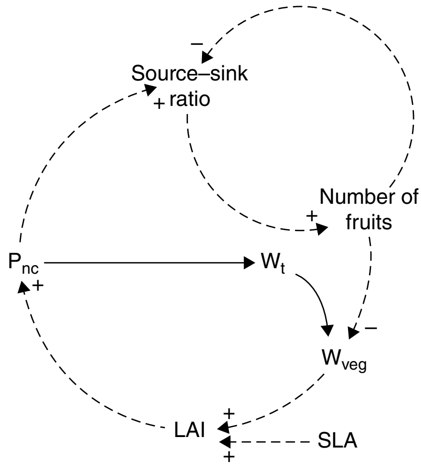

Fig. 4.2. A simplified representation of two important interactions (feedback mechanisms) between dry matter production and dry matter partitioning in an 'indeterminate' tomato crop: (+) positive influence; (–) negative influence. Solid lines represent carbon flow; dashed lines represent information flow. LAI, leaf area index; P_{nc}, crop net assimilation rate; SLA, specific leaf area; W_t, total crop dry mass; W_{veg}, vegetative crop dry mass.

in fewer fruits on the plant and hence decreased sink strength and increased source–sink ratio; and (ii) partitioning to the vegetative parts determining leaf area index (LAI) and hence future light interception and dry matter production. Fruit yield can be considered as the product of total biomass production and the fraction partitioned to the fruits (Fig. 4.1). A larger number of fruits per truss will on the one hand increase partitioning to the fruits. On the other hand, as at the same time less assimilates are partitioned to the vegetative part, larger fruit numbers will reduce LAI, the fraction of intercepted light, and hence total biomass production (Fig. 4.2). Conversely, individual fruit mass increases with decreased fruit number per plant, as competition for assimilates among fruits is reduced. This increase in individual fruit mass results from a higher average growth rate of individual fruits, as fruit growth period (time from anthesis until harvest ripe) is hardly affected by fruit load (see Chapter 3). At the commercial scale, fruit pruning is generally applied to obtain a certain desired average fruit weight. Because of these two counteracting effects, fruit yield shows an optimum response to fruit number per truss, as shown in Fig. 4.3 (Heuvelink and Bakker, 2003).

Crop growth and yield determining processes mentioned above will be considered in detail in this chapter, with emphasis on the influence of environmental factors and crop management such as light, CO_2, temperature, humidity, salinity, plant density, and leaf and fruit pruning. Guidelines on how to grow tomato in experiments with a controlled environment are provided by Schwarz *et al.* (2014).

GROWTH OF YOUNG PLANTS

Young plants usually show exponential growth with constant relative growth rate (RGR), as mutual shading of leaves is limited (Hunt, 1982):

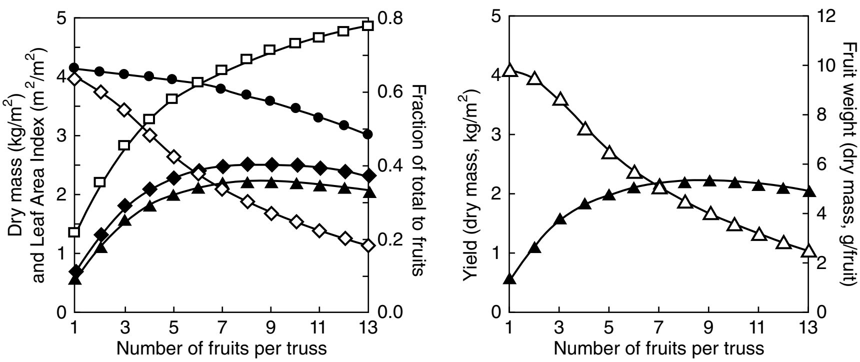

Fig. 4.3. Simulated (TOMSIM) effects of the number of fruits per truss for a round tomato cultivar on leaf area index (◇), the fraction dry mass distributed to the fruits (□) and on fruit dry mass production: total fruit including unripe fruits (◆), harvested ripe fruit (▲), total biomass (●) and average dry weight per fruit harvested (△).

$$W_{t2} = W_{t1} \times \mathrm{e}^{\mathrm{RGR}(t2-t1)}$$

where W_{t1} and W_{t2} represent plant dry mass at times $t1$ and $t2$, respectively. RGR can be separated in a 'photosynthetic term', net assimilation rate (NAR), and a 'morphogenetic term', leaf area ratio (LAR) (Hunt, 1982):

$$\mathrm{RGR} = \mathrm{NAR} \times \mathrm{LAR}$$

$$\mathrm{NAR} = 1/A_t \times \mathrm{d}W_t/\mathrm{d}t$$

$$\mathrm{LAR} = A_t/W_t$$

where A_t is plant leaf area at time t.

An evaluation of experimental design and computational methods in plant growth analysis is given by Poorter and Garnier (1996). LAR may be further separated into the leaf weight ratio (LWR), i.e. the ratio of leaf dry weight over total plant dry weight, and the specific leaf area (SLA), i.e. the ratio of leaf area over leaf dry weight. Variations in LAR are usually primarily caused by variations in SLA (Hunt, 1982).

Growth analysis has proved highly effective in studying a plant's reaction to environmental conditions. However, it has its limitation, especially when analysing crop growth, where mutual shading of leaves begins within a few weeks of emergence and is the major reason for a rapid decline of both RGR and NAR. Therefore, attempts to correlate these quantities with changes in the environment during the growing season have rarely been successful. Furthermore, NAR and LAR are often mutually dependent, showing a negative correlation, which hampers breeding for cultivars with a higher RGR. Finally, a comparison based on time-averaged growth parameters may be misleading and averaging over a dry weight interval is preferred, as growth rate is size dependent (ontogenetic effect).

BIOMASS PRODUCTION

Cumulative biomass production of a tomato crop basically follows a sigmoid pattern in time. A short exponential growth phase with a constant relative growth rate gradually evolves to a long phase of linear growth, with a more or less constant absolute growth rate. At ripe fruit stage, growth rate will decrease for 'determinate' tomato types, as a result of leaf senescence and reduction in sink strength negatively influencing photosynthesis. For 'indeterminate' tomato types the linear growth phase continues until the end of the cultivation, resulting in an expo-linear growth pattern. However, this type of growth pattern may not be observed, since towards the end of the cultivation plants are generally decapitated, giving the same response as 'determinate' types. Moreover, low light levels observed in late autumn (October and November) and changes in crop light interception as influenced by leaf area development may also reduce growth rate.

Leaf area and light interception

The amount of absorbed (intercepted) light by the canopy is a determining factor in tomato crop growth and biomass production and depends mainly on leaf area, the canopy structure and the distribution of the leaf area over ground area. The relationship between the fraction of light intercepted and LAI (m^2 leaf area per m^2 ground area) follows the Lambert-Beer law ($1 - e^{-k.\mathrm{LAI}}$). The constant in this relationship is called the extinction coefficient *k* and for tomato its value is about 0.7 for diffuse radiation. Hence, at LAI of 3, 90% of the incident light is intercepted. Modern greenhouse tomato cultivars have a lower *k* as a result of a changed morphology (e.g. leaf angles): for example, *k* decreased from 0.85 (in 1950) to 0.57 (in 2002), contributing to a higher light use efficiency resulting from a less heterogeneous vertical light distribution (Higashide and Heuvelink, 2009). For a processing tomato crop, at LAI of 4.5, *k* is around 0.45 (Cavero *et al.*, 1998), while for semi-indeterminate field-grown tomato *k* ranges from 1 early in the season to around 0.2 late in the season (Scholberg *et al.*, 2000). Estimated light interception for LAI of 4–5 is in that case about 50–60% during the period of maximum seasonal fruit development. These values are substantially lower than for greenhouse crops, because of wide row spacing, leading to a percentage ground area covered that could be as low as 55% (see below). In particular, internode length and leaf shape affect the vertical distribution of light in the canopy. A more spacious canopy architecture due to long internodes and long and narrow leaves led to an increase in simulated crop photosynthesis of up to 10% (Sarlikioti, 2011). To optimize the light interception by the canopy according to the incident solar radiation, a crop starts with a rather low plant density (e.g. 2.5 plants/m^2) to avoid any negative effect on fruit size or yield under low light (winter), and then in spring a side shoot is retained on every second plant (see Chapter 9). In this way, stem density is increased (e.g. 3.75 stems/m^2) towards the summer and a higher LAI can be maintained, resulting in a significant yield increase without additional planting cost. It was also reported that the fraction of light intercepted for a north–south orientation of rows was 10% (winter) up to 23% (summer) higher compared with an east–west orientation (Sarlikioti, 2011). LAI is also modulated by greenhouse tomato growers, since leaves are removed up to just above the truss to be harvested (see Chapter 9) or pruned according to the targeted leaf-to-fruit ratio.

For semi-determinate field-grown tomato, after an initial lag phase of 225°Cd (225 degree-days), LAI increased linearly with degree-days after planting (base temperature of 10°C), resulting in an average maximum value of 3.8 attained 11 weeks after planting (Fig. 4.4A) (Scholberg *et al.*, 2000). Near-optimal light interception within the row for field-grown tomato appears to be attained within 4–6 weeks at intra-row spacing of 45 and 60 cm, respectively (Scholberg *et al.*, 2000). Canopy closure between rows, however, does not occur practically and pesticide applications and harvesting operations may become difficult with row spacing under 1.8 m. Thus, based on a maximum observed canopy width of 1.0 m, the percentage ground area covered by the crop would be around 83%, 67% and 55% for row spacings of 1.2, 1.5 and 1.8 m, respectively (Scholberg *et al.*, 2000). For an optimum light interception and fruit yield of field-grown tomato crop, the LAI should be around 4–5 with a plant population of 10,000–60,000 plants/ha, according to the cultivars, soil fertility, growing and irrigation systems (raised bed, mulch, drip irrigation, sub-irrigation), and available solar radiation. Lower LAI values would reduce light interception and increase yield losses due to sunburn, while higher values may delay the onset of fruit production and reduce the effectiveness of foliar pesticides.

For greenhouse and field tomato, the amount and quality of light intercepted by a crop can be enhanced or modified by the presence of reflecting material on the ground, such as white plastic sheets and mulches. Bare soil reflects 10–20% of the photosynthetic active radiation (PAR) (400–700 nm), whereas white plastic sheets on the soil surface, a common practice in greenhouses, may reflect 50–80% of the PAR and increase crop photosynthesis over the whole season by at least 7% for LAI at 3 (Gijzen, 1995). This is especially important for the young crop, where a lot of light is transmitted by the crop and reaches the ground. For determinate type of fresh tomato, young plants grown with white mulch had ~3 times more axillary leaves than plants in black mulch, contributing to a 3.6 times greater ratio of axillary to main leaf area

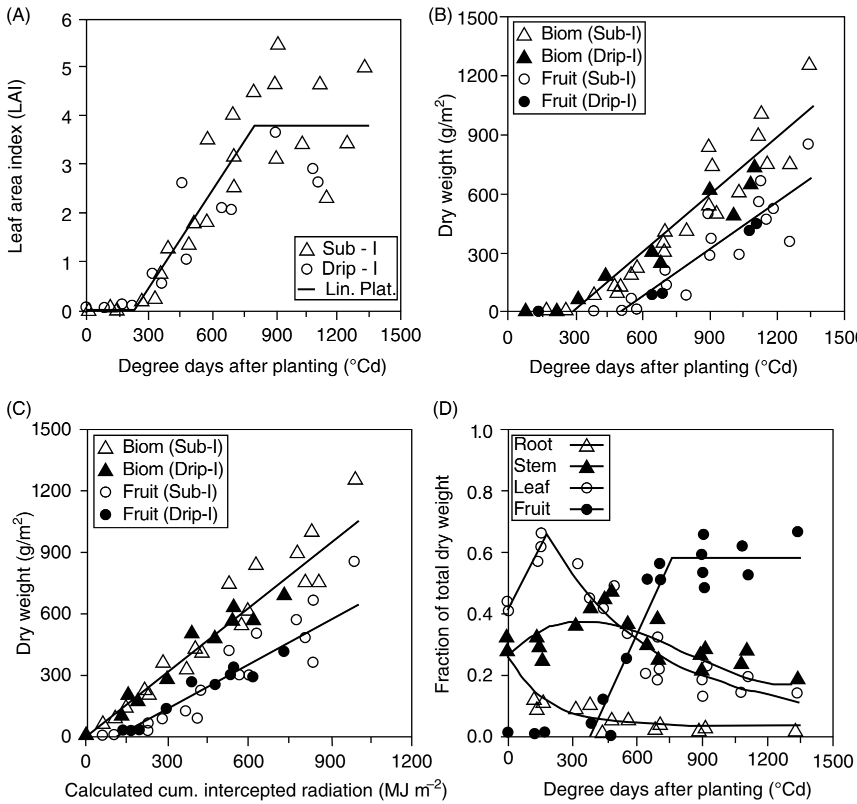

Fig. 4.4. (A) Leaf area index (r^2=0.74); (B) total dry weight (Biom) (r^2=0.91) and fruit dry weight (r^2=0.79) accumulation of field tomato grown in Florida (USA) as a function of thermal time; (C) estimated cumulative intercepted radiation (r^2=0.92 and r^2=0.84, respectively) with subirrigation (Sub-I) and drip irrigation (Drip-I); and (D) fraction of total dry weight accumulated in root (r^2=0.92), stem (r^2=0.50), leaf (r^2=0.93) and fruit (r^2=0.81). (Reprinted, with permission, from Scholberg *et al.* (2000), *Agronomy Journal* 92, 152–159.)

compared with plants grown with black mulch. However, leaf area for total leaves (main + axillary) and plant biomass was unaffected by mulch surface colour. This mulch effect is not maintained throughout plant development, due to the increased shading of the mulch by the canopy (Decoteau, 2007).

Photosynthesis

The leaf photosynthetic rate depends strongly on the amount of photosynthetic protein per leaf area, CO_2 conductance through the stomata and mesophyll conductance. While the light-harvesting and electron transport systems convert the free energy of absorbed light into useful metabolic forms (e.g. adenosine triphosphate (ATP), reduced ferredoxin (Fd), nicotinamide adenine dinucleotide phosphate (NADPH)), CO_2 assimilation is a key process in crop production. PAR supplies the energy for photosynthesis. The RuBisCO enzyme (ribulose-1,5-*bis*-phosphate-carboxylase-oxygenase) catalyses the incorporation of CO_2 (carboxylation), but RuBisCO also has affinity to O_2 (oxygenation), resulting in release of CO_2 (photorespiration). RuBisCO contains as much as 25–40% of the total leaf nitrogen (N) and so it is understandable that leaf nitrogen and leaf photosynthetic capacity are strongly correlated. N deficiency decreased maximal rate of gross photosynthesis ($P_{g,max}$) and the chlorophyll fluorescence parameter F_v/F_m, which is a quantitative measure of photochemical efficiency of photosystem II (PSII). The relationship between nitrogen status of the tomato plant and photosynthesis is covered in Chapter 6.

Photosynthetic capacity of leaves varies widely according to light, water and nutrient availability and these differences in capacity usually reflect RuBisCO content. Leaves in high light environments ('sun' leaves) have greater CO_2 assimilation capacities than those in shaded environments or lower in a canopy (Fig. 4.5) (Trouwborst *et al.*, 2011) and this is reflected in the larger allocation of nitrogen-based resources to photosynthetic carbon reduction. Sun leaves have a high stomatal density, are thicker and have a higher ratio of RuBisCO to chlorophyll in order to utilize the larger availability of photons (and

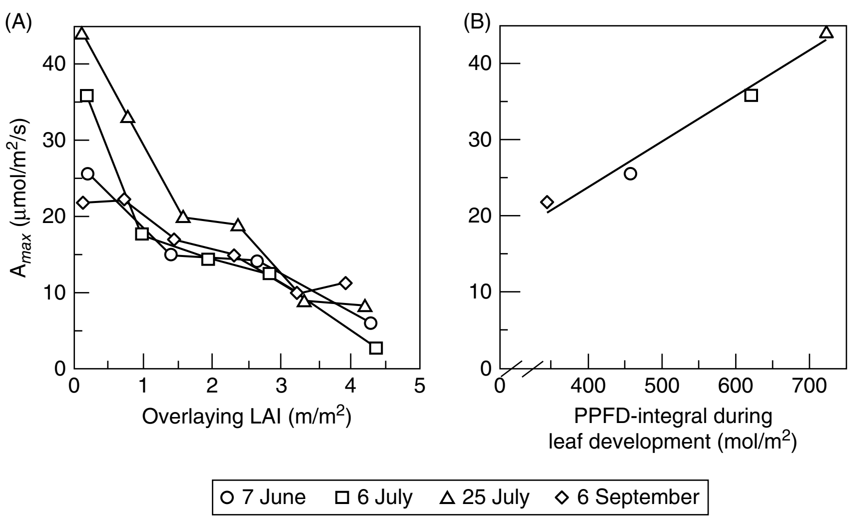

Fig. 4.5. (A) The effect of overlaying leaf area index (LAI) on the A_{max} of leaves in vertically grown tomato plants measured on four dates in summer and (B) the effect of the PPFD-integral received during 21 days of leaf development on the A_{max} of the youngest mature leaves. Reprinted from Trouwborst *et al.*, 2011, by permission of Taylor and Francis Ltd (http://www.tandfonline.com).

hence ATP and NADPH). Shade leaves are larger and thinner, but have more chlorophyll per unit leaf dry weight than sun leaves. Shade leaves achieve a lower maximum rate of assimilation.

To uncouple leaf age from photosynthetic photon flux density (PPFD), tomato plants were grown horizontally, hence PPFD was similar for all leaves. Photosynthetic capacity and chlorophyll contents were higher in late spring than in winter but were hardly affected by leaf age (Trouwborst *et al.*, 2011). Shading fully developed mature leaves dramatically decreased their $P_{g,max}$ and chlorophyll contents within a few days. Therefore, during the normal 70-day lifespan of tomato leaves in commercial cultivation, the decrease in PPFD within the canopy, and not leaf-ageing, is the most important factor causing reduction in $P_{g,max}$ with canopy depth. The lower $P_{g,max}$ of older leaves is attributed to a reduced content of RuBisCO rather than a lower RuBisCO activity.

In a tomato canopy of LAI 8.6, the uppermost 23% of the total leaf area assimilated 66% of the net CO_2 fixed by the canopy (Acock *et al.*, 1978). In contrast to individual leaves, the light saturation for canopy photosynthesis does not occur for a tomato crop, as part of the canopy does not reach light saturation. In conditions where the rate of photosynthetic CO_2 assimilation exceeds the capacity of sink organs (e.g. high light and CO_2 concentration), there is an accumulation of carbohydrates in the leaves which might trigger a down-regulation of photosynthesis involving a repression of several photosynthetic genes encoding for RuBisCO and thylakoid proteins.

Truss peduncle, pedicels, calyxes and green fruit may contribute to a certain extent to photosynthesis. An early study reported that fruit contributes by its own fixed carbon 10–15% of the carbon skeletons required (Tanaka *et al.*, 1974). At light saturation, the photosynthetic rate in young truss peduncles reached 5 mg CO_2/h/g dry weight (DW) compared with 3.5 mg CO_2/h/g in older ones, while the gross photosynthetic rate per green fruit surface area was 15–30% of that in the leaves (Czarnowski and Starzecki, 1990). For small fruit (10 g fresh weight (FW)), $P_{g,max}$ reached 0.43 μmol CO_2 /kg/s, which was similar to a 3 cm^2 leaf (Xu *et al.*, 1997). As the fruit size increased, $P_{g,max}$ decreased to a negligible value and no net photosynthesis was found in a fruit over 40 g FW because of high dark respiration. Except for young fruit, respiration always exceeds gross photosynthesis in tomato fruit, even under high light. The accumulation of dry matter in a tomato fruit is thus dependent on the import of leaf assimilate. Although fruit photosynthesis is not necessary for fruit energy metabolism or development, it is essential for properly timed seed development and therefore may confer an advantage under conditions of stress (Lytovchenko *et al.*, 2011).

Whereas steady-state photosynthetic responses to environmental factors have been extensively studied, knowledge of dynamic modulation of photosynthesis remains scarce and scattered. However, incident irradiance on plant leaves often fluctuates, causing dynamic photosynthesis. SL provided by light-emitting diodes (LEDs) can be switched on and off rapidly; hence, growers using LEDs could use fluctuations in energy prices to determine when to switch their lighting on and off, resulting in a more dynamic irradiance. Dynamic photosynthesis is separated into sub-processes related to proton and electron transport, non-photochemical quenching, control of metabolite flux through the Calvin cycle (activation states of RuBisCO and RuBP regeneration, and post-illumination metabolite turnover) and control of CO_2 supply to RuBisCO (stomatal and mesophyll conductance changes) (Kaiser, 2016). Increases in ambient CO_2 concentration and temperature (up to about 35°C) enhance rates of photosynthetic induction and decrease its loss, facilitating more efficient dynamic photosynthesis. Depending on the sensitivity of stomatal conductance, dynamic photosynthesis may additionally be modulated by air humidity. However, RuBisCO activase and stomatal conductance are main targets for improvement of photosynthesis in fluctuating irradiance.

Light use efficiency

Quantum yield or LUE, expressed as mol CO_2 fixed per mol photons absorbed, can be defined at leaf level but also at crop canopy level. Its value depends on light intensity, atmospheric CO_2 concentration and humidity, the composition of solar radiation (higher LUE at diffuse radiation; see 'Diffuse light' pp. 111–112) and root environment (soil moisture, salinity). At ambient CO_2 concentration, leaf LUE is about 0.05 mol/mol, with a maximum of 0.08 mol/mol without photorespiration, i.e. at very high CO_2 levels. A combination of CO_2 enrichment and SL for greenhouse crops grown in winter in northern regions could have a synergistic effect in increasing LUE at crop level. A high temperature, however, increases (photo)respiration and consequently decreases LUE. At ambient CO_2 concentration, LUE decreases by about 15% when temperature rises from 15°C to 30°C. In greenhouse-grown tomato plants, LUE may be higher than in field-grown plants because of the ultraviolet (UV)-protective epidermis of field plants. Higher LUE of modern greenhouse cultivars (3.3 to 3.5 g/MJ PAR compared with 2.4–2.5 g/MJ PAR) results from a decrease in the light extinction coefficient and an increase in leaf photosynthetic rate (Higashide and Heuvelink, 2009). For the humid tropics of Central Thailand, in ventilated greenhouses with polyethylene (PE)-film roofs and PE-net walls, LUE of 2.6 g/MJ was reported, which is similar to regions at greater latitude with much lower global radiation (Kleinhenz *et al.*, 2006). For field-grown fresh tomato, LUE of 2.1 g/MJ (PAR) was observed (Scholberg *et al.*, 2000) (Fig. 4.4C), while 2.4 g/MJ is generally used for processing tomato (Cavero *et al.*, 1998).

BIOMASS PARTITIONING

The total dry matter production and partitioning within a tomato plant differs considerably between tomato genotypes and depends on the developmental stage. In young tomato plants, 8% (high N supply) to 24% (low N supply) of total dry matter was distributed to the roots (Nagel *et al.*, 2001), while 4% was allocated to the roots in producing tomato plants under ample nutrient supply (Khan and Sagar, 1969). For a long-season crop, the total yield is determined by the balance between the vegetative and reproductive growth and consequently the assimilate partitioning. In a tomato plant, assimilate availability (source strength) is usually lower than assimilate demand (sink strength), which is shown by increased fruit size when some of the fruits are removed at an early stage. Increased light intensity and CO_2 concentration improve source strength, whereas sink strength primarily depends on temperature. Hence, competition occurs between vegetative and generative plant organs, among trusses and among fruits within a truss.

Only biomass allocated to the fruits contributes to yield (Fig. 4.1); therefore, biomass allocation has a direct impact on tomato crop production. For a year-round greenhouse crop, 72% of total dry mass was allocated to the fruits (Table 4.1) (De Koning, 1993), while a harvest index (HI) of 69% (plant without side shoots) was reported (Cockshull *et al.*, 1992) compared with 53–71%, with an average of 58%, for field-grown semi-determinate tomato (Scholberg *et al.*, 2000) (Fig. 4.4D) and 57–67% for processing tomato (Cavero *et al.*, 1998; Hewitt and Marrush, 1986). High-yielding crops typically have HI values of about 65%. Higher and lower values may be observed according to the number of trusses harvested, the crop management and the season (e.g. winter versus spring/summer/autumn). In an indeterminate plant, the fresh weight gain by fruit accounts for almost 85% of the plant gain, because fruits accumulate more water than other organs (Table 4.1). A certain balance between vegetative (future production potential) and generative growth (short-term productivity) should be maintained, as sufficient but not too much new leaf area has to grow for future light interception and biomass production. Besides its influence on total fruit mass, partitioning also influences individual fruit mass in tomato (Fig. 4.3).

Although the mechanism by which a plant partitions its resources between the different organs is of both theoretical and practical interest, it is still not fully understood. It is generally agreed that sinks play an important role

Table 4.1. Total performance of an early tomato crop (De Koning, 1993).

	Fresh weight		Dry weight	
	kg/m^2	%	kg/m^2	%
Fruits	51.7	84.2	2.96	71.5
Leaves	6.6	10.7	0.76	18.2
Stem	3.1	5.1	0.43	10.3
Total	61.4		4.15	

in partitioning, and in a tomato plant the fruits are the most important sinks. Biomass allocation in tomato is both dynamic and complex. It involves the transport of assimilates from sources to sinks. Hence sources, the transport path and sinks may influence allocation. Evidence for this comes from a range of studies. Changes in the leaf-to-fruit ratio, modification of leaf photosynthetic activity by shading, or altering of the activity of some leaf metabolic enzymes have all had repercussions in fruit yield and/or sugar accumulation (Luengwilai *et al.*, 2010).

Hormonal signals, such as cytokinins, abscisic acid, the ethylene precursor 1-aminocyclopropane-1-carboxylic acid and the auxin indole-3-acetic acid may coordinate assimilate production and biomass partitioning (Pérez-Alfocea *et al.*, 2010). Hormonal regulation of source–sink relations during the osmotic phase of salinity, which is independent of the ionic effect, affects whole-plant energy availability to prolong the maintenance of growth, root function and ion homeostasis and could be critical to delay the accumulation of sodium (Na^+) or any other ion to toxic levels (Pérez-Alfocea *et al.*, 2010). Initial growth maintenance requires regulation of stomatal conductance and source–sink relations to avoid premature stomatal and metabolic inhibitions of photosynthesis, and subsequent oxidative damage and senescence.

Shoot/root ratio

Despite the fact that it has been challenged, the functional equilibrium is a concept that helps in understanding the partitioning of assimilates between shoot and root. According to this paradigm, partitioning between shoot and root is regulated by an equilibrium between root activity (water or nutrient absorption) and shoot activity (photosynthesis):

$$W_s/W_r \propto S_r/S_s$$

where W_r is root mass, W_s is shoot mass, S_s is specific photosynthesis rate of the shoot and S_r is specific absorption rate of the root. Observations of environmental effects on shoot/root ratio in tomato are well explained by this theory. For example, shoot/root ratio is lower for plants grown in higher irradiance, which increases specific shoot activity. Similarly, shoot/root ratio is higher for plants grown at higher root temperature or improved nutrition, which improves specific root activity. Increasing the temperature of the root zone from 14°C to 26°C increased the quantity of water absorbed during a day by 30%, the rate of absorption of nitrogen (N), potassium (K) and magnesium (Mg) by 21–24% and the rate of absorption of calcium (Ca) and phosphorus (P) by 45% and 64%, respectively (Dorais *et al.*, 2001a).

The functional equilibrium paradigm is useful in understanding partitioning in young, vegetative tomato plants. However, it is not easily applied to fruiting plants. The shoot/root ratio increases as plants grow and in the early fruiting phase root growth often ceases completely. It later resumes and thereafter the ratio of vegetative shoot-to-root fresh weight remains essentially constant (4:1).

Source strength

The source strength (availability of assimilates) has no direct influence on assimilate partitioning, as different assimilate availability affects growth rate of all organs to the same relative extent. For example, despite a large reduction in assimilate availability at high plant density (plants grown at 1.6, 2.1 and 3.1 plants/m^2 reached a total dry mass of 596, 478, 340 g/plant, respectively), biomass allocation was not influenced (Heuvelink, 1995b). Indirect effects, through the number of sinks (sink strength), were in this experiment excluded by fruit pruning. However, at very low assimilate availability, priority of the apex over an initiating inflorescence has been found. Under such conditions the inflorescence, or part of its flowers, may abort, while the growth of young leaves continues.

Transport path

Supply of assimilates from leaves to trusses in a multi-truss tomato plant is rather localized, the three subtended leaves of a truss being the principal suppliers. A truss together with the three leaves immediately below it has been regarded as a sink–source unit, though this relationship is not absolute. As fruits attract assimilate rather locally, the supply of assimilate to the apex and the roots may be mainly confined to a few upper leaves and the bottom leaves, respectively (Ho, 1996). Within each truss, the fruits on one side tend to receive more assimilate from leaves on the same side of the stem, even if this distribution is not absolute. Removing a truss at anthesis results in yield increases on some of the remaining trusses closest to the one removed (Fig. 4.6). These observations seem to suggest that the transport path (phloem resistance) plays an important role in biomass allocation in tomato plants. However, a study in double-shoot tomato plants where on half of the plants no trusses were removed from one shoot and all trusses were removed at anthesis from the other shoot (100–0), whereas on the other plants every other truss was removed from both shoots (50–50), showed that biomass allocation was the same for both treatments (Fig. 4.7). No significant influence of distance between source and sink on partitioning was found. Hence, biomass allocation may be considered to originate from one common assimilate pool. This does not exclude the possibility that in intact plants trusses are fed predominantly by the leaves nearby and it also does not conflict with previous observations, which can be easily explained without assuming a 'distance effect' on assimilate partitioning. Trusses closest to the one excised get the highest yield increase. Earlier-initiated trusses have a shorter growth period left to take profit from removing a truss and later-initiated trusses miss a larger part of the period

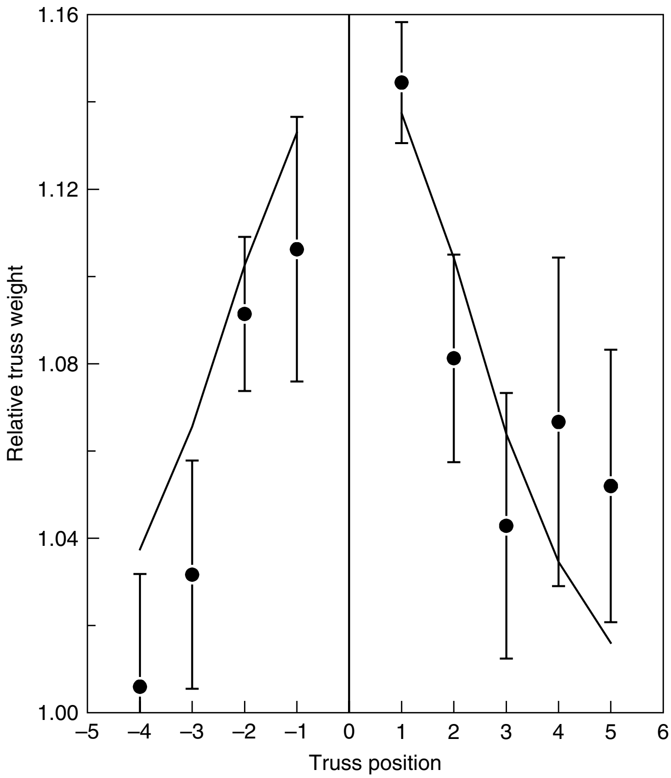

Fig. 4.6. Relative truss weight over controls for trusses above (positive numbers) and below (negative numbers) an excised truss (0) on tomato plants. No influence of position on the plant on sink strength of a truss or vegetative unit was assumed. Vertical bars indicate standard error of mean. Measurements (•) reprinted, with permission, from Slack and Calvert, 1977, *Journal of Horticultural Science* 52, 312, and simulation lines (TOMSIM).

where removal of the truss plays a role. During one fruit growth period (about 60 days at 20°C) after expected anthesis date of the excised truss, its removal no longer plays a role. Furthermore, trusses closest to the excised truss exhibit highest sink strength in the period where excision has the largest influence on total sink strength, i.e. the period where the highest sink strength of the excised truss would have occurred. In conclusion, transport path plays only a minor role in biomass allocation in tomato.

Sink strength

Partitioning may be analysed according to the concept of sink strengths. The term sink strength is used to describe the competitive ability of an organ to

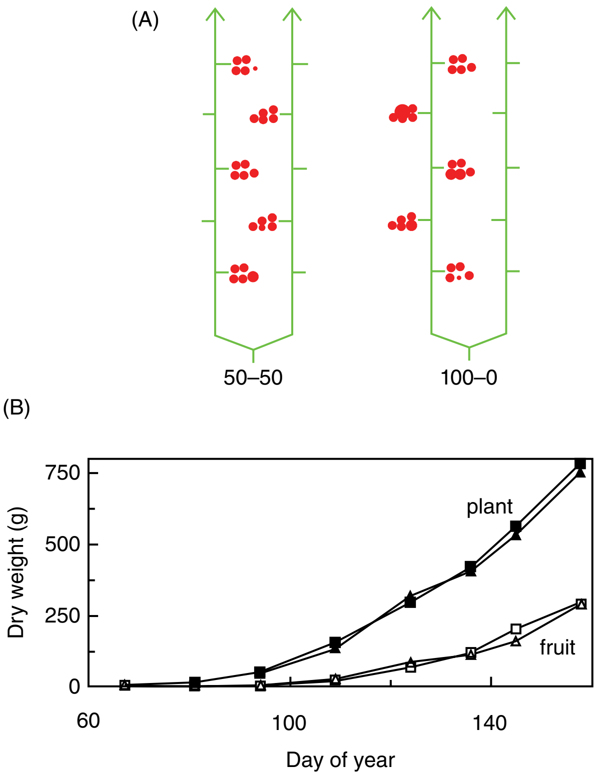

Fig. 4.7. (A) Schematic presentation of two pruning treatments on tomato plants with two equal stems: removal of every second truss at anthesis on each of the two stems (50–50) or removal of all trusses at anthesis from one stem and no truss removal from the other stem (100–0). (B) Total (closed symbols) and fruit (open symbols) dry weight increase for pruning treatment 50–50 (squares) and 100–0 (triangle).

attract assimilates. Sink strength can be quantified by the potential growth rate of a sink, i.e. the growth rate under conditions of non-limiting assimilate supply, and depends on sink activity and size. Whereas sink activity is determined by processes such as phloem transport, metabolism and compartmentation, sink size is determined by the cell number (Ho, 1996).

Since sink strength determines dry matter distribution, the fraction of dry matter partitioned to the fruits (F_{fruits}) may be calculated as generative sink strength (SS_{gen}) divided by total plant sink strength, the latter being the sum of generative and vegetative sink strength (SS_{veg}):

$$F_{fruits} = \frac{SS_{gen}}{SS_{gen} + SS_{veg}}$$

Generative sink strength is the integral of sink strength of all trusses growing on the plant. Sink strength of a truss is proportional to the number of fruits per truss. However, at high fruit numbers per truss, this is no longer so, as the distal fruits show a reduced sink strength (De Koning, 1994). Nevertheless, cell size measured on proximal fruits was not significantly affected by fruit load treatments in one of two studied genotypes (Prudent *et al*., 2010).

The order of priority in assimilate partitioning changes from the order of root > young leaf > flower in flowering plant to that of fruit > young leaf > flower and root in fruiting plants (Ho, 1996). The low sink strength of flowers might be due to low cell division activity, as enhanced cell division activity in the ovary caused by hormonal treatment (cytokinin and gibberellic acid) may generate an improved sink strength. The role of cell division and cell enlargement in fruit sink strength and fruit growth is further discussed in Chapter 3.

Sink strength of a fruit is primarily determined by fruit developmental stage. The fraction partitioned into the fruits shows a saturation-type response with time (Fig. 4.4D): in the early stages only a small proportion is partitioned into the fruits, as only a few small fruits are competing with vegetative growth. Once fruit harvest has commenced, a steady-state constant partitioning is reached, assuming equal numbers of fruit for each truss. Fruits are competing with vegetative parts as a whole, as partitioning within the vegetative parts is fairly constant.

Leaf sink strength

The effect of the second leaf removal of each vegetative unit between two successive trusses when it was only 1–3 cm long was simulated by Xiao *et al*. (2004) (Table 4.2). As expected, the reduced vegetative sink strength by leaf pruning increased partitioning to the fruits from 66% to 74%. However, yield was hardly affected, as leaf pruning resulted in a lower LAI and therefore lower light interception and total biomass production. In contrast, when leaf

Table 4.2. Average LAI, total and fruit dry weight, and fraction partitioned to the fruit in a simulation study for a greenhouse tomato crop (January–September). Dry matter partitioning is simulated based on sink strengths of plant organs. Leaf pruning (removal of one out of each three young leaves) was simulated by reducing the sink strength of each vegetative unit by 1/3. Lowest, old leaves from a vegetative section were removed 1 week before fruits on the corresponding truss above the section were harvest-ripe. Seven fruits per truss were assumed. When delayed leaf removal was applied, removal of old leaves was delayed by 2 weeks compared with the control (Xiao *et al*., 2004).

Treatments	LAI (m^2/m^2)	Total dry weight (g/m^2)	Fruit dry weight (g/m^2)	Fruit fraction (%)
Control	2.4	3190	2093	66
Leaf pruning	1.7	2887	2125	74
Leaf pruning and delayed old leaf removal	2.3	3201	2362	74

pruning was combined with delayed picking of older leaves to obtain the same LAI as in the control, a yield increase of 13% was predicted. Hence, a tomato cultivar with two instead of three leaves between trusses would improve yield, when combined with measures to keep LAI sufficiently high (Heuvelink *et al.*, 2005). On the other hand, it was observed that leaf-to-fruit ratio (LFR) higher than 1 had a negligible effect on fruit growth but could increase fruit solid content as well as the sucrose concentration of the phloem sap. Specifically, when LFR increased from 0.2 to 1, fruit dry weight increased linearly, but it saturated beyond 1. In contrast, the contents of dry matter and soluble sugars on a fresh weight basis increased linearly from LFR of 0.2 to 3 (Jan and Kawabata, 2011).

Fruit sink strength

In developing tomato fruit, sucrose metabolizing enzymes may regulate sucrose unloading and sink strength and thus fruit dry matter accumulation. Acid invertase may not play an important role in the regulation of assimilate import by the tomato fruit (Dorais *et al.*, 1999). However, over-expression of sucrose-phosphate synthase (SPS) in field tomato fruit increased sucrose synthase (Susy) activity by 27%, and 70% more sucrose was unloaded in transformant fruit (20 days after anthesis) compared with the untransformed control. Acid invertase and ADP-glucose pyrophosphorylase (ADP-Glc PPase, or AGPase) activities remained at similar levels or were slightly lower than in the untransformed control. Unexpectedly, the repression of Susy activity in the fruit (antisense cDNA of the tomato fruit specific Susy, TOMSSF) did not affect the unloading capacity when compared with the untransformed plants. Hence, four sucrose turnover cycles (I, sucrose degradation and resynthesis in the cytosol; II, sucrose degradation in the vacuole and resynthesis in the cytosol; III, sucrose degradation in the apoplast and resynthesis in the cytosol; and IV, starch degradation and synthesis in the amyloplast) may control sink activity of tomato fruit (Dorais *et al.*, 1999). Thus, sink utilization (i.e. respiration, cellular structure, growth and storage) determines the sucrose import rate. Sucrose metabolizing enzymes may affect the unloading rate but they are not the main regulatory factors.

Substrate cycles of sucrose and starch may provide metabolic flexibility and help to maintain the fruit as a carbon sink (Luengwilai and Beckles, 2009). Studies using transgenic lines whereby AGPase was over-expressed and suppressed provided evidence that starch plays a more direct role in determining total soluble sugars and yield (Beckles *et al.*, 2012). Indeed, partitioning of carbon to starch may increase sink strength in green fruit and then, when it is degraded during ripening, adds to the pool of sugars imported from the phloem. Approaches to manipulate fruit sink strength in order to improve harvest index and thereby crop yield and quality have been discussed earlier (Herbers and Sonnewald, 1998), while biochemical factors contributing to tomato fruit sugar content have been reviewed by Beckles *et al.* (2012).

Source–sink ratio

For several cultivars, the altered source–sink ratio by leaf or fruit pruning did not influence the light-saturated photosynthesis of young, fully expanded leaves measured under CO_2-limited conditions (Matsuda *et al.*, 2011). Although soluble sugars accumulated in leaves under high source–sink conditions, the amount of RuBisCO was not affected by the altered source–sink balance. This study supports previous works suggesting that a decreased sink capacity did not lower leaf photosynthetic rate under CO_2-limited conditions (Heuvelink and Buiskool, 1995). However, photosynthetic acclimation did occur under elevated CO_2, when plant sink strength was reduced by about 70% (Qian *et al.*, 2012).

It was estimated that total assimilate demand is twice the assimilate availability, averaged over a whole growing season (De Koning, 1994), while a calculated source–sink strength ratio of 0.3 was reported by Bertin (1995). This difference may have a genetic background, as these authors did not use the same cultivar. For greenhouse crops, the pattern of source–sink ratio over time for three types of tomato (cherry, round and beefsteak type) was initially above 1 (Li *et al.*, 2015). This occurs because in the young crop there are only a few sink organs – young leaves and perhaps one truss with small fruit, whereas the crop already intercepts a lot of the light (a relatively high leaf area). In this period supplementary light is not useful, as the plant is sink limited. During most of the growing season, however, the source–sink ratio is lower than 1 (Fig. 4.8), so higher PPFD will stimulate growth and yield.

FRUIT DRY MATTER CONTENT

Fruit dry matter content (dry matter as percentage of fruit fresh weight) decreases during fruit development (see Chapter 3) and is larger in summer than spring or autumn. Generally, 4–7.5% of tomato content is dry matter; this is

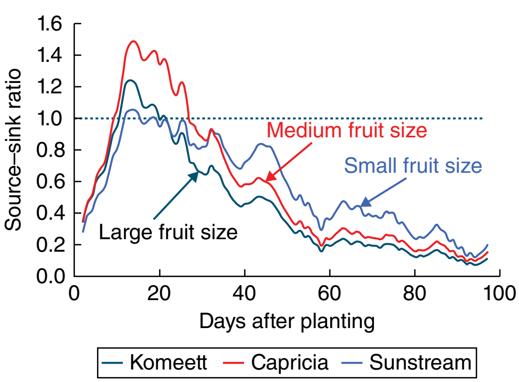

Fig. 4.8. Calculated source–sink ratio over time for three tomato cultivars with standard fruit load. Lines are moving averages over 5 days. Dashed horizontal line represents a source–sink ratio of 1. Fruit set started between 20–30 days after planting for all three cultivars (Li *et al.*, 2015).

dependent on genotype (e.g. higher for cherry tomatoes than beefsteak tomatoes) and could reach 17% for wild tomato species (Dorais *et al.*, 2001a). Dry matter content correlates positively with a better taste, as normally sugars represent about 50% of the dry matter in fruits, and a higher sugar concentration correlates with a better taste (see Chapter 5). For some Japanese tomato cultivars, a strong positive correlation between fruit dry matter content and the soluble solids content in fruits has been observed (Higashide *et al.*, 2012). Dry matter content of harvest ripe fruit is slightly higher for fruits grown at non-limiting assimilate supply, probably due to extra storage of carbohydrates. It also increases with temperature (De Koning, 1994). A higher solids content of tomato fruit for processing is an important breeding goal. Unfortunately, there is often an inverse relationship between fruit size, total yield and fruit total solids content. Dry matter content of the fruit increases with the salinity of the root environment (Fanasca *et al.*, 2007). This is explained as a 'concentration' effect, because fruit enlargement (fresh weight increase) is suppressed by limited water uptake as a result of salt stress.

INFLUENCE OF ENVIRONMENTAL FACTORS

Light

As for many other crops, a linear relationship between cumulative intercepted PAR, dry mass production and yield has been reported. The slope of this relationship is the LUE, expressed in g dry mass per MJ of intercepted PAR (see above). In Dutch tomato production it has been observed that fresh tomato yield per unit of light is lower in autumn than in spring. Two main reasons are a lower leaf area index and lower greenhouse CO_2 concentration in autumn compared with spring.

A rule of thumb for estimating the effects of light on production often used in practice is the 'one-percent rule', stating that 1% reduction in light will reduce production by 1%. For tomato, this value varies between 0.7% and 1% (Marcelis *et al.*, 2006). The effect on fruit fresh weight is stronger than on plant dry weight and the relative effect of radiation decreases at low temperatures, low CO_2 concentrations and at high light levels. Despite its simplicity, the one-percent rule often gives close estimates of the consequences of light loss on tomato yield. A light requirement equal to or higher than 30 mol/m^2/day (1 MJ/m^2 natural PAR = 4.6 mol/m^2 = 71.9 klxh = 6640 ft-c, while for HPS 1 MJ/m^2 PAR = 5 mol/m^2 = 118 klxh = 10,970 ft-c; 1 J = 1 W/s; 1 kWh = 3.6 MJ) is reported for a tomato culture, while a light integral of 4.8– 6.0 mol/m^2/day is generally favourable for tomato seedling production, which corresponds to a light intensity of 83 µmol/m^2/s during a photoperiod of 16–20 h. Under low light conditions, initiation of the first inflorescence is delayed, as more leaves are initiated prior to the inflorescence, and reduced flower development and flower abortion is observed. In commercial practice, fruit set could temporarily be improved under low light by removing some young leaves (competing sink organs for assimilate).

Light can play an indirect or direct role in regulating source–sink relationships involved in allocation of photo-assimilate within the growing plant. The light intensity received by the plant affects the quantity of assimilates available to the plant organs and thus their degree of sink competition. Light also directly stimulates tomato fruit growth, due to mechanisms other than photosynthesis. This can be interpreted as a direct influence of light on tomato fruit sink strength. Fruit illumination stimulates cell division but has no detectable effect on fruit size, suggesting that effects of cell division are compensated by effects on cell expansion (Okello *et al.*, 2015). In field tomato crops grown in summer sunlight over mulches, it was reported that an upwardly reflected far-red:red (FR:R) ratio higher than the ratio in incoming sunlight would signal the growing plant to partition more of its new photosynthate to the shoot and fruit growth, while an FR:R ratio lower than that in incoming sunlight would favour partitioning to the root (Kasperbauer and Kaul, 1996).

Supplemental lighting (SL) and photoperiod

For northern countries, the use of SL for greenhouse crops to secure flowering and fruit set in the winter is successful when there is shortage of assimilates (lower source than sink strength; see 'Source–sink ratio', p. 106). Early works showed that SL during the winter increased yield by 70–106% compared with natural light and sustained a minimal weekly yield of 1 kg/m^2 from November to February (Dorais and Gosselin, 2002; Dorais *et al.*, 2017). Increasing SL from 100 to 150 μmol/m^2/s gave an additional 20–36% yield increase, which has been related to a higher number of fruits instead of a higher fruit size. The use of SL for greenhouse-grown fruit vegetables, including tomato, has been detailed in recent reviews (Lu and Mitchell, 2016; Dorais *et al.*, 2017).

High-pressure sodium (HPS) gaseous discharge lamps are most commonly used but LEDs are receiving more and more attention. Advantages of LED lighting are as follows (Morrow, 2008; Persoon and Hogewoning, 2014; Lu and Mitchell, 2016; Dorais *et al.*, 2017).

- More efficient in transferring electricity into light. LEDs (95% red, 5% blue) have reached an efficiency of 2.3 μmol/J electricity, whereas for the best HPS lamps this is 1.85 μmol/J, hence LED lamps are about 35% more efficient. Efficiencies reported by LED manufacturers vary between 1.4 and 2.7 μmol/J electricity, or even 3.1 umol/J for red LEDs.
- Adjustable spectrum (light quality) that allows optimization for photosynthesis and plant form and function.
- Hardly any radiative heat emission, so lamps can be placed close to the plants (e.g. interlighting).
- Very long life (> 50,000 hours).

- Less light pollution (human-eye light sensitivity depends on colour; yellow most sensitive).
- Good safety characteristics.

Because growers are used to HPS lamps, they sometimes feel that transition to LEDs gives disadvantages because the head of the plant needs additional heating to maintain an acceptable development rate. On the other hand, the 'coldness' of LED light has the advantage that addition of light is separated from addition of heat, which makes it possible to bring the lamps between the crop for interlighting (see next section). The main limitation for the application of LED light in greenhouses is the high investment costs (Persoon and Hogewoning, 2014; Nelson and Bugbee, 2014).

Canopy interlighting

Conventionally, commercial greenhouses apply HPS lamps as an SL source above the crop canopy. Plants are known to benefit from light distributed evenly throughout the crop canopy. In the high-wire tomato cultivation system with high plant density, most of the SL is intercepted by the upper part of the plant canopy. A recently developed SL technique to overcome this problem is interlighting, where some of the lamps are placed within the crop canopy to give light to the middle or lower part of the canopy. When grown under top lighting, leaves low in the canopy have a low photosynthetic capacity because they acclimate to low light conditions. However, with interlighting this acclimation happens to a lesser extent and the lower leaves stay more productive. Interlighting reduces loss of light due to crop reflection towards the sky, which corresponds to approximately 6–7% of the incident irradiance. It also reduces loss of light due to crop transmittance towards the floor, which can be in the order of 5–10% (LAI of 3–4 with an extinction coefficient of 0.75). Interlighting with low-wattage HPS lamps was first tested experimentally with success in Finland and Norway, where increases in annual production in the order of 10–20% were found. The popular twin-row system (narrow row) often used in tomato production in northern regions, together with the high operating temperature of HPS lamps (> 200°C) resulting in a significant near-infrared (NIR) heat radiation, makes HPS unsuitable for interlighting.

In order to take advantages of both HPS and LED SL, hybrid systems have shown great potential for greenhouse crops. For example, an HPS and LED hybrid system resulted in a 20% higher yield for two tomato cultivars compared with top HPS lighting only (Moerkens *et al.*, 2016). However, production under LEDs was lower than HPS when four lighting treatments (PPFD 170 μmol/m^2/s) were compared: (i) top lighting with HPS or (ii) LED; (iii) top hybrid lighting with HPS in combination with LED lighting (50/50) above the crop; and (iv) HPS above the crop in 50/50 combination with LED between the canopy (interlighting) (Dueck *et al.*, 2012b).

To optimize the intra-canopy lighting system, the positioning of LED lamps should also be considered. By using a three-dimensional (3D) light model in combination with a 3D model of a tomato crop to simulate the 3D plant structure and light distribution for an interlighting system, LEDs shining slightly upward (20 degrees) increased light absorption and LUE relative to horizontally beaming LEDs, which could avoid loss of light to the ground (De Visser *et al.*, 2014).

Using LEDs as light sources for interlighting raises questions about the optimal light spectrum within the crop. Using narrow-band LED light sources provides possibilities for lighting with efficient spectra for photosynthesis and plant development. There is little variation in the photosynthetic effectiveness of photons of different wavelengths; though red light has the highest quantum efficiency (McCree, 1972), blue light plays a key role in controlling stomatal regulation (Hlavinka *et al.*, 2013), whereas the ratio of red to far-red light is known to control stem elongation (Franklin and Whitelam, 2005). The presence of blue light in the LED spectrum delayed stomata closure during water deficit (Kotiranta *et al.*, 2015). It was reported for high-wire greenhouse tomato that interlighting with varying red (627 nm), blue (450 nm) or far-red (730 nm) ratio altered leaf photosynthesis and stomatal properties but did not affect plant productivity expressed by fruit number and total fruit fresh weight (Gomez and Mitchell, 2016), which can be explained by the presence of natural irradiance that dilutes the spectral effects of LED lighting. Although the 'optimal' light spectrum for interlighting in tomato is not known, a general conclusion from sole-source light-quality research is that 5–10% of total PPFD needs to be blue. Furthermore, the optimum light spectrum for various plant growth processes such as leaf and fruit growth may be different, as manipulating light spectral distribution with LEDs in the vertical profile of the canopy has a large influence on plant growth and development (Guo *et al.*, 2016).

Photoperiod

It is important to take into account that a tomato plant grown under SL needs a minimum dark period of about 6 h (Dorais and Gosselin, 2002). When exposed to continuous light (CL), tomato plants show a set of characteristic symptoms, of which interveinal mottled chlorosis starting at the leaf/leaflet base is the most distinctive. It was hypothesized that CL implies continuous energy supply for photosynthesis (energy component) and continuous signalling to the photoreceptors (signalling component); both could play a role inducing injury (Vélez-Ramírez, 2014). The CL-induced injury could be photo-oxidative damage, early senescence and/or photosynthetic down-regulation (feedback inhibition) (Dorais *et al.*, 1995). A dominant locus on chromosome 7 of wild tomato species (known to be tolerant to CL) that confers CL tolerance and the type III light-harvesting chlorophyll a/b binding protein 13 (CAB-13) gene has been identified as a major factor responsible for the tolerance (Vélez-Ramírez, 2014). Introgressing the tolerance into modern tomato hybrid lines resulted in up to 20% yield increase. Hence, it was proposed that CL-induced injury in tomato arises from retrograde signals that counteract signals derived from the cellular developmental programme that promote chloroplast development, such that chloroplast development cannot be completed, resulting in the chlorotic phenotype (Vélez-Ramírez, 2014).

Diffuse light

Solar light is composed of a diffuse component and a direct component. Diffuse light arises from the scattering of light by molecules or larger particles in the atmosphere and comes from many directions simultaneously; direct light arrives in a straight line from the sun without being scattered. Many studies have reported that plants, including tomato, use diffuse light more efficiently than direct light (Mercado *et al.*, 2009; Li *et al.*, 2014). Under Dutch conditions, diffuse glass cover increased year-round tomato production by 8–11% (Dueck *et al.*, 2012a). Only diffuse covering materials, developed in recent years, increase the diffuseness of light without affecting total light transmission (Hemming *et al.*, 2014). Under PE the fraction of diffuse light is higher than under glass; however, glass generally obtains a higher total light transmission.

Diffuse light creates a more homogeneous light profile (both vertically and horizontally) in the canopy than direct light. It exhibits a lower light extinction coefficient than direct light (Li *et al.*, 2014), hence diffuse light penetrates deeper into the canopy. Diffuse light also results in a more homogeneous light distribution as less sunflecks occur (Fig. 4.9). High light levels usually lead to

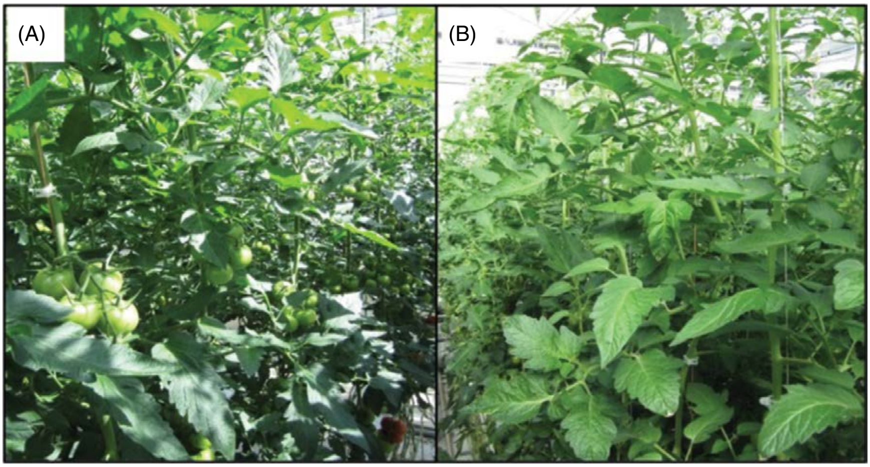

Fig. 4.9. Light distribution in a tomato canopy in a conventional clear glasshouse (direct light, A) and diffuse glasshouse (diffuse light, B) on a clear day. Light is more homogeneously distributed under diffuse cover (B) compared with clear cover (A) where many sunflecks in the middle and lower part of the canopy occur. (Photograph courtesy of Wageningen UR Greenhouse Horticulture.)

photosynthetic saturation and a decrease in LUE, which often occurs on the top canopy leaves under direct light. Diffuse light effectively mitigates the light saturation constraint by more evenly distributing light among all leaves in the canopy and therefore leads to a more efficient use of light. Diffuse light also results in a lower leaf temperature (3–5°C) and less photo-inhibition of top leaves in the middle of clear days (Li *et al.*, 2014).

Besides the direct effect of a more homogeneous distribution of diffuse light described above, there is also an indirect effect resulting from acclimation. Since diffuse light penetrates deeper into the canopy, the lower leaves receive on average a higher light intensity, which leads to a higher total nitrogen and chlorophyll content and hence a higher photosynthetic capacity in these lower leaves. Acclimation to diffuse light also includes acclimation of leaf morphology, which affects light interception and, consequently, photosynthesis. Tomato plants grown under diffuse light showed a lower specific leaf area (SLA), which indicates thicker leaves, as well as a higher leaf area index (LAI) mainly caused by a greater leaf width.

A simulation study showed for Dutch greenhouse conditions that diffuse glass increased crop photosynthesis by 7.2%; this increase mainly resulted from a better horizontal light distribution (33% of the total effect), whereas improved vertical light distribution and acclimation were each responsible for 21–23% of the total effect (Li *et al.*, 2014).

Carbon dioxide

CO_2 is the substrate for photosynthesis, and a higher concentration increases the rate of diffusion of CO_2 into the leaf and therefore gross leaf and crop photosynthesis. It also increases photosynthesis, because of suppressing photorespiration (see above). Photosynthetic rate, growth and yield show a saturation-type of response to CO_2. The relative increase in crop growth and yield (X, % per 100 μmol/mol) caused by additional CO_2 at certain CO_2 concentrations (C, μmol/mol) can be roughly estimated for indeterminate tomato grown under normal greenhouse conditions by the following rule of thumb: $X = (1000/C)^2 \times 1.5$ (Nederhoff, 1994). In (semi-)closed greenhouses, increased rates of photosynthesis due to the higher CO_2 concentrations are primarily responsible for the yield increase of 10–20% (De Gelder *et al.*, 2012).

For several species, including tomato, increases in RGR at elevated CO_2 were correlated with the relative increase in NAR and partly counteracted by a decrease in LAR, because of decreased SLA. Total fresh and dry weight, plant height and stem thickness of young tomato plants increased with elevated CO_2 without affecting the root-to-shoot ratio (Li *et al.*, 2007). However, Wang *et al.* (2009) observed an increased root-to-shoot ratio under elevated CO_2 (800 μmol/mol versus 350 μmol/mol), which agrees with the functional equilibrium (see above). Increased assimilation rate, decreased stomatal conductance and transpiration rate as well as higher total dry mass, fruit set and higher yield were observed at 700 μmol/mol CO_2 concentration compared with 500 μmol/mol (Mamatha *et al.*, 2014). Raising CO_2 concentration to 1000 μmol/mol increased yield by up to 84% compared with 400 μmol/mol (Khan *et al.*, 2013). CO_2 concentration does not affect appearance rate of leaves and trusses and has no direct effect on dry matter allocation, as sink strength rather than source strength determines the assimilate partitioning (see 'Biomass Partitioning', pp. 98–106). Indirectly, however, CO_2 enrichment in tomato plants increases fruit set (Fig. 3.5) and thus dry matter allocation to the fruits.

From experimental observations, increasing CO_2 concentration from 360 to 720 μmol/mol increased the photosynthetic rate of young tomato by 55% (Li *et al.*, 2007), while CO_2 increase from 400 to 1000 μmol/mol increased the photosynthesis rate of the lower foliage by up to 100% (Bencze *et al.*, 2011). An increase of 100 μmol/mol in CO_2 concentration could reduce stomatal conductance of tomato by about 3% (Nederhoff, 1994). Despite the increased stomatal resistance at elevated CO_2, the intercellular CO_2 concentration is kept high under CO_2 atmosphere enrichment which maintains high photosynthesis. No significant effect of CO_2 concentration on average leaf temperature was established, except under very low radiation (Nederhoff, 1994).

Acclimation

Although photosynthesis shows an initial higher short-term response to CO_2 enrichment, the long-term response often declines markedly (Besford, 1993) (Fig. 4.10). The long-term CO_2 acclimation of C_3 plants is related to a substantial reprogramming of gene expression (Fukayama *et al.*, 2009) in response to metabolic changes. Environmental, genetic or management practices that limit the development of sink strength predispose plants to a greater acclimation of photosynthetic capacity and lessen the stimulation of photosynthetic carbon uptake by growth at elevated CO_2 concentration (Foyer *et al.*, 2012). In fact, elevated CO_2 concentrations in a semi-closed greenhouse did not cause feedback inhibition in a high-producing tomato crop, because the plants had sufficient sink organs (fruits) to utilize the extra assimilates (Qian *et al.*, 2012). Different responses observed on young and mature plants and under different environmental conditions (e.g. light, photoperiod, CO_2, temperature) clearly showed the influence of the source-to-sink balance on the acclimation of photosynthetic capacity to elevated CO_2 and therefore degree of stimulation of CO_2 uptake. Although genetic factors play an important role in photosynthetic response to high CO_2, the down-regulation of transcripts encoding RuBisCO and other proteins associated with CO_2 fixation and an up-regulation of those encoding proteins involved in RuBP regeneration and starch synthesis is a common response to long-term growth at elevated CO_2 concentration (Leakey *et al.*, 2009; Foyer *et al.*, 2012).

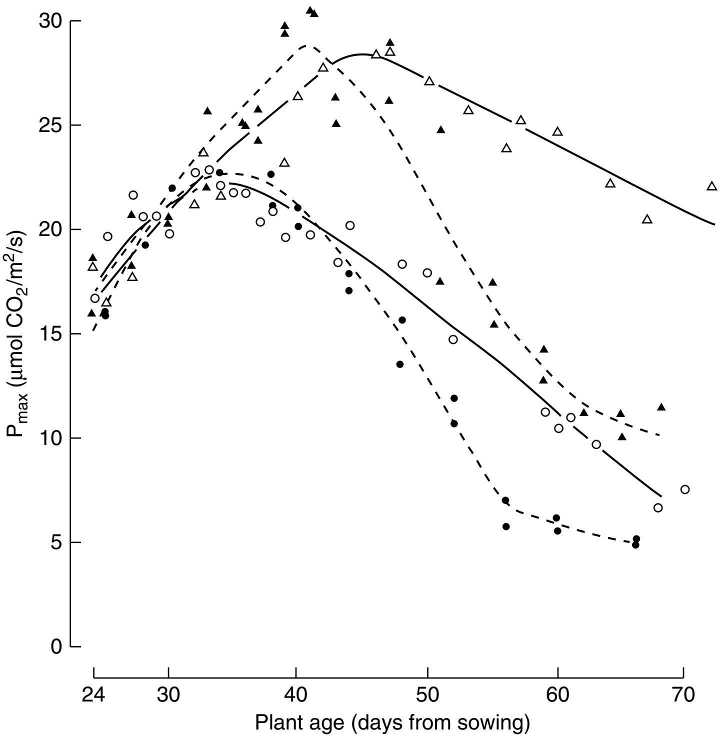

Fig. 4.10. Light-saturated rate of photosynthesis, P_{max} (1500 µmol/m²/s) of the unshaded fifth leaf of tomato plants at various stages of development. Plants grown in 340 µmol/mol CO_2 (○, △) or in 1000 µmol/mol CO_2 (●, ▲) and measured in 300 µmol/mol CO_2 (○, ●) or in 1000 µmol/mol CO_2 (△, ▲). (Reprinted from Besford, 1993, *Vegetatio* 104/105, 444, with permission from Kluwer Academic Publishers.)

Growing environment

The higher rates of photosynthesis and assimilation production of plants grown under CO_2 enrichment is also related to enhanced nutrient demands and a reduced C/N ratio (Li *et al.*, 2007). Although RuBisCO acclimation at high CO_2 may save nitrogen, the increase of nitrogen use efficiency typically observed is driven by enhanced CO_2 uptake rather than by the saving and redistribution of leaf nitrogen (Leakey *et al.*, 2009). Acclimation of plants to elevated CO_2 also improves the water use efficiency by reducing stomatal conductance and transpiration (Leakey *et al.*, 2009; Takagi *et al.*, 2009; Bencze *et al.*, 2011). Besides, under elevated CO_2 levels, tomato plants may suffer greater damage from herbivorous insects because of their reduced resistance and tolerance caused by the suppression of the jasmonic acid signalling pathway, an important hormone involved in plant defence against chewing insects (Guo *et al.*, 2012). Increase in the potential crop losses of field tomato due to higher weed competition under elevated CO_2 may also occur (Valerio *et al.*, 2013).

Temperature

Temperature primarily influences tomato yield because it influences plant development (see Chapter 3) through processes such as leaf initiation rate, leaf area development, pollen quality and hence fruit set. Besides, temperature has a direct influence on plant metabolism (e.g. photosynthesis, respiration, sink activity) and thus affects crop growth and yield. Yield potential reduces at temperatures above 26°C, with fruit set being one of the first processes that is negatively influenced by supra-optimal temperatures (see Chapter 3). The effect of temperatures around the optimal one on the growth of closed canopies is mainly through maintenance respiration. The rate of maintenance respiration more or less doubles with a 10°C rise in temperature. In the long term, temperature effect on crop maintenance respiration rate is small. At cool temperatures a heavier crop will compensate for a lower respiration rate per unit of biomass, whereas at warmer temperatures biomass is reduced, which in turn compensates for the increase in respiration rate per unit of biomass (Heuvelink, 1999).

Average fruit size decreases with temperature, being a consequence of increased truss appearance rate and accelerated fruit development (including reduction of the ripening time; see Chapter 3). Hence at higher temperatures an almost similar amount of assimilates has to be distributed over a larger amount of fruit, resulting in a lower average fruit weight when temperature is increased from 17°C to 26°C.

The optimum temperature for vegetative growth is 18–25°C, while the truss appearance rate increases almost linearly when temperature increases from 17°C to 27°C. At temperatures of 17–23°C, the ratio between vegetative and generative growth over a long growth period is independent of the average temperature (De Koning, 1989). The optimal temperature for anthesis is 18–20°C.

Chilling temperatures

As tomato is a tropical species, it is sensitive to chilling temperature. No growth occurs below 12°C and long-term exposure to these low temperatures causes chilling injury. For field crops, low-temperature exposure in combination with high irradiance, as found during the early growth season in California, causes a rapid and severe inhibition of photosynthesis. Several elements that contribute to this inhibition have been identified and all may ultimately arise from the photosynthetic production of highly reactive oxygen species (ROS) (Apel and Hirt, 2004). ROS formation occurs when the photosynthetic light-driven electron transport chain exceeds demand by the slowly turning Calvin cycle resulting in photo-inhibition, which is manifested in damage to the D1 protein of photosystem II. Under moderately low temperature and light as observed in several European countries during spring, photo-inhibition may not occur, but due to cessation of growth and development a reduced demand for photo-assimilates occurs, resulting in carbohydrate leaf accumulation and changes of the starch and soluble sugar partitioning. Consequently, sucrose synthesis in the cytoplasm will cease and may cause feedback inhibition on the reduction of phosphoglyceric acid in the Calvin cycle. Moreover, Calvin cycle turnover rates will decrease for thermodynamic reasons (Venema *et al.*, 2005). As negligible demand for NADPH occurs under low temperature, the photochemical quenching decreases (Havaux, 1987) and the non-photochemical quenching through the xanthophyll cycle increases. Alternatives routes such as photorespiration and cyclic electron transport around PSI are also involved in protecting the photosynthetic apparatus from over-excitation. Upon long-term chilling, the specific activity of RuBisCO declines in parallel with photosynthesis rates in tomato (Brüggemann *et al.*, 1994). Due to the low turnover rates of tomato RuBisCO (Besford *et al.*, 1985), particularly at low temperatures, inactivated enzyme molecules accumulate in the chloroplast, which increases the plant's susceptibility to over-excitation by light and low temperature. When irreversible RuBisCO inactivation occurs, Calvin cycle rates decrease and plants are not able to benefit from better temperatures (Venema *et al.*, 2005). For example, exposing young tomato plants during 9 days at high day and low night temperatures (25°C/9°C compared with 25°C/15°C) caused a reduced capacity for CO_2 assimilation, which decreased the utilization of NADPH (Liu *et al.*, 2012). Specifically, this reduction was accompanied by stomatal limitation of CO_2 supply and decline in RuBisCO activity at the transcriptional level. Soluble sugars and starch content and sucrose synthase activity increased under low night temperatures (Qi *et al.*, 2011).

Respiration – a vital component of plant metabolism, providing energy and carbon skeletons for biosynthesis, cell maintenance and active transport in plants – is one of the most important metabolic processes in chilling acclimation (Shi *et al.*, 2013). It has been observed that plants capable of maintaining a similar respiration rate under different temperatures are better adapted to chilling (Atkin *et al.*, 2006). Flexible changes in both the plant mitochondrial electron transport chain and cytosolic glycolysis cooperatively control chilling tolerance in tomato plants (Shi *et al.*, 2013). For chilling-sensitive species such as tomato, the circadian rhythm controlling the activity of SPS and nitrate reductase (NR), key control points of carbon and nitrogen metabolism in plant cells, is delayed by chilling treatments (Jones *et al.*, 1998). The mistiming in the regulation of SPS and NR, and perhaps other key metabolic enzymes under circadian regulation, underlies the chilling sensitivity of photosynthesis in tomato. Moreover, it has been observed that low temperature (6°C for 5 days) altered the membrane lipid composition of tomato chloroplast and decreased the number of granal thylakoids but had no effect on the total content of protein (Novitskaya and Trunova, 2000). The possible role of variation in membrane lipid composition as a causal factor in chilling sensitivity of several species has been noted by several authors (Somerville, 1995; Venema *et al.*, 2005). On the other hand, tomato under chilling conditions appears incapable of optimal regulation of stomatal conductance, resulting in high transpiration and wilting.

Suboptimal temperatures

Under field conditions, photosynthesis decrease was observed when day temperature was reduced from 25°C to 16°C (Venema *et al.*, 1999a, b). A reduction in the night temperature from 14°C to 6°C, combined with a day temperature of 19°C, did not affect net photosynthesis on a leaf area basis, while photosynthesis on a fresh leaf weight basis decreased under lower night temperatures, due to increasing leaf thickness (Van de Dijk and Maris, 1985). The RGR on tomato plants is reduced at suboptimal averaged temperatures and at low night temperatures, which is correlated with the LAR decrease. NAR is not affected by night temperature, while LAR reduction under low night temperatures is correlated with the SLA reduction. Suboptimal temperatures produce thicker leaves (reduced SLA) due to larger cells and starch accumulation (Venema *et al.*, 1999 a, b). This is especially of importance for young plants, where thicker leaves result in less light interception and consequently a lower relative growth rate. Under suboptimal day/night temperatures of 17°C/14°C, suppression of the shoot growth rates and leaf area expansion compared with optimal 22°C/18°C temperatures was not due to restriction in the rates of net CO_2 assimilation but rather related to hormonal signalling such as abscisic acid, auxin and salicylic acid (Ntatsi *et al.*, 2013). Temperature can also modulate stomatal responses to water deficits and to ABA (Correia *et al.*, 1999), as observed in low-temperature-tolerant wild species (Venema *et al.*, 2005).

In greenhouse tomato production, energy input is an important aspect regarding sustainability (economics and CO_2 emission). Within the cultivated tomato, there is only little genetic variation in response to temperature, which hampers breeding for equal production and quality at lower temperatures. However, in wild tomato species (e.g. *Solanum pennellii*, *S. hirsutum*) low-temperature tolerance is present (Fig. 4.11). It has been shown that the decrease in SLA at lower temperature in cv. 'Moneymaker' was related to the accumulation of non-structural carbohydrates (soluble sugars + starch),

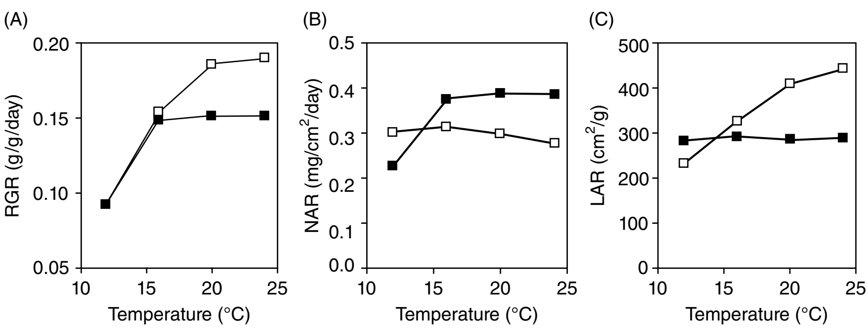

Fig. 4.11. Effect of temperature on (A) RGR, (B) NAR and (C) LAR of seedlings of *S. lycopersicum* 'Moneymaker' (□) and *S. pennellii* LA716 (■). (J.H. Venema, A. Van der Ploeg and E. Heuvelink, unpublished.)

which was much less pronounced in more cold-tolerant related wild tomato species (Venema *et al.*, 1999a, b).

Supra-optimal temperatures

Over a wide range, temperature has only a minor effect on gross photosynthesis of tomato. Temperature affects mainly the assimilate demand by the fruit (see 'Biomass Partitioning', pp. 98–106) without changing photosynthesis rate in the range of 15–25°C. Full acclimation of the photosynthetic apparatus of mature leaves to altered temperature requires at least several days. Stomatal conductance increases exponentially with temperature over the range of 15–35°C, even when assimilation rate decreased at high measurement temperature (Bunce, 2000). Below these ranges, photosynthesis increases with temperature because of the increased rates of the light reaction and carboxylation, but net CO_2 assimilation is depressed by the increase in dark respiration and photorespiration.

Based on detailed tomato canopy CO_2 gas exchange measurements at supra-optimal temperatures and high CO_2 levels (300, 600, 900 and 1200 μmol/m^2/s for 20–33°C and 400, 700 and 1000 μmol CO_2/mol), the temperature that maximized gross crop photosynthesis (P_{gc}) was predicted, showing a clear shift to higher optimum temperatures at elevated CO_2 levels (Fig. 4.12) (Körner *et al.*, 2009). Compared with leaf photosynthesis, crop photosynthesis had a lower temperature optimum (the difference could be several degrees), and the shift in optimum temperature from a low to a high CO_2 level was lower for a canopy compared with a leaf. Therefore, optimizing the leaf photosynthetic rate in model-based greenhouse climate control would not result in optimum crop photosynthesis.

Supra-optimal temperatures can cause severe damage to the photosynthetic apparatus. Generally, high temperatures reduced photosynthesis by changing the structural organization of thylakoids, resulting in the loss of grana stacking or its swelling (Wahid *et al.*, 2007) and down-regulation of PSII activity (Ogweno *et al.*, 2008). Decreases in RuBisCO and in PSII activities are the main factors involved in photosynthesis limitation resulting from supra-optimal temperatures (Zhang *et al.*, 2012). In tomato genotypes differing in heat tolerance, reduction in the CO_2 assimilation rate was not related to stomatal limitation (Camejo *et al.*, 2010) but rather the result of effects on the Calvin cycle and PSII functioning, while an increased chlorophyll *a*:*b* ratio and a decreased chlorophyll-to-carotenoids ratio was observed in the tolerant wild genotypes as sun-type chloroplasts. The heat-sensitive step of the sensitive genotype appeared to be the maximum rate of electron transport as a consequence of the activation of alternative pathways. However, when plants of a heat-sensitive genotype were adapted during 30 days to 35°C, net rate of photosynthesis was not negatively affected by 20 min at 35°C compared with the 25°C control plants (Camejo *et al.*, 2010). As fruit set declines under excessive temperature, lower fruit sink strength may decrease the export of photo-assimilates from

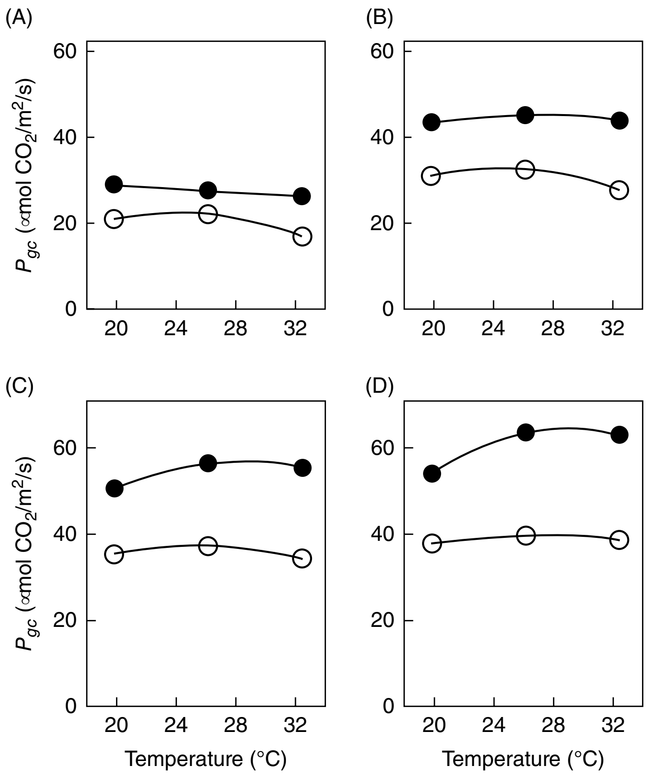

Fig. 4.12. Measured and fitted (negative-exponential) gross crop photosynthesis (P_{gc}) values for tomato as a function of temperature at two CO_2 concentrations (400 (○) or 1000 (●) µmol CO_2/mol) at four photosynthetic photon flux densities (A: 300, B: 600, C: 900 or D: 1200 µmol/m^2/s). Data were fitted using quadratic functions. Standard errors (±SE) were smaller than the symbols used. Reprinted from Körner *et al.*, 2009, by permission of Taylor and Francis Ltd (http://www.tandfonline.com)

source (leaf) to sink (fruit) organs, resulting in photosynthesis feedback inhibition and starch accumulation in the leaf (Zhang *et al.*, 2012).

Temperature and partitioning

High temperature enhances early fruit growth at the expense of vegetative growth. At high temperatures, the rate of plant development (new leaves and trusses) is higher (see Chapter 3). Therefore, growing young plants (only a few trusses have appeared) at high temperatures will increase partitioning into the fruits, as it increases the number of growing fruits on the plant. The strong assimilate demand by the growing fruits at the higher temperatures causes not only reduced leaf growth but also delay of growth of newly set fruits and even flower abortion. As a consequence, after some time total sink strength of the fruits is low and the plant recovers vegetatively and good-quality flowers develop. Subsequently, these flowers set fruit and form strong sinks, resulting in the onset of a second cycle of strong fruit growth. This leads to a more or less pronounced alternation of fruit growth and vegetative growth (De Koning, 1989). These effects of temperature, however, should not be confused with a direct effect of temperature on dry matter allocation.

Literature data are not conclusive on a possible direct effect of temperature on partitioning in tomato. For example, partitioning towards the fruits was only improved a little when young producing tomato plants were exposed for 3 weeks at 24°C compared with 19°C (Heuvelink, 1995a). However, when a similar experiment was conducted for a period of 2 weeks during the anthesis of the sixth truss, the fraction of dry matter partitioned to the fruits was 0.68 and 0.80 at 19°C and 23°C, respectively (De Koning, 1994).

Temperature integration

The long-term average temperature, rather than the day–night temperature regime, determines crop growth and yield in producing tomato crops. Consequently, temperature integration up to 12 days with an amplitude of 6°C does not reduce yield once the canopy (LAI) is fully developed (De Koning, 1990); and under high CO_2 enrichment (1000 μmol/mol) yield is hardly influenced by temperatures ranging between 16°C and 24°C (Heuvelink *et al.*, 2008). This great ability of a mature tomato crop to integrate (compensate within a certain period) temperature offers good possibilities for saving energy in greenhouse cultivation (see Chapter 9). For example, in spring, energy can be saved by allowing the greenhouse to heat up above the desired long-term average temperature by solar radiation during daytime, which is compensated for by reduced heating during the night. Similarly, in periods with high wind speed (high energy loss) the temperature set-point may be below the desired average, to be compensated for by a higher than average set-point in periods with low wind speed, reducing total heating costs. These control possibilities are included in greenhouse climate computer software in modern facilities.

Humidity

Humidity (vapour pressure deficit (VPD) 0.2–1.0 kPa) has little effect, or higher humidity increases photosynthesis. Pollination, fruit growth and development of tomato are generally not influenced in this VPD range (Picken, 1984), while an increase of VPD from 1 to 1.8 kPa could reduce plant growth of several crops (Hoffman, 1979). Humidity can influence leaf photosynthesis through its effect on stomatal conductance: low humidity may result in water stress and (partial) stomatal closure. Under extreme conditions of low humidity (1.5–2.2 kPa), there is a reduction of growth due to stomatal closure and therefore reduced photosynthesis. Long-term high relative humidity is known to induce abnormal stomatal functioning, bigger stomata and changed stomatal density in several plant species, including tomato (Fanourakis *et al.*, 2013; Arve and Torre, 2015). Carbon assimilation per se appears not to be greatly affected by humidity in the range of 0.2–1.0 kPa unless leaf or plant transpiration exceeds the water supply and water stress results (Grange and Hand, 1987). Hence, photosynthesis is decreased by low humidity only in high electrical conductivity (EC)- and/or water-stressed plants.

High humidity is often prevented in greenhouses, as it increases the risk of diseases (e.g. botrytis), especially in combination with a heterogeneous temperature distribution such that water vapour condensates on the coldest parts. It may also reduce yield; for example, 18–21% reduction at a VPD of 0.1 kPa (high humidity) compared with a VPD of 0.5 kPa, which may be related to a diminution of the leaf area due to a calcium deficiency in the foliage (Holder and Cockshull, 1988) and a reduced fruit growth rate (Bakker, 1990). High humidity may also reduce the ovule fertilization rate, due to a more difficult release of the pollen. On the other hand, high humidity (misting) has been used to partially mitigate the negative effect of high salinity in the root environment, by reducing the transpiration rate and thus the water demand (Li and Stanghellini, 2001; Cuartero *et al.*, 2006).

Under Mediterranean conditions, VPD higher than 2 kPa during the hottest hours of the day reduces fruit growth and fruit fresh weight, probably due to a reduction of the fruit-stem water potential gradient. Under those conditions total yield is also reduced by a decrease in fruit size (higher dry mass content). Moreover, there is an increase in the number of fruit affected by blossom-end rot (BER), due to a high foliage transpiration rate, thereby limiting the supply of xylem sap (calcium) to the fruit. At high salinity (10 dS/m), depressing plant transpiration by 35% under the same solar radiation reduced the incidence of BER from 20% to 2% of the total yield and might enlarge fruit size, thereby improving fruit quality and yield value (Stanghellini *et al.*, 1998). Low VPD did not affect the enhanced organoleptic quality of fruit grown under these high EC levels.

Salinity

In practice, water and salinity stresses are believed to favour generative development in tomato. Young plants are often stressed to stimulate fruit growth on the first truss. There is no proof that water and salinity stress directly influences partitioning, but influence in an indirect way by improving fruit set is well known. Improved fruit set leads to more fruits, and a higher fruit number on the plant will increase partitioning towards the fruits (Fig. 4.3). According to Dorais *et al.* (2001b), at an EC of 6 dS/m, biomass and dry matter partitioning among fruit, vegetative parts and roots were not affected by salinity. Nevertheless, EC of 10 dS/m reduced plant dry weight by 19% as compared with an EC of 2 dS/m but did not influence dry matter partitioning. An EC of 17 dS/m, however, slightly reduced dry matter distribution to the fruits. On the other hand, for young fruiting plants (first flower anthesis until harvest of the second truss) exposed to a salinity of 8 dS/m, it was reported that salinity increased the sink strength of tomato fruits due to an increase in the fruit sucrose synthase activity, while both forms of invertase were not affected (Saito *et al.*, 2009). Although salinity reduced photosynthesis rate per unit leaf area, increased transport and distribution of photo-assimilates to the fruits was measured.

Depending on temperature, cultivar, the severity of salinity and the duration of salinization, the number of trusses per plant as well as the number of flowers per truss might be influenced by salinity. For example, ECs of 4.6–8 dS/m reduced fruit yield because of a reduction in fruit size, whereas an EC of 12 dS/m reduced both the number and size of fruit. Salinities higher than 2.3–5.0 dS/m resulted in an undesirable yield reduction, while ECs of 3.5–9.0 dS/m improved tomato fruit quality (Dorais *et al.*, 2001a, b). In general, the yield and individual fruit weight of cultivars with smaller fruit size would be less affected by high salinity than cultivars with larger fruit.

Reduction of growth under salt stress is due to both an osmotic stress (days to weeks) and then an ionic stress (weeks to months), which provoke nutritional imbalance and accumulation of toxic ions such as sodium. Reduction of fruit growth and the final fruit size by an osmotic effect is due to a lower plant water potential, which will reduce the water flow into the fruit and therefore the rate of fruit expansion. A growth reduction of 6–10% for each 1 dS/m increase is expected (Adams, 1991). Under salinity, stem internode length, plant height and leaf area of tomato are reduced (Najla *et al.*, 2009). It has been proposed that shoot growth inhibition under salt stress may be regulated by hormones or their precursors (Pérez-Alfocea *et al.*, 2010). For example, root-localized induction of cytokinin biosynthesis modified both shoot hormonal and ionic status, thus ameliorating salinity-induced decreases in growth and yield under moderate salinity (Ghanem *et al.*, 2011). Abscisic acid and the ethylene precursor 1-aminocyclopropane-1-carboxylic acid (ACC) concentrations increased in roots, xylem sap and leaves after 1 day and 15 days, respectively, of salinization. The auxin concentration decreased in the leaves while it accumulated in the roots. The auxin/cytokinin ratio in the leaves and roots may explain both the salinity-induced decrease in shoot vigour and the shift in biomass allocation to the roots (Albacete *et al.*, 2008; Lovelli *et al.*, 2012). Functional evidence about the role of metabolic and hormonal inter-regulation of local sink processes in controlling tomato fruit sink activity, growth and yield under salinity have been shown (Albacete *et al.*, 2014).

Increasing water salinity (EC = 0.5 up to 15.7 dS/m) negatively affected LAI, radiation use efficiency (RUE) and above-ground dry weight accumulation, resulting in lower total and marketable yield (De Pascale *et al.*, 2015). Although the smaller leaf area of salinized plants was largely responsible for reduced RUE, approximately 50% of this reduction was accounted for by processes other than changed crop architecture, such as an increased stomatal resistance, increased mesophyll resistance and other impaired metabolic functions that may occur at high salinity. Interestingly, plants grown on long-term salinized soils (exposed to > 20 years of salinization) had a slightly higher RUE at a given EC of soil extract than plants grown on one-season salinization soil, indicating that permanent modifications of the soil physical properties may trigger physiological mechanisms of adaptation.

A proportional decrease in leaf photosynthesis rate as salinity increased has been reported (Romero-Aranda *et al.*, 2001). However, a decline in crop photosynthesis is primarily the result of a reduced leaf area (Heuvelink *et al.*, 2003). Photosynthetic performance of tomato under salinity varies among genotypes and may be related to: (i) their capability for toxic ion exclusion and maintenance of appropriate essential macronutrient concentrations in leaves; and (ii) metabolic and diffusion limitations (Nebauer *et al.*, 2013). Stomatal closure and decreased mesophyll conductance (g_m) initially limit photosynthesis due to reduced CO_2 availability but subsequently this low CO_2 level can lead to leaf metabolism changes (Nebauer *et al.*, 2013). The mechanisms that down-regulate g_m under salinity are still unknown, though they may involve regulation of proteins such as aquaporins or carbonic anhydrase (Nebauer *et al.*, 2013). Metabolic limitations are related to a reduction in active RuBisCO as well as limited RuBP regeneration due to an inadequate ATP or NADPH supply of enzymatic activities.

Mitigating salinity effects

Three cultural techniques that have proved useful in tomato to overcome, in part, the negative effects of salinity are: (i) treatment of seedlings with drought or NaCl, which ameliorates the adaptation of adult plants to salinity; (ii) mist (increasing humidity) applied to tomato plants, improving vegetative growth and yield in saline conditions; and (iii) grafting tomato cultivars on to appropriate rootstocks (Cuartero *et al.*, 2006). In addition to enhancing tolerance of tomato against salinity, rootstocks reduce infection by soil-borne pathogens and relieve nutrient, water, thermal and pollutant stresses (Colla *et al.*, 2010; Schwatz *et al.*, 2010). Elevated CO_2 concentration may also alleviate the negative effect of salinity on photosynthesis and plant growth (Maggio *et al.*, 2002; Takagi *et al.*, 2009). This effect was promoted under high sink-to-source activity, probably owing to improvement of oxidative stress as well as the water status through stomatal closure at high CO_2 concentration (Takagi *et al.*, 2009).

A significant consequence of climate change is the increased frequency of stress combinations that plants are exposed to, especially of abiotic factors such as salinity in combination with pathogenic microorganisms. Recent work (Kissoudis *et al.*, 2017) has revealed the differential influence of ethylene and abscisic acid signalling pathways on tomato resistance to combined powdery mildew and salt stress.

CROP GROWTH MODELS

As seen in the previous sections, tomato crop growth is a complex phenomenon. Simulation models seem appropriate here, as these models have proved to be successful tools in prediction and explanation of the behaviour of such a complex, dynamic system as a growing crop. Models are powerful tools to test hypotheses, to synthesize knowledge, to describe and understand complex systems and to compare different scenarios. Furthermore, with a model, conditions that cannot be attained experimentally can be evaluated, which may be important in theoretical studies. Models may be used in decision support systems, greenhouse climate control and prediction and planning of production. A model is a set of equations that represent the behaviour of a system. Often descriptive and explanatory models are distinguished. Descriptive models, also called statistical, regression, empirical or black-box models, reflect few or none of the mechanisms that are the cause of the behaviour of a system, whereas explanatory models consist of a quantitative description of these mechanisms and processes. Crop models can also be distinguished into process-based (functional) models and structural (architectural) models. The former represent plant processes such as photosynthesis and leaf area expansion, whereas the latter focus much more on developmental aspects (initiation of new organs, morphogenesis). A more recent development is the combination of both approaches into so-called functional–structural plant models (FSPM).

An overview of photosynthesis-based models for tomato is provided by Marcelis *et al.* (1998). Two of these models are TOMGRO (Gary *et al.*, 1995) and TOMSIM (Heuvelink, 1999), which include most of the processes described before (e.g. leaf and crop photosynthesis, light interception, biomass increase and dry mass partitioning) and the developmental processes (e.g. truss appearance rate, fruit growth period) presented in Chapter 3. More recently, process-based tomato crop growth models have been presented by: (i) Boote and Scholberg (2006), who showed results obtained with the crop simulation model CROPGRO-tomato with full C, N and water balance (Fig. 4.13); (ii) Marcelis *et al.* (2009), who concluded that their mechanistic crop model accurately simulates yield, growth, development and water and nutrient relations of greenhouse-grown tomato in different climate zones; (iii) Wang *et al.* (2012), who developed (by analysing the quantitative relationships between growth, partitioning index and harvest index and physiological development time) a mathematical model to estimate total dry matter accumulation, shoot dry matter partitioning and yield for processing tomatoes; and (iv) Katerji *et al.* (2013), who showed that the FAO AquaCrop model could adequately simulate final biomass and yield of tomato crops growing under non-stressed and moderate water stress treatments, whereas yields were poorly simulated under severe water stress treatments.

FSPMs for a tomato crop have also been presented (Kang *et al.*, 2011), analysing tomato fruit set based on the quantitative link between source–sink

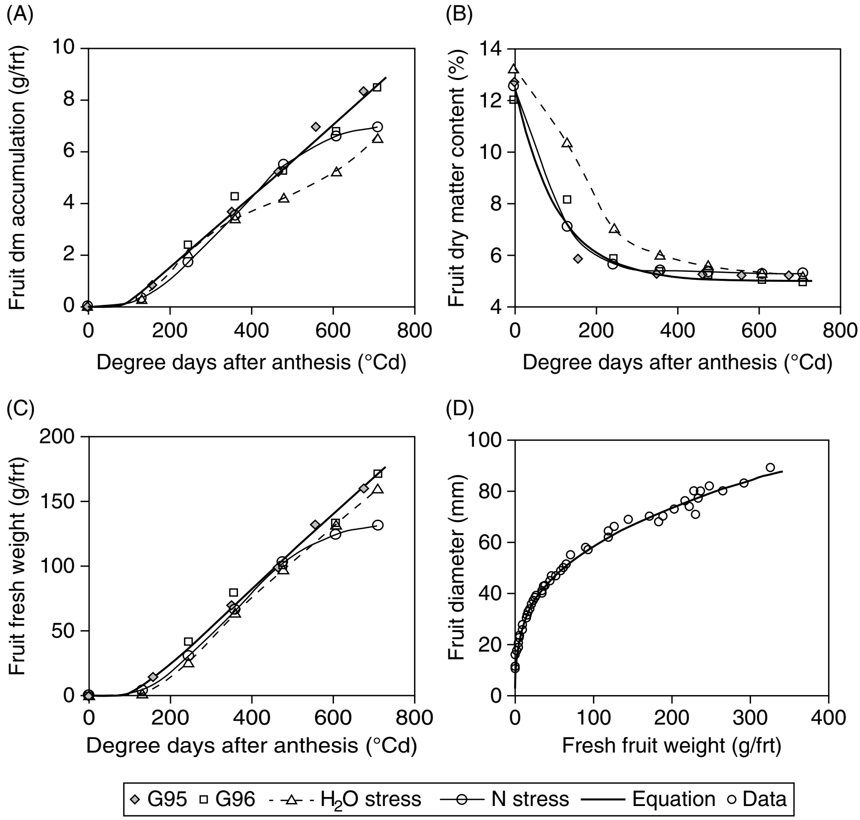

Fig. 4.13. (A) Dry matter accumulation, (B) dry matter concentration and (C) fresh weight of tagged early tomato fruits versus degree-days after anthesis for four treatments (G95 = Control 1995; G96 = Control 1996; H_2O stress = irrigation was cut off 3 weeks prior to flowering of cluster used for fruit development, and fruit yield was 57.2 versus 74.3 t/ha, although there was a shallow water table; N stress = no N-fertilizer applied, the crop accumulated 30 kg compared with 140 kg N/ha for well fertilized crop, and yield was 22.9 versus 79.1 t/ha; and (D) Fruit diameter as a function of fresh fruit weight. (Reprinted, with permission, from Boote and Scholberg, 2006.)

ratio and fruit set probability with the GreenLab FSPM. The impact of architectural traits on total light absorption and biomass production under dynamic conditions (e.g. up to 20% shoot dry mass increase by altering leaf angle) was demonstrated in a dynamic FSPM (Chen *et al.*, 2014) and earlier with a static tomato FSPM (Sarlikioti, 2011).

A novel knowledge-and-data-driven modelling (KDDM) approach for simulating tomato plant growth that consists of two sub-models has been developed (Fan *et al.*, 2015). A GreenLab model was adopted as the knowledge-driven sub-model and the radial basis function network as the data-driven sub-model. In comparison with the knowledge-driven model and data-driven model alone, the KDDM approach presented several benefits in predicting tomato yields. It effectively addressed situations of missing variables or data and showed good flexibility in maximally utilizing domain knowledge and data.

CONCLUSION

Genetics, environmental factors and crop management all affect tomato crop growth and yield. Yield is determined by the canopy photo-assimilation, assimilate distribution within the plant and fruit dry matter content. One or several of these yield components can be manipulated directly or indirectly by light (intensity, day length, spectrum, diffusiveness), CO_2, temperature, humidity and/or water, and nutrient availability during cultivation. Cultural practices such as planting density, leaf and fruit pruning also play a role. Better understanding of the rate-limiting steps of the accumulation of dry matter in tomato fruit would identify critical physiological processes that can be improved by plant breeding or genetic engineering. Plant breeding may provide a long-term solution for yield and quality improvement under suboptimal growing conditions such as low or high temperature, high salinity, or low light levels. Improvement can also be made by a better integration of growing conditions and cultural practices for an optimally balanced plant. In this context, crop modelling has become a major research tool in horticulture. A number of descriptive and explanatory models have been developed. These models can be used for yield and quality prediction, climate and crop management, as well as economical and policy analysis, and for teaching concepts.

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5

Fruit Quality

Nadia Bertin

INTRODUCTION

Quality of fleshy fruit is a complex trait including multiple variables. While the commercial quality relies mainly on external attractiveness (e.g. colour, shape, size), firmness and shelf-life, the organoleptic quality depends on physical (texture or firmness) and biochemical traits determining the overall taste and flavour. On the other hand, the health benefits rely on the composition in vitamins and antioxidant compounds (lycopene, β-carotene, ascorbic acid and polyphenol) as well as minerals (potassium, calcium, phosphorus, magnesium), whereas the sanitary quality is defined by residues of pesticides or other unhealthy compounds such as allergens, mycotoxins, antibiotics, environmental persistent pollutants and pathogenic microorganisms. In past decades, genetic improvement mainly favoured the producers and distributors, by derivation towards resistant productive cultivars and long-life products, whereas consumer preferences were generally overlooked (see Chapter 2). At the same time, yields have steadily increased by improvement of technical crop management in horticulture, so that intensive production systems launched on the market homogeneous, firm but tasteless products suitable for large distribution networks. Recently, the social demand for tasty, healthy fruits rich in vitamins and antioxidant compounds, but also for environmentally friendly production of fruits free of pesticides and residues, has given rise to new research concerning these traits. Meanwhile, fresh fruits have been recognized as a major source of vitamins and antioxidants and as important components of human diet and welfare on account of their nutritional value. Thus, increasing their consumption became a worldwide priority, in particular to limit the risks of chronic diseases and nutritional deficiencies. Tomato also represents a major economic issue as it is the second highest vegetable (first fruit) consumed worldwide (see Chapter 1). This changing social and environmental context led to the search for new cultivars adapted to more sustainable modes of production and able to maintain yield and produce high quality fruits.

This chapter focuses on main quality traits from the consumer's point of view, which are fruit size or fresh weight, colour, taste, flavour, texture and nutritive value. Quality standards have been defined that give quantitative market values. However, there are as many quality standards as there are consumer types. A large scale study (Causse *et al.*, 2010) revealed that preferences are quite homogeneous across European countries (The Netherlands, France and Italy) and that within each country two main factors discriminate the consumers of fresh tomatoes: the first is based on overall fruit flavour (aroma, juiciness, taste); the second is made up of textural components such as firmness, 'meltiness' and crunchiness.

Fruit quality is evaluated at harvest but it is elaborated in the course of fruit development, resulting from many interacting growth processes and metabolic activities, which are regulated according to the fruit ontogenetic programme and by environmental conditions (Génard *et al.*, 2007). Fruit taste, texture, overall flavour and nutritive value are mainly determined by the amount of dry matter and its composition in sugars, acids, cellulose and proteins, antioxidant compounds and minerals as well as by the ratio between sugars and acids and their dilution by water. At harvest, these traits are highly variable among fresh cultivated tomatoes: fruit fresh weight may vary 200-fold whereas the dry matter, sugar, acid or vitamin C contents may vary by four- to fivefold. After harvest, quality continues to evolve in relation to postharvest storage conditions and ripening stage at harvest. Despite increasing knowledge on the main metabolic pathways and molecular regulations occurring at plant and fruit levels, the improvement of fruit quality and the reconciliation of both yield and quality traits is still a challenge. Indeed, due to antagonistic relationships between fruit size and composition in sugars, the most tasty varieties are cherry or cocktail tomatoes, which represent only a minor part of the market, while the taste of large tomato fruit varieties is rather poor. In order to gain insight into the formation and variations of fruit quality in relation to genetic and environmental conditions, a preliminary step is to understand the processes underlying each trait and the development of integrative approaches to fruit quality to disentangle the complex network of interactions among these processes in response to external or internal stimuli. For this reason, in recent decades understanding fruit quality in relation to gene × environment × management (G×E×M) interactions and analysing correlations among yield and quality traits have been key objectives for fruit physiologists and ecophysiologists in view of crop management and cultivar adaptation to specific environments.

This chapter is divided into five parts. The first describes major quality traits of tomato fruit and the underlying processes. The second and third parts review some of the effects of environmental and genetic factors involved in quality variations. A few physiological disorders that may negatively impact the marketable yield are presented in the fourth part. Finally, an integrative approach for optimizing fruit quality is discussed in the last part.

QUALITY CHARACTERISTICS AND UNDERLYING PROCESSES

Fruit size and shape

Size and shape of tomato fruits are major traits of quality exhibiting a huge genetic variability (Tanksley, 2004). Fruit shape is determined mainly by the genetic make-up and it has been strongly diversified during breeding. Fruit size largely varies in response to G×E×M interactions. The increase in fruit volume results from the development of pericarp tissue, which generally accounts for more than two-thirds of total fruit weight (Ho, 1996). The increase in pericarp volume is achieved through two important processes: (i) the production of new cells, which is limited to a more or less short period (10–25 days after anthesis) of development; and (ii) cell expansion, which generally proceeds until start of maturation (Fig. 5.1). Although most of the fruit volume increase occurs during the expansion phase, several authors have provided evidence that final size is highly correlated to the number of cells determined early during the division phase (Fig. 5.2) (Bertin *et al.*, 2003). After cell division stops, the increase in tissue volume results from cell growth by increase in cytoplasmic volume and expansion through vacuolation, and from biophysical constraints related to epidermal extensibility (Thompson, 2001). Cell expansion is supported by the pressure of cell contents and constrained by cell wall properties (Cosgrove, 1997). The decrease in cell turgor and water potential resulting from cell wall relaxation and loosening enables water to enter the cell and to stimulate expansion (Lockhart, 1965). Water enters the fruit through xylem (15–20% of water influx) and phloem (75–80% of water influx) tissues following the stem-to-fruit gradient of water potential, which is generated by the gradient of osmotic potential between source and sink tissues, linking cell expansion to sugar metabolism and subcellular compartmentalization. Cessation of cell division and increase in cell size have been found to be closely linked to the switch of the complete mitotic cycle to an incomplete cycle without mitosis, so-called endoreduplication, which leads to polyploid cells in pericarp tissue where C-values range from 2C to 512C (Joubès and Chevalier, 2000). In tomato, large endoreduplicated cells are located in the mesocarp with DNA contents up to 256C (eight endocycles) or even 512C (nine endocycles) in cherry tomatoes as well as in large-fruit cultivars (Bertin *et al.*, 2007). The functional role of endoreduplication remains controversial. Among a large diversity of tomato lines, Cheniclet *et al.* (2005) reported a positive correlation between endoreduplication and cell size of pericarp tissues. By contrast, absence of correlation has been reported for mutants affected by the number of endocycles (Leiva-Neto *et al.*, 2004), in transgenic lines over-expressing a cell-cycle inhibitor (De Veylder *et al.*, 2001), or in response to changes in growth conditions affecting cell size (Bertin, 2005). Many theories have been developed to explain the link between cell size and nuclear size or DNA content, such as the

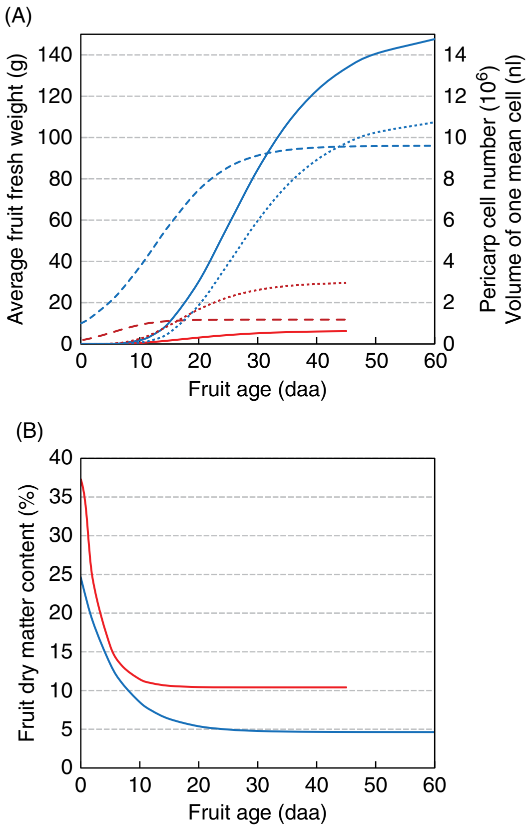

Fig. 5.1. Dynamics of (A) fruit fresh weight (solid lines), pericarp cell number (dashed lines) and mean cell volume (dotted lines) and (B) fruit dry matter content in cherry (red) and large-fruited (blue) tomatoes. The curves were fitted on several sets of experimental data. (N. Bertin, unpublished data.)

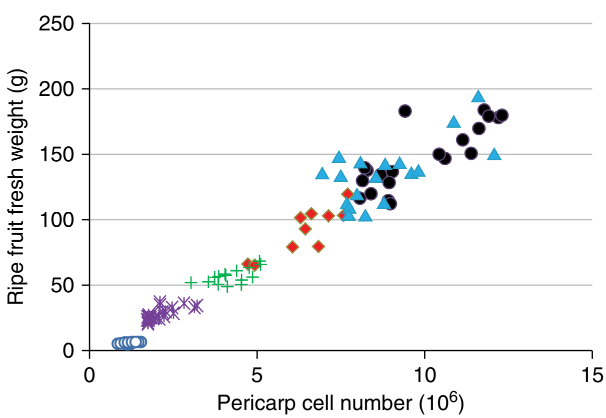

Fig. 5.2. Relationship observed at maturity between fruit fresh weight and cell number in a QTL-NILs population of tomato. Each point represents an individual fruit and each colour is a different genotype. (Bertin *et al.*, 2009; reprinted by permission of the Society for Experimental Biology.)

nuclear–cytoplasmic ratio theory, but the molecular basis of this correlation remains poorly understood (Chevalier *et al.*, 2011).

The onset of fruit ripening coincides with the rapid slowdown of cell expansion and the onset of intensive metabolic transformations. Tomato is a climacteric fruit and ripening is associated with ethylene production and cell respiration peaks in both attached and detached tomato fruits. As ripening progresses, fruit colour changes from green to red as chloroplasts are transformed into chromoplasts, chlorophyll is degraded and carotenoids accumulate. Fruit softening and textural changes occur as the fruit cell wall is partially disassembled by enzymes and the ripe flavour develops as specific volatiles increase and the sugar–acid balance alters (p. 42, this volume). Ethylene biosynthesis in pre-climacteric immature and mature-green tomato fruits and ripening climacteric tomato fruits is by the conventional pathway of methionine to SAM to ACC to ethylene (Abeles *et al.*, 1992). Exposure of mature-green fruit to endogenous levels of ethylene hastens the onset of the climacteric and ripening. Once the ethylene concentration within the fruit surpasses a 'threshold' level, it will promote its own biosynthesis (i.e. positive feedback), and autocatalytic ethylene production will cause a rapid increase in production and accumulation within the tissues (Abeles *et al.*, 1992). The atmosphere within a tomato fruit is effectively isolated from the surrounding atmosphere by an impermeable skin and cuticle; about 95% of gas exchange occurs through the stem scar. Therefore, once ethylene has started its positive feedback climacteric rise, few external treatments can modulate its synthesis. Reduced temperatures and lowered oxygen atmospheres slow overall metabolism but ripening will continue, albeit at a slower pace. However, certain inhibitors of ethylene action (e.g. 1-methylcyclopropane (1-MCP), ethanol vapours) appear to stop reversibly ethylene-enhanced fruit ripening at almost any stage of ripeness (Saltveit and Sharaf, 1992).

Fruit texture

Texture is a main trait of quality determining the end-use value of fruits, whether intended for fresh market or for industrial processing. In the case of tomato, texture not only influences purchasing and consumer acceptance but also has a significant impact on whole organoleptic quality, shelf-life and transportability (Seymour *et al.*, 2002), and it strongly interferes with flavour and aroma perception (Causse *et al.*, 2003). Texture is a complex trait, implying several components such as firmness, 'meltiness', mealiness, juiciness or crunchiness (Harker *et al.*, 2002). It can be evaluated by sensory analysis (Szczesniak, 2002), or it can be objectively measured by instrumental methods, including mechanical measurements, magnetic resonance imaging, or sonic and ultrasonic techniques (Abbott, 1999). The most common mechanical methods are compression and puncture tests (Barrett *et al.*, 2010), which mainly evaluate fruit or tissue firmness and elasticity and usually correlate well with sensory evaluation (Causse *et al.*, 2002).

Mechanisms underlying fruit texture are complex. Extensive work has focused on the molecular and biochemical mechanisms that lead to fruit softening during ripening, when the decline in fruit firmness coincides with the dissolution of the middle lamella, resulting in the reduction of intercellular adhesion, depolymerization and solubilization of hemicelluloses and pectic cell wall polysaccharides (Toivonen and Brummell, 2008). These events are accompanied by the increased expression of numerous cell wall-degrading enzymes, including hydrolases, transglucosylases, lyases and other wall-loosening proteins such as expansins. However, fruit texture might be already determined during the fruit growth period (Chaïb *et al.*, 2007), involving various mechanisms. Several reports have examined the importance of fruit anatomical and histological properties (Barrett *et al.*, 1998). At the fruit scale, proportion and thickness of the different tissues are determinants for fruit texture (Bourne, 2002). At the tissue scale, the cellular structure is likely involved in fruit mechanical properties, such as firmness (Aurand *et al.*, 2012). Moreover, cell turgor (Shackel *et al.*, 1991), transport of solutes among cell compartments (Almeida and Huber, 1999), chemical and mechanical properties of cell walls (Rosales *et al.*, 2009), cuticle structure and loss of water by transpiration (Saladié *et al.*, 2007) also contribute to textural properties. A recent study (Aurand *et al.*, 2012) noted that tomato firmness and stiffness measured by puncture tests correlate with both morphological (locular number), histological (cell size) and biochemical (dry matter and soluble sugar content) fruit traits.

After harvest, texture evolves rapidly, while membrane and cell wall breakdown occurs in relation to turgor loss and to enzyme-orchestrated cell wall loosening. This evolution may critically threaten the distribution of the production, with dramatic economic consequences.

Fruit composition in sugars and acids

Soluble sugars (glucose, fructose and sucrose) and organic acids (mainly malic and citric acids) are major osmotic compounds accumulated in tomato fruit. Both the absolute amounts and the balance between sugars and acids are responsible for fruit sweetness and sourness and contribute to their overall flavour (Davies and Hobson, 1981). Tomato fruit is made up of about 90–95% water and 5–10% dry matter, of which about 50% is represented by sugars and 15% by organic acids and amino acids (Fig. 5.3). The fruit structure and proportion of the different tissues may influence its taste, since jelly tissues contain more acid and less sugars than pericarp tissue. Metabolic pathways of sugar and acid syntheses and links between enzymatic activities and product accumulation in fruits have been well documented (reviewed by Etienne *et al.*, 2013, for acids and by Beckles *et al.*, 2012, for sugars).

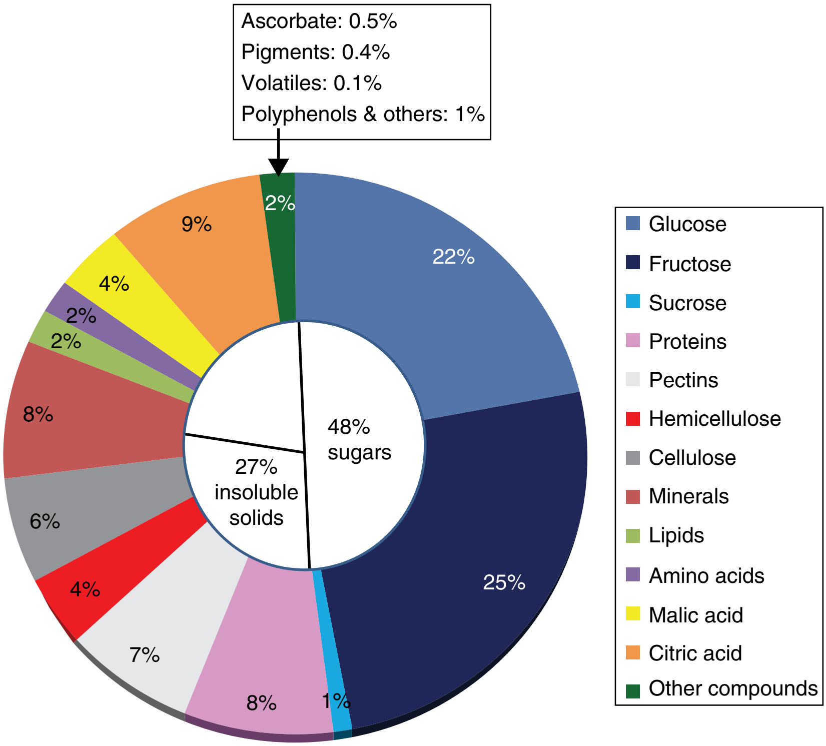

Fig. 5.3. Composition of tomato fruit dry matter. (Adapted from Davies and Hobson, 1981, by permission of Taylor & Francis Ltd (http://www.tandfonline.com).)

Tomato fruit shifts from partially photosynthetic to truly heterotrophic metabolism. Rare quantitative studies indicated that carbon fixed by the fruit itself contributes between 10% and 15% of that required for fruit growth (Tanaka *et al.*, 1974). Other studies based on transgenic lines or mutants confirmed that fruit photosynthesis is not necessary for fruit energy metabolism or development but suggested an important role for photosynthesis in the initiation of fruit development, or in properly timed seed development (Lytovchenko *et al.*, 2011). Thus most of the carbon required for growth is imported from leaves into fruit through the phloem tissues. Sucrose is the main form of carbon import in tomato fruit. It is either metabolized in the apoplast by a cell wall invertase or directly transported into fruit cells (Fig. 5.4). The transport occurs via symplastic (through the plasmodesmata) or apoplastic loading, involving hexose transporters. Although symplastic loading of sucrose is believed to predominate in young fruit, apoplastic loading has been suggested to occur throughout fruit development (Zanor *et al.*, 2009). Part of the imported carbon is consumed through respiration, providing energy for maintenance and structural growth, and

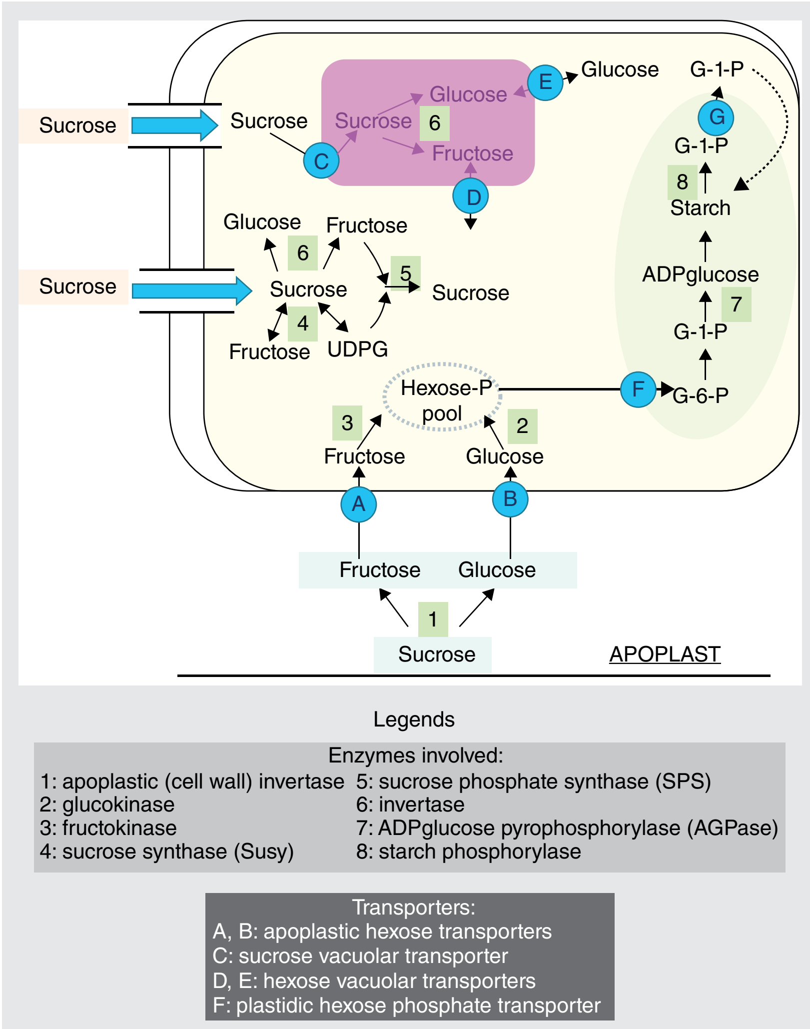

Fig. 5.4. Carbohydrate metabolism in developing tomato fruit. Sucrose may be imported directly via the symplast or may be inverted in the apoplast to hexoses, which are then imported into the cell. Both sucrose and hexoses may be stored in the vacuole. The flux of sucrose to starch occurs early in fruit development from anthesis to ~20–25 days post-anthesis (DPA). Here, sucrose metabolism via sucrose synthase and hexokinases dominates. Hexose phosphate intermediates are then imported into the plastid for the synthesis of starch. As the fruit ripens, Susy and hexokinase activity declines relative to invertase and the apoplastic import of hexose becomes more significant with storage of sugars in the vacuole. Starch biosynthesis is minimal and active degradation of the starch occurs, which may add to the sugar content available for storage. (Beckles *et al.*, 2012, reprinted by permission of Cambridge University Press.)

the rest is stored in different forms (mainly starch, sucrose, glucose, fructose). While the pools of hexoses increase until maturation, a transient pool of starch particularly located in the columella, the placenta and the inner and radial pericarp (Schaffer and Petreikov, 1997) is filled in the early phase of fruit expansion and peaks around 10–25 days after anthesis (Fig. 5.5). Starch accumulated during this period is completely hydrolysed afterwards and maximum starch levels correlate well with final levels of soluble sugars (Ho, 2003). In the mature fruit, glucose and fructose are present at an approximately equimolar ratio (Davies and Hobson, 1981), while sucrose represents a small portion of the soluble sugars (generally less than 5%) in most commercial cultivars (Ho, 1996).

Malic and citric acids are the main organic acids accumulated in tomato fruit. They determine fleshy fruit acidity, as measured by titratable acidity and/or pH. The pH of tomato juice currently ranges between 4 and 4.5 but it does not correlate with acid content, which fluctuates over a much larger range. The perception of fruit acidity is due mainly to citric acid. Even though some organic acids are supplied by the sap, variations in fruit acidity primarily result from the metabolism of malate and citrate in the fruit itself (Fig. 5.6). Etienne *et al.* (2013) demonstrated that accumulation of malate and citrate is the result of complex interactions between metabolism and vacuolar storage. The first steps of acid synthesis, i.e. malate and oxaloacetate synthesis, take place in the cytosol and require the fixation of CO_2 on

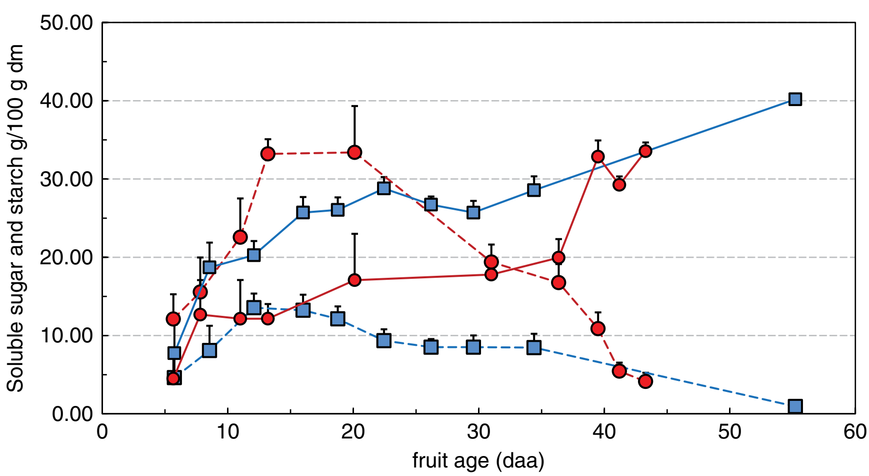

Fig. 5.5. Dry matter composition in total soluble sugars (solid lines) and starch (dashed lines) during development of cherry (red) and large-fruited (blue) tomato fruit. Vertical bars indicate 95% confidence intervals. (Adapted from Bertin *et al.*, 2009, by permission of the Society of Experimental Biology.)

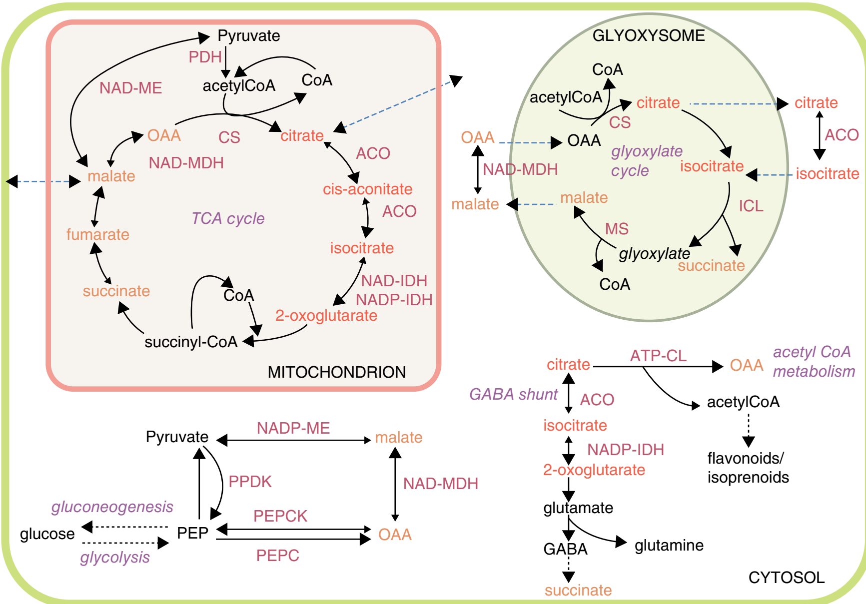

Fig. 5.6. Citrate and malate metabolic pathways in fruit mesocarp cells. Only the enzymes described in the source are shown. ACO, aconitase; ATP-CL, ATP-citrate lyase; CS, citrate synthase; ICL, isocitrate lyase; MS, malate synthase; NAD-MDH, NAD-malate dehydrogenase; NAD-ME, NAD-malic enzyme; NAD-IDH, NAD-isocitrate dehydrogenase; PDH, pyruvate dehydrogenase; PEPC, phosphoenolypyruvate carboxylase; PEPCK, phosphoenolpyruvate carboxykinase; PPDK, pyruvate orthophosphate dikinase. The probable direction of reversible reactions is indicated by the large arrow. Dashed blue arrows indicate malate and citrate transport. Names in orange are dicarboxylates and names in red are tricarboxylates. (Etienne *et al*., 2013; reprinted by permission of the Society for Experimental Biology.)

a carbon skeleton derived from hexose catabolism. Both cytosolic pH and malate concentration play an important role in the regulation of malate synthesis (Etienne *et al*., 2013). Then malate can be converted to citrate or to other tricarboxylates or dicarboxylates via several pathways in the mitochondria, leading to fruit acidity fluctuations. While acid synthesis and conversions through different pathways occur in different cell compartments (cytosol, mitochondria and glyoxysomes), large amounts of acids are accumulated mainly in the vacuole after transport through the tonoplast, which depends on both vacuolar pH and electric potential gradient across the tonoplast. Then, during fruit ripening, the cytosolic degradation of organic acids through the gluconeogenesis pathway promotes the accumulation of soluble sugars (Etienne *et al*., 2013).

Fruit ascorbate (vitamin C)

Ascorbic acid or ascorbate (AsA) plays important roles in plants. For instance, it is involved in cell division and cell wall synthesis, and in the interaction of plants with the environment, pathogens and oxidizing agents (Smirnoff, 2000). It is also an important micronutrient and an essential antioxidant in the human diet. In tomato, vitamin C exists in two water-soluble, biologically active forms: ascorbate (reduced form) and dehydroascorbic acid (oxidized form). Both forms are present in all cell compartments of tissues undergoing active growth and development, and the total amount of vitamin C ranges from 20 to 60 mg/100 g fresh weight among tomato species and cultivars (Table 5.1) (CTIFL, 2011). The biosynthesis pathway of AsA in plants and the numerous enzymes involved have been well described (L-galactose pathway) (Wheeler *et al.*, 1998). AsA is synthesized from D-glucose both in leaves and in fruits. Once synthesized, AsA can rapidly be oxidized, as result of its antioxidant function, and so the recycling pathway (reduction of the oxidized form) also plays an important role in maintaining AsA levels and redox status in plant cells (Massot *et al.*, 2013). Thus the AsA pool size in fruit results from its import (or the import of precursors) from leaves, its synthesis and recycling within the fruit and its export, as well as its degree of oxidation or degradation. During fruit development, fruit AsA content decreases during the periods of cell division and expansion, probably because of dilution effects, then increases during ripening concomitantly to the hexose content (Massot *et al.*, 2010). Many studies have found good correlations between fruit soluble sugar and AsA contents. However, recent works have demonstrated that AsA synthesis in tomato fruit is not limited by sugar content (Massot *et al.*, 2010) and that the *in situ* synthesis prevails over AsA transport from leaves (Gautier *et al.*, 2009).

In the postharvest period, ascorbic acid content shows significant losses during storage but the content of vitamin C shows remarkable stability when expressed as the sum of ascorbic acid and dehydroascorbic acid. Moreover, it has been shown that mature-green and breaker fruits that have been ripened with ethylene lose less vitamin C by the time they reach the red-ripe stage than fruits allowed to ripen without added ethylene. However, both are lower than in fruits ripened on the plant. Interestingly, monodehydroascorbate reductase (MDHAR) activity plays an important role in the maintenance of ascorbate levels in fruit after chilling injury induced by storage below 10°C. Furthermore, under these conditions, an increased fruit MDHAR activity and a lower oxidation level of the fruit ascorbate pool are correlated with decreased loss of firmness (Stevens *et al.*, 2008).

Fruit carotenoids

Carotenoids are also of importance in the human diet due to their antioxidant properties (Dorais *et al.*, 2008). Besides their interest for humans, carotenoids

Table 5.1. Composition of different ripe tomatoes. (Cantwell, 2010.)

Tomato type	Fruit weight (g)	Red colour, Hue	Soluble solids (%)	Sugar (mg/mL)	Titratable acidity (%)	Vitamin C (mg/100 g)	Lycopene (mg/kg)
Campari	53.1	44.2	6.3	31.4	0.58	40.5	63.0
Cherry	20.3	45.5	4.2	28.9	0.31	54.0	84.6
Grape	5.0	51.3	5.6	29.5	0.51	47.1	49.1
Grape	6.2	41.7	4.2	39.6	0.35	61.7	98.0
Orange Cluster	111.5	71.5	4.6	26.1	0.33	29.2	4.2
Round Cluster	102.1	43.2	7.6	20.1	0.62	26.9	53.6
Round Cluster	119.8	44.6	3.8	15.3	0.44	26.0	44.8
Round Greenhouse	231.2	45.9	4.5	22.5	0.36	30.4	28.0
Round Greenhouse	179.4	47.7	4.7	25.0	0.44	20.4	42.5
Roma	94.8	42.1	4.3	24.0	0.27	22.8	46.4
Roma	84.5	45.2	6.2	20.2	0.67	24.3	44.4
Romanita	20.5	41.3	6.3	32.9	0.44	45.9	70.3
LSD 0.05	6.7	2.4	0.3	5.0	0.08	8.0	7.5

play fundamental biological roles in plants (for instance, in light harvesting in photosynthetic membranes, protection of the photosynthetic apparatus from excessive light energy, pigment-protein complexes in thylakoids, precursors of abscisic acid, and attractants to pollinators, thus involved in seed dispersal).

Carotenoids are C_{40} isoprenoid molecules formed in plastids, alongside the differentiation of chloroplasts into chromoplasts. Carotenoid accumulation is spatially and temporally regulated in the chloroplasts and chromoplasts of fruits and flowers. All carotenoids are synthesized from geranylgeranyl diphosphate (GGPP). Then isomerization, cyclization, hydrogenation and oxygenation events give rise to the chemical variability of carotenoids, whose accumulation results from the balance between synthesis and catabolism through enzymatic or non-enzymatic oxidative cleavage (review by Fanciullino *et al.*, 2014). The most abundant carotenoid in tomato fruit is lycopene, followed by phytoene, phytofluene, zeta-carotene, gamma-carotene, beta-carotene (precursor of vitamin A), neurosporene and lutein. Lycopene and β-carotene are the principal ripe fruit pigments of tomato responsible for fruit colouration, and lycopene represents about 80–90% of total carotenoid content of ripe fruit. In tomato, total carotenoid concentration increases between 10- and 50-fold during fruit ripening, with a concomitant decrease in chlorophyll (Fraser *et al.*, 1994). Reported contents in lycopene and β-carotene range from, respectively, 0.4 to 10 mg/100 g fresh weight and from 0.6 to 1.4 mg/100 g fresh weight (CTIFL, 2011) (Table 5.1). The primary mechanism that controls the increase in carotenoid content is based on the differential regulation of expression of genes encoding enzymes of the carotenoid biosynthetic pathway, such as the phytoene synthase and phytoene desaturase gene (Fraser *et al.*, 1994). The carotenoid biosynthetic pathway has been well described and several regulatory mechanisms have been suggested, including ethylene, light, availability of substrates and metabolic sequestration (Ronen *et al.*, 1999; Fanciullino *et al.*, 2014). However, carotenoid accumulation and activities of many enzymes involved in the biosynthesis do not correlate during fruit development.

The ability of the fruit to synthesize lycopene and β-carotene is almost the same for harvested mature-green fruits as it is for fruit ripening on the plant (Table 5.2). Exposure of harvested mature-green fruits to ethylene stimulates

Table 5.2. Composition of tomato fruit at different stages of ripeness. Fruit were harvested at the mature-green, breaker or red-ripe stage of ripeness. (From Cantwell, 2000.)

Stage of ripeness	Soluble solids (%)	Reducing sugars (%)	pH	Titratable acidity (%)	β-carotene (µg/g)	Lycopene (µg/g)	Ascorbic acid (mg/100 g)
Mature-green	2.37	0.81	4.20	0.28	0.0	0.0	12.5
Breaker	2.42	0.85	4.17	0.39	0.40	0.52	18.0
Red-ripe	5.15	1.62	4.12	0.43	4.33	48.3	22.5

normal ripening with the synthesis and accumulation of lycopene and β-carotene, often to a greater extent than for harvested fruits left to ripen without ethylene stimulation (Saltveit, 1999).

Fruit colour and appearance

During ripening, the most obvious external changes are associated with the loss of chlorophyll and the accumulation of lycopene, which first becomes apparent at the blossom end of the fruit and progresses toward the stem end. Therefore during ripening the fruit can be partially green and red. Once ripe, however, high-quality fruits have uniform red distributed over the entire surface of the fruit. Under proper conditions of temperature and humidity, tomato fruits progress through six well defined stages to the red-ripe stage (Fig. 5.7). These stages are: (1) mature-green, (2) breaker, (3) turning, (4) pink, (5) light-red and finally (6) red-ripe; and they are based almost entirely on the external colour change of the fruit from green to red (i.e. destruction of chlorophyll and synthesis of lycopene). At the mature-green stage (no external red colouration), fruits have reached about 80% of their final size and acquire the ability to continue to develop and ripen normally after harvest. While fruits of most cultivars change from a uniform green to red as they ripen, a few cultivars have fruits that turn yellow (reduced or incomplete synthesis of lycopene) or are variegated in colour.

Fruit polyphenols

Polyphenols are a widespread family of phytochemicals with diverse biological functions in plants, for instance in response to various biotic or abiotic stress factors. They are also bioactive health compounds involved in the prevention of cancer and cardiovascular diseases related to their potent antioxidant activity as well as hepatoprotective, hypoglycaemic and antiviral activities (Slimestad and Verheul, 2009). Polyphenols are mainly synthesized from the phenylalanine produced by the shikimic acid pathway. Tomatoes are an important source of phenolic compounds, mostly restricted to their skin and exocarp. About 50 compounds have been reported in fruit, among which phenolic acids, phenylpropanoids, coumarins and flavonoids play an important role in fruit quality (Slimestad and Verheul, 2009). Chlorogenic acids and flavonoids are the main polyphenols in tomato. According to the flavonoid database of the US Department of Agriculture (USDA) (Bhagwat *et al.*, 2014), red tomatoes contain on a year-average basis 15 mg flavonoids per kg of fresh weight (FW) but the total flavonoid content of different tomato types varies from 4 to 26 mg/100 g FW (Slimestad and Verheul, 2009). Naringenin (45%) is reported to be the main flavonoid, followed

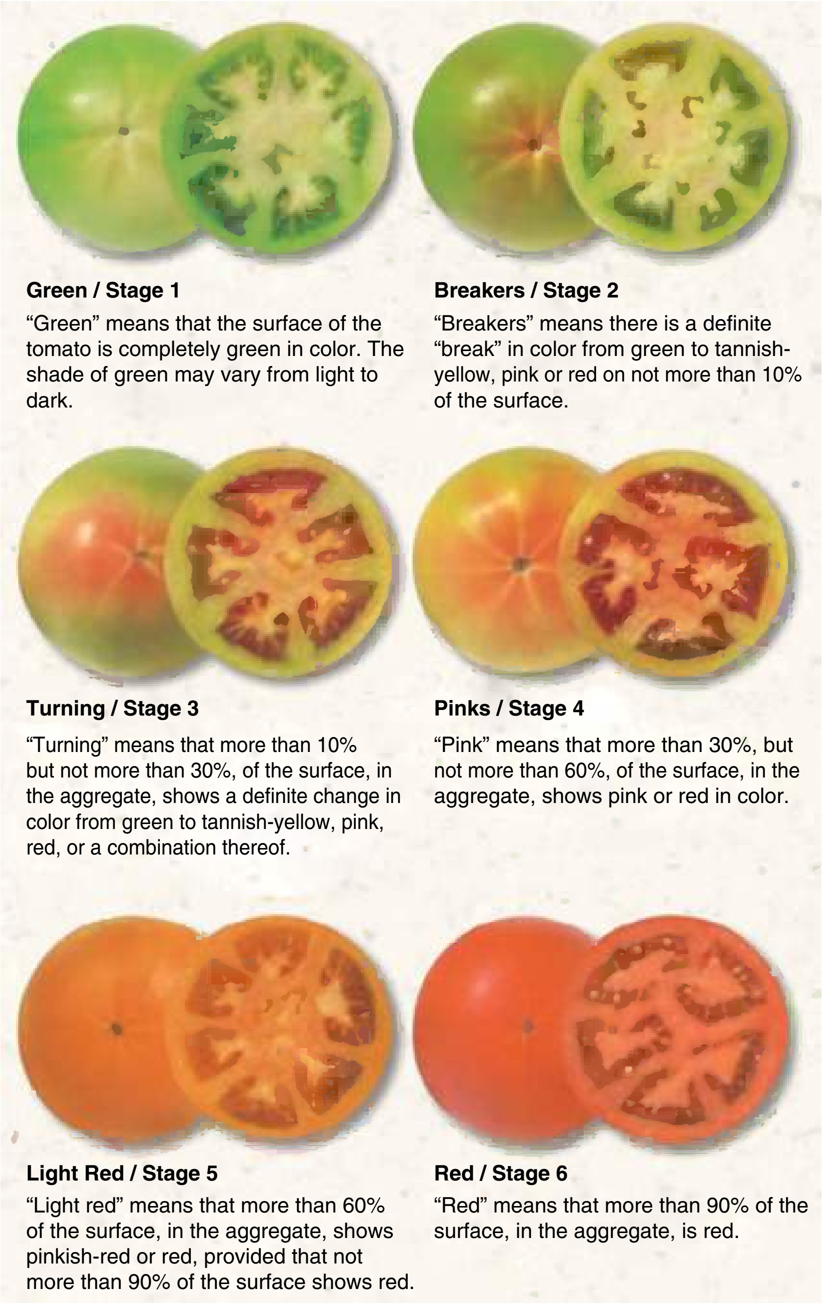

Fig. 5.7. Classification of fresh market tomatoes based on changes in external and internal colour and tissue softening. (USDA Tomato Ripening Stages.)

by quercetin (39%), myricetin (10%) and kaempferol (5%) (Slimestad and Verheul, 2009).

Fruit aroma and flavour

Flavour and aroma have often given rise to consumer complaints about the quality of tomato. Flavour results from complex interactions among sugars, acids and aroma volatile compounds (Baldwin *et al.*, 2008). The latter have been shown to affect human perception and preference of tomato fruit (Tieman *et al.*, 2012). Over 400 volatile compounds have been identified in fresh tomatoes, among which only 30 are present in concentrations over 1 ppb and thus have been long considered to contribute significantly to the perceptible flavour (Buttery, 1993) (Table 5.3). However, the presence of a molecule even at a relatively high level does not necessarily contribute to flavour or consumer liking. A study by Tieman *et al.* (2012) demonstrated that the near complete removal of the most abundant class of volatiles (C6 volatiles) in transgenic fruit, without affecting sugars, acids and other volatiles, impacts on flavour intensity but does not significantly impact on consumer preference. Thus odour thresholds alone are inadequate to predict the impact of particular volatiles on flavour. Moreover, the same authors demonstrated that aroma volatiles, such as geranial, make contributions to perceived sweetness independently of sugar concentration, suggesting a novel way to increase perception of sweetness. Volatile compounds are synthesized during ripening with a lesser contribution of the locular gel in the total production by the whole fruit (Maul *et al.*, 1998). Volatile components include acyclic, cyclic and heterocyclic hydrocarbons, alcohols, phenols, ethers, aldehydes, ketones, carboxylic acids, esters and lactones, as well as nitrogen, sulfur and halogen-containing compounds (reviewed by Lewinsohn *et al.*, 2001). The aroma of tomato is not directly related to the presence or absence of a single compound, but rather to synergism among components. (Z)-3-hexenal, hexanal, 1-octen-3-one, methional, 1-penten-3-one and 3-methylbutanal belong to the most odour-active aroma volatiles in fresh tomatoes (Krumbein and Auerswald, 1998) whereas the acyclic monoterpene alcohol, linalool, strongly influences the flavour of tomatoes (Buttery, 1993). Volatile compounds found in fruits are formed in different metabolic pathways (Klee and Tieman 2013) and derived from fatty acids (e.g. *cis*-3-hexenol and *n*-hexanal), aliphatic amino acids (e.g. 2-isobutylthiazole and guaiacol), phenolic compounds (e.g. eugenol and methyl salicilate), or longer terpenoids such as β-carotene and lycopene (e.g. β-ionone and geranylacetone). Although the biosynthetic pathways have been well described, the enzymes and genes controlling the production of aroma compounds are poorly known.

Fruits harvested mature-green and improperly ripened or ripe fruits stored at chilling temperatures do not produce the characteristic volatiles associated with high-quality tomatoes (Saltveit, 1999).

Table 5.3. Volatiles emitted by ripe fruits of two tomato lines, *Solanum habrochaites* LA1777 and *Solanum lycopersicum* LA4024 (Mathieu *et al.*, 2009; reprinted by permission of the Society for Experimental Biology.)

Volatile	*S. habrochaites*	*S. lycopersicum*	Ratio
(ng g^{-1} FW h^{-1})	LA1777	LA4024	LA1777/LA4024
β-Damascenone	0.18±0.07	0.01±0.00	30.31
Methyl jasmonate	0.69±0.55	0.03±0.01	23.53
β-Ionone	0.31±0.21	0.03±0.00	12.00
2-Methylbutanal	12.53±1.54	2.56±0.42	4.90
cis-2-Penten-1-ol	2.68±1.31	0.64±0.09	4.15
Isobutyl acetate	11.14±4.07	2.71±0.47	4.11
Geranylacetone	10.04±4.58	2.65±0.65	3.78
Phenylacetaldehyde	1.03±0.29	0.28±0.09	3.69
2-Methylbutanol	46.96±22.95	14.99±2.99	3.13
Benzaldehyde	14.90±7.45	4.93±0.81	3.02
1-Penten-3-ol	10.78±3.61	3.64±0.28	2.96
1-Penten-3-one	2.00±0.81	0.68±0.11	2.93
Pentanal	12.78±4.42	5.07±0.36	2.52
Methylsalicylate	3.15±2.98	1.26±0.28	2.50
Methyl benzoate	4.00±2.00	1.61±0.49	2.48
cis-3-Hexenal	169.21±117.83	69.13±13.94	2.45
2-Methoxyphenol	2.39±1.26	1.01±0.31	2.36
trans-2-Hexenal	4.47±1.53	2.52±0.72	1.77
Methional	0.24±0.18	0.14±0.02	1.70
Geranial	0.37±0.28	0.23±0.07	1.65
cis-3-Hexen-1-ol	63.87±32.25	45.12±9.39	1.42
trans-2-Pentenal	0.81±0.50	1.14±0.16	0.72
Benzyl alcohol	0.27±0.15	0.43±0.11	0.63
2-Phenylethanol	0.24±0.04	0.42±0.15	0.58
Hexyl alcohol	20.02±16.24	36.08±13.02	0.56
3-Methylbutanol	15.74±6.04	34.80±5.91	0.45
Hexanal	54.12±27.38	120.18±14.58	0.45
1-Pentanol	2.21±1.42	4.98±1.07	0.44
1-Nitro-2-phenylethane	0.42±0.28	1.25±0.30	0.33
trans-2-Heptenal	0.20±0.10	0.84±0.23	0.24
2-Isobutylthiazole	0.48±0.29	5.05±0.75	0.09
Isovaleronitrile	0.61±0.33	6.90±1.80	0.09
6-Methyl-5-hepten-2-one	0.27±0.10	6.31±1.45	0.04

ENVIRONMENTAL CONTROL OF FRUIT QUALITY

A wealth of descriptive studies have outlined the effects of environmental factors (temperature, light, CO_2) or cultural management (fruit pruning, fertilization, water and saline stress) on quality traits. Environmental factors may impact fruit quality through their effects on carbon fixation and distribution among sink organs, but also through stress-induced osmotic and turgor regulations, and stimulation of antioxidant metabolism in the case of induced oxidative stress. Because in many studies fruit composition is often reported on a fresh weight basis, it is not clear whether the reported enhancement of quality is due to an actual stimulation of the biosynthetic capacity and/or to dilution/concentration effects associated with changes in fruit size and water content (Koricheva, 1999).

Important traits of quality (size and composition) have been assessed mainly in response to water or salinity stress (Dorais *et al*., 2001; Guichard *et al*., 2001), partial root-drying (Zegbe *et al*., 2006), assimilate partitioning and carbohydrate compartmentation in fruit (Ho, 1996; Prudent *et al*., 2011) and fertilizers (Bénard *et al*., 2009). Generally, a moderate water or carbon stress promotes fruit quality through an increased accumulation of compounds involved in taste and health value (e.g. sugars and acids, aroma, vitamins, lycopene), but reduces fruit size and marketable yield (Ripoll *et al*., 2014). Compromises between quality and yield need to be found by manipulating the fruit environment to alter relevant physiological processes such as carbon metabolism and water relations. For instance, it has been demonstrated that both water content and chemical composition of tomato fruits can be manipulated by water and salt stresses with small loss of yield (Ehret and Ho, 1986; Dorais *et al*., 2001), especially if moderate stress is imposed during the mid to late stages of fruit development (Mizrahi *et al*., 1988; Ripoll *et al*., 2016). Under high electrical conductivity (EC), fruit size is inversely related to EC while the fruit dry matter content linearly increases with EC at a rate that depends on cultivars, environmental factors, composition of the nutrient solution and crop management (Dorais *et al*., 2001).

Most of the aroma and health-promoting compounds accumulated in tomato are regulated by environmental factors, in particular light and temperature (reviewed by Dorais *et al*., 2008; Poiroux-Gonord *et al*., 2010; Fanciullino *et al*., 2014). Low temperatures (above 10°C) during the growth period are generally favourable to the accumulation of ascorbic acid, phenolic compounds and carotenoids, whereas temperatures above 26°C decrease lycopene and β-carotene but promote rutin and caffeic acid derivatives (Gautier *et al*., 2008). Generally, good exposure or high light intensity is a positive factor for the accumulation of ascorbic acid, lycopene, β-carotene and phenolic compounds in tomatoes (Gautier *et al*., 2008; Truffault *et al*., 2015). For instance, seasonal variations in vitamin C levels have been described in relation to light intensity variations and sugar content, and fruits harvested during summer contain 8–50% more vitamin C compared with fruits harvested during early spring (Massot *et al*., 2010). Fruit shading experiments have shown that AsA accumulation is not limited by leaf photosynthesis or sugar substrate but strongly depends on the fruit irradiance itself (Gautier *et al*., 2008; Massot *et al*., 2010). Moreover, the red to far-red ratio, which increases fourfold in pericarp tissues during ripening, stimulates lycopene accumulation. This red-light-induced lycopene accumulation is regulated by fruit-localized phytochromes and their photo-equilibrium in response to changes in the spectral composition of the light that penetrates the pericarp during ripening (Alba *et al.*, 2000).

The effects of drought and high salinity on the accumulation of vitamins and secondary metabolites are more complex, and conflicting results have been reported (review by Poiroux-Gonord *et al.*, 2010; Ripoll *et al.*, 2014). This is due to the fact that environmental factors may have indirect effects related to photosynthesis and carbon allocation (precursor availability) and/or direct effects on their biosynthesis. For instance, drought stress may be regarded as a negative factor for the synthesis of secondary metabolites because it limits photosynthesis; but, on the other hand, water (or other abiotic factor) stress may exacerbate photo-oxidative stress, thus providing a positive stimulus for their synthesis, probably involving signals transmitted from leaves to fruits (Poiroux-Gonord *et al.*, 2010). High salinity has global positive effects on ascorbic acid, lycopene and β-carotene (Frary *et al.*, 2010), with strong genotype × environment (G×E) interactions (Gautier *et al.*, 2009), whereas conflicting responses with regard to phenolic compounds are reported.

Nitrogen depletion lessens the accumulation of phenolic compounds alongside a slight increase in ascorbic acid concentration (Bénard *et al.*, 2009).

The environmental control of tomato fruit texture has been mainly investigated in the postharvest period, while the effects of environmental factors during fruit development have been more rarely reported (Sams, 1999; Rosales *et al.*, 2009). Generally, tomato firmness (physical component of texture) decreases in summer, which is empirically attributed to high temperature and high vapour pressure deficit. Regarding the mineral nutrition, high EC and high magnesium (Mg) supply increase fruit firmness in summer production, while high calcium (Ca) supply reduces fruit firmness (Hao and Papadopoulous, 2003). The latter authors suggested that high EC and high Mg might increase fruit tissue rigidity, whereas high Ca might increase only tissue plasticity. Water deficit also induces significant variations in fruit firmness with contrasting effects (Ripoll *et al.*, 2014). A moderate water deficit decreases firmness measured by compression test but increases firmness measured by puncture test, which correlates well with firmness and crunchiness assessed by sensory evaluation. Contrasting effects reported in the literature likely result from different methods of texture evaluation, different stress intensity, strong G×E interactions, and from the complex interactions among the numerous processes involved in final fruit texture.

In the postharvest period, the effects of temperature during storage and atmospheric composition on fruit quality have been primarily investigated. In particular, low temperatures, used by retailers or consumers to extend fruit shelf-life, may trigger physiological disorders and loss of quality. The optimum storage temperature differs with the stage at which fruits are harvested, due to different influences on the enzymatic activity (reviewed by Passam *et al.*, 2007).

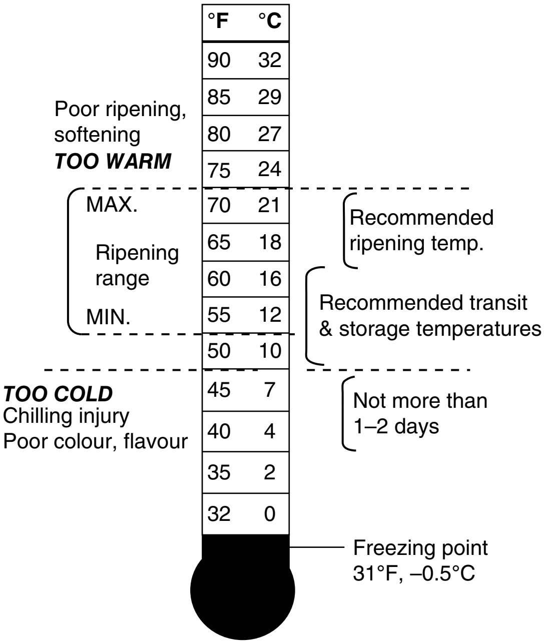

Fig. 5.8. Recommended temperature for tomato fruit storage. (Cantwell, 2010.)

It also differs between varieties. For instance, long or extended shelf-life varieties may not be able to attain full red colour if harvested too early (Cantwell, 2010). Although fully ripe tomatoes may be held at 2–5°C for a few days prior to consumption (not longer, since colour loss and softening may occur), fruits that are mature-green or at the turning or breaking stage should not be subjected to temperatures lower than 12°C, to avoid chilling injury and relative symptoms such as rubbery texture, watery flesh and irregular ripening (Fig. 5.8) (Stevens *et al.*, 2008; Cantwell, 2010). Fruits stored at 5°C were rated by sensory analysis as significantly lower in ripe aroma (attributed to a loss of the principal volatile components), sweetness and tomato flavour, and significantly higher in sourness, compared with those stored at 20°C (Passam *et al.*, 2007). Although low temperatures during storage lower the lycopene content of fruit (twice less after 10 days storage at 7°C compared with 15°C or 25°C) even when harvested at a mature-red stage, they do not necessarily reduce the total antioxidant capacity related to phenolics and ascorbate over a limited period (Passam *et al.*, 2007).

GENETIC CONTROL OF FRUIT QUALITY

Major components of tomato fruit quality, such as fruit size, appearance, firmness and shelf-life, are quantitative traits that have continuous variation in segregating populations (Causse *et al.*, 2002). They have been tremendously modified by breeding during the past 50 years, leading to a large diversity in size, shape and colour. For instance, domestication has increased fruit size up to 500-fold in cultivated species. On the contrary, the genetic improvement of organoleptic quality or health value has been considered only recently. The genetic control of interesting traits of quality relies either on a few natural mutations with tremendous effects or on the accumulation of numerous quantitative trait loci (QTLs) and genes with small individual effects. The multi-genic control of most traits of fruit quality, the co-localization of QTLs/genes with contrary effects, and the interactions among QTLs/genes and with the environment or cultural practices put a damper on the fast genetic improvement of tomato quality (Causse *et al.*, 2011). For instance, the antagonistic relationship between fruit size and taste represents a locking point for breeders. As a result, cherry tomatoes have the best flavour and higher contents in sugar and acids than large-fruited genotypes (Causse *et al.*, 2011). Although very few QTLs have been identified at the molecular levels, great advances are expected in the coming years by manipulating the components of tomato, thanks to progressing knowledge of the underlying genes, the release of the tomato genome sequence and the recent advances in sequencing and genotyping methods that allow identification of important genes and loci through analyses based on single nucleotide polymorphism (SNP) such as QTL mapping and genome-wide association study (GWAS) (Zsögon *et al.*, 2017). Many genetic engineering approaches have been already developed to modify the expression of genes controlling quality attributes, especially during ripening (e.g. for improving the taste, sugar-to-acid ratios, aroma and carotenoid levels in tomato) (Lewinsohn *et al.*, 2001; Guo *et al.*, 2012), but the use of genetic transformations for commercial purposes is highly regulated. On the other hand, the natural genetic variability within or among tomato species offers great potential for creating new improved varieties (Lin *et al.*, 2014). Until now, most genetic studies on QTL detection have been performed on populations derived from interspecific crosses between wild species and processing tomatoes. Numerous QTLs for fruit weight, shape, sugar and acid contents, firmness, volatiles and sensory attributes have been detected (reviewed in Labate *et al.*, 2007, and in Causse *et al.*, 2011). The introduction of five major regions bearing QTLs of quality has been successfully realized (Lecomte *et al.*, 2004) and fruit quality could be improved with favourable effects provided by cherry tomato (Causse *et al.*, 2002). Yet, interactions between genes or QTLs and environmental conditions should be considered (Albert *et al.*, 2015). Major mutations and genes for improving tomato fruit quality have been reported by Causse *et al.* (2011) (Table 5.4).

Fruit size and shape

Many QTLs of fruit mass have been detected. Six of them explain more than 20% of the phenotypic variance (Grandillo *et al.*, 1999), among which fw2.2

Table 5.4. Major mutations and genes for improving tomato fruit quality. (Reprinted from Causse *et al*. 2011, by permission of John Wiley & Sons.)

Quality trait	Mutation	Phenotype	Chromosome	Activity	Reference
Size and shape	*fw2.2 (QTL)*	Fruit weight	2	Transcription factor	Cong & Tanksley, 2006
	fas (fasciated)	Fruit shape	11	Transcription factor	Cong *et al*., 2008
	o (ovate)	Fruit shape	2	Transcription factor	Liu *et al*., 2002
	SUN	Fruit shape	7	IQD protein	Xiao *et al*., 2000
Sugar content	*Lin5*	Increased sugar content	9	Cell wall Invertase	Fridman *et al*., 2000
Vitamin C	*Vtc9.1*	Higher vitamin C	9	MDHAR	Stevens *et al*., 2008
Shelf life	*Rin*	Inhibited ripening (semi-dominant)	5	MADS-box transcription factor	Vrebalov *et al*., 2002
	nor (non ripening)	Inhibited ripening (semi-dominant)	10	Transcription factor	Moore *et al*., 2002
	Nr (Never-ripe)	Inhibited ripening (dominant)	9	C_2H_4 receptor	Wilkinson *et al*., 1995
	Cnr (Colourless non-ripening)	Inhibited ripening (dominant)	2	Epigenetic control	Thompson *et al*., 1999; Seymour *et al*., 2002
Colour	*B* (Beta)	Yellow fruits	6	Lycopene cyclase	Ronen *et al*., 2000
	og (old gold-crimson)	Higher lycopene content	6	Lycopene cyclase	Ronen *et al*., 2000
	Del (Delta)	Orange fruits	12	Lycopene cyclase	Ronen *et al*., 1999
	r (yellow flesh)	Yellow fruits	3	Phytoene synthase	Fray & Grierson, 1993
	t (tangerine)	Orange fruits	10	Carotenoid isomerase	Isaacson *et al*., 2002
	hp-2 (high pigment)	Higher lycopene content	12	DETI homolog	Mustilli *et al*., 1999
	hp-I (high pigment)	Higher lycopene content	2	DDBI light signaling	Liu *et al*., 2004
	Dg (dark green)	Higher lycopene content	12	DETI homolog; allelic to hp1	Levin *et al*., 2003
	y	Uncoloured epidermis	1	MYB transcription factor	Adato *et al*., 2009

controls 30% of fruit size variations and is involved in the control of cell division in the pre-anthesis period. Among 13 QTLs that contributed to fruit mass enlargement during the improvement of modern tomato cultivars, five are located at the distal end of the long arm of chromosome 2 (fw2.2, lcn2.1, fw2.1, fw2.3 and lcn2.2), whereas five others related to fruit mass (fw1.1, fw5.2, fw7.2, fw12.1 and lcn12.1) likely contributed to the enlargement of tomato fruits during domestication of wild species (Lin *et al.*, 2014).

Three major QTLs for fruit shape have been identified on chromosomes 2, 7 and 8; and two major QTLs for locule number, fasciated (*fas*) and locule number (*lc* or *lcn2.1*), are located on chromosomes 11 and 2, respectively. Other QTLs such as OVATE and SUN modify fruit shape more specifically, as they are responsible for elongated fruit shape (reviewed in Causse *et al.*, 2011).

Fruit texture

Forty-six QTLs controlling texture as assessed by touching (30 QTLs), by mechanical measurement (11 QTLs) and by sensory evaluation (five QTLs) were reported in several populations, with a few main clusters on chromosomes 1, 2, 4, 5, 9, 10 and 11 (Causse *et al.*, 2002; Labate *et al.*, 2007). At the molecular level, most studies on texture have focused on the mechanisms that lead to fruit softening during ripening and on the characterization of mutants affected in ripening-related genes; for instance, the ripening inhibitor (*rin*) gene, non-ripening (*nor*) gene, delayed-fruit deterioration (*DFD*) gene or colourless non-ripening (*Cnr*) gene (Seymour *et al.*, 2002; Giovannoni, 2004; Saladié *et al.*, 2007). On chromosomes 2, 5 and 10 the firmness QTL co-localized with the genes *rin*, *nor* and *Cnr*. Interestingly, the divergence between firm processing tomatoes and soft fresh tomatoes resides in SNPs located on chromosome 5 in a region where a major QTL for firmness has been detected (Lin *et al.*, 2014). However, no key determinants of fruit texture have been clearly identified, due to the high number of processes involved, including processes unrelated to cell wall loosening (Saladié *et al.*, 2007; Aurand *et al.*, 2012).

Fruit composition in sugar and acids

Increasing fruit sugar content is one main objective, of breeding programmes focusing on organoleptic quality. This is particularly true for tomato, as domestication has led to a loss of flavour, despite abundant literature and knowledge on the physiological and genetic factors controlling the main steps of sugar and acid metabolisms in tomato (Beckles *et al.*, 2012; Etienne *et al.*, 2013). The number of regions bearing QTLs for sugar and acid contents obtained in different populations is very high (Labate *et al.*, 2007), likely because of the high number of processes involved and strong G×E interactions (Prudent *et al.*, 2009; Albert *et al*., 2016). For instance, among 56 regions (95 QTLs) bearing QTLs for sugar content (or Brix), about 28 regions were found in several populations (the wild allele mostly increasing the trait) (Labate *et al*., 2007). Only a few regions are common to sugars and acids (Sauvage *et al*., 2014) while frequent colocations between sugars and fruit weight with opposite allelic effects have been detected (Grandillo *et al*., 1999; Prudent *et al*., 2009). At the molecular level, a high number (more than 60) of candidate genes involved in carbon metabolism have been identified (Causse *et al*., 2011). Among them, a few genes encoding major enzymes of sugar metabolism affect the final fruit sugar content. For instance, *Lin5*, encoding the apoplastic invertase, is involved in the control of total soluble sugar content due to an increased capacity to uptake the phloem unloaded sucrose (Baxter *et al*., 2005) and would explain a QTL for soluble compounds on chromosome 9 (Sauvage *et al*., 2014). On chromosome 10, a QTL controlling fructose content contains two cell wall invertase genes *Lin6* and *Lin8* (Albert *et al*., 2016). Similarly, *AgpL1*, *AgpL2*, *AgpL3* and *AgpS1* regulate the AGPase activity and starch synthesis during the expansion stage of fruit development, which constitutes a reservoir contributing to higher soluble sugar content at the ripe stage (Causse *et al*., 2011).

Fruit composition in health-promoting phytochemicals

Metabolic engineering of plants to produce novel compounds or to improve the production of existing compounds is possible by over-expressing one or more specific genes coding for enzymes that control key steps of the known biosynthetic pathways.

Several QTLs for fruit ascorbic acid have been identified in different populations with a range of phenotypic variation from 6 to 50 mg/100 g fresh weight (Stevens *et al*., 2007; Ruggieri *et al*., 2014). The ascorbic acid concentration in cells depends on its biosynthesis, recycling and degradation. Thus, efficient ways to manipulate ascorbic acid content might be achieved via regulatory genes of the synthesis or via the up-regulation of recycling. Two main genes, both on chromosome 9, co-localized with QTL for fruit ascorbic acid: *MDHAR* (monodehydroascorbate reductase) involved in the recycling and *GME* (GDP-mannose epimerase) involved in the synthesis of ascorbic acid (Stevens *et al*., 2007, 2008; Sauvage *et al*., 2014).

Several mutants of different genes of the carotenoid biosynthesis pathway have been identified (reviewed in Liu *et al*., 2003). For instance, the *yellow-flesh* (*r*) mutant corresponds to a loss of the phytoene synthase gene resulting in the absence of lycopene. Another single dominant gene, *Del*, encoding a lycopene ε-cyclase, changes the fruit colour to orange in the tomato mutant *Delta* as a result of accumulation of δ-carotene at the expense of lycopene (Ronen *et al*., 1999). Similarly *Beta* (*B*) is a partially dominant, single-locus mutation that causes an orange colour in the fully ripened fruit because of the accumulation of β-carotene at the expense of lycopene. In the wild type, β-carotene constitutes 5–10% of total fruit carotenoids, whereas in *Beta* it is 45–50% and can exceed 90% (Ronen *et al.*, 2000). The existence of *r*, *B* and *Del* in wild tomato species suggests a hypothetical scenario for the evolution of fruit colour in tomato (Ronen *et al.*, 2000). Other genes are involved in the carotenoid and other antioxidant contents. For instance, the *high pigment* (*hp*) mutant, a light-signalling gene not involved in the carotenoid biosynthesis pathway, also affects the carotenoids, flavonoids, ascorbic acid and sucrose contents in leaves and fruits (reviewed in Poiroux-Gonord *et al.*, 2013, and in Causse *et al.*, 2011). However, only a few candidate genes involved in the carotenoid biosynthesis pathway co-localize with colour QTL, indicating a complex regulation (Causse *et al.*, 2003; Liu *et al.*, 2003).

Concerning aroma formation, the complexity of the multiple biosynthetic pathways contributing to volatile composition has discouraged tomato breeders and there are relatively few examples of genetic improvement enhancing the profile or quantity of tomato fruit volatiles (Lewinsohn *et al.*, 2001; Tieman *et al.*, 2012). However, a few recent studies identified important loci that control aroma in tomato. Thirty QTLs spread over the genome (except on chromosomes 8 and 9) and affecting 24 different volatile compounds were mapped in a population of introgression lines derived from a cross between the cultivated tomato *Solanum lycopersicum* and its wild relative, *Solanum habrochaites* (Mathieu *et al.*, 2009). In a *Solanum pennellii* introgression line (IL) population, 25 loci were identified that significantly alter one or more of 23 different volatiles (Tieman *et al.*, 2006). More recently a GWAS analysis on 28 main volatiles outlined significant associations on chromosomes 2 and 4 where previous QTLs have been identified (Zhang *et al.*, 2015). Genes regulating the output of volatile synthesis pathways and associated with flavour-imparting volatiles in tomato have been reviewed by Klee and Tieman (2013). Moreover, a large natural diversity in volatile composition was reported in Heirloom populations of tomato (Table 5.5), which could be exploited with molecular-assisted breeding techniques (Tieman *et al.*, 2012).

FRUIT PHYSIOLOGICAL DISORDERS

A number of disorders affect the quality of fresh market tomato fruit and reduce the marketable yield. These disorders result from a combination of environmental, production or handling procedures, or are genetic in origin. A few of them are represented in Fig. 5.9.

Blossom-end rot

This disorder (BER) is associated with a local calcium deficiency in the distal fruit tissue, resulting from a misbalance between calcium supply/transport and

Table 5.5. Observed variations in flavour volatiles within *S. Lycopersicum* Heirloom varieties. (Reprinted from Current Biology 22(11), Tieman *et al.,* The chemical interactions underlying tomato flavor preferences, 1035-1039, Copyright 2012, with permission from Elsevier.)

	High	Low	Fold Difference	Median
1-penten-3-one	9.37	0.17	55	1.18
isovaleronitrile	68.45	0.58	117	7.63
trans-2-pentenal	5.16	0.31	17	1.23
trans-2-heptenal	2.71	0.09	30	0.42
isovaleraldehude	51.08	1.55	33	8.59
3-methyl-1-butanol	184.46	3.20	58	27.26
methional	1.616	0.012	137	0.07
isovaleric acid	0.953	0.004	262	0.09
2-isobutylthiazole	63.61	0.37	174	8.34
6-methyl-5-hepten-2-one	20.07	0.17	120	3.38
β-ionone	0.396	0.008	47	0.05
phenylacetaldehyde	1.90	0.00	654	0.24
geranylacetone	28.96	0.03	1095	1.22
2-phenylethanol	5.269	0.002	3142	0.05
isobutyl acetate	11.93	0.14	85	1.67
cis-3-hexen-1-ol	124.15	10.00	12	40.00
1-nitro-2-phenylethane	2.59	0.02	149	0.25
trans,trans-2,4-decadienal	0.30	0.00	211	0.02
2-methylbutanal	14.66	1.14	13	3.47
hexyl alcohol	84.03	0.99	85	13.86
guaiacol	8.09	0.03	290	0.77
hexanal	381.05	15.55	25	88.65
1-octen-3-one	0.312	0.017	18	0.07
cis-3-hexenal	399.66	8.29	48	71.09
methylsalicylate	14.16	0.00	3354	0.40
trans-2-hexenal	48.01	0.39	123	3.54
β-damascenone	0.1733	0.0020	86	0.01
2-methyl-1-butanol	115.69	1.93	60	15.08

Note: Volatile emissions were measured as ng/g fresh weight/hr.

requirement for growth, especially during the phase of rapid fruit expansion, which is the most sensitive period (Ho and White, 2005). Symptoms occur first in the internal part of the fruit and then extend to the external tissues. The blossom end of a green fruit develops a water-soaked area near the blossom scar. The area dries, turns brown as a consequence of cell death, and there is subsequent leakage of cell contents into the extracellular space. High temperature, light, vapour pressure deficit (i.e. low humidity) and salinity, irregular watering, plant vigour, high nitrogen fertilization and root pruning are conducive factors. The sensitivity is genotype-dependent. For instance, large-fruit

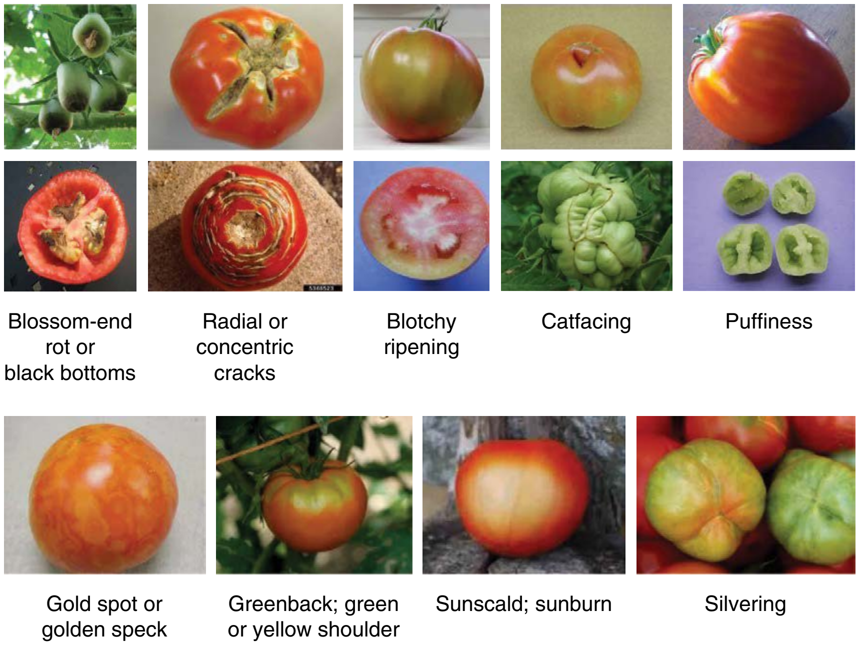

Fig. 5.9. Illustrations of main tomato disorders that impact negatively on marketable yield.

or plum tomatoes are much more sensitive than cherry or wild tomatoes. The incident of blossom-end rot increases significantly when the concentration of calcium in the fruit falls below 0.08% (dry weight) while the disorder seldom occurs at levels above 0.12%, and spraying Ca directly on to young fruits is recommended for the prevention of BER (Ho and White, 2005).

Cracking and russeting

Cracking occurs when the internal expansion is faster than the expansion of the epidermis and so the epidermis splits. Concentric cracks can develop around the stem end of the fruit, and/or radial cracks can develop that extend from the stem to the blossom end. Cracking can occur at all stages of fruit growth, but as the fruit matures they become more susceptible, especially as colour develops. A rapid influx of water and solutes, and reduced strength and elasticity of the tomato skin and pericarp wall contribute to the occurrence and severity of cracking. This disorder is not only unsightly but breaks in the epidermis also increase water loss and the entry of pathogens. Cracking can be minimized by the uniform application of water to avoid periods of water stress, adequate calcium nutrition and the selection of crack-resistant cultivars. Cracking and splitting are inherited tendencies and cultivars differ greatly in susceptibility. The development of numerous very fine cracks is another disorder, known as russeting or micro-cracks. Unlike cracking where the cracks extend several millimetres into the pericarp, russeting is the development of numerous fine cracks in the tomato skin. Microbial infection is not a significant problem but water loss is increased and visual appearance decreased (reviewed in Dorais *et al.*, 2004).

Blotchy ripening

Blotchy ripening leads to uneven ripening with hard grey to yellow patches, usually near the calyx end of the fruit, retaining chlorophyll and not accumulating sufficient lycopene to produce a normal red fruit. Blotchy areas contain less nitrogenous compounds, organic acids, starch, sugars and dry matter (Yahia and Brecht, 2012). Early and mid-season greenhouse crops are particularly susceptible. The cause of this physiological disorder and its relationship to 'grey wall' is not well understood. Improper plant nutrition (e.g. potassium and/or boron deficiency and high nitrogen levels, which promote excessive growth), insect feeding and environmental stresses (e.g. chilling, temperature above 30°C, low light intensity or high soil moisture) may contribute to the occurrence of this disorder (Yahia and Brecht, 2012).

Catfacing

Catfacing is a generic term used to describe a tomato fruit that has a gross deformity. In this case the differential growth of the various locules produces a convoluted shape in contrast to the smooth shape of most fruits. Symptoms of catfacing also include corky brown scarring, cracks and uneven ripening. Large-fruited tomatoes with high number of locules (more than five) are more prone to this problem than small-fruit varieties. Critical flower development or pollination may be responsible for this problem. Moreover, low temperature and light intensity during flowering and fruit set may exacerbate the occurrence of catfacing (CTIFL, 2011; Yahia and Brecht, 2012).

Puffiness

Puffy fruits are flat-sided or angular fruits in which one or more seed cavities (i.e. locules) are empty of some or all tissue. Puffiness is also known as boxiness or hollowness. It may be impossible to detect light puffiness until the fruit is cut and open cavities are observed between the seed gel area and the outer wall. Puffy fruits are less dense than good ones and so they can be separated by flotation in water. Growing conditions (too low or too high temperatures, high N, low light) that cause improper pollination, fertilization or seed development and genotype contribute to the occurrence of this disorder. Distal fruits of the inflorescence are more frequently affected (CTIFL, 2011; Yahia and Brecht, 2012).

Gold spot

Gold spot is described as yellow or white flecks distributed all over the fruit surface, and more especially near the calyx and shoulder of the fruit. The number can vary from a few to many. Flecks develop as small irregular-shaped green spots on the surface of immature fruit and turn gold as the fruit ripens. Gold spot is due to calcium oxalate crystals resulting from an excess of calcium. The disorder depends on genetic and environmental factors. Plum, Roma and saladette tomato types appear to be more susceptible than round tomatoes. The abundance of gold spot is increased by high humidity and high Ca fertilization (CTIFL, 2011; Yahia and Brecht, 2012).

Green shoulder

In this case the shoulders of a ripening fruit near the calyx remain green due to a high chlorophyll content, while the rest of the fruit turns red. While generally undesirable, this condition is actually preferred by consumers in some countries. Incorporating the 'uniform ripening' gene can eliminate this disorder in susceptible cultivars. Predisposing factors include exposure to excessive heat, high salinity and an inadequate supply of potash and phosphate fertilization (CTIFL, 2011; Yahia and Brecht, 2012).

Sunscald

Sunscald can be described as a yellow, hard area, usually on the shoulder of the fruit, which occurs when tissue temperature rises above 30°C and prevents the synthesis of red pigments and the softening of flesh. Fruit temperatures above 40°C are lethal and the exposed tissue will die, turn white, dry out and form a flat parchment-like covering over the affected area. Green fruits are more sensitive to solar injury than ripe ones. Damage usually occurs when fruits are suddenly exposed to sunlight, for instance after an over-pruning of leaves. Indeed, direct sunlight may increase the temperature of exposed fruit tissue by 10°C or more above ambient air temperatures. Plant architecture that shades the fruit during growth and bin covers that shade the harvested fruit during transport to packing facilities are the most effective ways to reduce sunscald (CTIFL, 2011; Yahia and Brecht, 2012).

Silvering

Silvering is represented by silver-green streaks that turn yellow as the fruit ripens. They are due to abnormal tissue development but the actual causes are unknown. Symptoms are more frequent in early crops with short day-periods or in case of thermal shocks (low temperature) (CTIFL, 2011).

OPTIMIZING FRUIT QUALITY THROUGH MODELLING APPROACH

As previously discussed, all the major traits of quality result from complex interactions and regulation loops among numerous processes, regulated at the plant and fruit levels. These processes are controlled by interactions between genetic, environmental and management (G×E×M) factors and the qualities of a given genotype under contrasted environments are hardly predictable. Significant variations in all quantitative traits of quality have been reported in response to G×E×M interactions (Causse *et al*., 2003; Bertin *et al*., 2009; Prudent *et al*., 2009). A multi-site experiment including 42 tomato genotypes revealed as much as 211% change in performance of some genotypes in a particular location. Lycopene was found to be most influenced by the environment, whereas total acidity was the least influenced (Panthee *et al*., 2012). Such interactions are difficult for breeders and producers to handle and multi-site tests are necessary for developing varieties that will perform consistently well across multiple environments. However, exploring all G×E×M combinations by experiments is an endless task. An alternative approach relies on the development of process-based simulation models (PBSMs) (Bertin *et al*., 2010; Kromdijk *et al*., 2014; Bertin and Génard, 2018). Indeed PBSMs allow complex traits, such as quality, to be unravelled by simulating interactions among the various components or processes that underlie the trait of interest. Moreover, G×E×M interactions are emergent properties of simulation models, i.e. unexpected properties generated by complex interconnections between subsystem components and biological processes. In this perspective, bottom-up (based on mechanistic knowledge of underlying processes), top-down (statistical regression to establish links between data and phenotype) or middle-out (combination of bottom-up and top-down) modelling approaches have been developed (Génard *et al*., 2007; Yin and Struik, 2010) (Fig. 5.10). In these models, the so-called component traits are characterized in terms of model parameters, which instead of the complex trait itself may subsequently be linked to underlying genetic variations (Fig. 5.11).

Several predictive models of the processes involved in the quality of fruit have been developed (Martre *et al*., 2011). Concerning tomato growth, we can report a model of cell division and endoreduplication (Bertin *et al*., 2007) and a model of cell expansion related to carbon and water fluxes

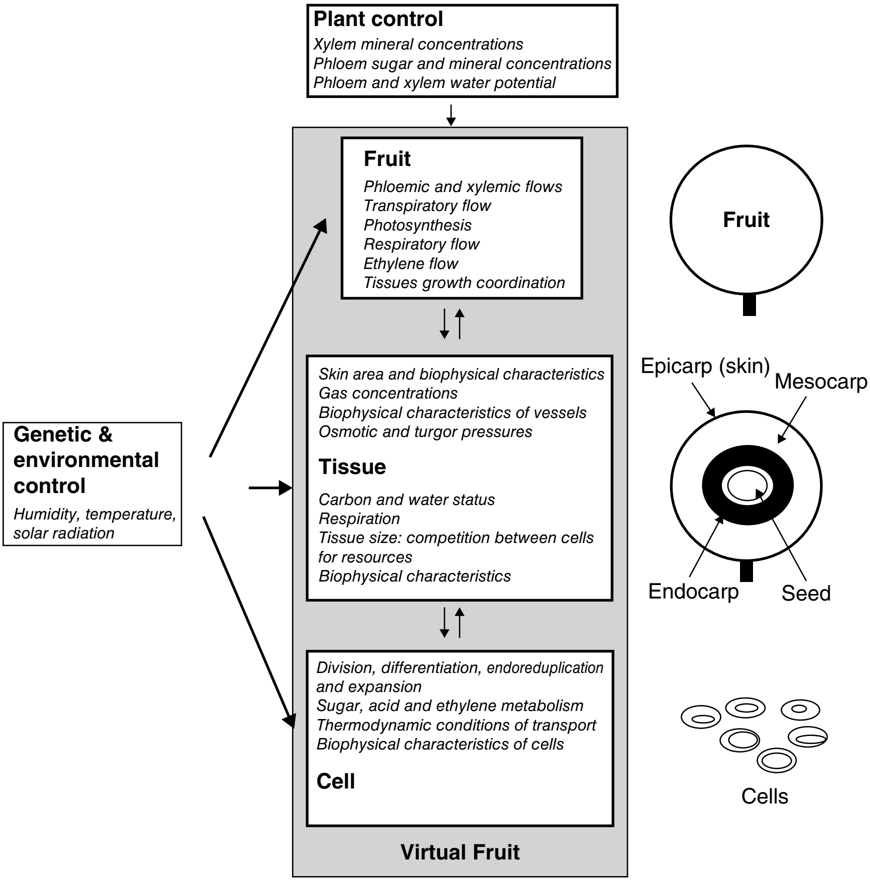

Fig. 5.10. Schematic presentation of the virtual fruit model organization in levels and objects (right). For each level, constraints on the lower level are given in the lower part of the box and initiating conditions are given in the upper part of the box. (Génard *et al.*, 2007, reprinted by permission of the Society for Experimental Biology.)

(Liu *et al.*, 2007). Concerning fruit composition, rare models of sugar and acid metabolism have been developed for peach (Génard *et al.*, 2003; Lobit *et al.*, 2003, 2006). These models could be easily transferred to tomato, taking account of species-dependent control of sugar and acid metabolisms. Several PBSMs that predict fruit quality as a function of the environment or crop management are now available. The next important steps to progress in this field will be: (i) the integration of the sub-models to consider the complex interactions and feedback regulations across the various organizational levels (Fig. 5.10); and (ii) the coverage of the genetic and molecular control on the modelled processes.

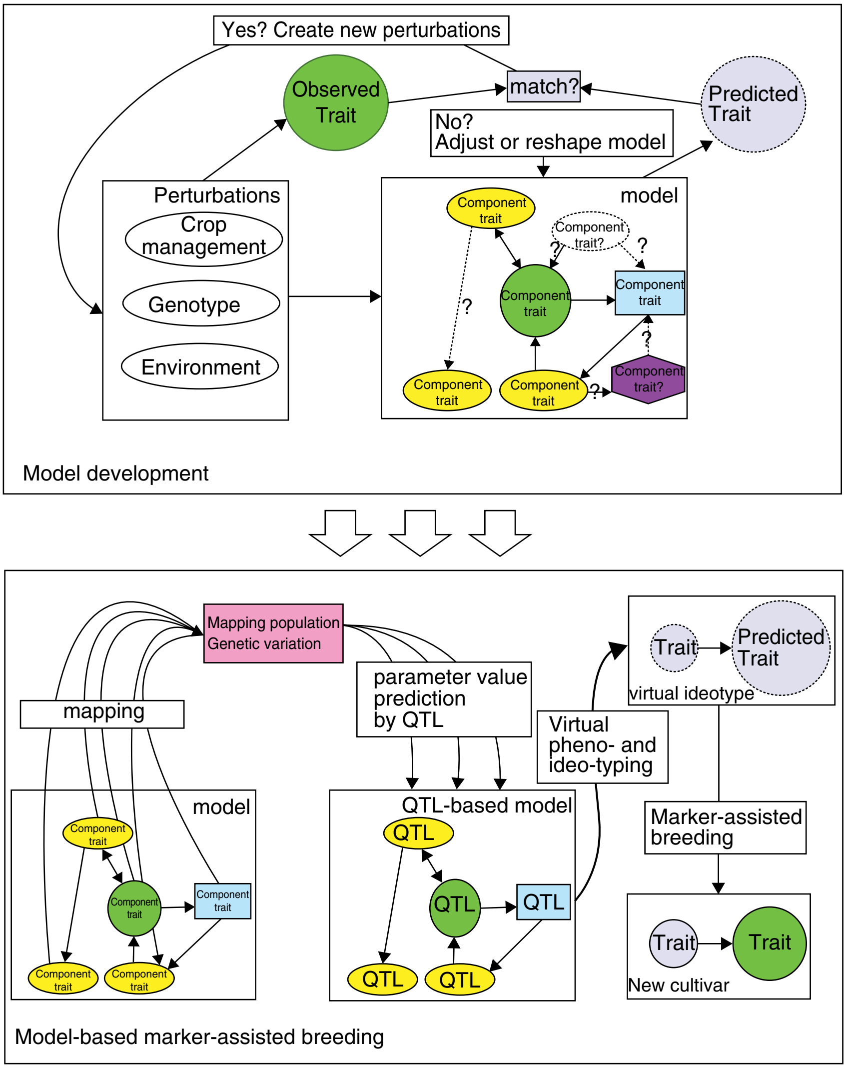

Fig. 5.11. Scheme of recommended workflow to use the combination of physiological modelling and breeding based on molecular markers. First, model development takes place by decomposing complex traits into components (characterized in terms of model parameters). To find reliable values for these model parameters, various perturbations are applied and the resulting data are used to evaluate or calibrate the model. Next, sensitivity analysis is applied to find influential component traits, which are selected for mapping on to marker-defined chromosomal regions. These model parameter values can subsequently be predicted by means of the identified correlations with one or various QTLs. The resulting QTL-based version of the model allows prediction of phenotypic traits based on QTLs and as such can also predict performance of novel QTL

Current fruit models have been developed either at cell or at organ level (one big cell model) (Génard *et al.*, 2007). Actually, the cell level is likely to be the elementary level for mechanistic modelling of fruit which will further allow linking of the fruit model with cellular models describing complex metabolic pathways and molecular regulatory networks. Therefore, modelling the way in which cell division and expansion progress together is crucial to understanding the emergence of specific morphological traits (Baldazzi *et al.*, 2012). Recently, models integrating cell division, cell expansion and DNA endoreduplication in the tomato fruit have been proposed (Fanwoua *et al.*, 2013; Baldazzi *et al.*, 2017). At a higher scale of integration, the virtual tomato model has been connected to a plant model describing water and carbon fluxes in the plant architecture, and the induced gradients of water potential and phloem sap concentration in carbon within the plant (Baldazzi *et al.*, 2013). Such integrated models centred on the fruit open new perspectives to integrate information on the molecular control of fruit cellular processes into the fruit model and to analyse the effects of G×E×M interactions on fruit quality. Yet the prediction of other traits of quality including the accumulation of healthy compounds such as vitamins and carotenoids still needs some developments.

Covering the genetic control in plant and fruit models is still far from satisfactory, despite promising approaches (Bertin *et al.*, 2010). Model parameters are considered as either generic parameters when they do not vary among genotypes, or as genotypic parameters (also called genetic coefficients) when they are genotype-dependent. Each set of gene or allele combination is represented by a set of parameters and the phenotype can then be simulated *in silico* under various environmental and management conditions. This implies that model parameters are usually specific to one genotype, restricting the validity range of the model itself. To overcome this limitation, the values of the genotypic parameters have to be predicted depending on combinations of QTLs (QTL-based models), alleles or genes (gene-based models) involved in the process that is modelled. Until now only a few genotypic parameters (i.e. allelic variants) have been advantageously introduced into simulation models. Regarding fruit quality, a QTL-based model of peach quality (Quilot *et al.*, 2005) and a model of tomato sugar composition (Prudent *et al.*, 2011) have been proposed. The model-based approach has clear benefits over traditional QTL mapping,

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Fig. 5.11. Continued.
combinations or different growth environments or management practices, i.e. virtual phenotyping. This procedure can be used to derive virtual ideotypes, which can be realized by means of marker-assisted breeding. In this way, trait improvement can be reached in a fast and efficient way; namely, more QTLs are identified; QTLs are likely to be more robust, and when the environment is known, it is more straightforward to identify which markers will give the largest trait improvement. (Kromdijk *et al.*, 2014, reprinted by permission of the Society for Experimental Biology.)

since more QTLs are usually identified that tend to be more stable and the importance of which can be ranked under varying conditions. However, its wider implementation is hampered by the lack of genetic information on the traits and processes simulated by models, and also by the low degree of functionality of current PBSMs, which should be further refined in order to bind model parameters and physiological components. An important issue of simulating G×E×M interactions is the design of ideotypes or QTL/gene combinations relevant to optimizing fruit growth and quality under specific conditions by multi-criteria optimization methods (Quilot-Turion *et al*., 2012; Génard *et al*., 2016; Constantinescu *et al*., 2016), adding genetic constraints to account for pleiotropic and linkage effects (Quilot-Turion *et al*., 2016).

CONCLUSION

The various components of tomato quality are involved in many plant biological functions as well as being essential for a healthy and hedonic human diet. Thus improving fruit quality as a whole is a top priority. The increasing knowledge of the genetic and molecular controls of many traits of quality, the development of high-throughput methods for plant and fruit phenotyping under varying environments and the development of integrative approaches to unravel G×E×M interactions should rapidly contribute to this objective. Interestingly, most of the chemical components that contribute to the health value of fruits are also involved in plant adaptation and defence against stress. Thus, meeting the social demand for high-quality fruits is likely to be favourable in facing the urgent need to reduce inputs of water, fertilizers and pesticides in intensive production systems. Yet, finding viable compromises between yield and quality remains a challenge. In tomato this should be facilitated by the large genetic resources, such as near-isogenic lines, mutants, transgenics and mapping populations presenting interesting variability in size and composition of fruit.

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6

Irrigation and Fertilization

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INTRODUCTION

This chapter deals with the basic principles of water and fertilizer management in tomato, as well as discussing the practical application of these topics in both open field and greenhouse production of the crop, with recommendations for soil and soilless conditions. Where applicable, a clear distinction between these two production environments is made on the particular aspects of irrigation and plant nutrition. However, there is a great degree of overlapping information that is equally pertinent to greenhouse and field production. Decisions on the use of certain irrigation and fertilization practices will depend on the specific growing conditions (e.g. soil, soilless, tomato types and cultivars, cultural practices) of each grower, as described elsewhere in this book. The basic principles on irrigation and fertilization cover the influence of water quality, irrigation requirements, plant nutrition and fertigation on tomato growth and yield. Sonneveld and Voogt (2009) provided specific information on water and fertilizer management in greenhouse crops (see also Chapter 9). Fertilization in organic tomato production is discussed in Chapter 11.

WATER QUALITY AND IRRIGATION REQUIREMENTS

Salinity and pH

High-quality irrigation water is a prerequisite for any production system. Salinity level and water pH are two of the main parameters used by growers for crop management. Salinity is expressed by the electrical conductivity (EC) of a solution, in units of mS/cm (millisiemens per centimetre) or dS/m (decisiemens per metre). These units represent a quantitative measure of the ability of a material to allow the movement of an electric charge (conductance). Measurements of EC are related to the total concentration of electrolytes in solution (dissolved salts form ionic particles, each with positive or negative charge). Hence, diluted salts contribute greatly to solution conductance. Soil EC can also be correlated with soil texture, cation exchange capacity and soil moisture (Grisso *et al.*, 2009). Irrigation water EC is generally used as an indicator of total fertilizer concentration.

EC measurements are one of the simplest, fastest and least expensive tests available for growers. The test allows growers to have an indicator of the total concentration of ions in solution but it does not discriminate among ions in solution or their concentrations. This is especially important in soilless culture, given the low buffer capacity of most soilless media. Adequate EC levels in tomato production are generally managed between 1.6 and 5.0 dS/m, depending on the cultivar, environmental conditions and production practices (Dorais *et al.*, 2001). Too high EC in the rhizosphere can negatively affect stomata conductance and leaf areas and therefore plant photosynthetic rate, water and nutrient uptake, dry matter accumulation, and fruit development and quality (see Chapter 4).

High salt concentrations in solution outside the root tissue create an equal or higher osmotic potential than inside the plant tissue. Under this condition, water will move out of the plant tissue by osmosis, causing stress by dehydration and eventual wilting (Fig. 6.1). Stomatal conductance decreases (about

Fig. 6.1. Tomato plant subjected to heat and salinity stress. Leaves also show symptoms of calcium deficiency.

30%) but stomata are not completely shut down, leading to desiccation and loss of turgor of the cells (Wu and Kubota, 2008). Salinity in solution tends to increase as water is depleted by transpiration. Furthermore, depending on the concentration of ions in solution, plants can be affected by nutritional imbalance generally associated with sodium (Na^+) and chloride (Cl^-) uptake, competitive uptake, and disorders in transport and partitioning of solutes within the plant.

Nitrogen ($N\text{-}NO_3^-$) uptake decreases under high EC concentrations, possibly due to an interaction between nitrate (NO_3^-) and Cl^- at the ion transport sites. Furthermore, membrane depolarization has been related to high Na^+ concentrations (Dorais *et al.*, 2001). Under high EC, water and nutrient uptake decreases, including NO_3^-, which regulates nitrate reductase transcription, translation and activation in higher plants, thus reducing its activity. Debouba *et al.* (2007) and Martinez and Cerda (1989) showed that nitrate reductase activity in tomato plants decreased by 40% when grown with 100 mM of NaCl in solution.

The effect of high salinity depends on the specific nutrient under consideration. A high NaCl concentration will reduce tomato yield regardless of phosphorus (P) concentrations in the soil solution. However, high P concentrations can enhance manganese (Mn) uptake and/or reduce zinc (Zn) utilization. Tomato plants grown with 3 dS/m of NaCl did not suffer reduced potassium (K^+) uptake. However, under 5 and 8 dS/m of NaCl, K^+ uptake was reduced by 27% and 36% (Adams and Ho, 1995). Calcium (Ca^{2+}) can also be displaced by high Na^+ concentration around cell membranes, triggering additional physiological disorders.

Hot, dry conditions coupled with high salinity could cause rapid wilting and permanent damage. However, the outcomes of high EC in tomato production change based on the part of the plant under examination, environmental conditions, cultivar, ions in solution and time of day. For example, tomato grown at 17 dS/m (high EC) compared with 2 dS/m (low EC) resulted in a reduction of fruit growth rate during daytime, with no effect on growth rate at night (Ehret and Ho, 1986). Meanwhile, the effects of high salinity on fruit yield are extensively reported to be detrimental. Total yield reduction under high EC levels are associated with decreased fruit size and number, and development of blossom-end rot, a physiological disorder related to low Ca concentration, inadequate water supply and gene expression under stress conditions (see Chapter 5) (Van Ieperen, 1996).

On the other hand, high EC levels can improve organoleptic and nutraceutical features of tomato fruit and help with plant conditioning (see Chapter 5) (Adams, 1991; Mitchell *et al.*, 1991). Under high EC conditions, starch accumulation and sucrose synthase enzyme activity are built-up in fruit, which generally results in improved fruit quality (Dorais *et al.*, 2001). Furthermore, controlled high EC condition might positively affect tomato production systems. During the early stages of plant establishment, high EC may strengthen cell walls in plants. The same study reported that fruit cuticle thickness increased, while turgor pressure and firmness decreased, making the fruit less susceptible to cracking and allowing for a longer shelf-life (Dorais *et al.*, 2001). High EC concentration has also been associated with better colour indexes and reduction of uneven ripening. It is important to note that this would vary according to cultivars.

Tomato crops could be grown in nutrient solutions containing 100 ppm Cl^- without too much difficulty, but it is recommended to use water sources with < 50 ppm Na^+ and < 70 ppm Cl^- (OMAFRA, 2001). Adjusting the EC normally allows modification of plant water availability and maintenance of a healthy balance between vegetative and reproductive parts of plants (Holder and Christensen, 1989). Low-quality water with relatively high concentrations of salts, such as those containing dissolved Na^+, Cl^- and sulfate (SO_4^{2-}), should be utilized with caution in substrate and soil culture systems, as long as root zones can be leached and salts removed as water moves down. It is a common practice in many hydroponic systems without water recirculation to flush media using frequent irrigations to prevent salt accumulation. However, environmental protection measures push towards recirculation of the nutrient solution, which puts much higher demands on the irrigation water and the purity of the used fertilizers (e.g. Beerling *et al.*, 2014).

Acceptable EC levels for irrigation water range between 0.75 dS/m in the early crop stages and 1.5 dS/m during harvest. Detailed water quality standards for greenhouse production are presented in Table 6.1. Regardless of the production system, irrigation water should be tested at least twice a year, or every time a new source of water (e.g. well, river, pond or municipal system) is selected for crop production. Additionally, it is important to acknowledge that other salinity sources could also be present in everyday operations, such as nutrient ions dissolved in surface waters. Nutrients that have leached or drained from adjacent farms or landscapes could lead to high amounts of nitrates (NO_3^-), sulfates (SO_4^{2-}) and phosphates (PO_4^{3-}) in surface waters.

For soilless-grown greenhouse crops, the concentration of fertilizers is an important factor in the irrigation. When radiation and air temperature increase, the water uptake rate increases at a higher rate than nutrient uptake. Thus, the concentration of the remaining nutrient solution in the root zone and salinity rise. If this effect is not controlled, both water and nutrient uptake

Table 6.1. Minimum standards for greenhouse crop irrigation water (OMAFRA, 2001).

Water class	EC[a] (dS/m)	Sodium (ppm)	Chloride (ppm)	Sulfate (ppm)
1	< 0.5	< 30	< 50	< 100
2	0.5–1.0	30–60	50–100	100–200
3	1.0–1.5	60–90	100–150	200–300

[a]Electrical conductivity.

may be reduced, resulting in wilting and slower growth. Generally, this tendency is overcome by adjusting the EC of the supplied nutrient solution in accordance with the climate.

The pH of the irrigation water is not so much of a problem in soil-grown crops but it is crucial in soilless cultivation. A high pH is a risk for micronutrient availability and can cause precipitation and clogging of irrigation systems. The most common pH range for irrigation water is from 6.5 to 8.4. A pH outside the adequate range may cause low availability of certain nutrients or allow for dilution of toxic ions derived from the soil. Low salt content in irrigation water (EC $<$ 0.2 dS/m) may also lead to a pH outside the normal range, due to the low buffering capacity of the solution. One of the major problems of abnormal water pH is the effect on the irrigation equipment. High pH in water may be corrosive for pipelines, sprinklers and control equipment.

Soil buffering capacity will allow for slow changes in medium solution pH. Trying to change initial pH of the water source is impractical, because any introduction of an amendment will only achieve results in a few instances. However, the nutrient solution pH, used for fertilization, could be adjusted as long as this is done on a regular basis. In the case of production in soils, a most effective approach is to try to adjust the soil pH through practices such as liming or addition of elemental sulfur (S).

Environmental conditions affecting irrigation

A basic principle of irrigation is that the total amount of applied water during irrigation should be equal to the crop requirement plus the volume lost through diverse processes, such as evaporation and leaching. The irrigation requirement is a function of crop type, growing stage and evaporation (Dukes *et al.*, 2012). The volume of water lost through evaporation from the soil and plant surfaces and through transpiration from leaves is termed evapotranspiration (ET). Evaporation is the process by which water is converted from liquid to vapour, while transpiration causes water loss through the leaf stomata. ET is affected by environmental conditions including solar radiation, air temperature and relative humidity, as well as wind speed.

As water evaporates from soil or a wet surface, energy transfer occurs. During evaporation, water absorbs approximately 540 calories/g from the surrounding air in a process called evaporative cooling (Albregts and Howard, 1985). Eventually, the overlying air becomes saturated with water until air temperature decreases below the dew point (temperature where air is saturated with water) and condensation begins. When both processes – evaporation and condensation – are equal, the relative humidity (RH) of the air will be 100%. High temperatures, wind and solar radiation tend to increase evaporation, while high RH has the opposite effect. Humidity can be expressed as absolute humidity (g/m^3), vapour pressure (kPa), specific humidity (g water/kg air) or RH (ratio between the mass of water vapour in the air and the maximum mass it can hold at that temperature, i.e. saturation). The amount of water vapour that a given volume of air can hold is dependent on temperature, almost doubling for every 10°C rise in temperature. Transpiration rate increases as the difference between the fully saturated atmosphere inside the leaf (100% RH) and the water vapour content outside the leaf increases. Mathematically, this difference is described as vapour deficit or vapour pressure deficit (VPD). Since water vapour exerts pressure, VPD is expressed in units of pressure, such as millibars (mbar) or kPa. The transpiration rate also depends on the stage of growth, as well as the soil water status. Transpiration is reduced when VPD is too low and leaves may appear thicker and larger. Stems are also thick but root systems may appear weak and plants are more susceptible to disease (OMAFRA, 2001).

A greenhouse irrigation system should be able to provide at least 8 l/m^2 per day (80,000 l/ha per day). A mature tomato plant may use between 2 and 3 l/day (OMAFRA, 2003). Most of this water (90%) is used for transpiration and the rest for growth. Water use efficiency is much higher in greenhouses than in field production. To produce 1 kg of fresh tomatoes, between 15 l and 60 l of irrigation water are required, depending on growing conditions and crop management (Stanghellini *et al.*, 2003). In a survey of water productivity of growing systems for tomato production in Spain, it was demonstrated that productivity of greenhouse growing was about four times higher than in the field. For a given growing system (either field or greenhouse), the level of 'management' of the farm could make a difference of two-and-a-half times again. In short, an 'excellent' greenhouse grower is able to produce ten times more tomatoes with 1 m^3 of water than an 'average' field grower (Stanghellini, 2014).

Irrigation scheduling

Soil and soilless production: irrigation based on evapotranspiration

The reference ET (ETo) is defined as the ET rate from an extended surface covered with green grass or alfalfa 8–15 cm tall, actively growing and completely covering the ground without limiting water supply (Doorenbos and Pruitt, 1977). There are several methods to assess reference values for specific production regions, including the Penman-Monteith, the Thornthwaite, the Hargreaves and the Hamon methods. In open field situations, the former is often considered the standard for estimation of water requirements for crops and it is expressed as water volume per surface area over time. The water use of the crop is obtained by multiplying the crop coefficients (Kc), which are obtained from crop water use during different growth stages, and the ETo. For mature tomato, the Kc during mid-season is between 0.75 and 0.90. This method is widely used for scheduling irrigation programmes for field-grown crops, including tomato.

Soil and soilless production: irrigation based on solar radiation

Solar radiation is the main factor in calculating crop irrigation requirements. Within a day, transpiration closely follows radiation (Fig. 6.2). When radiation is higher, leaf temperatures increase, which in turn increases VPD and transpiration. Because of this close relationship between incoming solar radiation and water loss through transpiration, computerized irrigation in commercial greenhouses is frequently based on received solar radiation. Net radiation is generally equal to maximum ET, except for unusually hot and windy days, when ET can be 30% higher than net radiation (Tanner *et al.*, 1960).

In modern, well controlled greenhouses, the adjustments to seasonal and even day-to-day fluctuations of radiation and temperature are part of the standard irrigation strategies. However, these adjustments will also take into account crop stage, as well as the overall strategy of EC control with regard to crop development and fruit quality management. At low humidity (high VPD), transpiration may be excessive, stressing the plant. Indications of excessive transpiration in greenhouse tomatoes include small thin leaves and stems but a strong root system (OMAFRA, 2001).

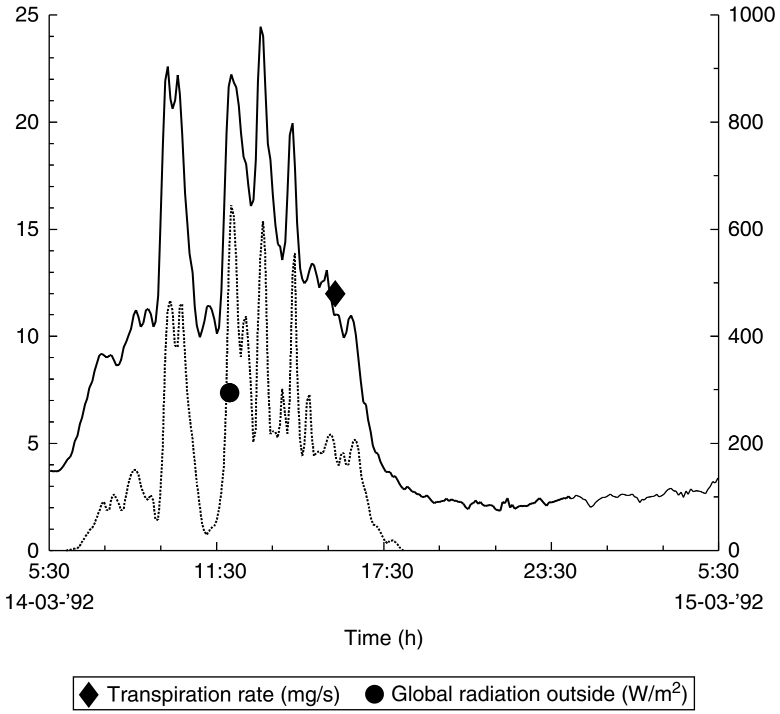

Fig. 6.2. Transpiration during daytime strongly correlates with solar radiation (greenhouse tomato crop in The Netherlands). (Reprinted, with permission, from *Journal of Experimental Botany* 45, 56, Van Ieperen and Madery, 1994.)

Inside greenhouses, VPD cannot be completely controlled. However, raising temperatures, venting and/or air movement will generally increase VPD, whereas misting (fogging) will generally decrease VPD. The frequency of watering is also a tool in crop management. With frequent irrigation, plants experience less stress because water status is more constant. In managing plant growth in soilless growing systems, stress resulting from either periodic water deficits or high salinity levels results in reduction of plant biomass, chlorophyll content and total yield with increased electrolyte leakage (Kirnak *et al.*, 2001).

Soilless production: irrigation based on drainage

Irrigation management in soilless media may be based on the amount of water draining out (also referred to as over-drainage or percentage leach). Irrigation will stop when a pre-established drainage percentage is collected from a collective sample of the greenhouse. More specifically, the nature of this strategy is based on the necessity of system uniformity and security for the grower that the growing media is completely wet.

Some overwatering may be necessary under specific conditions, especially in greenhouse drip irrigation systems, plant-to-plant differences and differences between locations. For example, outer rows may require more water than centre rows, given the higher radiation received from the lateral walls of the structure and therefore lack of plant-to-plant shade. Another example: water movement by capillarity may be slower compared with the reference flow in growing media with big particle size (e.g. perlite). Therefore, in order to assure complete water coverage in the medium, growers may intentionally allow drainage of water and fertilizer. Recommendations for irrigation drainage in soilless media change depending on the medium, season and the crop stage. Even though irrigation based on drainage allows for economic losses of water and fertilizer, it assures that, after every irrigation cycle, the medium is completely wet again. If slabs are too dry, root hairs will be damaged, EC may become excessive, nutrient uptake may be reduced and plant growth will become harder. In this case, plants would have thin stems and small dark leaves. On the other hand, excessively wet slabs will decrease aeration, increase the potential for root disease and result in softer vegetative growth and poor rooting.

By matching the amount of water provided to plant demands, fertilizer runoff and water use can be minimized. In addition to adverse environmental impacts, provision of water in excess of plant requirements can reduce the amount of air available in the root zone (Adams, 1999). In the field, this is most frequently a problem with poorly draining soils or those with subsurface hardpans. In soilless culture, low oxygen levels generally occur in hot weather because the available oxygen in the solution decreases as the root zone temperature rises. Root zone temperatures above 25°C are said to be deleterious for greenhouse tomato production. Low root-zone oxygen can occur in either the field or the greenhouse but it is most common in peat-based greenhouse substrates, because they have a higher water-holding capacity (i.e. drain less freely than stone wool or perlite). In peat, overwatering can lead to iron deficiency, which must be corrected with a reduction in amount of watering and an addition of iron to the feed (Adams, 1999).

Moisture monitoring and sensors

In hydroponic systems, drain measurement equipment and weighing gutters are commonly used to monitor irrigation. An irrigation sensor is any instrument that can measure some type of physical or chemical factor related to the water content of the root medium. It is important to keep in mind that all sensors have a measurement error margin. There are three key components that need to be considered to reduce this error: accuracy, repeatability (also called precision) and resolution. The ideal sensor would combine high accuracy with repeatability and adequate resolution, a combination that will generally result in increased price (Van Iersel *et al.*, 2013).

Monitoring soil or medium moisture is one of the most important steps in preventing water stress and eventual wilting of tomato. Its importance is accentuated by the role irrigation plays in nutrient distribution and uptake, salt leaching and temperature control in open fields and inside protective structures. When designing a comprehensive water monitoring system, both open field and greenhouse growers should consider three basic steps: (i) determination of irrigation efficiency; (ii) use of moisture sensors; and (iii) adjusting according to growing stages.

Firstly, after the water use of the crop has been determined, the efficiency of the delivery of the irrigation system (e.g. drip, subsurface, sprinkler) must be verified to compensate for the water losses due to delivery. For instance, most drip irrigation systems have irrigation efficiencies between 80% and 95%, in comparison with approximately 60% from sprinkler irrigation. Secondly, the correct use of water monitoring equipment allows growers to save water and to improve crop nutrition. There are several types of monitoring sensors: two of the most common are those that measure water tension and probes that measure volumetric water content. Tensiometers are the most common among the first group, giving a measurement of the tension at which water is retained in the soil or medium pores. An increase in soil moisture causes a decline in soil tension readings. These instruments are pressurized plastic cylinders with a porous ceramic cup in the lower part, a gauge on the top and an inner chamber filled with water. When the tube is inserted in the soil, water is suctioned out of the tensiometer through the porous ceramic cup and a negative pressure is created inside the tube, which is measured as negative pressure or tension. These devices are easy to use and relatively inexpensive but require routine maintenance to eliminate algae growth in the inner chamber and they can only be placed in fixed locations throughout the planted fields. In most soils, typical water tension at field capacity may be between 5 and 15 centibars (cbar). However, depending on the texture of the soil, more or less water would be needed to reach this desired tension. For instance, a sandy soil may need to have a volumetric water content of around 10–14%, whereas the same tension will be reached in a clay soil with 21–26%. This is due to the higher number and volume of micropores available in clay as opposed to sandy soil. Soilless media operate under similar principles; on the other hand, volumetric probes give a direct measurement of soil moisture and can be classified in two groups: capacitance sensors and time domain reflectometry (TDR) sensors. Although both are based on different principles, they provide instantaneous measurements of soil moisture with a small margin of error. One advantage of these sensors, particularly the TDR, is their practicality, because a single unit can be used to measure moisture in multiple fields. However, they cost more than simpler units such as tensiometers. Finally, a comprehensive irrigation monitoring system requires constant adjustments and supervision. For example, irrigation could be suspended for relatively long periods of time after heavy rains in open field production to avoid overwatering the crop. In protected culture, very cloudy or cold days reduce the ETo and thus the needed irrigation volumes, as described above. Table 6.2 describes alternative irrigation sensors.

FERTILIZATION AND FERTIGATION

Nutrition principles and essential nutrients

The goal of applying fertilizer for plant growth in soil or soilless media is to match the amount provided as fertilizer to the nutrient uptake of the crop as closely as possible. Currently, one of the main concerns of producing crops under intensive irrigation and fertilization is the risk of polluting both above- and below-ground water sources. Using appropriate fertilization programmes in combination with correct irrigation strategies reduces excessive leaching and runoff, thus limiting their environmental impact. The objective of this section is to describe the general nutrition principles and most common fertilization practices for tomato production.

Essential nutrients are elements needed to complete the plant cycle. These elements are irreplaceable for a vital reaction or structure. Essential nutrients must be differentiated from 'beneficial elements', which are elements that could improve growth, specific functions or plant structures. There are 17 essential elements for plant growth and development: hydrogen (H), oxygen (O), carbon (C), nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), sulfur (S), iron (Fe), manganese (Mn), zinc (Zn), boron (B), copper (Cu), molybdenum (Mo), chlorine (Cl) and nickel (Ni). The first three are freely obtained from the atmosphere and water. The remaining nutrients are divided into two groups (macronutrients and micronutrients) based on their absorbed volumes. The macronutrients are N, P, K, Ca, Mg and S, whereas Fe, Mn, Zn,

Table 6.2. Alternatives for soil water measurement methods used for irrigation scheduling. (Maughan *et al.*, 2015.)

Method	Type of measurement	Quality of measurement	Cost
Porous block	Electrical resistance of a porous block	Requires calibration to soil water content. Sensitive to temperature and soil salinity	Data logger/reader \$500 to \$1000, then \$40 per sensor (2–5 per site)
Neutron probe	Neutron thermalization as a function of water content	Estimates water in a larger area than most soil measure methods. Requires calibration to soil water content	Neutron Probe (\$10,000) plus \$15 per access tube. Regular dosimetry analysis and record keeping
Frequency domain probe	Electromagnetic soil capacitance as a function of water content	Accurate after calibration to soil water content. Relatively insensitive to soil water salinity levels	Data logger or reader \$500 to \$3000, then \$280+ per sensor (2–5 per site). Access tube installation kit \$2600
Heat dissipation	Uses heat buffering capacity of soil and thermal conductivity of soil water	Requires calibration to soil water content. Quite accurate and relatively insensitive to soil water salinity levels	Data logger/reader \$500 to \$3000, then \$155 per sensor (2–5 per site)

B, Cu, Mo, Cl and Ni are considered micronutrients. Several other elements are beneficial, such as cobalt (Co), sodium (Na) and silicon (Si) (Barker and Pilbeam, 2007). Some of the most important functions and deficiency and toxicity symptoms of the most absorbed nutrients are discussed below.

Nitrogen

This element is part of the building blocks for amino acids, enzymes, nucleic acids and energetic compounds. Under hot, bright growing conditions, N must be adequate for plants to grow rapidly, but high N levels encourage vegetative growth, which can be detrimental to reproductive growth under low light. Typically, N levels are kept relatively low until fruit set to promote reproductive growth. In soilless culture, the ratio of N to other nutrients is closely controlled to encourage either reproductive or vegetative growth, depending on the grower's perception of crop needs. Tomato plants have different absorption rates of N at various stages of growth. The N absorption by tomato plants during the formation of the first two to three flower trusses is about the same as for K. As the fruit load increases, so does the K uptake, resulting in a K:N uptake of 1.5:1 or 2:1. Excessive pruning, use of fertilizer and water under low light conditions early in the production cycle will result in an overly vegetative plant prone to disease and poor flower development, poor fruit set and size (OMAFRA, 2001). Lower N feed during this phase helps to control plant growth and makes the plant more generative. Plant growth can also be slowed by increasing the K:N ratio in the fertilizer, increasing EC by irrigating less frequently and in lower amounts, and reducing RH.

N limitation reduces chlorophyll concentration and thus absorption and light harvesting (De Groot *et al.*, 2003). Deficiencies of N are sometimes hard to detect without a well fertilized control for comparison. Growth may be reduced overall, so the plants are stunted, but leaves may look healthy, except for being a paler green than normal. Symptoms appear first on the lower leaves and continue to be more pronounced there, as N is a mobile element, moving from older to younger tissue. Symptoms at the top of the plant include pale flowers rather than deep yellow and a main stem that is thin at the top. The whole plant has a spindly appearance. Rather than the foliage being lush or succulent, leaves are small, erect and hard. Over time, the whole plant can turn yellow, flowers drop and fruits remain small. Plants may mature early, but fruit yield and quality decline (Guidi *et al.*, 1997).

Most cases of toxicity caused by N produce dark green leaves, sometimes thickened and brittle. At the top of the plant, stems remain thick and new leaves may curl into a ball. Clusters of flowers are large but fruit set may be poor. Although leaf growth is initially promoted, it is eventually restricted under excessive N. Plants may be more susceptible to diseases and insects (Jones, 1999). The applied N form is also an important consideration in guarding against toxicity. Tomato is much more sensitive to N in the ammonium form than in the nitrate form, especially under low light. The early symptoms of ammonium toxicity are small chlorotic spots on the leaves, which later turn necrotic (brown and dead). Spot size may increase, covering the entire interveinal area, giving the leaf a scorched appearance, and leaf margins may roll up. Lesions may also occur on the stems. As the plant matures, the vascular tissue at the base of the plants begins to deteriorate, with wilting occurring during periods of high atmospheric demand, followed by plant death (Jones, 1999).

Cation–anion ratio in tomato tissue is generally balanced. However, fertilization programmes with high ammonium (NH_4^+) or nitrate (NO_3^-) concentration will result in pH changes in plant tissue and nutrient solution in both soil and soilless conditions. Ion uptake in tomato plants is accompanied by a corresponding exchange from the root. Therefore, the uptake of nitrate causes excretion of hydroxyl or bicarbonate, while ammonium uptake is balanced by exchanging hydrogen with the nutrient solution or soil solution (Kirkby and Mengel, 1966). Nitrate in plant tissues is reduced to nitrite, and during this process hydrogen carbonate (HCO_3^-) and carbon dioxide (CO_2) are produced, increasing the organic acid content in plant tissue. Part of the generated HCO_3^- and CO_2 diffuses out of the root tissue and increases the nutrient solution pH. On the contrary, ammonium uptake will lead to the production of H^+, which will lower the pH of plant tissues and reduce the accumulation of organic acid anions (Kirkby and Mengel, 1966). It is important to regulate nitrate and ammonium in tomato fertilization from a practical pH management standpoint.

Phosphorus

Some of the functions of P in plants are as part of the energy compounds, such as adenosine diphosphate and triphosphate (ADP and ATP). It is a basic component of enzymes, proteins, phospholipids and nucleic acids. Although P is used in much smaller quantities than N and K, it must also be provided continuously. Initially, P is important for early fruit growth, especially under cool soil conditions when P uptake decreases, but later it is necessary for vegetative growth and fruit set. Phosphorus is stored well in soil but is easily leached in peat media and less available at high pH.

Deficient plants show symptoms on the lower leaves and stems. Plants appear stunted and upper leaf surfaces may turn unusually dark green. A characteristic red or purple colour on the undersides of the leaves (including the veins) and stem appears later, starting on the older leaves. Leaves are small and curved slightly downwards. Plants become slender, with thin stems, and cluster development is poor. P limitation reduces the functioning of photosystem II (De Groot *et al.*, 2003). Roots become brown and develop few lateral branches. Low rooting-zone and air temperatures can also reduce P uptake, and cause the purple pigmentation typical of P deficiency (Besford, 1980; Hosier and Bradley, 1999). Toxicities due to P excess are uncommon but can occasionally be seen as slow plant growth, with visual symptoms related to Zn deficiency. These symptoms can be severe, as large sections of the leaves turn light brown, giving a burned appearance. This is particularly noticeable on anaerobic rooting conditions (Jones, 1999).

Potassium

This element functions in the enzyme activation, protein synthesis and in the adjustment of the osmotic potential of tissues, including stomata. It is required for best fruit quality and to regulate growth; as a major nutrient with a positive charge, K balances the negative charges of organic acids produced within the cell and those anions such as sulfates, chlorides and nitrates. Potassium levels are particularly important at transplanting, to prevent subsequent plant growth problems and future ripening disorders. The ratio of N:K is also important in controlling growth. Tomato plants extract 250–500 kg K/ha per growing season, with 2/3 of all K allocated to the fruit when fully mature (Besford and Maw, 1974).

Major deficiencies are shown as reduced growth and chlorotic spots in older leaves, especially along the leaf margins, which could degenerate to necrotic areas. Potassium deficiency is first expressed as dark green foliage, which later turns purplish brown. Marginal chlorosis and necrosis appear first on the lowest leaves, then progress up the plant. Like N and P, K is phloem-mobile and young leaves are the last affected. Chlorosis almost always occurs first at the margins of older leaves, which often curve downwards. Later chlorosis moves into the interveinal areas towards the centre of the leaf and necrosis of leaf margins follows. In advanced stages, the small veins lose their colour, older leaves become severely scorched and drop, young leaves turn yellow and remain small, plant growth is restricted and fruit ripening is uneven. Postharvest quality is also poor (Jones, 1999). Interveinal chlorosis is also symptomatic of low Mg and the two deficiencies are sometimes confused.

Compared with Mg deficiency, K deficiency is more likely to occur along the leaf margins and is also more likely to develop into necrotic spots. Problems with fruit quality, such as blotchy ripening, boxy fruit and even to some extent greenback, are associated with low levels of K and some cases can be counteracted with high-K feeds (OMAFRA, 2001). Plant biomass can be reduced by Mg deficiency, induced by heavy K fertilization. Kabu and Toop (1970) reported a reduction of stem tissue Mg under high K fertilization in greenhouse tomato. Ions of similar size and valence interact, causing deficiencies. Cellular bidding sites of the plant cannot distinguish between K, Ca and Mg ions. Thus, media concentration of these ions will influence the rate of adsorption and competition. Uptake of K can also be reduced on anaerobic rooting conditions and low root-zone temperatures. Although K toxicities are rare, very high rates of K may induce Ca or Mg deficiency or salinity damage. Reductions in yield occur at very high levels of K, when K:N ratio in the liquid feed is too high, or when both N and K are too high.

Calcium, magnesium and sulfur

Calcium is a basic component of cell walls and it is involved in cell elongation and division, and in enzyme and hormone activity. Its role in the formation of structures gives tissues rigidity and strength. Deficiencies of this nutrient occur in younger leaves and fruit, including disorders such as blossom-end rot and burning of leaf tips and edges. Magnesium is the centre of the chlorophyll molecule, thus it has a direct influence on the green of leaves. It is also related to the many enzymatic reactions in the plant together with K, Na and NH_4 cations. Deficiencies are first noticed in the older leaves as mottling. Both Ca and Mg are cations absorbed mostly by mass flow and need to be balanced to avoid blocking each other's root uptake. In soils, it is desirable to have Ca:Mg proportions between 4:1 to 5:1 (Hao and Papadopoulos, 2004). Sulfur is required as being a component of the essential amino acids cysteine and methionine, the formation of S bonds in some hormones, vitamins, glucosides, sulfolipids and coenzymes. Deficiencies are rare – though in field crops they are increasing lately due to the reduction of atmospheric S deposition, mainly from burned fossil fuels – and resemble those caused by lack of N.

Micronutrients

Iron is part of several enzymes, as well as a component of proteins during respiration (Römheld and Nikolic, 2007). It is necessary for the formation of chlorophyll and it serves as an activator of respiration and photosynthesis. Its deficiency can be induced by heavy Mn fertilization, while low pH allows for higher solubility and availability of Fe. Iron deficiency symptoms include chlorosis of the interveinal part of young leaves and in some cases death of the entire plant (Lohry, 2007).

Manganese is an enzyme activator and it is used in the synthesis of certain amino acids and lignin (Humphries *et al.*, 2007). It participates in chlorophyll formation along with Fe. Conversely, high Mn fertilization may result in Fe deficiency. Symptoms of Mn deficiency generally include interveinal chlorosis, with darker colour next to the veins (Lohry, 2007).

Zinc is a structural constituent of several essential enzymes, including some that are needed for synthesis of histidine, glutamine and asparagine with more than 80 Zn-containing proteins reported. High concentration of Ca and P can affect Zn uptake. Within plant tissues, Zn translocation to shoots could be inhibited by high bicarbonate concentration. Zinc deficiency symptoms include decreased stem length and fruit bud formation, mottled leaves and interveinal chlorosis (Lohry, 2007).

Boron is needed for the formation of reproductive structures and metabolite transport across membranes, and it is required for root elongation and protein and nucleic acid metabolism. Boron deficiency results in a wide range of nutritional disorders. Symptoms of B deficiency include internode shortening, discoloured buds and youngest leaves and flower, and fruit dropping. Cells may keep dividing but plant structures may not differentiate in its absence (Lohry, 2007).

Copper activates several enzymes and improves C fixation and electron transport. Deficiency symptoms include chlorotic leaves with blue-green margins and drop of flowers and fruits (Lohry, 2007). Molybdenum is essential for the nitrate and nitrogenase reductase enzymes and it plays a role in the oxidation of sulfite to sulfate. Chlorine is required for the activity of at least four essential enzymes in plants and it is an essential cofactor in the photosynthetic process. Nickel has been shown to be needed by certain non-redox and redox enzymes in several plant species.

Nutrient requirements and sufficiency

For field-grown tomatoes, a clear knowledge of the nutrient requirements is needed to design an effective fertilization programme. This requirement will depend intrinsically on the crop type and its yield potential (Phillips *et al.*, 2009) and it could be supplied from three main sources of nutrients: (i) atmospheric deposition and inputs; (ii) the natural soil supply; and (iii) the applied fertilizers. The former is important in the case of the essential elements C, H and O, which are in most cases under non-limiting conditions, and for S deposition in the form of acid rain (Ceccotti *et al.*, 1998). Nonetheless, the soil supply and added fertilizers provide the majority of the crop nutrient requirements. For both bare ground and under plastic mulch production, it is suggested that growing media pH values initially be 6.0–6.5 and that the soil be tested before pre-plant fertilizers are added. If pH needs to be increased, pre-plant lime or dolomite application could be used. On the contrary, when pH must be reduced, application of elemental S would be necessary. In order for the soil reaction to occur, a period of 2–3 months might be needed from the time of the amendment application to planting. As mentioned above, plant nutrient uptake may affect the pH of the nutrient solution. It is necessary to keep a record of pH changes in nutrient solution for both soil and soilless conditions. This will allow the fertilization programme to be adjusted for sudden changes in pH levels. Many greenhouse growers keep a daily record of pH in soilless production. Open field production allows for a longer period of time in between measurements, given the buffer properties of the soil. However, this will depend on the soil's physical and chemical properties.

Based on soil test results, N, P, K, Mg and Ca are usually added at pre-planting, even if the crop is to be fertigated later. Pre-plant application can be broadcast and disked in or side-dressed in bands 10 cm deep and 10 cm from the row at planting. The application can also be split between the two methods of incorporation. For greenhouse crops, there are different methods to establish tomato seedlings, namely stone wool blocks or containers filled with growing media (e.g. bags, pots and troughs). In most cases, nutrient solution should have a pH of 5.5–6.5 and an EC of 1.5–2.0 dS/m to maximize nutrient availability and avoid salinity toxicity. After the transplants are established, the EC of the nutrient solution could reach 3.0–3.5 dS/m, based on available light, rate of growth, plant vigour, available moisture and temperature regime (OMAFRA, 2001).

It is important to clarify that EC levels are commonly measured at three different points within the production system: (i) at the entry point (emitter); (ii) media or soil solution extraction; or (iii) drainage water (in soilless production). Levels of EC tend to change depending on the measurement point. Entry EC is usually a result of the fertilization programme. Within the growing medium, EC level changes will depend on the fertilization programme, water status of the media, plant uptake and ion content in the medium. Values of EC in the drainage water will be related to the remaining ions of the solution, after plant and media uptake and water losses are taken into account. Depending on the cultivar, soil-grown tomatoes are moderately sensitive to soil salinity, compared with other vegetables, with a maximum threshold for yield loss of 2.5 dS/m and a 10% yield decrease per dS/m above this threshold (Maas, 1984). Where salinity is high, but not as extreme, fruit quality and soluble solids may increase with increasing soil salinity, just as in soilless systems.

While seasonal tomato nutrient absorption will vary according to particular growing conditions, nutrient removal will strongly depend on the growing habits of the cultivars. For instance, determinate tomato varieties are mostly planted in open fields and last between 3 and 4 months from transplanting to the end of commercially feasible harvests. On the other hand, indeterminate cultivars are produced for 10–16 months under protective structures, such as greenhouses and high tunnels (Table 6.3). In both cases, removal of N, P and K are the largest on a per-weight basis. Other nutrients, such as Ca and Mg, need to be closely monitored to maintain an appropriate ratio, as previously described.

Monitoring the nutritional status of the tomato plants is an essential task, regardless of the production system. Several methodologies exist to assess this nutritional status. These methods range from very simplistic to sophisticated approaches. Regardless of the chosen methodology, none provides a 100% accurate prediction or adjustment of the fertilization programme and, in many cases, a combination is preferred for different essential elements. Additionally, the nature and functions of each one of the essential elements for crop production along with the specific conditions of the farm and crop will dictate the necessary adjustments of the applied nutrient rates and thus the fertilization programme.

The two most common methods to determine the nutritional situation of tomato are: (i) petiole sap concentration; and (ii) diagnostic plant part analysis. These methods could be used alone or in conjunction with the others to manage programmes for crops that have multiple moments of fertilizer application (i.e. drip-applied nutrients and split side-dressing applications). Petiole sap concentration is a real-time measurement of ions flowing through the plant vascular system and serves as a quick diagnostic tool for determining deficiencies and adjusting fertilization programmes, especially for NO_3^- and K (Hochmuth, 1994). These tests are accurate and inexpensive but sufficiency values must be determined and validated for each crop prior to analysis.

The use of diagnostic plant parts has been the most typical 'in-season' diagnostic method for decades. The most recently expanded mature leaf is

Table 6.3. Nutrient removal by tomato grown in the field and greenhouses. (Halliday and Trenkel, 1992; Jones, 1999; OMAFRA, 2001; Sonneveld and Voogt, 2009.)

Nutrient (kg/ha)	Field crop (yield of 40–50 t/ha)	Greenhouse crop (yield of 500 t/ha)
N	100–150	1000
P_2O_5	20–40	500
K_2O	150–300	1500

collected, dried and ground for nutrient analysis. Some of the advantages of this method are its accuracy and broad range of nutrients that can be detected with one sample. However, it needs to be performed before or when deficiencies are barely suspected, because the samples need to be dried and may be sent to commercial laboratories for analysis. A potential limitation of this method is that the whole process could last from 5 to 15 days, which could be too late to correct deficiencies. The nutrient sufficiency contents in mature tomato leaves and acceptable leaf petiole concentrations of NO_3-N and K are presented in Tables 6.4 and 6.5. These ranges serve as reference values for tomato production and should be confirmed depending on the growing conditions.

Fertilization programmes

Nutrient rates

Accurate management of a fertilization programme seeks to reduce nutrient losses to the minimum and to enhance plant absorption. When designing an effective fertilization programme, the rate, sources, timing and placement of fertilizers must be considered (Roberts, 2007; Bruulsema *et al.*, 2009). It is widely known that a single nutrient programme is unpractical because of the differences across locations, soils, seasons, cultivars, fertilization practices and production techniques. Instead, a fertilization programme relies on in-farm conditions (e.g. irrigation and available equipment), as well as external situations (e.g. fertilizer prices and environmental regulations). For open field production in soil, fertilization programmes need to estimate the soil nutrient supply potential to avoid over-fertilization. In some cases, the rate of fertilizer may be higher than needed to compensate for the absence of nutrients in the soil and restore soil fertility (Phillips *et al.*, 2009). In greenhouse and high tunnel production where inert media are used, all the nutrients must be applied through the irrigation lines. A common way of managing nutrient levels in this production system would be by EC measurements. The exception could be when some organic growing amendments (e.g. composted substrate, peat moss) are utilized due to their potential nutrient contributions.

Determination of the adequate nutrient rate is also dependent on the assessment of the possible nutrient losses. These losses could be due to biological and mineral immobilization, volatilization, leaching and runoff. Biological and mineral immobilization occurs when soil microbes, such as bacteria, use C atoms as energy source or build mineral blocks as part of the mineralization process. A typical example of this process is N fixation by microbes in the presence of a high C:N ratio (Bengtsson *et al.*, 2002). Volatilization of ammonia (NH_3) and nitrous oxide (N_2O) are two cases of N losses in soils due to microbial activity (Reddy *et al.*, 1979; Snyder *et al.*, 2007). Nutrient leaching and runoff are two physically driven processes that could cause severe nutrient depletion in soils. Heavy NO_3^- leaching to ground waters is one of the main causes of eutrophication in

Table 6.4. Plant nutrient relative sufficiency concentration on a dry weight basis in newly opened mature leaves of tomato foliar tissue. (Maynard and Hochmuth, 2007; Olson *et al.*, 2012.)

	N	P	K	Ca	Mg	S	Fe	Mn	Zn	Cu	B	Mo
Growth stage	(%)						(mg/kg)					
5-leaf stage	3.0–5.0	0.3–0.6	3.0–5.0	1.0–2.0	0.3–0.5	0.3–0.8	40–100	30–100	25–40	5–15	20–40	0.2–0.6
First flower	2.8–4.0	0.2–0.4	2.5–4.0	1.0–2.0	0.3–0.5	0.3–0.8	40–100	30–100	25–40	5–15	20–40	0.2–0.6
Harvest	2.0–3.0	0.2–0.4	1.5–2.5	1.0–2.0	0.25–0.5	0.3–0.6	40–100	30–100	20–40	5–10	20–40	0.2–0.6

Table 6.5. Testing guidelines for tomato leaf petiole fresh sap NO_3-N and K. (Sanders, 2004; Olson *et al.*, 2012.)

	Fresh petiole sap concentration (mg/kg or ppm)	
Development stage/time	NO_3-N	K
First buds	1000–200	3500–4000
First open flowers	600–800	3500–4000
Fruit 50 mm diameter	400–600	3000–3500
First harvest	300–400	2500–3000

rivers and lakes. Runoff of P and S may occur from fertilized fields and have been reported as another source of pollution into ground waters and lakes (Moore *et al.*, 2000; Bates *et al.*, 2002). Fertilizer management practices that enhance crop absorption of available and applied nutrients are the basis for sustainability of production systems and the efficacy of fertilization programmes. This is directly linked to the amount of nutrients applied for crop production. Different production systems will utilize varying techniques to achieve sustainability. In open field production, the soil will allow for a certain level of error in the fertilization programme, while in soilless production the nutrient solution will be the primary source of chemical change in the medium. Closed soilless systems with water recirculation will depend 100% on the nutrient solution and the potential adjustment that growers apply. A compilation of macronutrient rates for tomato production from selected references is presented in Table 6.6.

Fertigation systems and scheduling

Fertigation is the most common method for applying fertilizers and water at the same time and it requires precise scheduling to maximize the efficiency of nutrient and water absorption. In this section, fertigation scheduling, injection equipment and fertilizer sources are discussed. Examples of scheduling of greenhouse and open field tomato production are presented in Tables 6.7 and 6.8.

Under protected culture, all fertigation systems have their own characteristics, which will affect the functionality of the system. There are two major micro-irrigation methods for vegetable crops: drip irrigation and microsprinklers. Drip irrigation is the most used and inexpensive of the two methods. For drip irrigation systems, the type, number and distance between drip emitters, the injection system, the number of valves and the uniformity coefficient are some of the parameters to consider when calculating the water discharge. Knowing this total discharge will allow the application of the correct water volumes and calculation of fertilizer rate. Other important factors to consider are as follows.

Table 6.6. Ranges of macronutrient rates for maximum growth and yield of tomato types in nutrient-deficient soils from selected research reports. (IMCC, 1984; Kaniszewski *et al*., 1987; Baselga-Yrisarry *et al*., 1993; Seliga and Shattuck, 1995; Andersen *et al*., 1999; Sainju *et al*., 2001; Maynard and Hochmuth, 2007; Sanchez, 2007; Olson *et al*., 2012.)

Optimum range (kg/ha)	Comments
Nitrogen (N)	
134–202	Determinate tomato, pre-plant fertilization, open field, drip irrigation
90–180	Determinate tomato, broadcast split in three doses
90–135	Determinate tomato, pre-plant incorporation
150–225	Determinate tomato, without and with irrigation
50–150	Processing tomato
198	Determinate tomato
220	Tomato, drip or seepage irrigation
88–99	Fresh tomato fresh in mid-Atlantic, USA
220	Fresh tomato in Florida, USA
154–176	Fresh tomato in New England, USA
Phosphorus (P)	
53	Determinate tomato
220	Tomato, drip or seepage irrigation
55–83	Tomato, sandy soil, very low to medium P level
88–176	Fresh tomato fresh in mid-Atlantic, USA
110–165	Fresh tomato in Florida, USA
55–220	Fresh tomato in New England, USA
Potassium (K)	
370	Determinate tomato
220–242	Tomato, drip or seepage irrigation
88–176	Fresh tomato fresh in mid-Atlantic, USA
110–248	Fresh tomato in Florida, USA
110–275	Fresh tomato in New England, USA

Table 6.7. Fertigation schedule (kg/ha) for indeterminate tomatoes grown in stone wool slabs. (OMAFRA, 2001.)

Growth stage	N	P	K	Ca	Mg	Fe	Mn	Zn	B	Cu	Mo
Slab saturation	200	50	353	247	75	0.8	0.55	0.33	0.5	0.05	0.05
4–6 weeks after planting	180	50	400	190	75	0.8	0.55	0.33	0.5	0.05	0.05
Normal feed	190	50	400	190	65	0.8	0.55	0.33	0.5	0.05	0.05
Heavy fruit load	210	50	420	190	75	0.8	0.55	0.33	0.5	0.05	0.05

Table 6.8. Suggested fertigation schedule (kg/ha) for field-grown tomato plants in the south-eastern USA on low-K soils. (Sanders, 2004.)

Days after planting	Daily N	Daily K_2O	Cumulative N	Cumulative K_2O
Pre-plant			56	140
0–14	0.56	0.56	64	148
15–28	0.78	1.58	75	170
29–42	1.13	2.25	90	202
43–56	1.69	3.35	114	249
57–77	2.81	5.60	173	367
78–98	3.38	6.75	237	495

- Type of solid pipes and drip emitters. These include the lateral pipes and single-wall drip tapes that deliver water to the emitters. Drip tapes usually work at pressures between 1 and 3 bar and deliver 1–2 l/h per emitter. These are very susceptible to emitter obstruction due to debris or algae and do not possess a great deal of uniformity when located in slopes. A variation of this type of tape is the double-walled drip tape, which could be used for more than one season. All these drip tapes could be autocompensated and/or anti-drainage. The former type allows uniform water volume distribution at a minimum working pressure, whereas the latter stops irrigation when pressure drops below certain threshold.
- Average emitter delivery. The average discharge per emitter could change depending on environmental conditions, pressure and distance from the pumping station. There is a very practical method to determine the average emitter delivery, using emitters at the beginning and end of each line and selecting several lines along the feeding lateral lines. Then coefficient of uniformity is determined by dividing the lowest output by the average volume. Coefficients around 90% are excellent.
- Number of drip lines. The number of tapes per bed is important to create a wet front as soon as possible, to avoid overuse of water pumps. Also, the number of lines per planted area will help to determine the total water volume needed for a given irrigation event.
- Distance between emitters. Most drip tapes in the market have emitters between 10 cm and 30 cm apart. The distance between plants may dictate the appropriate emitter distance, with short in-row distances requiring closer emitters than widely spaced plants.

Besides the irrigation set-up, the most common types of fertilizer applicators for fertigation are: (i) suction injectors (commonly known as 'Venturi'); (ii) constant-concentration injectors; and (iii) computerized injection nozzles. Suction injectors use a physical principle to generate differential pressure in the main water line, which creates suction into a secondary line that pulls fertilizer from the injection tanks. The advantages of this system are that these devices are inexpensive and easy to install and manage, as long as water pressure remains relatively constant. The disadvantage is that fertilizer injection could be variable when water pressure and temperature fluctuate, which could lead to irregular nutrition. Constant-concentration injectors possess a piston pump or diaphragm that draws liquid fertilizer from a non-pressurized container and mixes it with pressurized water into the main lines. These injectors can be either electric or hydraulically driven and require a minimum working pressure. They are more expensive and require more maintenance than the 'Venturi' but are more efficient and precise. A more expensive but precise method is the computerized injection nozzles. These use computerized consoles that can be programmed to deliver the selected fertilizer volume at the right time into the water stream. Whether a simple or a sophisticated method is used to fertigate, it is recommended that application rates should be checked periodically to ensure proper rates.

With regards to fertilizer sources, especially in protected culture, there are two main formulations: (i) solid compound salts; and (ii) liquid fertilizers. In many parts of the world, it is common to mix solid fertilizer (hydrosoluble) salts with water in large tanks. This method facilitates fertilizer transport to remote areas. However, it requires the constant use of labour to produce the mix and several fertilizer tanks to avoid precipitation and volatilization of some materials. Additionally, injection water must have a relatively high temperature (to allow the nutrients to stay in solution) and low salt content. All these factors increase the chances for human error during the process. During the application of hydrosoluble fertilizers, it is common to have several tanks to avoid combination of reactive sources. For example, a concentrated tank with Ca^{2+} will react with concentrated phosphatic and sulfatic sources, forming either calcium phosphate or calcium sulfate, which can form solid deposits that eventually will clog the irrigation emitters. For this reason, growers will have one tank with a Ca source (e.g. calcium nitrate and potassium nitrate) and another with sulfates (e.g. sulfate of potash and magnesium sulfate) and phosphates (e.g. monoammonium phosphate and monopotassium phosphate). Typically a third tank will be used for phosphate for practicality in the daily changes in the fertilization programme. A fourth tank may contain acids, such as phosphoric acid for pH levelling and line-cleaning purposes.

Liquid fertilizers contain predesigned concentrations of nutrients, as either a single or compound salt, that are delivered directly through the irrigation stream without further mixing. This method is more costly than soluble salts, because it requires the transport of the fertilizer already dissolved in water. However, it minimizes the risks for human error during application and the costs of maintenance and purchase of large tanks and injection equipment. Similarly, these sources reduce precipitation of nutrients, provided that temperatures do not drop below 10°C.

Foliar sprays

Foliar sprays should be avoided because of the potential damage to the foliage from spraying in hot or bright weather and because leaves do not absorb nutrients well, but they are sometimes used under emergency conditions while the situation around the root zone is corrected. Foliar sprays of urea dissolved in water at 2.5 g/l can be applied to correct a severe N deficiency. For P deficiency, foliar sprays with potassium or ammonium phosphate are possible but are not recommended, as they can cause serious leaf damage. For K deficiency, the crop may be sprayed with a solution of 20 g/l potassium sulfate. To correct Ca deficiency quickly, plants can be sprayed with a solution of 2–7 g/l or a 0.3% calcium chloride solution. Such sprays are rarely beneficial in correcting blossom-end rot, since as discussed above Ca absorbed through the leaves cannot move readily into the fruit. Hence, only the Ca that lands on the fruit surface is utilized (Adams, 1999). Magnesium deficiencies can also be corrected with sprays of magnesium sulfate (Epsom salts). Roosta (2011) reported that foliar spray of all elements significantly increased plant fruit number and yield in the aquaponics in order of K > Fe > Mn > Zn > Mg > B. In the hydroponics, foliar application of K, Mg and Zn increased fruit number and yield of plants compared with control.

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7

CROP PROTECTION: PEST AND DISEASE MANAGEMENT

Gary E. Vallad, Gerben Messelink and Hugh A. Smith

ECONOMIC AND CULTURAL IMPORTANCE OF TOMATO

According to the Food and Agriculture Organization (FAO) of the United Nations (FAOSTAT database), nearly 177 million metric tons (t) of tomatoes were produced globally from 4.8 million hectares in 2016, an average yield of 37 t of tomatoes for every hectare of land in production throughout the world. The level of tomato production throughout the world demonstrates the cultural importance of this New World vegetable (see also Chapter 1). The efficiency of tomato production, represented here as the average yield of fruit per hectare of land, varies greatly throughout the world, ranging from 1.5 t/ha for Somalia to 650 t/ha for greenhouses in The Netherlands. This vast discrepancy is further illustrated by the fact that Somalia produces tomatoes on an estimated 11,200 ha of open field production, compared with only 1750 ha of mostly greenhouse production in The Netherlands (FAOSTAT database). It is logical to assume that much of the discrepancy in production efficiency observed globally is directly related to the level of key inputs that have become standard for modern agriculture, including infrastructure (irrigation systems and protected structures), machinery, fertilizer and the assorted pesticides utilized for the management of weeds, pests and diseases. It is paramount that food production keeps up with global population growth; and an expanding middle-class population in many developing areas of the world is likely to increase the demand for tomatoes. In a society that is becoming more globally aware of limited environmental resources, there is a need not only to improve production through increased inputs in developing countries, but also to ensure that these inputs are properly managed to maximize yields and quality while minimizing unwanted environmental impacts.

The goal of this chapter is to review pests and diseases important to global tomato production, giving an update of basic and applied research on the management of some pests and diseases with some considerations for greenhouse and open field production. Special emphasis is also placed on integrated pest management and the biological control of key pests in greenhouse production.

IPM AND BIOLOGICAL CONTROL CONCEPTS

To meet both the worldwide demand for food security and environmental safety, agriculture must increase food production and quality while decreasing its ecological footprint. Ensuring sustainability and competitiveness with reduced pesticide inputs is a major challenge but this can be achieved by integrated pest management (IPM). IPM is a sustainable approach to managing pests that combines available biological, cultural, physical and chemical tactics to minimize the economic, health and environmental risks associated with crop production. Biological control of pests and diseases through the use of natural enemies such as predators, parasitoids, pathogens and other antagonists to reduce pest densities and reduce disease can be an important component to any IPM system.

BIOLOGICAL CONTROL OF PESTS

Several types of biological control can be distinguished. Classical biological control refers to the introduction of natural enemies to a new area where they did not originate or do not occur naturally. This is usually done by government authorities. Augmentation biological control is the release of mass-reared natural enemies to obtain an immediate control of pests. This type of biological control is a commercialized method, with natural enemies sold by professional companies. Finally, biological control can be achieved through actions that protect and stimulate the performance of naturally occurring natural enemies, which is referred to as conservation biological control.

IPM practices between field and greenhouse tomato crops are fundamentally different. Augmentation biological control in greenhouse crops is much better developed than in field crops for several reasons. Firstly, greenhouse crops have a higher economic value with higher production levels, which means that more money can be spent on control measures. Releases of natural enemies in field crops are often considered to be too expensive. Secondly, the number of pest species and diseases is often lower in greenhouse than in field crops. Particularly in modern greenhouses in northern Europe, problems with nematodes and several soil diseases are solved by growing plants on hydroponics. Thirdly, natural enemies released in greenhouses are confined largely to the crop under production, whereas those released in the field may disperse to surrounding weeds or fields, thereby reducing or eliminating their effectiveness. Fourthly, airborne diseases, such as *Botrytis*, can be minimized by climate management in modern greenhouses. This reduces the need for spraying fungicides and thereby interference of fungicides with natural enemies. Finally, a major drive for switching to biological control in greenhouse tomatoes has been the introduction of bumblebees for pollination. Using bumblebees is not compatible with many insecticides, thus forcing growers to use non-chemical pest control measures.

Field-grown tomatoes benefit more from conservation biological control than from augmentation biological control. Various techniques of habitat modifications, such as flowering strips or mulching, can be applied to stimulate natural enemies (Barbosa, 1998). Although augmentation biological control is well developed in greenhouse crops, it might be enhanced by similar conservation tools that are applied in field crops for both released and naturally occurring natural enemies. One example is the use of so-called 'banker plants' in greenhouses that support the reproduction of natural enemies by supplying alternative prey or food (Huang *et al*., 2011). The system of biological control strongly depends on the type of natural enemies used. These will be discussed below.

Biological control with generalist predators in greenhouse crops

Biological control of pests can be achieved using specialist and generalist natural enemies. For a long time, biological control was mainly focused on specialist natural enemies, because they are well adapted to their prey (van Lenteren and Woets, 1988). However, they often cannot persist in a crop when prey are scarce or absent. Repeated introductions are usually needed to control pests, which may involve problems with timing, costs and quality of the natural enemies. In general, generalist predators establish better in crops by feeding on alternative food sources and they can potentially control several pest species. Inoculating crops with generalist predators allows them to quickly detect and control pests that invade a crop, which benefits the resilience of cropping systems against pests. A drawback of generalist predators can be their interference with other natural enemies. Yet, biological control programmes in greenhouse crops are increasingly based on releases of generalist predators and they appear to be successful (Messelink *et al*., 2012).

Tomato plants are particularly suitable for generalist predatory bugs of the family Miridae, mostly species of the genera *Nesidiocoris*, *Dicyphus* and *Macrolophus*. These predators are zoophytophagous, which means that they feed and survive both on plant sap and on prey. They feed on multiple pest species such as whiteflies (Gerling *et al*., 2001), aphids (Alvarado *et al*., 1997), thrips (Riudavets and Castañé, 1998), spider mites (Hansen *et al*., 1999) (Fig. 7.1), leaf miners (Arnó *et al*., 2003) and lepidopteran species, including *Tuta absoluta* (Urbeja *et al*., 2009) (Table 7.1). Moreover, they are well adapted to the hairy tomato surfaces both morphologically (slim body, long slender legs, elongated curved claws) and behaviourally (locomotion) (Voigt *et al*., 2007). In the Mediterranean area, several species spontaneously invade tomato crops and these invasions can be supported by offering suitable host plants in greenhouse surroundings (Perdikis *et al*., 2011). Some of these species are mass produced on a commercial scale. The most important species are currently *Nesidiocoris tenuis* and *Macrolophus pygmaeus* (Rambur) (formerly identified

Fig. 7.1. Two-spotted spider mite adult and eggs on tomato. (Photograph courtesy Hugh Smith.)

as *Macrolophus caliginosus* Wagner) (Fig. 7.2) (van Lenteren, 2012), and in Canada *Dicyphus hesperus* has been released (Sanchez *et al.*, 2003). These predators are usually released at the start of a cropping cycle and can even be introduced on seedlings at plant nurseries (crop inoculation). Population increases can be supported by applying alternative food sources, such as pollen or sterilized eggs of the flour moth *Ephestia kuehniella* (Vandekerkhove and De Clercq, 2010). One of the risks in using zoophytophagous predators is that high densities in absence of prey may cause crop damage (Castañé *et al.*, 2011). However, such risks may be managed by supplying supplemental food sources in the crop, such as sugars (Urbaneja-Bernat *et al.*, 2012). By inoculating crops with generalist mirid predators, supporting their development and managing their behaviour, crops will increase in resilience against pests. This approach appears to be more effective than releasing natural enemies as 'biopesticides' in response to pest infestations. Nevertheless, additional releases of specialist natural enemies remain indispensable for effective biological control programmes. The most important specialist natural enemies will be discussed below.

Biological control with specialist natural enemies in greenhouse crops

Plant-feeding mites

Spider mites (*Tetranychus urticae* Koch and *T. evansi* Baker and Pritchard), tomato russet mite (*Aculops lycopersici* (Massee)) and broad mite (*Polyphagotarsonemus latus* (Banks)) are pests of tomatoes. Spider mites are larger (0.3–0.5 mm) than russet mites (0.15–0.2 mm) and broad mites (females 0.2–0.3, males half as large), which can only be seen with at least 14× magnification. Mites have piercing–sucking mouthparts and puncture plant cells to extract sap. They establish on the undersides of leaves but can be found on all plant surfaces when infestations are high. Spider mites produce webbing and chlorotic abraded patches on leaves where they feed in high numbers. Russet mite feeding produces a silvery chlorotic appearance to the upper surface of tomato leaves.

Table 7.1. Pests and their commercially available natural enemies in greenhouse tomatoes.

Pest group	Pest species	Natural enemy
Plant-feeding mites	Two-spotted spider mite *Tetranychus urticae*	Specialist predatory mite *Phytoseiulus persimilis*, predatory midge *Feltiella acarisuga* Generalist predators *Macrolophus pygmaeus* and *Nesidiocoris tenuis*
	Red spider mite *Tetranychus evansi*	*Phytoseiulus longipes*
	Tomato russet mite *Aculops lycopersici*	Limited effect of generalist phytoseiid mites
	Broadmite *Polyphagotarsonemus latus*	Limited effect of generalist phytoseiid mites
Aphids	Potato aphid *Macrosiphum euphorbiae*	Parasitoids *Aphidius ervi* and *Aphelinus abdominalis*
	Foxglove aphid *Aulacorthum solani*	Parasitoids *A. ervi* and *A. abdominalis*
	Peach aphid *Myzus persicae*	Parasitoid *Aphidius matricariae*, *Aphidius colemani* and *A. ervi*
	All species	Predatory midge *Aphidoletes aphidimyza*, generalist predators *M. pygmaeus* and *N. tenuis*
Psyllids	Tomato/potato psyllid *Bactericera cockerelli*	Predatory beetle *Delphastus catalinae*
Whiteflies	Tobacco or silverleaf whitefly *Bemisia tabaci*	Parasitoids *Eretmocerus eremicus* and *Eretmocerus mundus*, generalist predators *M. pygmaeus* and *N. tenuis*
	Greenhouse whitefly *Trialeurodes vaporariorum*	Parasitoids *Encarsia formosa* and *E. eremicus,* generalist predators *M. pygmaeus* and *N. tenuis*
Mealybugs	Greenhouse mealy bug *Pseudococcus viburni*	No effective natural enemies available, limited use of predatory beetle *Cryptolaemus montrouzieri*

Continued

Table 7.1. Continued.

Pest group	Pest species	Natural enemy
	Other species: *Phenacoccus madeirensis*, *Phenacoccus solenopsis*, *Phenacoccus solani*	No effective natural enemies available
Leaf miners	Tomato leaf miner *Liriomyza bryoniae*	Parasitoids *Diglyphus isae* and *Dacnusa sibirica*, generalist predators *M. pygmaeus* and *N. tenuis*
Caterpillars	Tomato leaf miner moth *Tuta absoluta*	Egg parasitoid *Trichogramma achaeae* (limited use)
	Tomato hawk moth *Lacanobia oleracea*	Egg parasitoid *Trichogramma evanescens* (limited use)
	Tomato looper *Chrysodeixis chalcites*	Egg parasitoid *T. evanescens* (limited use)
	Cabbage looper *Trichoplusia ni*	Larval parasitoids are effective, but not mass-produced
	All	Generalist predators *M. pygmaeus* and *N. tenuis*
Thrips	Several species but mainly western flower thrips *Frankliniella occidentalis*	Generalist predators *M. pygmaeus* and *N. tenuis*
Plant damaging bugs	Southern green stink bug *Nezara viridula*	No natural enemies available

Fig. 7.2. *Macrolophus pygmgeus*, whitefly predator. (Photograph: Gerben Messelink.)

Broad mite causes bronzing of the underside of tomato leaves and terminal shoots. Tomato russet mite feeds primarily on solanaceous plants but spider mites and broad mite have a broad host range. Under favourable conditions (24–27°C), tomato russet mite and broad mite can complete their life cycle in about 1 week. The life cycle of spider mites is longer, though they can complete their development in about 1 week at 30°C.

Biological control of spider mites with the specialist predatory mite *Phytoseiulus persimilis* is one of the cornerstones of biological control in greenhouse crops (Hussey and Bravenboer, 1971). Spider mites in tomato can also be controlled effectively by this predator but it does not perform as well as in other crops. The reason is that it is hampered not only by entrapment in the sticky exudates of glandular trichomes, but also indirectly by the secondary plant metabolites in its prey (Koller *et al.*, 2007). Predators that are reared on tomato get better adapted to these plant traits and perform better than predators reared on bean (Drukker *et al.*, 1997). Commercial producers of natural enemies have adopted this knowledge and nowadays sell special tomato strains of *P. persimilis.*

A related species of *T. urticae* damaging tomato is *Tetranychus evansi*. This mite is common in Africa and South America but crop infestations have been observed in southern Europe and North America as well. This species seemed to be a real threat, because *P. persimilis* is not able to deal with the dense webbing this spider mite produces and therefore is not able to control this pest. The related predatory mite *Phytoseiulus longipes* seems to be more adapted to this prey and is able to control this mite on tomato plants (Furtado *et al.*, 2007).

Biological control of the much smaller tomato russet mite *Aculops lycopersici* and the broad mite *Polyphagotarsonemus latus* is not well developed. Several species of phytoseiid mites are excellent predators of broad mites in other crops (e.g. sweet pepper) (van Maanen *et al.*, 2010) and leaf disc experiments show that several predatory mites can prey on russet mites (Brodeur *et al.*, 1997). However, none of these mites so far performs well on tomato plants, probably because they are strongly hampered by the glandular trichomes. An interesting plant response to russet mites is the local degradation of these glandular hairs. Once these trichomes are shrivelled, it appears that predatory mites are able to control the russet mite (van Houten *et al*., 2013). This indicates that a locally applied 'overkill' of predatory mites may suppress russet mite densities but they may not be able to prevent plant infestations as they do not establish on tomato plants.

Whiteflies

The primary whitefly pests of tomato are *Bemisia tabaci*, Gennadius biotype B (also known as the silverleaf, sweetpotato or tobacco whitefly, and sometimes referred to in the literature as *Bemisia argentifolii* Bellows and Perring) and *Trialeurodes vaporariorum* Westwood, the greenhouse or glasshouse whitefly.

Whitefly adults are small (1.0–1.3 mm) insects with waxy white wings, a yellow body and red eyes. Whitefly females typically lay eggs on the underside of the leaf and usually lay between 100 and 300 eggs during the 3–6 weeks that they are alive (Thompson, 2011). Eggs are oblong and attached to the leaf surface on the basal end. Eggs are pearly or greenish white when first oviposited and darken to a coffee colour. Whiteflies pass through four nymphal instars prior to becoming an adult. The first instar crawler possesses six legs and searches for a feeding site after emerging from the egg. Subsequent nymphal instars lack legs and are immobile. Whitefly nymphs are oval in shape, yellowish and translucent. The red eyes of the pharate adult are visible in the latter stage of the fourth instar. This is referred to as the red-eyed nymph or pupa, although technically, as hemimetabolous insects, whiteflies do not possess a pupal stage.

B. tabaci and *T. vaporariorum* can be distinguished from each other in the adult stage by the manner in which they hold their wings. *B. tabaci* holds the wings 'roof-like' over the body, and with sufficient magnification the yellow abdomen can be seen between the wings from above. *T. vaporariorum* holds the wings 'fan-like' over the body, and the abdomen is not visible from above. The fourth instar nymph is also easily distinguished between the two species and is the stage commonly used to identify whitefly species. The fourth instar of *T. vaporariorum* has a marginal fringe of waxy rods where the dorsal and lateral surfaces meet. By contrast, the fourth instar of *B. tabaci* is largely flush with the surface of the leaf and has few marginal setae.

The developmental time for egg to adult on tomato at 25°C is 20.5 days for *B. tabaci* (Yang and Chi, 2006) and 21 days for *T. vaporariorum* (Dorsman and van de Vrie, 1987). *B. tabaci* is a pest of field-grown tomato in warmer regions. *T. vaporariorum* is a problem in protected agriculture in temperate regions, in field and protected production under Mediterranean conditions, and in the tropics at higher elevations. In the tropics, *B. tabaci* is common up to about 1000 m above sea level and *T. vaporariorum* predominates at 800 m above sea level and in growing areas at greater elevations (Caballero, 1996). In some regions, both species are present on tomato (Dovas *et al*., 2002; Arnó *et al*., 2006).

Whiteflies have a piercing–sucking stylet and are phloem feeders. When high numbers of whitefly feed on a plant, loss of phloem sap can debilitate the plant. Whiteflies produce honeydew, a sugar-rich excretion that accumulates on the surface of leaves and provides a substrate for sooty moulds. Sooty moulds are Ascomycete fungi that contaminate crops, reducing photosynthesis and quality. Whiteflies primarily reduce yield in tomato by transmitting viruses. *B. tabaci* can transmit several types of viruses but the most damaging viruses transmitted by *B. tabaci* to tomato are in the genus *Begomovirus* (Family *Geminiviridae*) (Lapidot and Polston, 2010). There are over 80 recognized whitefly-transmitted geminiviruses of tomato (Varma *et al.*, 2011). Tomato yellow leaf curl disease is associated with several viruses that produce indistinguishable symptoms: stunting, small distorted yellow leaves, upward curling. Globally, *Tomato yellow leaf curl virus* (TYLCV) is the most important. The virus can be acquired by both immature and adult whiteflies and is transmitted in a persistent circulative manner. That is, the whitefly acquires the virus by feeding, can transmit the virus after a latent period of 8–24 hours and may retain the virus for life (Cohen and Nitzany, 1966). Whiteflies also induce the systemic disorder known as irregular ripening, which is caused by feeding of nymphal *B. tabaci* (Schuster, 2001).

Resistant varieties are available for management of TYLCV (Fig. 7.3). Producers of field tomatoes rely primarily on chemical control, applying both systemic and contact materials to manage adults and nymphs. *T. vaporariorum* is a vector of *Tomato infectious chlorosis virus* (Closteroviridae) but is less important as a virus vector in tomato than *B. tabaci*. Reflective plastic mulches

Fig. 7.3. TYLCV-susceptible tomato variety (left) next to TYLCV-tolerant variety under high virus conditions. (Photograph courtesy Hugh Smith.)

are used to reduce whitefly infestations early in the season (Csizinsky *et al.* 1997, 1999). Important sources of TYLCV and other whitefly-transmitted viruses include crop residues, asymptomic hosts such as pepper and resistant varieties, and certain weeds. Crop hygiene, particularly the destruction of old crop residues, is an important control measure for reducing viral inoculum.

Biological control of the greenhouse whitefly *T. vaporariorum* in greenhouses started in the UK with the parasitoid *Encarsia formosa* Gahan (Speyer, 1927) and this wasp is still of major importance in biological control programmes in greenhouse crops (van Lenteren, 2012). Females prefer to lay eggs in the third and fourth instar larvae of whiteflies. Parasitized greenhouse whiteflies can easily be recognized by pupae that have turned black. The related parasitoid *Eretmocerus mundus* is more effective in the control of *B. tabaci* and *Eretmocerus eremicus* can be used against both whitefly species (Greenberg *et al.*, 2002) (Table 7.1). Low greenhouse temperatures (driven by energy saving) can restrict the efficacy of these wasps. Trimming of older leaves before adult wasps have emerged from parasitized whitefly pupae may hamper their population increases. To promote the survival of parasitoid populations, an adapted leaf trimming strategy is now advised. Biological control of whiteflies has been very successful in other crops such as cucumber and sweet pepper (Messelink *et al.*, 2008; Calvo *et al.*, 2009). However, as mentioned before, predatory mites so far do not perform well on tomato, because they can get trapped in the sticky hairs on the stems on tomato. Thus in practice, whitefly control is mainly based on a combination of mirid bugs and parasitoids.

Aphids

Aphids are small soft-bodied, pear-shaped insects with piercing–sucking mouthparts. They are phloem feeders. They possess two small tube-like structures, called cornicles, on the fifth or sixth abdominal segments. In temperate regions, aphids pass through an egg stage and four to five nymphal stages before becoming an adult. In warm regions and in protected structures, aphids tend to be parthenogenetic and viviparous, that is, they reproduce without mating and give birth to nymphs rather than eggs. These adaptations enable aphid populations to grow rapidly. Aphid adults may be winged or wingless, depending on environmental conditions. The most important species in both field and greenhouse production are the potato aphid *Macrosiphum euphorbiae* (Thomas), the foxglove aphid *Aulacorthum solani* (Kaltenbach), the green peach aphid *Myzus persicae* (Sulzer) and the cotton or melon aphid *Aphis gossypii* Glover. Each of these pests has a broad host range. *M. persicae* and *A. gossypii* adults tend to be smaller (1.8 mm) than *M. euphorbiae* and *A. solani* (3.0 mm). *M. euphorbiae* has both green and pink biotypes attacking tomato, but colour can be variable within aphid species. Other characteristics are more useful for identification, which requires magnification (Blackman and Eastop, 2000). Aphids in greenhouses are more problematic in temperate climates than in subtropical climates where temperatures are probably too high for many species. Aphid infestations tend to initiate on the underside of upper canopy tomato leaves, but under favourable conditions aphids can be found on most plant parts. They debilitate the plant by extracting sap in high amounts, producing honeydew and sooty mould. Aphid feeding can produce wilting, necrotic spots on leaves and leaf distortion. Aphids vector damaging viruses to tomato, including *Tobacco etch virus*, *Potato Y virus* and *Cucumber mosaic virus*. Control of aphids in the field involves the use of insecticides, reflective plastic mulches and crop hygiene. There are presently few options for managing aphid-vectored viruses with resistant tomato varieties.

In greenhouses, aphids can be controlled by specialist parasitoids of the genera *Aphidius* and *Aphelinus* (Table 7.1). The most important parasitoid that is able to parasitize all aphid species in tomato is *Aphidius ervi*. However, the efficacy of parasitoids can by strongly disrupted by hyperparasitoids, which are secondary parasitoids that can develop on primary parasitoids. Aphids can additionally be controlled by the specialist gall midge *Aphidoletes aphidimyza*. The larvae of this midge are impressive predators of all kinds of aphids. It is quite common for natural enemies of aphids to enter greenhouses during summer. Naturally occurring enemies such as parasitoids, chrysopids and syrphids can contribute significantly to the control of aphids.

Thrips

Species of thrips attacking tomato around the world are primarily in the family Thripidae and include the western flower thrips *Frankliniella occidentalis* (Pergande), tobacco thrips *F. fusca* (Hinds), flower thrips *F. tritici* (Fitch) and *F. schultzei* (Trybom). These species are found primarily in the flower, although feeding damage to foliage (Fig. 7.4) and oviposition damage on fruit are common at high thrips densities. The onion thrips *Thrips tabaci* (Lindeman), the greenhouse thrips *Heliothrips hemorrhoidalis* (Bouché), the chilli thrips *Scirtothrips dorsalis* Hood and *Echinothrips americanus* Morgan also attack foliage. Thrips are small insects and possess fringed wings as adults. Adult western flower thrips measure 1.2–1.9 mm. Most thrips found on tomato range in colour

Fig. 7.4. Thrips feeding damage on tomato leaf. (Photograph courtesy Hugh Smith.)

from pale yellowish or orange-brown to dark brown. Thrips females insert eggs partially into plant tissue. Thrips attacking tomato pass through two larval stages, a pre-pupal stage and a pupal stage. The two pupal stages are mobile but do not feed. Pupation usually occurs in the soil. Thrips have asymmetrical mouthparts that pierce and extract plant juices from outer cells. Thrips feeding abrades the surface of leaves and flowers, and scarred tissue eventually turns brown and dry. Thrips often leave small blackish-green faecal pellets in feeding areas. Oviposition on tomato fruit results in stippling and haloing on the fruit surface. In addition to direct damage to the fruit, thrips cause crop losses in tomato by vectoring viruses in the Tospoviridae, including *Tomato spotted wilt virus* and *Ground nut ring spot virus*. Tomato spotted wilt is acquired by the larvae but is transmitted by adults feeding on foliage. Several varieties of tomato resistant to *Tomato spotted wilt virus* are available. Flower thrips, particularly western flower thrips, are very difficult to manage using insecticides, in part because of the rapid development of resistance.

Tomato psyllid

The tomato psyllid, *Bactericera* (*Paratrioza*) *cockerelli* (Šulc), is a pest of tomato and other solanaceous crops, including aubergine, potato and pepper. Adults are 2.5–2.75 mm in length with transparent wings held roof-like over the body. Newly emerged adults are amber in colour and turn dark brown or black with a banded appearance over time. Eggs are oval and attached by a thin stalk to the leaf. Eggs are oviposited on both upper and lower leaf surfaces and on apical leaves. Nymphs are elliptical and flat. They are usually found on lower leaf surfaces and tend not to move. First instar nymphs are yellowish-orange and subsequent instars tend to be yellowish-green. Nymphs and adults feed on phloem with piercing–sucking mouthparts and produce large amounts of whitish granular excrement containing honeydew and wax. Tomato psyllid is responsible for a physiological disorder known as 'psyllid yellows', which can significantly reduce yields. Psyllid yellows produces an array of symptoms in affected tomato plants, including reduced growth, chlorotic, reddish and purplish discoloration of leaves, upward cupping of leaves, shortened swollen internodes and premature plant decline. The tomato psyllid vectors the bacterial pathogen *Candidatus* Liberibacter solanacearum, also known as *Ca*. L. psyllaurous, which causes stunting and reduced yields in tomato. The tomato psyllid is adapted best to temperatures in the range of 21–27° C. At 23°C, tomato psyllid needs approximately 27 days to develop from the egg stage to the adult stage, passing through five nymphal instars. Egg hatch is suppressed at temperatures above 32°C (List, 1939). The tomato psyllid is found in North America west of the Mississippi, in southern parts of Canada and in Mexico, Central America and New Zealand.

Insecticides representing several modes of action are used to manage the tomato psyllid and suppress transmission of *C*. Liberibacter solanacearum (Butler and Trumble, 2012). Management of alternative hosts such as solanaceous weeds aids in the suppression of the pest. Natural enemies of *B. cockerelli* include predators of soft-bodied insects such as ladybird beetles and lacewings; the parasitoids *Tamarixia triozae* (Burks) (Eulophidae) and *Metaphycus psyllidis* Compere (Encyrtidae); and entomopathogens including *Beauvaria bassiana* (Balsamo), *Isaria fumosorosea* (Wize), *Metarhizium anisopliae* (Metschnikoff) and *Verticillium lecanii* (Zimmerman).

Leafminers

The leaf miner species attacking tomato are *Liriomyza bryoniae* (Kaltenbach), *L. huidobrensis* (Blanchard), *L. sativae* Blanchard and *L. trifolii* (Burgess) (Diptera: Agromyzidae). Leaf miner adults are small (approximately 2 mm) flies with black and yellow markings. Females oviposit eggs into the surface of the leaf. *L. huidobrensis* typically oviposits on the underside of the leaf, and larvae mine near the central or lateral leaf veins. The other species tend to oviposit on the upper surface of the leaf and larval mines do not follow a specific pattern. The three larval instars feed on mesophyll tissue between the upper and lower plant surfaces, exiting the leaf and dropping to the soil to pupate. In warm weather, each species needs about 2 weeks to develop from egg to adult on tomato. In addition to the damage caused by leaf mines, females cause 'stippling' damage by puncturing holes in leaves with their ovipositor to feed on plant juices. Selective insecticides can be used to manage leaf miners. However, the use of broad spectrum insecticides tends to make leaf miner infestations worse by suppressing the parasitic Hymenoptera that naturally keep leaf miners in check.

The tomato leaf miner *L. bryoniae* is the dominant leaf miner species in greenhouses. The parasitoids *Dacnusa sibirica* and *Diglyphus isaea* are two larval parasitoids applied worldwide for biological control of several species of leaf miners (van Lenteren, 2012). *D. sibirica* is most effective in the early season, while *D. isaea* provides control during the summer (Minkenberg, 1990). Additional control can be achieved by naturally occurring parasitoids, such as *Opius pallipes*.

Caterpillars

Tomato is attacked by many types of caterpillar from the seedling stage until harvest. Cutworms such as *Peridroma saucia* (Hübner) and *Agrotis ipsilon* (Hufnagel) attack seedlings at the soil level and are most active at night. Poisoned baits, the reduction of crop residues and weed suppression around the edges of the field contribute to the suppression of cutworms. Feeding by loopers is usually confined to foliage. Loopers attacking tomato include *Autographa californica* (Speyer), *Chrysodeixis chalcites* (Esper), *Pseudopludia includens* (Walker) and *Trichoplusia ni* (Hübner). Loopers can generally be controlled using insecticides, including *Bacillus thuringiensis* products that are compatible with biological control. Some loopers serve as alternative hosts for parasitoids such as *Hypersoter exiguae* (Viereck) that attack caterpillars directly

Fig. 7.5. *Spodoptera eridania* (lower right) and damaged tomato. (Photograph courtesy Hugh Smith.)

damaging fruit. Many species of *Spodoptera* (armyworms) attack tomato foliage and will in some instances attack green or ripened fruit. These include *Spodoptera exigua* (Hübner), *S. littoralis* Boisd., *S. eridania* (Cramer) (Fig. 7.5), *S. ornithogalli* (Guenée) and *S. frugiperda* (J. E. Smith). Armyworms lay their eggs in masses on the undersides of leaves, usually covered with a fuzzy layer of scales. Insecticides representing several modes of action are available to suppress armyworms, which are more easily managed in early instars than in later stages. Most caterpillar species are attacked by a range of predators and parasitoids in the egg, larval and pupal stages. The tomato hornworm, *Manduca quinquemaculata* (Haworth), and the tobacco hornworm, *M. sexta* (Linnaeus), are often suppressed by natural enemies.

The most threatening and damaging caterpillar is the leaf miner moth, *Tuta absoluta*. As discussed above, biological control with generalist mirid bugs has been very successful in the Mediterranean area. Specialist natural enemies can, if necessary, be added to the control measures against this pest. A commonly used species is the egg parasitoid *Trichogramma achaeae* (Desneux *et al.*, 2010). Other *Trichogramma* species can be used against noctuid moth pests (Table 7.1) but they are only applied on a small scale. Most caterpillars are controlled by sprayings with biopesticides based on *Bacillus thuringiensis.* Additional control can be achieved by birds that enter greenhouses in summer.

Stink bugs and leaf-footed bugs

Stink bugs and leaf-footed bugs are hemimetabolous insects that pass through an egg stage and five nymphal instars before becoming an adult. These insects have piercing–sucking mouthparts. Both nymphs and adults can feed on foliage, stems and blossoms of tomato but the most serious damage they inflict is on fruit. Stink bug damage on tomato fruits produces a small point surrounded by a lighter patch that either becomes yellow or remains green as the fruit ripens. Stink bug feeding can produce white thickened tissue inside the fruit. Feeding by leaf-footed bugs produces malformed, discoloured fruits. Stink bugs attacking tomato include the southern green stink bug, *Nezara viridula*

(Linnaeus), the brown stink bug, *Euschistus servis* (Say), and the consperse stink bug *Chlorochroa sayi* (Stål) (Hemiptera: Pentatomidae). Stink bugs have similar life histories (Capinera, 2001). The southern green stink bug is a cosmopolitan pest with a broad host range that prefers crucifers and leguminous crops, including cover crops such as *Sesbania* and *Indigofera* spp. The optimal temperature for *N. viridula*'s development is approximately 30°C; it needs about 35 days to develop from egg to adult during warm months. Females lay pale or pinkish-yellow barrel-shaped eggs in hexagonal clusters. Colour changes and colour patterns on nymphal *N. viridula* are complex (Capinera, 2001). The adults are green and shield-shaped.

Leaf-footed bugs attacking tomato include *Leptoglossus phyllopus* (Linnaeus) and *Phthia picta* (Drury) (Hemiptera: Coreidae). Females oviposit oval metallic brown eggs in rows on leaves or stems. Nymphs range in colour from orange to reddish brown. Adults are about 20 mm long and brown. *L. phyllopus* has a white band across the wing covers and has flattened tibia. Other hemipteran pests that cause sucking damage to tomato fruit include *Lygus* spp. and the tomato bug, *Cyrtopeltis modesta* (Distant) (Miridae).

Weeds, particularly solanaceous and leguminous weeds, can serve as alternative hosts for stink bugs and leaf-footed bugs attacking tomato and can be important sources of infestation. Stink bugs are attacked by a range of natural enemies, including scelionid and eupelmid egg parasitoids and tachinids that parasitize late nymphal instars and adults. Predators attacking stink bugs include *Geocoris* spp. (big-eyed bugs), nabid damsel bugs, predatory stink bugs (Pentatomidae), green lacewings (Chrysopidae), fire ants (*Solenopsis invicta* Buren) and birds.

ECONOMICALLY IMPORTANT DISEASES OF TOMATO

A disease can be broadly defined as any departure from normal growth. Tomatoes are afflicted by a broad array of diseases caused by parasitic nematodes, fungi, oomycetes, bacteria, viruses and viroids. Although often referred to as diseases, any disorder caused by a physiological (see Chapter 5 for fruit physiological disorders) or genetic abnormality related to environment or nutrition (see Chapter 6 for nutritional disorders) will not be covered here.

Disease management requires knowledge of the pathogen's life cycle and an understanding of those environmental factors that govern pathogen interaction with the host. In plant pathology, the concept of the disease triangle is often used to describe the three factors necessary for disease development: (i) a susceptible host plant; (ii) a virulent pathogen; and (iii) adequate environmental conditions for infection and growth of the pathogen. Disease management utilizes tactics to disrupt the interaction between the host and pathogen, and the environment to prevent or slow disease development.

The most logical disease management tactics for tomato production are those that eradicate or exclude the pathogen from the environment, avoid production in those areas where the pathogen exists, or avoid production when the environmental conditions are most favourable for disease development. Sanitation should be the base of any integrated disease management strategy. Most pathogens of tomato are able to survive in the residues of crops, related weed species and on volunteer plants, creating a bridge from one production cycle to the next. Cultural practices that ensure the timely destruction of plant residues and sanitization of pots, tools and other production surfaces and implements used during tomato production will limit the carry-over of pathogens from one crop to the next. A number of tomato pathogens are also seed-borne or can lie as a latent infection without any apparent symptoms present in infected tissues until conditions become more favourable for growth and disease development. As such, testing or screening of tomato seed, transplants or grafted materials can greatly limit the introduction of pathogens into the production environment, especially for greenhouse production.

Genetic resistance to a number of important pathogens is available in tomato (see Chapter 2), which provides the most efficient and economical means for managing many diseases. Resistance is usually limited to a specific portion of the pathogen population, referred to as a race, which can limit usefulness in some production areas as the deployment of resistance often leads to the selection of virulent isolates within the pathogen population over time. In addition, many resistance traits are introduced from wild tomato relatives, which during the breeding process can also introduce other genetic factors that can have an undesirable effect on important horticultural traits such as plant architecture, yield or fruit quality – a process known as linkage drag. These undesirable traits often discourage broader commercial adoption of certain resistance traits. While great gains have been made with the development of resistant tomato varieties through traditional breeding methods, the availability of genome sequence for the conventional tomato (The Tomato Genome Consortium, 2012) and wild *Solanum* species (Labate *et al.*, 2014; The 100 Tomato Genome Sequencing Consortium, 2014) have given tomato breeders access to a whole new arsenal of molecular tools, such as genotype by sequencing (GBS) (Poland and Rife, 2012), that will not only accelerate the development of varieties with improved resistance but will also help limit linkage drag. The advent of several genome-editing strategies, such as clustered regularly interspaced short palindromic repeats (CRISPR) (Brooks *et al.*, 2014), as well as classic plant transformation strategies, could potentially allow for the targeted disruption of genes that confer susceptibility (Li *et al.*, 2012) or the introduction of resistance genes from other plant genera (Horvath *et al.*, 2012).

Tomato is truly a global crop (see Chapter 1), playing an important role in the diets of millions. With an ever increasing global demand comes an increase in the movement of plant materials in the form of seed and fruit, aiding in the spread of numerous tomato diseases throughout the world. In addition, increased market demand and reduced availability of arable land throughout the world has increased production in protected structures, such as greenhouses. This review will cover common diseases of importance to the worldwide production of tomatoes. Each disease will be accompanied by a brief description of symptoms, the biology of the causal agent and some reference to control measures.

COMMON FOLIAR DISEASES OF TOMATO

Anthracnose

Anthracnose is a fungal disease caused by several species of the genus *Colletotrichum*, to include *C. coccodes* (Wallr.) Hughes, *C. gloeosporioides* (Penz.) Sacc. and *C. dematium* (Pers. Ex Fr.) Grove (Baxter *et al.*, 1985). Spores of the fungus survive on plant debris and are easily splashed to lower tomato foliage by rain. Foliar symptoms are fairly inconspicuous, consisting of small round necrotic lesions surrounded by a faint yellow chlorotic halo, but do develop a target-like appearance as they expand, similar to other common foliar fungal diseases. Symptoms of anthracnose are most commonly associated with ripening tomato fruit, beginning as small water-soaked lesions that gradually enlarge, becoming sunken and remaining water soaked beneath the surface as the fruit ripens. Fruits often develop cracks within the lesions leading to secondary infections by soft rot bacteria, such as *Erwinia* spp., leading to rapid fruit decay. Eventually, specialized fungal structures known as acervuli develop, causing the centre of the lesion to turn black. Moisture causes spores (conidia) to be discharged in a pink gelatinous mass from acervuli, allowing splash dispersal of spores to other fruits and foliage. One species, *C. coccodes*, also infects roots, causing brown lesions and root rot in which the fungus can produce copious quantities of microsclerotia; this disease is known as black dot root rot (Dillard and Cobb, 1998). The fungus can survive in the soil as acervuli or microsclerotia embedded in plant tissues (Blakeman and Hornby, 1966; Farley, 1976). Anthracnose is primarily managed through the timely destruction of plant debris at the end of season to prevent carry-over of inoculum, avoiding rotations with other solanaceous crops, with a 2–3-year rotation between solanaceous crops, avoiding fields with poor drainage, proper plant spacing to promote airflow in the canopy, and making preventive fungicide applications after fruit set when conditions are conducive for the disease. Disease development is favoured by temperatures above 18°C and periods of extended leaf wetness of 6 h or more (Dillard, 1989).

Bacterial spot

Bacterial spot of tomato is caused by a diverse genetic group of pathogenic bacteria representing four distinct species, *Xanthomonas vesicatoria*, *X. euvesicatoria*, *X. perforans* and *X. gardneri* (Jones *et al.*, 2004; Potnis *et al.* 2015), that are a major problem wherever tomatoes are grown. The distribution of these bacteria varies throughout the world. However, outbreaks of exotic xanthomonads have been identified in recent years in several major tomato/pepper-producing regions in the western hemisphere (Jones *et al.*, 2005). *X. gardneri* has become a significant problem in Brazil (Quezado-Duval *et al.*, 2004) and Canada, and more recently throughout Midwestern production areas of the USA (Ma *et al.*, 2011). Several physiological races of bacterial spot have been identified, mostly associated with the various species, with race 1 corresponding to *X. euvesicatoria*, race 2 to *X. vesicatoria*, and races 3, 4 and 5 to *X. perforans* (Stall *et al.*, 2009).

Bacterial spot is most problematic during warm, humid weather and is often initiated by episodes of wind-driven rain. On the leaf, infection begins when the bacterium enters the plant through natural openings and wounds, where it multiplies within plant tissues. Within 3–4 days, initial symptoms of water-soaked lesions are typically observed on lower leaf surfaces. If ideal environmental conditions persist, lesions enlarge and coalesce, causing extensive leaf chlorosis and defoliation. Once established, the disease can still cause significant losses even in the absence of rain. Under conditions of high relative humidity such as a heavy dew or fog, the disease can spread around the leaf margin (presumably by the infection of hydathodes) and cause a general blighting that can again lead to premature leaf drop even when only moderate disease pressure exists. All above-ground plant tissues are susceptible to the disease. Fruit lesions are particularly problematic for growers, since they not only detract from the appearance of the fruit but also offer a site for other microbes to enter the fruit; both scenarios impact the marketability of the fruit. Fruit lesions begin as small raised blisters on the fruit surface that are a lighter green than the rest of the fruit. As the lesions enlarge, they turn brown to black and develop a layer of scab-like tissue.

Disease is managed by avoiding successive crops in the same field, using clean seed, producing disease-free transplants and keeping fields free of weeds and volunteer tomatoes. Disease management in transplant and field production has relied heavily on the use of copper-based fungicides, which has led to the establishment of copper-tolerant strains in many areas of the world (Ritchie and Dittapongpitch, 1991; Martin *et al.*, 2004). The use of bacteriophages, certain rhizobacteria and chemical elicitors of systemic acquired resistance have shown efficacy in managing bacterial spot (Louws *et al.*, 2001; Obradovic *et al.*, 2004; Byrne *et al.*, 2005; Huang *et al.*, 2012).

Bacterial speck

Caused by *Pseudomonas syringae* pv. *tomato* (Okabe) Young, Dye, & Wilkie, this gram-negative aerobic bacterium occurs in nearly every region of the world where tomato is produced. It is considered a cool season pathogen that is easily dispersed by rain and in water. Disease development is favoured by low temperatures (18–24°C) and high humidity. Initial symptoms consist of small water-soaked lesions that become brown and necrotic with a prominent yellow halo surrounding the lesion, typically less than 1 mm in diameter. Under high disease pressure, lesions can coalesce and lead to blighting of foliar tissues. All above-ground tissues are susceptible to the pathogen, including stems, petioles, peduncles, sepals and fruit. Developing fruit lesions become sunken, with tissues surrounding the lesion exhibiting a darker green coloration, and typically less than 1 mm in diameter, though larger lesions can develop under favourable conditions.

Pseudomonas syringae pv. *tomato* is a successful epiphyte of many common weed species and capable of surviving on crop residue for extended periods of time. Two races (race 0 and race 1) have been identified. Disease is managed by avoiding successive crops in the same field, using clean seed, producing disease-free transplants and keeping fields free of weeds and volunteer tomatoes (McCarter *et al.*, 1983). Disease management in transplant and field production has relied heavily on the use of copper-based fungicides, which has led to the establishment of copper-tolerant strains in many areas of the world. Similar to bacterial spot, the use of chemical elicitors of systemic acquired resistance and rhizobacteria have demonstrated efficacy in managing bacterial speck (Louws *et al.*, 2001; Graves and Alexander, 2002; Ji *et al.*, 2006).

Bacterial canker

Caused by the gram-positive, non-spore-forming and aerobic bacterium *Clavibacter michiganensis* ssp. *michiganensis* (Smith) Davis *et al.*, bacterial canker is a devastating disease of tomato production throughout the world, especially in commercial greenhouse settings (Gleason *et al.*, 1993). Early symptoms consist of an upward curling and necrosis of leaflet margins, and a wilting of leaflets that is often unilateral; leaf petioles initially remain turgid (Fig. 7.6). Although symptoms typically begin in the lower leaves and progress acropetally, the pathogen is easily introduced through wounds or pruning cuts and can then spread downwards through the plant. Eventually the wilt progresses, leading to plant death. Vascular tissues within the stems, and especially within nodes, display a yellow to reddish brown discoloration. The pith tissues of wilting plants also become discoloured and break down. These symptoms of wilt and vascular discoloration of stem and pith tissues are easily confused with bacterial wilt. However, bacterial streaming from the cut stems of bacterial canker-infected plants is limited and consists of yellow ooze when squeezed.

Occasionally, leaves can develop off-white blister-like spots that are surrounded by a dark ring of necrotic tissue. The veins of these infected leaves typically darken, with the leaflets turning yellow as the bacterium spreads. Fruit symptoms do not always occur but consist of a raised brown centre surrounded by an opaque white halo, usually 3–6 mm in diameter. The fruit lesions, often

Fig. 7.6. Foliar symptoms of bacterial canker on tomato. (Photograph courtesy Gary Vallad.)

referred to as bird's-eye spot, tend to give the fruit a scabby appearance and are very characteristic of the disease.

Infected seed and transplants are the most important sources of inoculum. Other sources include plant debris, weed hosts, volunteer plants and contaminated stakes and tools. Like many pathogenic bacteria, the disease is easily spread through contaminated water via rain splash, equipment, or various hand operations such as staking, tying, pruning and grafting. Plants from infected seed may die, or show few to no symptoms, and infected transplants may take 3–6 weeks to display symptoms – all factors that increase the difficulty of management and emphasize the need for strict sanitation practices to limit spread in greenhouse production environments (Chang *et al.*, 1991, 1992; Hausbeck *et al.*, 2000; Xu *et al.*, 2010; Tancos *et al.*, 2013).

Candidatus Liberibacter solanacearum

This phloem-limited unculturable bacterium is primarily transmitted by the tomato/potato psyllid *Bactericera cockerelli* (Homoptera: Psyllidae) or indirectly through grafting. The bacterium is closely related to *Candidatus* L. asiaticus, *Candidatus* L. africanum and *Candidatus* L. americanus, the reported causal agents of huanglongbing (commonly referred to as citrus greening) of citrus and other members of the family Rutaceae (Secor *et al.*, 2009; Levy *et al.*, 2011; Lin *et al.*, 2011). *Candidatus* L. solanacearum has been found in several members of the Solanaceae, causing zebra chip in potato (*Solanum tuberosum*). Symptoms on tomato consist of a spiky chlorotic apical growth, a general mottling of leaves, plant stunting and fruit deformation. Because of its unculturable state, detection requires the use of PCR primers specific to the16s rRNA gene, which has also led to the detection of *Candidatus* L. solanacearum in several other Solanaceae to include tamarillo (*Solanum betaceum*), cape gooseberry (*Physalis peruviana*) and peppers (*Capsicum* spp.) (Liefting *et al.*, 2009). Management of the psyllid vector is the only means currently available for managing *Liberibacter*.

Early blight

Traditionally this disease was attributed primarily to the fungus *Alternaria solani* (Ell. & Mart.) L.R. Jones & Gout, but recent work has identified additional *Alternaria* spp. capable of causing early blight symptoms on tomato, including *A. tomatophila* (Weir *et al.*, 1998; Simmons, 2000; Rodrigues *et al.*, 2010) and *A. grandis* (Rodrigues *et al.*, 2010). Symptoms of early blight consist of light to dark brown circular lesions that develop characteristic alternating light and dark concentric rings as the lesion expands over time (Fig. 7.7). Yellow chlorotic tissue may surround the lesions and the entire leaf may turn yellow as lesions expand and coalesce. The disease typically starts on older leaves but may also develop on stems and fruit. Severe stem lesions can girdle and kill plants. Fruit infections usually begin near the top of the fruit at the calyx, or where the stem attaches to the fruit, and will also develop concentric rings

Fig. 7.7. Foliar symptoms of early blight. (Photograph courtesy Gary Vallad.)

as they enlarge. Infection of seedlings and transplants often leads to a collar rot that can quickly spread and kill young plants. Disease development is favoured by conditions with mild temperatures (24–29 °C) and frequent rains or heavy dews that promote spore germination. Under ideal conditions with free moisture, spores germinate within 2 h, with developing visible lesions within 2–3 days. The fungus produces spores that are dispersed with air currents to other susceptible plants. In addition, *A. solani* can produce specialized structures called chlamydospores that can survive in soil for up to a year (Patterson, 1991). The fungus survives between seasons on seed, infected plant debris, on volunteer tomatoes and other solanaceous weeds, stressing the importance of sanitation and crop rotations for managing the disease. Proper fertilization is also critical to maintain vigorous growth, since older senescent tissues are most susceptible to the fungus. Varieties with resistance or tolerance to the disease are commercially available (Gardner and Panthee, 2012). Finally, the timely application of protective fungicides is often necessary for effective management, especially when environmental conditions favour disease.

Grey leafspot

Caused by three species of the fungus *Stemphylium* (*S. botryosum* f.sp. *lycopersici*, *S. solani* Weber and *S. floridanum* Hannon and Weber), grey leafspot is a devastating foliar disease of tomato wherever susceptible varieties are grown under warm humid environments (Weber, 1930; Hannon and Weber, 1955; Rotem and Bashi, 1977; Camara *et al.*, 2002; Nasehi *et al.*, 2014). Foliar symptoms consist of small random brownish-black lesions that can be round to oblong in shape. The lesion centres are grey with dark brown to black margins and the lesion centres may crack. Infected leaves with numerous or larger lesions will yellow and drop. Grey leafspot primarily affects the foliage; only rarely will it affect stems or petioles when optimum conditions for disease exists. Infection of seedlings can also lead to defoliation without any obvious yellowing. Although this disease does not affect fruit directly, premature defoliation can expose fruit to sunlight and rain, leading to sunscald and raincheck. *Stemphylium* species affect a broad range of weed and plant species and will overseason on residues. Ensuring the timely destruction of plant residues in the field, and disinfecting pots, tools and production surfaces in the greenhouse, will limit the carry-over of pathogens from one crop to the next. Effective commercial resistance exists and is the primary means of managing grey leafspot (Andrus *et al.*, 1942; Hendrix and Frazier, 1949; Bashi *et al.*, 1973; Blancard and Laterrot, 1986).

Grey mould

Caused by the fungus *Botrytis cinerea* Fr., grey mould is typically a minor disease that can affect all above-ground tomato tissues in field and greenhouse production. Botrytis is considered an opportunistic pathogen of tomato, colonizing tissues wounded during production activities or injured by insect activity. The disease is typically associated with mature plants that have a dense canopy and often begins during cooler weather, not requiring prolonged periods of moisture beyond what is available within the canopy (Wilson, 1963; Verhoeff, 1968; Yunis *et al.*, 1990a,b; Elad and Yunis, 1993; O'Neil *et al.*, 1997). Symptoms typically begin on older senescent tissues and consist of necrotic tissue that is covered with a grey to brown velvety mass of fungal sporophores. After periods of high humidity these sporophores can produce copious amounts of spores that are easily liberated as a cloud of spores when the infected tissue is moved. The fungus may also produce irregularly shaped dark brown to black sclerotia within infected tissues. The fungus can start on senescing leaf tissues that can progress into younger leaf tissues, leaf petioles and finally stem infections. Stem infections can also be initiated through pruning wounds, at graft union wounds, or wounds created from rubbing against string or stakes used to keep plants upright. Stem infections can girdle the plant and cause wilting of upper foliage. The fungus can also infect senescent flower petals, which can lead to the infection of the developing fruit, a direct fruit infection if the petals remain attached to the fruit; as such resulting infections occur at either the stem or blossom end of fruit (Fig. 7.8). Infected petals can even lead to stem infections if the petals get lodged within stem–petiole junctions. Direct infection of young fruit from spores can cause the appearance of a necrotic fleck accompanied by a characteristic white halo referred to as a ghost spot as fruit ripen. These ghost spots are sites where the fungus had penetrated the epidermis of the young fruit and then died. Successful infections of green mature or ripening fruits leads to a typical soft rot with watery, sunken, whitish lesions that rupture in the centre and eventually affects the entire fruit. Infected fruits, if not rogued, can lead to large postharvest losses during shipments. *B. cinerea* is a weak pathogen of numerous plants and a common saprophyte of decaying plant tissues, making sanitation through the removal or destruction of senescent and dead plant tissues the primary means of reducing the threat of the disease. Regardless, spores are windborne and easily disseminated over long distances. The fungus can be controlled with the preventive

Fig. 7.8. Botrytis fruit rot of tomato. (Photograph courtesy Gary Vallad.)

application of fungicides prior to the formation of dense canopies. In addition, the liming of acidic soil can limit disease by improving calcium levels in plants (Stall *et al.*, 1965). In addition to proper pruning and plant nutrition, the ability to manipulate environmental variables can greatly aid control of grey mould in protected structures, by minimizing the microclimatic conditions necessary for the fungus to infect and spread (Elad *et al.*, 1988; Yunis *et al.*, 1990a,b; Elad and Yunis, 1993).

Hairy root disease

Hairy root disease, also referred to as 'crazy root' or 'root mat', is mostly associated with the hydroponic production of tomato, aubergine and cucurbits, though the disease can occur in soil as well. The disease consists of an extreme proliferation of roots that promotes strong vegetative growth at the expense of fruit yield (reduced fruit set and fruit size) (Bosmans *et al.*, 2017). In addition, plants with hairy root disease are more prone to other diseases caused by weak secondary pathogens, such as *Pythium* and *Pseudomonas* species (Weller *et al.*, 2006). The disease is caused by several members of the genus *Agrobacterium* that carry a root-inducing (Ri) plasmid, which is the sole pathogenicity factor for the agrobacterium. Similar to the transfer of tumour-inducing (Ti) plasmids associated with members of *A. tumefaciens* that cause crown gall disease, symptoms of hairy root are the result of the natural transformation of root cells by a portion of the Ri plasmid (the T-DNA), which encodes genes for the production of specific opines and increased sensitivity to plant auxins. The opines are a specialized class of carbohydrates that can only be utilized by the specific agrobacterium (Dessaux *et al.*, 1998; Wetzel *et al.*, 2014), while the increased sensitivity to auxins leads to the prolific root growth. Incidentally, since the Ri plasmid confers pathogenicity, it has been associated with several members of what is also referred to as rhizogenic *Agrobacterium* biovar 1 and biovar 2. Members of rhizogenic *Agrobacterium* biovar 1 are most commonly associated with hairy root disease in hydroponic production, while biovar 2 are most commonly associated with the disease in soil (Bosmans *et al.*, 2015). Both *Agrobacterium* biovars are heterogenic groups that contain tumorogenic strains of *A. tumefaciens*, rhizogenic strains of *A. rhizogenes* and avirulent strains of *A. radiobacter*. Recent changes in classification (Lindström and Young, 2011) led to the redesignation of *Agrobacterium* biovar 2 to *Rhizobium rhizogenes,* and *Agrobacterium* biovar 1 to *Agrobacterium tumefaciens* species complex until additional genomic information is provided to further delineate this heterogeneous complex.

There are no treatment options once plants are transformed with the T-DNA from the Ri plasmid, and the causal rhizogenic *Agrobacterium* biovar 1 strains agents easily persist within irrigation systems. Therefore, management relies on tactics to eliminate the pathogen from the system or through changes in cultural practices to reduce the effect of the disease (Bosmans *et al.*, 2017). Various cultural practices, including changes in substrate (Aghdak *et al.*, 2016), nutrient solution, water cycles, and other practices to increase light and oxygen, may help to manage hairy root disease but still require further investigation (Bosmans *et al.*, 2017). Chemical disinfection of water sources, irrigation systems and the recirculated nutrient solutions has utilized a variety of techniques to include the application of ultraviolet light (Pozos *et al.*, 2004) and biocidal compounds, such as benzalkonium chloride, cetyltrimethylammonium bromide, quaternary ammonium, sodium hypochlorite and hydrogen peroxide. The latter is most preferred, as it degrades into benign secondary products, eliminating any residue concerns, and does not corrode infrastructure. Effective reduction of rhizogenic *Agrobacterium* populations requires high levels of hydrogen peroxide, since many of the strains are catalase-positive and can tolerate hydrogen peroxide (Bosmans *et al.*, 2015). This requires growers to monitor levels throughout irrigation systems to maintain effective hydrogen peroxide concentrations of 50 ppm for catalase-negative strains to as high as 100 ppm for catalase-positive strains (Bosmans *et al.*, 2016).

Late blight

Caused by the fungal-like oomycete *Phytophthora infestans* (Mont.) de Bary, late blight is one of the most destructive foliar pathogens of tomato and potato. On tomato, the disease affects foliage, stems and fruit. Foliar infection begins as a water-soaked lesion that becomes brown and necrotic as the lesion expands. Under cool, humid conditions, white mycelium and sporulation can often be seen on the underside of the lesion and along the lesion margin. Often the healthy leaf tissue surrounding the lesion has a light green appearance. Rapidly expanding lesions can have a zonate appearance. Tissues rapidly blight under favourable conditions as leaf lesions coalesce or the fungus progresses to the leaf petiole. The rapid necrosis gives the blighted tissues a distinctive light brown, shrivelled appearance as if they were burned or exposed to frost, especially if tissues do not become colonized by other saprophytic organisms. Stems and petioles can also be directly infected with lesions progressing in a manner similar to foliar infections but can be girdled by the fungus, leading to wilting of associated foliage. Fruit infections begin as dark green greasy spots on the fruit that eventually turn light brown or greyish as the fruit begins to ripen. Fruit lesions can also have a zonate appearance as the lesions expand, and white mycelium and sporulation appear on the lesion surfaces when cool, humid conditions exist.

The pathogen typically survives from season to season on infected fruit (cull piles) and infected volunteer plants of tomato and potato, requiring living tissue to overwinter in most temperate climates. However, in some areas of the world, strains of both mating types of the fungus are present, allowing the fungus to produce a specialized thick-walled structure called an oospore that can survive in the soil or plant debris. Disease development can be rapid, especially when conditions favour sporulation; optimum conditions are 91–100% relative humidity and temperatures of 18–22°C, though sporulation can still occur to as low as 3°C and as high as 26°C. Sanitation in the form of destroying cull piles and volunteers is critical for management, in addition to using clean transplants and timely applications of preventive fungicides when conditions are favourable for disease (Becktell *et al.*, 2005; Kamoun and Smart, 2005; Cooke *et al.*, 2011; Nowicki *et al.*, 2012; Fry *et al.*, 2013; Arora *et al.*, 2014).

Leaf mould

The fungus *Passalora fulva* (Cooke) U. Braun & Crous (syn. *Cladosporium fulvum* Cooke and *Fulvia fulva* (Cooke) Ciferri) is responsible for leaf mould (Crous and Braun, 2003). Foliar symptoms of leaf mould typically begin on older leaves, consisting of pale green or yellow patches on the upper leaf surfaces that eventually turn a bright yellow. The underside of the affected leaves will be covered with patches of an olive-green mould; often the tissue surrounding the mould will be a pale green or white compared with unaffected tissue. Affected leaves will eventually curl and drop. The disease is usually limited to older leaves but can also develop on younger leaves and can even affect petioles, peduncles, stems, blossoms and fruit under favourable conditions. Affected fruit will have a black dry rot on the stem end of the fruit.

The disease is more common in greenhouse production but can occur in the field as well and is favoured by relative humidity above 85% and temperatures of 22–24°C, though symptoms can develop at 4–32°C. Sanitation and limiting humidity are key to controlling the disease in greenhouse production. Spores are easily dispersed in the air and the fungus can survive as a saprophyte in the soil on leaf tissue. Therefore, removing and destroying diseased plant debris, in addition to keeping the relative humidity below 85% (see Chapter 9), can limit the disease in greenhouse production. Staking and pruning plants can also help to reduce relative humidity by improving air movement through the canopy. Several genes conferring host resistance are known and widely deployed (Rivas and Thomas, 2002, see Chapter 2).

Powdery mildew

Several fungi are known to cause powdery mildew on tomato: *Leveillula taurica* (Lev.) G. Arnaud (asexual stage = *Oidiopsis taurica*) and two species of *Oidium* (*O. neolycopersici* and *O. lycopersici*) for which the sexual stage remains unknown (Kiss *et al.*, 2001). The two *Oidium* species cause the more classic form of powdery mildew consisting of white patches of mycelium and spores on upper and lower surfaces of foliar tissue, including stems, petioles, peduncles and blooms, and even on fruit. Infected tissue may initially darken, taking on a purple hue before turning yellow, and eventually develop into a necrotic lesion. With older infections, the mycelium will usually turn greyish as well. Most importantly, since *Oidium* grows epiphytically on the leaf surface, the mycelium, conidiophores and conidia are easily observed. Conidia of *O. neolycopersici* are ellipsoid–ovoid in shape and form singly or occasionally in small chains of two to three spores from simple conidiophores, whereas *O. lycopersici* produces chains of three to five elliptical to doliform-shaped spores (Jones *et al.*, 2001; Kiss *et al.*, 2001).

In contrast, mycelium of *O. taurica* (the asexual stage of *L. taurica*) grows endophytically beneath the leaf epidermis, with conidiophores and conidia developing through the stomata in the lower leaf surface. Symptoms are typically confined to mature leaves, starting as pale green irregularly shaped spots on the upper leaf surface. Lesions become yellow and necrotic and have an angular, vein delimited appearance as they expand. Since the fungus grows within the leaf tissue, mycelium and conidia are limited to the oldest portions of the lesions on the underside of the leaf and only present when conditions are favourable (relative humidity > 95%). *O. taurica* is dimorphic, producing pointed or lanceolate primary conidia and cylindrical secondary conidia with rounded ends. The sexual stage, *L. taurica*, produces globose structures with spiked appendages called cleistothecia that bear specialized ascospores that are the products of sexual reproduction. *L. taurica* has a broad host range but appears to be more associated with tomato production in drier climates (Correll *et al.*, 1987). Regardless of the causal pathogen, powdery mildew is managed through the repeated application of foliar fungicides.

Septoria leaf spot

Caused by *Septoria lycopersici* Speg., Septoria leaf spot is a destructive disease of tomatoes throughout the world. Although the disease affects all above-ground tissues, symptoms usually start on the lower leaves after fruit set as small circular lesions with dark brown margins and tan to greyish centres. Foliar lesions are often surrounded by a narrow band of chlorosis and the lesion centres are dotted with small black fungal structures called pycnidia. Fruit infections are rare but stem, petiole and calyx infections produce smaller lesions that may not produce pycnidia. Each pycnidium exudes massive amounts of spores within a gelatinous mass in the presence of moisture. These spores are readily splash-disseminated by rain or overhead irrigation. Symptoms typically progress up the plant over the season.

The disease is most problematic in temperate production areas, where long periods of leaf wetness, high relative humidity and temperatures of 20–25°C occur (Pritchard and Porte, 1924). The fungus can be seed-borne and also can survive between production seasons on infected tomato debris, or susceptible weeds, and on stakes and cages used to support tomatoes during production. Control is based on crop rotation and field sanitation, acquiring clean seed, limiting leaf wetness or the handling of wet plants, improving air flow within the canopy and the timely application of protective fungicides.

Target spot

Target spot is caused by the fungus *Corynespora cassiicola* (Burk & Curt.) Wei., a common foliar pathogen and saprophyte of diverse plants in tropical and subtropical environments. Initial foliar symptoms of target spot on tomato consist of small pinpoint water-soaked lesions that appear on the upper leaf surface. Lesions eventually turn dark to tan brown and can easily be mistaken for symptoms of bacterial spot or speck. However, as the lesions increase in size, the distinctive tan centre surrounded with darkened circular bands develops within the lesion (Fig. 7.9). These lesions are often surrounded by a narrow band of chlorosis. Expanding lesions can coalesce, leading to the rapid collapse of leaflets. Similarly, these expanding lesions can girdle petioles and stems, leading to a rapid blight of affected foliar tissues. Tomato fruits are also quite susceptible

Fig. 7.9. Foliar symptoms of target spot caused by *Corynespora cassiicola*. (Photograph courtesy Gary Vallad.)

to the disease. On ripe fruits, large brown circular lesions develop with pale brown centres that often crack. Lesions on green immature fruits begin as small, dark brown, sunken lesions that can quickly develop into craters as they expand, especially as the fruit ripens. Lesions also render fruits prone to secondary infections by soft rot organisms. Resistance to target spot has been identified and reportedly conferred by a single recessive gene (Bliss *et al.*, 1973) but little progress has been made to further incorporate this resistance into commercial cultivars.

C. cassiicola is a diverse pathogen with a broad host range that includes more than 280 plant hosts documented in over 70 countries (Farr *et al.*, 2007). In addition, studies have revealed that more diversity exists among global isolates of *C. cassiicola* than previously recognized, including isolates recovered from hosts in Florida (Dixon *et al.*, 2009). Of the 50 isolates from American Samoa, Brazil, Malaysia, Micronesia and the USA tested for pathogenicity on eight known hosts of *C. cassii*cola, 23 were rated as moderately to highly virulent on tomato. Of these 23 isolates, 14 were originally isolated from tomato and the remaining nine from diverse hosts such as basil, bean, cucumber, lantana, papaya, pumpkin and sweet potato. Most of the isolates were pathogenic on several of the hosts tested, but one isolate, GU28 from Guam, was only pathogenic on tomato. More importantly, these 23 isolates clustered into four of the five phylogentic clades based on variation from ribosomal DNA internal transcribed spacer (rDNA ITS), ga4, caa5 and act1 sequences (Dixon *et al.*, 2009). This agrees with an earlier study showing that diversity exists among isolates of *C. cassiicola* regarding host range, with some isolates exhibiting some level of host specificity (Onesirosan *et al.*, 1974).

COMMON CROWN AND ROOT DISEASES OF TOMATO

Bacterial wilt

Bacertial wilt is a serious soilborne disease caused by the gram-negative bacterium *Ralstonia solanacearum* that affects many solanaceous crops, including tomato. The disease begins as the pathogen enters roots through wounds and natural openings, colonizing plant xylem vessels. As the pathogen progresses acropetally, it produces copious amounts of polysaccharides that plug vascular tissues and lead to a wilt of foliar tissues. Initial wilting symptoms can be subtle, beginning with the upper apical tissues that will often recover at night. However, as symptoms progress the entire plant will develop an irreversible wilt that eventually leads to death. Bacterial wilt is often referred to as a 'green wilt' as symptoms often progress rapidly with little or no chlorosis of foliar tissues, unlike vascular wilts caused by species of *Fusarium* or *Verticillium*. Infected plants also develop adventitious roots on the lower stem. Vascular tissues of infected plants develop a yellow to light brown discoloration that becomes dark brown as the disease progresses and spreads upwards through crown and stem tissues. Pith tissues within the crown and stems also develop the brown discoloration and begin to break down, making the stems soft and easy to squeeze. Submerging the cut stem of an infected plant in water allows the bacterium to stream with a milky white appearance and can serve as a quick diagnostic to distinguish this disease from those caused by other vascular pathogens. Strains of *R. solanacearum* are quite variable and have historically been divided into several races and biovars based loosely on host range and ability to utilize certain carbohydrates, respectively (Denny and Hayward, 2001; Denny, 2006). The pathogen was reclassified into four phylotypes and sequevars, based on the phylogenetic analysis of several loci among global strains, which has corresponded with ancestral relationships and geographical origins of strains (Alvarez, 2005; Fegan and Prior, 2005; Prior and Fegan, 2005). Unlike most bacterial plant pathogens, *R. solanacearum* can survive in the soil in the absence of a host plant; and it has a broad host range of over 200 species of cultivated plants and weeds in 33 plant families, making disease management through crop rotation and fallow programmes impractical. In addition, *R. solanacearum* is easily disseminated through contaminated water, soil, machinery and plant material, including transplants. Chemical fumigation (see Chapter 8) can reduce disease development but typically fails to give season-long control. Conventional and biologically based pesticides offer little practical control of the disease. Breeding efforts throughout the world have led to the development of tomato varieties with partial resistance to bacterial wilt, though the performance of these varieties has varied when trialled at different locations. Grafting susceptible tomato varieties to resistant rootstocks has gained global interest and offers a viable option for production in problematic areas (Rivard *et al.*, 2012).

Fusarium wilt and Fusarium crown and root rot

Fusarium wilt caused by *Fusarium oxysporum* f.sp. *lycopersici* (FOL) and Fusarium crown and root rot caused by *Fusarium oxysporum* f.sp. *radicis-lycopersici* (FORL) are both serious soilborne diseases of tomato throughout the world. In older plants, the initial symptoms of Fusarium wilt consist of a characteristic yellowing and wilting of lower leaves (Fig. 7.10). This yellowing and wilting quickly spreads further up the plant, often limited to only one side of the plant, and eventually leads to plant death. A distinct yellow to dark brown vascular discoloration extending from roots and crown up through the stems of symptomatic tissues is a key characteristic of the disease.

Fusarium crown and root rot occurs in both field and greenhouse tomatoes. Initial symptoms consist of a yellowing of lower leaves, which eventually become necrotic and wither. These symptoms move up the plant and, depending on the age of the plant, can result in stunting or even a sudden collapse of the plant. A yellow to brown discoloration can be found in the vascular tissues of

Fig. 7.10. Initial foliar symptoms of Fusarium wilt of tomato. (Photograph courtesy Gary Vallad.)

infected plants but, unlike Fusarium wilt, is typically only found in the roots, crown and lower stem of the plant. Fusarium crown and root rot can be introduced to new production fields by means of infected seeds, transplants, soil and compost (Jarvis, 1988; Hartman and Fletcher, 1991; Menzies and Jarvis, 1994). Once introduced, this polycyclic pathogen can be disseminated via root-to-root contact, dispersal of airborne conidia, water flow and fungus gnats of the genus *Bradisya* (Rowe *et al.*, 1977; Jarvis, 1988; Hartman and Fletcher, 1991; Gillespie and Menzies, 1993; Rekah *et al.*, 1999), making control of this disease more difficult.

Fusarium crown and root rot is favoured by cool weather (below 20°C) in both field and greenhouse tomatoes, and is commonly associated with winter and spring tomato plantings. The host range of FORL consists of at least 36 other plant species (Menzies *et al.*, 1990). In contrast, FOL thrives in warm weather (Jones *et al.*, 1991) and consists of three races that are host-specific to tomato. Of these three races, race 3 was shown to have a better over-seasoning ability than races 1 and 2. A major vegetative compatibility group (VCG) was previously identified in Florida (VCG 0033), which belongs to race 3 (Gale *et al.*, 2003). Epidemiologically, as with FORL, the sporulation on tomato stems and aerial dissemination of conidia of FOL have been reported to cause serious outbreaks (Katan *et al.*, 1997); although the relative importance of aerial dissemination compared with soilborne inoculum in field epidemics has never been fully substantiated, it may be more important in greenhouse operations where constant plant pruning (see Chapter 9) creates wounds for the fungus to potentially colonize. Other dissemination methods include contaminated seeds, tomato stakes, soils and transplants (Jones *et al.*, 1991).

Although FOL and FORL may be differentiated in the field based on symptomology and disease development, morphologically the two causal pathogens are indistinguishable from each other and from non-pathogenic isolates of *F. oxysporum*. Most modern field tomato varieties carry genetic resistance to races 1 and 2 of FOL but only a few varieties have resistance to FOL race 3 and to FORL. Long rotations with non-susceptible crops, deep tillage practices to bury inoculum and infested debris, and pre-treating soils with effective chemical fumigants are all practices that can help reduce inoculum levels in subsequent crops. When possible, removing and destroying infected plants can help to reduce the amount of inoculum released into the soil.

Nematodes

Nematodes are microscopic roundworms that live in water or the films of moisture that coat soil particles. Several genera of nematodes are reported parasites of tomato, each varying in its below-ground life cycle and the symptoms associated with infected root tissues. Foliar symptoms typical of nematode infections include chlorosis, stunting, wilting, premature senescence and reduced fruit yields; these symptoms are most severe with early plant infections. Nematode infections can also increase plant vulnerability to soilborne plant pathogens, such as *Verticillium* and *Fusarium* spp. (Back *et al.*, 2002). Economically, the most important group of parasitic nematodes on tomato are various species of *Meloidogyne* that cause swelling of roots at the sites of infection known as galls or knots. These knots are the result of nematode feeding and cannot be rubbed off. As such, *Meloidogyne* spp. are commonly referred to as 'root-knot nematodes' based on the resulting root symptoms. Genetic resistance to *Meloidogyne* spp. is conferred by the *Mi* gene (Williamson, 1998; Jacquet *et al.*, 2005). Other nematode species reported to parasitize tomato include *Belonolaimus longicaudatus* (sting nematode), *Rotylenchus reniformis* (reniform nematode), *Pratylenchus* spp. (root lesion nematodes), *Nacobbus* spp. (false root-knot nematodes), *Globodera* spp. (potato cyst nematodes), *Tylenchorhynchus* spp. (stunt nematodes), *Xiphinema* spp. (dagger nematodes) and *Trichodorus* spp. (stubby root nematodes). These nematode species vary in their life cycles and can infect a broad range of plants, making field management extremely difficult. In addition, they are easily moved throughout production areas with soil on farm machinery, infested plant materials, animals, farmyard manure and in surface runoff water (Noling and Ferris, 2012).

Southern blight

Southern blight is caused by the basidiomycete fungus *Sclerotium rolfsii* Sacc. (*Athelia rolfsii* (Curzi) Tu & Kimbrough) (Aycock, 1966). Besides tomato, the fungus has a broad host range that includes hundreds of plant species and is most problematic in subtropical and tropical regions of the world. Southern blight can develop at any stage of plant growth, favoured by moist and aerobic soil conditions and high temperatures of 30–35°C. Symptoms begin as a water-soaked lesion on the stem at or just below the soil line that turns into a brown to black open lesion. The lesion can expand rapidly, girdling the stem at the crown and leading to a rapid green wilt of all upper foliage. Under humid conditions, a mat of robust white mycelium is often visible on top of the soil surrounding the plant, extending to the stem lesion and sometimes even several centimetres up the stem. Sometimes this mycelial mat may not develop until the plant is dead and it can often extend to neighbouring plants, allowing the fungus to infect multiple plants in a patch or down a plant row. After a few days, numerous tan to black sclerotia, 1–2 mm in diameter, will develop on the mycelial mat. Any fruits exposed to infested soil or contacting the lower stem near an infection point are also prone to infection by *S. rolfsii*. Fruit infection begins as a yellowish, sunken, water-soaked lesion that expands, leading to fruit collapse within 3–4 days with white mycelium and developing sclerotia filling the cavity. In addition, the fungus is a good saprophyte capable of colonizing and producing sclerotia on various host substrates. Sclerotia are the primary inoculum; they can survive in the soil and on plant debris for years and are easily disseminated through the movement of soil or plant debris. Although the perfect basidial stage of this fungus (*A. rolfsii*) has been induced *in vitro*, its role in disease remains unclear (Punja, 1985).

Because of the longevity of sclerotia in the soil and the broad host range of the fungus, control of southern blight in problematic fields can be difficult, especially when conditions favour rapid disease development. Long rotations with non-susceptible grass crops, deep tillage practices to bury inoculum and infested debris, and pre-treating soils with effective chemical fumigants are all practices that can help to reduce inoculum levels in subsequent crops. When possible, removing and destroying infected plants can help to reduce the amount of inoculum released into the soil.

Verticillium wilt

Caused by the soilborne fungi *Verticillium albo-atrum* Reinke & Berthold and *V. dahliae* Kleb., Verticillium wilt is a vascular wilt disease affecting over 200 plant species and is quite common to most tomato production regions throughout the world. Symptoms commence as a wilting of plants, beginning with the lower leaves. The affected lower leaves develop a marginal and interveinal chlorosis. Often, the interveinal chlorosis will form characteristic angular V-shaped lesions on affected leaflets. These symptoms progress acropetally, sometimes only affecting one side of the plant, but gradually lead to wilting of the entire plant and death. A light brown to yellow vascular discoloration is evident in the stems of infected plants, especially in the lower stems, but does not extend into the pith of stem tissues or into petioles. These symptoms are easily confused with other vascular wilt diseases, such as Fusarium wilt. However, the progression of wilt symptoms, chlorosis and vascular discoloration is slower and less pronounced compared with Fusarium wilt. Sometimes symptoms of Verticillium wilt are subtle enough to be mistaken for inadequate irrigation (Klosterman *et al.*, 2009; Nunez, 2012).

Both *Verticillium* species are widely distributed, though *V. dahliae* is the most prevalent pathogen. Each is capable of surviving in the soil in infested plant material, or freely as dark resting mycelium in the case of *V. albo-atrum*, or as microsclerotia in the case of *V. dahliae*. Both structures are melanized allowing both *Verticillium* species to persist in soil for years, which in addition to their broad host range makes these fungi difficult to manage. Long rotations with non-susceptible grass crops, deep tillage practices to bury inoculum and infested debris, and pre-treating soils with effective chemical fumigants are all practices that can help to reduce inoculum levels in subsequent crops. When possible, removing and destroying infected plants can help to reduce the amount inoculum released into the soil. Tomato varieties with the *Ve* gene are resistant to race 1 isolates of *V. dahliae* and *V. albo-atrum*, though race 2 isolates of *V. dahliae* are widely distributed throughout the world (Bender and Shoemaker, 1984; Dobinson *et al.*, 1996; Klosterman *et al.*, 2009; Nunez, 2012).

COMMON TOMATO DISEASES CAUSED BY VIRUSES AND VIROIDS

Tomatoes are susceptible to numerous viruses and viroids that cause a myriad of symptoms, including stunted or distorted growth, mottling of leaves, chlorosis, necrosis and plant death, depending on the exact virus/viroid. Viruses are stretches of nucleic acid (RNA or DNA) with a protein coat (Zaitlin and Hull, 1987). Viroids, on the other hand, consist solely of a small stretch (a few hundred nucleobases) of circular, single-stranded, non-coding RNA. While the genomes of viruses can encode numerous proteins necessary for pathogenesis and viral replication, viroids do not encode any proteins; but some viroids have catalytic activity functioning as ribozymes. Evidence suggests that viroids disrupt plant host processes through a process known as RNA interference (Itaya *et al.*, 2001; Di Serio *et al.*, 2014).

Diseases caused by viruses and viroids can be extremely difficult to control and have the potential to cause extensive crop losses. Most viruses of tomato are transmitted by insect or arthropod pests and are typically managed by controlling these pest populations and by eliminating potential viral reservoirs such as infected tomatoes and other susceptible host plants from the production environment. Other viruses and all viroids are transmitted through mechanical means, such as physical handling of the plant during production by humans or machinery, or through plant-to-plant contact, and are typically seed-borne. These mechanically transmitted viruses and viroids pose a significant threat to greenhouse tomato production since plants are handled on almost a daily basis (see Chapter 9), facilitating rapid spread of the virus or viroid. In addition, these mechanically transmitted viruses and viroids are capable of surviving on contaminated work surfaces, tools and machinery for extended periods of time, emphasizing the importance of strict sanitary measures to exclude these viruses and viroids from the production environment and to limit their spread upon introduction. Proper identification is crucial for controlling virus or viroid outbreaks, as knowledge of the disseminating vector(s) and potential viral/viroid reservoir(s) is critical for directing management tactics (Verhoeven *et al.*, 2004; Pearson *et al.*, 2006; Adkins *et al.*, 2012). The following section is not intended to be an exhaustive review of the numerous viruses/viroids that afflict tomatoes, but rather serves to give a brief overview of the most important viruses in each virus family related to specific vectors and finally viroids.

Whitefly-transmitted begomoviruses – *Tomato yellow leaf curl viruses* (TYLCV)

Begomovirus is the largest genus belonging to the viral family *Geminiviridae*, and like other members of this family consists of single-stranded circular DNA genomes encapsidated within quasi-isomeric virion particles. Begomoviruses can be further divided into monopartite or bipartite forms, depending on whether their genomes consist of one or two circular ssDNA molecules. Tomato production throughout the tropical and subtropical regions of the world is besieged by numerous begomoviruses coincidental with the spread of the whitefly, *Bemisia tabaci*. Begomoviruses, once acquired, are transmitted by whiteflies in a persistent circulative non-propagative manner (Czosnek and Laterrot, 1997; Abhary *et al.*, 2007; Lefeuvre *et al.*, 2010).

One of the most important and severe begomoviruses afflicting tomato is *Tomato yellow leaf curl virus* (TYLCV) (Navot *et al.*, 1991; Pico *et al.*, 1996). Typical symptoms of TYLCV consist of severe stunting of infected plants with shortened internodes, and erect terminal and axillary shoots (Fig. 7.3). Developing leaves fail to expand and they develop leaflets that are reduced in size and abnormally shaped, with the leaf margins being extremely chlorotic and cupping upwards. Those plants infected early as seedlings are severely stunted and fail to bear any fruit. Fruits set on older plants after infection will ripen regularly, but subsequent fruits will be undersized, with the plant gradually stopping fruit set altogether (Fig. 7.3). Flowers are unaffected by TYLCV but abort and drop as the severity of symptoms progresses.

Perhaps one of the most alarming features of begomoviruses is their ability to develop variants through mutations, recombinations and pseudorecombinations, which has resulted in rapid global diversification. This diversification has led to a number of variants capable of causing symptoms of 'leaf curl' or 'yellow leaf curl' throughout the world. Similar events are unfolding for another important bipartite begomovirus known as *Tomato mottle virus*, which is also transmitted by whitefly but unlike TYLCV can be mechanically transmitted as well. Several genes conferring resistance to TYLCV have been identified and one gene, *Ty-1*, has been deployed in commercial tomato production (Lapidot *et al.*, 1997; Lapidot and Friedmann, 2002). Unfortunately, tomato varieties with resistance to TYLCV can still be infected and propagate the virus (though symptoms are greatly reduced), serving as reservoirs for other susceptible plants.

Thrip-transmitted tospoviruses – *Tomato spotted wilt virus* (TSWV)

Tomato spotted wilt virus belongs to the genus *Tospovirus*, the only plant-infecting genus within the *Bunyaviridae* family of viruses. The genomes of tospoviruses consist of a single-stranded RNA with negative polarity that is divided into three segments termed small 'S' (2.9 kb), medium 'M' (5.4 kb) and large 'L' (8.9 kb). Altogether, the three linear segments are 17.2 kb in size. Tospoviruses are quite complex compared with other plant viruses, as each of the three ssRNA segments is individually encapsidated and enveloped together in a membrane derived from the host plant. All tospovirus members, including TSWV, are transmitted by thrips in a persistent propagative manner. Although at least ten species of thrips can transmit tospoviruses, *Frankliniella occidentalis*, the western flower thrips, is considered to be the most important, due to its global distribution and ability to transmit most tospoviruses (Roselló *et al.*, 1996; Pappu *et al.*, 2009; Webster *et al.*, 2015).

Symptoms of TSWV can vary considerably depending on the strain of the virus and the age of the tomato plant at the time of infection. Young leaves of infected tomato plants develop a bronze discoloration and develop numerous small dark purple to black spots that can coalesce into streaks that can extend down to petioles and even the terminal ends of stems. Early on, these spots can look somewhat similar to fresh lesions of bacterial spot. However, the spots associated with TSWV are typically located along leaf veins and do not show through both sides of the leaf. Infected plants may also develop a lop-sided growth habit or even exhibit symptoms of wilt. Some new growing tips in severely infected plants may die back and the terminal end of stems can develop streaks. Tomato plants infected early may not develop any fruit, but those infected after fruit set develop fruits with chlorotic ring spots. On green fruits these ring spots are often subtle, with the centre of the spots being slightly upraised, but as the fruits ripen these ring spots become more pronounced in coloration with multiple concentric rings (Adkins *et al.*, 2012).

TSWV is unique compared with other tospoviruses, having a broad host range that includes over 1000 plant species and causing significant economic damage to many agronomic and horticultural crops. TSWV may be ubiquitous in some areas where it is able to infect many weeds, landscape plants and even native plants, allowing it to bridge tomato production seasons easily. For these reasons, most management strategies for TSWV rely on limiting or excluding the thrips vector and the use of resistant varieties (Saidi and Warade, 2008; Adkins *et al.*, 2012; Gao *et al.*, 2012).

Aphid-transmitted *Alfalfa mosaic virus, Cucumber mosaic virus* and potyviruses

Cucumber mosaic virus (CMV) is a positive-sense, single-stranded, tripartite RNA virus that is vectored by several aphids in a non-persistent manner and can also be transmitted mechanically. CMV is the type member of the genus *Cucumovirus* in the family *Bromoviridae*, with worldwide distribution and a host range that includes over 800 other crop and weed species (Roossinck, 1999). Symptoms vary but typically consist of severely stunted plants with pronounced distortion of leaves, which become elongated but never expand and are often referred to as 'shoe-string' or 'strapped'. Many different strains of CMV have been documented on tomato and other hosts (Adkins *et al.*, 2012).

The aphid-transmitted potyviruses *Potato virus Y* (PVY) and *Tobacco etch virus* (TEV) belong to the family *Potyviridae* and consist of flexous filamentous rod-shaped viral particles containing a linear positive-sense, single-stranded monopartite RNA genome. Both PVY and TEV are vectored by aphids in a non-persistent, non-circulative manner and can also be mechanically transmitted through sap. Symptoms of PVY can vary greatly, depending on environmental conditions, tomato variety and virus strain, but typically consist of a dark green banding of veins along the leaf, with mild mottling, mosaic and some distortion of the leaves. As symptoms progress, interveinal necrosis can develop on leaves and leaflets often curl downwards, giving the plant a droopy appearance. PVY typically does not affect fruit, but plants infected early are stunted, with reduced yield (Adkins *et al.*, 2012).

Symptoms of TEV consist of mottling and wrinkling of the leaves, vein clearing and the characteristic necrotic lines or etching of the leaves. Fruits from TEV-infected plants are mottled and often undersized. TEV affects peppers and other members of the Solanaceae family, and many common weeds can serve as a reservoir for this virus (Adkins *et al.*, 2012).

Mechanically transmitted tobamoviruses – *Tobacco mosaic virus* (TMV) and *Tomato mosaic virus* (ToMV)

Both TMV and ToMV are members of the genus *Tobamovirus* within the family *Virgaviridae*, consisting of positive-sense RNA viruses with coat protein forming straight rod viral particles. TMV and ToMV are typically grouped together because of their high similarity in host range and serological attributes. The most common symptom of disease by either virus is leaves with mottled areas of light and dark green. These affected leaves are typically curled, reduced in size and somewhat malformed; the entire plant may be somewhat stunted in growth. Fruit symptoms typically consist of uneven ripening and reductions in size and number, though fruits can develop yellow rings at high temperatures or develop an internal browning (similar to greywall) in mature green fruit; other forms of fruit disfiguration can occur occasionally. Overall, plant symptoms are greatly influenced by the viral strain, tomato cultivar, time of infection relative to plant growth stage, temperature and light quality. Symptoms can be exacerbated by high temperatures and low light intensity (Adkins *et al.*, 2012). However, tomato varieties with the *Tm-2* and *Tm-22* resistance genes can express a severe necrotic reaction if exposed to TMV or ToMV at high temperatures (Hall, 1980; Fraser and Loughlin, 1982).

Due to their coat protein structure, TMV and ToMV are probably the most persistent and infectious viruses in the plant world. Both viruses are mechanically transmitted and are spread by human activities. Those activities common to tomato production such as grafting, transplanting, staking, tying, pruning and harvesting disseminate the virus with ease (Broadbent and Fletcher, 1963; Lanter *et al.*, 1982). Since TMV can survive in tobacco products, the use of tobacco products near tomato production should be discouraged and personnel should wash hands to inactivate the virus before and during plant-handling procedures. Both viruses can also be seed-borne and so treating seeds with specialized disinfectants or heat is encouraged (Broadbent, 1965). Problematic fields should be avoided, as TMV and ToMV can survive in plant debris, including roots, for prolonged periods of time (Adkins *et al.*, 2012).

Mechanically transmitted *Pepino mosaic virus*

Pepino mosaic virus (PepMV) is a positive-sense, single-stranded, monopartite virus that is a member of the family Alphaflexiviridae. PepMV was first described in Peru in 1974 on pepino (*Solanum muricatum*). PepMV was not reported until it was rediscovered in tomato crops in Europe in 1998/1999. Ever since, PepMV has spread rapidly in greenhouse tomato production and is currently found throughout Europe and North America (Ling, 2008), likely introduced through contaminated seed (Hanssen *et al.*, 2010). PepMV induces a wide range of symptoms on tomato, such as mosaic, leaf distortion, nettle-like heads, single yellow spots, interveinal chlorosis and fruit discoloration. Schenk *et al.* (2010) showed that tomato plants can be protected against PepMV by a preceding infection with an attenuated isolate of this virus. Infection with the attenuated isolates alone affected neither bulk yield nor quality of the harvested tomato fruits.

Mechanically transmitted viroids – *Potato spindle tuber viroid* (PSTvd)

Viroids are unique to the plant kingdom consisting of only a highly complementary, circular, single-stranded RNA molecule ranging from 246 to 467 nucleotides in length. These infectious RNA particles replicate through a process called rolling circle synthesis using the host plant RNA polymerase II enzyme, which is normally associated with messenger RNA synthesis in the host. Although viroids do not code for any proteins, many act as ribozymes, having catalytic properties to cleave and ligate replication intermediates of themselves. Due to their lack of any proteins, viroids cannot be detected using serological methods and therefore require molecular-based methods for detection and identification. Viroids are divided into the *Avsunviroidae* and *Pospiviroidae* families. The first identified and best characterized viroid of tomato and potato, *Potato spindle tuber viroid* (PSTvd), is the type member of the *Pospiviroid* genus that includes several other tomato-infecting viroids: *Tomato chlorotic dwarf viroid*, *Mexican papita viroid*, *Tomato planta macho viroid*, *Citrus exocortis viroid*, *Tomato apical stunt viroid* and *Columnea latent viroid* (Hammond and Owens, 2006).

Initial symptoms of PSTvd in tomato consist of reduced growth and chlorosis of the terminal branches of the plant. As the symptoms progress, terminals become more stunted with a loss of flower and fruit initiation and a severe chlorosis that is often associated with a pronounced reddening or purpling of the affected foliage. Leaf tissues often become brittle and some plants will die, while others may partially recover (Owens and Verhoeven, 2009).

Many viroids, like PSTvd, are seed-transmitted, but all are easily spread mechanically through tools and handling of plant materials, which makes them quite problematic in greenhouse production systems. Seed treatments are ineffective at eliminating viroids and no plant resistance or chemical treatments are available for controlling viroids once a plant is infected. In addition, many viroid outbreaks have been linked to contaminated, non-symptomatic ornamental species (Verhoeven *et al.*, 2010). Therefore, strict sanitary measures must be in place to exclude viroid-contaminated materials from greenhouse production facilities and to eradicate viroid-infected vegetative materials during an outbreak.

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8

Production in Open Field

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INTRODUCTION

Tomato open field production throughout the world has changed dramatically in the past 3 decades from mostly bare ground plantings to a larger share of polyethylene-mulched fields, with the bulk of production in the latter system occurring in developed countries. Worldwide production (see Chapter 1) of fresh tomato focused primarily on beefsteak, cherry, grape and round cluster types, whereas for the processing industry Roma cultivars are utilized the most. The worldwide tomato growing area has increased during the past decade, with the highest increases occurring in Africa and Asia (FAO, 2018). This might be partly attributed to increased awareness of the importance of vegetable consumption for human diets. This chapter discusses the most important field procedures for managing the crop, including soil preparation and disinfection, planting techniques, crop management and harvesting.

SOIL PREPARATION

Bare ground and plastic-mulched production

Throughout the world, bare ground cultivation is widely used for processing tomatoes, as well as in subsistence low-input tomato systems and small gardens. Although this method reduces the initial investment and production costs, it results in lower yield and likely net income than polyethylene-mulched crops. Additionally, in many parts of the world bare ground production is viewed as an 'environmentally friendly' approach to managing annual vegetable crops, including tomato. In this production system, soil fumigation is mostly not utilized because of lack of plastic mulching, which is essential to retain gases for long enough to be effective against soilborne diseases, pest arthropods and weeds. This operation in addition to installation of drip irrigation usually increases the carbon footprint on farms.

The advantages of using plastic-mulched beds for tomato production have been widely documented. Plastic mulches increase soil moisture, tomato earliness, growth and quality by keeping fruits off the ground, while reducing fertilizer leaching and soil compaction (Grassbaugh *et al.*, 2004; McGraw and Motes, 2007). Schonbeck and Evanylo (1998a) indicated that early yields were generally the highest with black plastic mulch and lowest with organic mulches, and that mulching affected early and total tomato yields by regulating soil moisture and temperature regimes. Moreover, mulch colour plays a significant role in tomato response, especially for radiation and temperature control. The most common mulch colours are black, white and silver (often called 'metalized'). Mulch colour determines its energy-radiating patterns and its influence on the plant microclimate (Lament, 1993). Research has found that white or silver mulches reflect more total light and a lower ratio of far-red to red radiation than black or red mulches, while soil temperatures are warmer under black and red mulches (Decoteau *et al.*, 1989). Tomato studies in the Middle East suggested that the highest early yields were obtained from the silver, white-on-black and black-on-white plastic mulches in comparison with bare ground cultivation, and total marketable yields were significantly increased by all the mulch treatments (Suwwan *et al.*, 1988). Decoteau *et al.* (1989) suggested that mulch colour affects spectral balance and quantity of light, and root zone temperatures. Schonbeck and Evanylo (1998a), working with processing tomato, determined that black mulch increased soil temperatures by 1–2°C but sometimes resulted in lower soil moisture levels in early summer in the mid-Atlantic USA, probably by hindering penetration of rainfall. Diaz-Perez and Batal (2002) found that raising soil temperature is one of the benefits of using plastic film mulches under cool to moderate conditions but that some mulch types could reduce plant growth by excessively raising temperatures, especially during the summer months in the south-eastern USA. The same study indicated that the degree of soil warming is correlated with reflectivity of the mulch, with black mulch having the lowest light reflectance and silver the highest (Diaz-Perez and Batal, 2002). Additionally, it has been reported that black and clear mulches raise soil temperatures, whereas white and metalized mulches raise or lower it (Tarara, 2000). This is mostly because of the chemical composition of the plastic layers and their absorbance and reflection of certain wavelengths within the solar spectrum.

Other studies suggested that mulches may play an important role in sustainable vegetable production because of their potential to reduce pests, such as weeds and insects, and to protect the soil from surface crusting and erosion (Schonbeck and Evanylo, 1998b). Irizarry *et al.* (1966) found that black mulch suppressed the growth of 102 different weed species, though not of nutsedge (*Cyperus rotundus*) in tomato fields. Other studies reported that the incidence of *Tomato yellow leaf curl virus*, which is transmitted by whiteflies (*Bemisia* spp.), was reduced when using silver mulch (Suwwan *et al.*, 1988). This has been attributed to possible disorientation of flying adults when approaching the highly reflective mulch-covered surfaces. Greer and Dole (2003) determined that aluminum foil and aluminum-painted mulches are effective at repelling insect pests, especially aphids and thrips.

Soils and tillage

Regardless of the tomato type, the crop adapts to a very wide range of soil conditions and types. While high soil organic matter content improves general aeration and water retention and is beneficial to most crops, tomatoes are well adapted to a variety of soils. Soil texture can be from very coarse sands to fine, heavy clays, whereas optimum soil pH ranges between 6.0 and 6.5. Soils with poor drainage usually cause severe root-growth limitations and reduced yields.

Soil preparation depends on the production system to be utilized. The overall goal of this activity is to provide the correct tilth to allow rapid root formation and expansion, regardless of direct seeding or transplanting techniques. For bare ground culture, between three and six tillage passes may be necessary to obtain the desired soil structure prior to transplanting. In contrast, growing tomato under plastic-mulch culture may require additional passes because of the need to form well pressed and levelled beds that facilitate soil fumigation, laying drip irrigation lines and covering with plastic mulch in timely fashion. The soil type (i.e. mineral and organic fraction of the soil) will dictate the depth of tillage and needed implements for this activity. Soils with compacted subsoil may need additional tillage to avoid waterlogging and improve root penetration. Therefore, deep chiselling should be used to loosen lower topsoil horizons. This activity should also be performed with new land where no row crops have been planted previously (e.g. pastures). Passing once or twice in perpendicular fashion is recommended during the dry part of the year. Chisels may need to remove soil as deep as 60 cm.

On the other hand, conventional tillage usually occurs in the top 45 cm of the soil and when the soil is moist above field capacity. It requires turning over the surface soil with disc ploughs or shallow harrows. Thomas *et al.* (2001), working with processing tomato, indicated that as many as four to six tillage passes are often employed to prepare the seed bed prior to transplanting, including mouldboard ploughing, multiple passes with a disc or field cultivator, and formation of raised beds. The same study compared diverse tillage procedures and found yield increases for tomato fruit harvested from conventional tillage, disked and zone-tillage systems in comparison with no-tillage, which delayed crop maturity (Thomas *et al.*, 2001). Sainju *et al.* (2000) stated that ploughing increased fresh and dry fruit yield and increased stem and leaf dry weight in comparison with no-tillage.

Bedding differs greatly in fields used for bare ground production compared with those where the crop will be grown under mulching. In bare ground culture, raised beds that are slightly higher at the centre are formed using a ditch plough. The beds are 25–40 cm high and 80–100 cm wide at the base and are usually spaced 1.5 m apart. The soil is generally loose and the beds may be reshaped later during the season if either a rotary tiller or cultivator is used or during the application of side-dressing (banded) fertilizer. In this case, beds should not be too narrow or too short, to allow enough soil to remain around the roots and the base of the stem after use of cultivators that drag soil down the rows. This also applies to regions with high rainfall, where hydric erosion of beds is likely to occur. Raised bedding has several advantages in comparison with planting in flat ground, including increased drainage and aeration, the promotion of lateral root development, and easier hand weeding and harvesting. When mulching is used, beds are usually pressed (using a trapezoidal or rectangular bedding press) to increase compaction and formation of micropores, which facilitates lateral water and soil fumigant movement.

Soil fumigation

Growers producing tomatoes under plastic mulch often rely on soil fumigation to manage soilborne pests, including disease-causing fungi, bacteria, nematodes and weeds. For many years, commercial tomato growers in North America and Europe used the mix of methyl bromide plus chloropicrin as their fumigant of choice. However, methyl bromide was phased out in compliance with the Montreal Protocol, which classifies this fumigant as an ozone-depleting molecule (Albritton *et al.*, 1998; US Environmental Protection Agency, 1999).

Ensuing fumigant research identified a handful of alternatives to methyl bromide, but none of them provided its consistent broad-spectrum efficacy, which forced scientists to focus on an integrated, multi-strategy approach (Table 8.1). Most of the published articles have studied the efficacy of the fumigants 1,3-dicloropropene, chloropicrin, dazomet, metam sodium, metam potassium and dimethyl disulfide on soilborne disease, nematodes and weeds (Santos and Gilreath, 2006). In many instances, the combination of these molecules with herbicides to enhance weed control has been examined, as well as the influence of application methods and mulches on fumigant activity, retention and performance. Alternatives are less flexible than methyl bromide with regard to application methods and conditions to enhance efficacy. Most commercial applicators use gas knives, chisel ploughs, large sweeps and S-shaped tine harrows or combinations to deliver the fumigants in-bed prior to mulching. Adding low-permeability mulch to a fumigation programme improves fumigant retention and increases pest exposure, thus improving control and reducing yield loss (Santos and Gilreath, 2006).

Table 8.1. Summary of maximum use rate and relative effectiveness of various soil fumigants for nematodes, soilborne diseases and weed control in Florida. (Modified from Noling *et al*., 2012.)

Fumigants	Maximum use per ha	Relative pest activity		
		Nematodes	Diseases	Weeds
Chloropicrin	335 kg	None to poor	Excellent	Poor
Metam sodium	700 l	Fair to good	Poor to erratic	Fair to excellent
1,3-dichloropropene	170 l	Good to excellent	None to poor	Poor
1,3-dichloropropene + chloropicrin	330 l	Good to excellent	Good to excellent	Poor to fair
Metam potassium	570 l	Fair to good	Poor to erratic	Fair to excellent
Dimethyl disulfide	570 l	Good to excellent	Good to excellent	Fair to excellent

SEEDLINGS AND TRANSPLANTING

Seedling and transplant production

Tomato seeds and transplants are the two most common propagation means. However, the crop may also be propagated by cutting and grafting. Direct seeding is frequently used in home gardens and small cropping areas. Seeds must be sown between 1 cm and 3 cm deep and they require thinning to obtain the adequate planting density after germination. The soil texture and structure will play a major role on seedling emergence. Loose soil is preferred and, in some instances, soil amendments such as vermiculite and polymers may be applied to the planting holes to facilitate seedling emergence (Hoyle, 1983). A variety of environmental conditions are needed for optimum tomato seed germination but the crop is well adapted to survive extreme conditions. Under optimal conditions, seedlings break through the soil surface about 4 days after sowing (Hanson and Chen, 1998). Over-irrigation may cause seed drowning and hinder germination.

Transplants offer several advantages over direct seeding of tomato. In the USA and Mexico, transplanting is necessary during the cool part of the season to give plants a 4–6-week advantage over direct seeding, which allows fruit set before high temperatures occur. A good quality transplant is a compact plant with well developed root mass. Transplanting (Fig. 8.1) reduces the seed required per surface area, especially considering the high cost of hybrid seed. The cost of hybrid tomato seed can range from US$770/kg to over US$4000/kg (McGraw *et al*., 2007). The amount of seed needed to plant 1 ha in the field will depend on the planting density and seed presentation (e.g. pelletized and

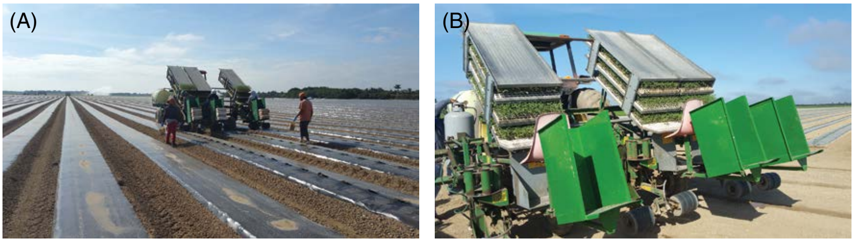

Fig. 8.1. Transplanting seedlings grown in polystyrene trays: (A) overview; (B) detail of planting machine. (Bielinski M. Santos.)

'naked' seed). Motes and Roberts (2007) suggested that 28 g of tomato seed could generate approximately 10,000 plants, equivalent to 100–140 seeds/g. Other authors indicated that 30–60 g of seed could produce enough transplants for 1 ha, assuming planting densities of 3000–6000 plants/ha, which reduced the seed requirement compared with direct field seeding (McGraw *et al.*, 2007). Further reports determined that approximately 84 g of seed are required to produce 10,000 seedlings (Boyhan and Kelley, 2010; Ivors and Sanders, 2010).

For transplant production, seeds are often sown in multi-cell trays inside growth chambers, greenhouses or shade houses. These structures provide controlled growing conditions, such as moisture, light and temperature, for proper seed germination and seedling development. For this operation, a clear knowledge of the specific environmental conditions for transplant production is needed. For instance, Liptay (1992) determined that growth inhibition was noticeable with as little as 15 min of daily exposure to air circulation in a growth chamber, while it increased with continuous exposure. Optimum day temperatures for germination are 21–27°C, while temperatures of 16–18°C are preferred at night (Rutledge *et al.*, 1999).

The tray size and length of holding time in the cells also influence transplant vigour and performance. Varying the cell size alters the rooting volume of the plants, which can greatly affect plant growth (NeSmith and Duval, 1998). Although there are many spatial configurations for planting trays throughout the world, it has been suggested that square cells with 1.5 cm sides produce high-quality and easy-to-handle transplants (Boyhan and Kelley, 2010; Ivors and Sanders, 2010). Cell depth is often 8–12 cm but Motes and Roberts (2007) suggested that the space for growing transplants should be cells of 5 cm × 5 cm. Larger cells may not be economical whereas smaller cells may limit root growth too much. Marr and Jirak (1990) found that holding tomato transplants in plug trays for more than 4 weeks resulted in reduced early and total yields. Diverse studies have suggested that seedlings need 5–7 weeks to grow and the seed should be planted 6–12 mm deep in the tray cells (Rutledge *et al.*, 1999; Motes and Roberts, 2007; Boyhan and Kelley, 2010; Ivors and Sanders, 2010).

Although there are no specific nutrient recommendations for raising tomato seedlings, it is widely recognized that excessive nitrogen combined with densely planted trays results in weak, stretched transplant stems with poorly developed roots. Nutrient levels to feed seedlings in greenhouses also have an influence on the performance of the transplants. Varying nitrogen rates have the largest effect on growth and phosphorus is required in very small quantities, while potassium can be varied with very little effect on transplant performance (Liptay *et al.*, 1992). After the transplants are 15–20 cm tall and have three to five true leaves, they should be hardened-off by reducing water and lowering temperature, which increases chances for survival (Boyhan and Kelley, 2010; Ivors and Sanders, 2010).

On a minor scale, grafting is another method for propagating tomato. Grafting also helps as a disease and nematode cultural control method in some parts of the world. In most cases, resistant rootstocks of wild or improved tomato types are utilized to graft desired scions. Extensive studies have been conducted on the subject. Grafting has been demonstrated to be an effective management tool for organic growers in the south-east USA to reduce risk of crop loss resulting from soilborne diseases for heirloom tomato (Rivard and Louws, 2008). Khan *et al.* (2006) determined that grafted tomato plants were more vigorous than the non-grafted ones, regardless of the environment (greenhouse or open field), and had higher yields in both systems. The study found that tomato quality was not affected by the procedure.

Field transplanting and planting densities

Most of the open field production of tomato occurs with determinate cultivars. However, there are many indeterminate cherry, grape and heirloom cultivars planted on mulched beds in many parts of the world. Because of all these variations, the in-row and between-row spacings vary widely. Additionally, the final market purpose also influences these practices. For instance, processing tomato is planted at high densities because the final objective is to obtain large volumes of fruit as opposed to high quality of individual fruits. In contrast, beefstake tomato requires wider spacings to reduce intraspecific competition for space, nutrients and water and to improve fruit size, as described by Elattir (2002), who found that the average fruit weight decreased when the plant density increased.

Open field production of beefsteak tomato in Florida and California often uses between-row distances of 1.5–1.8 m, whereas in-row distances range between 45 cm and 60 cm. These distances provide field densities of approximately 14,800–9250 plants/ha. Obasi (2007) found that spacing of 50 cm × 50 cm resulted in higher marketable fruit yield than spacing of 75 cm × 50 cm in beefsteak tomato. Another study determined that a density of 91,200 plants/ha increased yield by 40% in comparison with a density of 30,400 plants/ha (Elattir, 2002). Processing tomatoes are planted at in-row distances of

0.3–0.6 m. In contrast, grape, cherry and heirloom indeterminate cultivars are planted at densities as low as 7500 plants/ha, using between-row distances of 1.8–2.1 m.

In bare ground production fields, planting holes are made either by hand or mechanically using a hole-puncher on a tractor, whereas in mulched beds the latter system is preferred. Planting holes should be slightly deeper than the length of the rootball to allow complete covering of the roots and promote adventitious root growth. Vavrina *et al.* (1996) indicated that the depth of planting has a profound effect on tomato yield. In that study, it was determined that setting transplants as deep as the cotyledon leaves improved extra-large fruit number and maturity in the first harvest in comparison with transplants set just at the top of the rootball. This may be attributed to more vigorous plant stands due to larger exposure of transplant roots and the lower stem section to essential growing resources, leading to increased conventional and adventitious roots.

CROP MANAGEMENT

Environmental requirements

Clear knowledge of the optimum environmental conditions to produce tomato is needed to maximize net returns. Light is a main determining factor for crop yield (see Chapter 4). Although growers in open fields cannot manipulate some environmental factors, such as temperature and light, there are production practices that help to ameliorate the impact of adverse conditions, such as fertilization and irrigation. A more detailed discussion of these two factors is available in Chapter 6.

With regard to air temperature, tomato optimum growth occurs at 17–23°C and it stops at a maximum of 33°C and a minimum of 12°C (Swiader *et al.*, 1992). Sato *et al.* (2001) found that mean daily temperatures over 25°C reduced fruit and seed set in a determinate tomato cultivar. Fruit set is reduced in most cultivars when there are average maximum day temperatures above 30°C and night temperatures above 20°C during anthesis (Swiader *et al.*, 1992). Low temperatures and irradiation cause improper ovary development, malformation of the flowers, unviable pollen, fruit puffiness and blotchy ripening (Rylski *et al.*, 1994).

Tutoring and tying

Most tomato types need to be tutored to support leaf and branch weight, as well as to prevent fruit from touching the soil and therefore avoid diseases and mechanical injury. Tying and staking also facilitate harvesting and allow more uniform application of pesticides and foliar fertilizers (Rutledge *et al.*, 1999;

Ivors and Sanders, 2010). Kemble *et al*. (2004) indicated that stringing should be done when the foliage is dry, to prevent spread of common bacterial diseases such as bacterial leaf spot (*Xanthomonas* spp.).

There are several training methods for tomato, which vary in cost and complexity depending on the type of production operation. The simplest is the unsupported method widely used for processing tomato. Other tomato types are grown with stakes and twine, cages, wires or a combination of these (McGraw *et al*., 2007). A simple trellis for tomato consists of stakes or small poles buried at intervals in the planting beds and using nylon-based twine to create a cage around the plants (Konsler and Gardner, 1990). Strings run horizontally alongside the rows and plants are typically tied between four and eight times, depending on the tomato type. Determinate beefsteak and Roma tomatoes are tied firstly before the plants are 30 cm tall and this is successively repeated after every 20–25 cm. In contrast, indeterminate cherry, grape and heirloom tomatoes may require to be tied up to ten times (Rutledge *et al*., 1999; Cornell Cooperative Extension, 2007; Ivors and Sanders, 2010). Staking (Fig. 8.2) could improve fruit readiness by a week and result in higher yields overall than for non-staked tomato (Obasi, 2007). Bhardwaj *et al*., (2012), working with Roma tomato, determined that staking tomatoes increased the marketable fruit yield by 25% compared with no staking.

Pruning

This field operation is performed to remove unwanted lateral branches. Depending on cultivar, this practice usually occurs 2–4 weeks after transplanting

Fig. 8.2. Staked tomatoes with full-bed polyethylene mulch. (Teresa P. Salamé-Donoso.)

by hand-removing shoots from ground level up to the primary fork below the first flower cluster. Most growers of round tomatoes in Florida and California and in Mexico perform shoot pruning on their crops during the early part of the growing season. From the standpoint of disease management, it is suggested that early shoot pruning could be a potential practice to reduce bacterial spot infection, because it reduces the amount of foliage near the soil that could serve as an initial point of entry for the bacterium and it changes the architecture of plant canopies, thus changing air and moisture flow between the leaves (Carlton *et al.*, 1994).

It has been suggested that pruning can also hasten early fruit production (Rutledge *et al.*, 1999; Cornell Cooperative Extension, 2007; Ivors and Sanders, 2010). In spite of these reports, some studies have shown that shoot pruning is unnecessary for some determinate tomato cultivars (Kemble *et al.*, 1994; Santos, 2008). However, other studies established that shoot pruning increased early yield but reduced total yield (Sikes and Coffey, 1976; Carlton *et al.*, 1994). Navarrete and Jeannequin (2000) found that when shoot pruning was performed every 21 days, tomato stem diameter, vigour and fruit number and weight decreased. Vallad and Santos (2010) showed that light shoot pruning did not improve tomato yield of total and extra-large marketable fruit. At the same time, light pruning did not reduce the severity of bacterial spot on two round tomato cultivars. Heavy pruning (i.e. pruning plants twice) reduced seasonal marketable yields in comparison with non-pruned plants (Vallad and Santos, 2010). Because this activity is performed by hand, growers are urged to evaluate the cultivar characteristics to determine its feasibility. In the USA, pruning beefsteak tomato once during the season could cost $125–200/ha in labour, based on grower estimations and direct observations of production fields.

Weed control

Managing weeds in tomato fields will depend on the production system and tomato type. Weeds compete with the crop for essential factors such as water, nutrients and light. In bare ground production, weeds must be controlled both between tomato rows and between plants. When producing with plastic mulches and fumigation, weed control practices vary significantly from the bare ground system. This occurs partly because, in addition to fumigation, the plastic mulch acts as a physical barrier for weed seedling growth. Thus, weeds mostly occur in the planting holes and between the rows. The tomato type also plays an important role in deciding when to control weeds. For instance, processing tomato requires less herbicide applications or cultivator passes than staked tomato types because (i) the former tends to cover the ground faster and reduce competition from weeds than the latter, and (ii) the production cycle is a lot shorter for processing tomato.

Typical weed management in economically important open field tomato production can be divided into five control strategies or methods: (i) manual; (ii) mechanical; (iii) chemical; (iv) cultural; and (v) integrated. Manual weed control consists of removing or cutting weeds using bare hands or basic implements, such as hoes, knives and machetes. It is typical of small home or community gardens and subsistence agriculture. It is extremely effective when weeds are small and causes little injury to the crop, but it is time consuming and particularly costly in developed countries. Mechanical weed control uses fuel- or animal-powered equipment, such as ploughs, cultivators, mowers and harrows, to remove weeds from the soil. It is typically used to control weeds between the rows and around the fields. It can be very effective for large tomato fields but it requires purchasing or renting equipment to pull the implements. Frequent and shallow cultivations in bare ground systems are preferred, to minimize damage to tomato roots (Ivors and Sanders, 2010). This method is also practical for removing weeds late in the season in bare ground, non-staked tomatoes, rendering it useless for most determinate beefsteak tomato crops, as well as indeterminate cherry, grape, cluster and heirloom tomato cultivars after staking.

Chemical control of weeds is based on herbicides, which are designed to disrupt vital metabolic functions or structures of the target species. There are several pre- and post-emergence herbicides on the market and their availability changes from country to country. Herbicide usage in open field production is critical in many parts of the world, such as the USA and Mexico, to manage troublesome weeds. Typically, the lowest dose of the recommended rates is used in sandy soils or soils with high organic matter content. It is recommended to select a herbicide, or herbicide combination, based on the specific weeds present in each field, because not every herbicide controls every weed. In bare ground culture, pre-emergence herbicides are applied to the soil after beds are made to prevent grass, broadleaf and *Cyperaceae* growth. This practice is supplemented by a single or multiple herbicide applications during the season to control weeds in the row middles with selective or shielded non-selective herbicides. When plastic mulching is used, a great deal of the seasonal weed control is achieved with fumigation. However, in-bed application of selective herbicides may be needed to control troublesome weeds that penetrate the mulch, such as *Cyperus rotundus* and *C. esculentus*. Other herbicides may be needed for row middle application as described for bare ground culture. A list of selected herbicides labelled for control of important weeds is presented in Table 8.2.

Cultural weed control depends on practices that enhance the competitive ability of the crop and/or that reduce the interference of weeds. Examples of these practices are using transplants free of weeds, growing healthy and vigorous plants through manipulation of fertilization and irrigation, using appropriate bed preparation and recommended in- and between-row spacing, and managing other pests such as diseases and insects (Culpepper, 2010). Integrated weed management uses combinations of two or more control methods. It is widely considered a sustainable method to manage weeds in

Table 8.2. Selected herbicides labelled for troublesome weed control in tomato fields in Florida. (Modified from Olson *et al*., 2012.) Mention of a herbicide does not constitute recommendation for its use. Specific label recommendations should be followed and consulted before using.

Active ingredient	Herbicide group	Controlled weeds and comments
Pre-plant and/or pre-emergence		
Carfentrazone	Protoporphyrinogen oxidase inhibitors	Emerged broadleaves
EPTC	Lipid synthesis inhibitors	Annual broadleaves and grasses, and yellow/purple nutsedge. Labelled for transplanted tomatoes grown on low-density mulch
Flumioxazin	Protoporphyrinogen oxidase inhibitors	Annual broadleaves and grasses
Halosulfuron	Acetolactate synthase inhibitors	Broadleaves and yellow/purple nutsedge. Only two applications per season
Lactofen	Protoporphyrinogen oxidase inhibitors	Broadleaves. Apply to row middles only with shielded or hooded sprayers. Will cause excessive injury if applied to green foliage or fruit
S-metolachlor	Chain fatty acid synthesis inhibitors	Annual broadleaves, grasses and yellow nutsedge. Apply to row middles
Napropamide	Chain fatty acid synthesis inhibitors	Annual broadleaves and grasses. For direct-seed or transplanted tomatoes. Apply to well worked soil that is dry enough to permit thorough incorporation to a depth of 5 cm
Oxyfluorfen	Protoporphyrinogen oxidase inhibitors	Broadleaves and some grasses and sedges. Must have a 30-day treatment–planting interval for transplanted tomatoes
Paraquat	Photosystem I electron diverters	Emerged broadleaves and grasses. Apply as a pre-plant burn down treatment
Pelargonic acid	Unknown	Emerged broadleaves and grasses. Apply as a pre-plant burn down treatment
Pendimethalin	Microtubule assembly inhibitors	Grasses. Should not be applied under plastic mulch
Rimsulfuron	Acetolactate synthase inhibitors	Annual broadleaves. Needs rainfall or irrigation for activation
Trifluralin	Microtubule assembly inhibitors	Annual broadleaves and grasses
Post-emergence		
Carfentrazone	Protoporphyrinogen oxidase inhibitors	Emerged broadleaves. Hooded application to row middles only

Continued

Table 8.2. Continued.

Active ingredient	Herbicide group	Controlled weeds and comments
Clethodim	Acetyl CoA carboxylase inhibitors	Perennial and annual grasses
DCPA	Synthetic auxins	Annual grasses and certain broadleaves
Halosulfuron	Acetolactate synthase inhibitors	Small seeded broadleaves and nutsedges
Lactofen	Protoporphyrinogen oxidase inhibitors	Broadleaves. Apply to row middles only with shielded or hooded sprayers
S-metolachlor	Chain fatty acid synthesis inhibitors	Annual broadleaves, grasses, and yellow nutsedge. Apply to row middles
Metribuzin	Photosystem II inhibitors	Emerged, small broadleaves and grasses weeds. Apply after transplants are established or direct-seeded plants reach the five true-leaf stage
Paraquat	Photosystem I electron diverters	Emerged broadleaves and grasses. Apply as a pre-plant burn down treatment
Pelargonic acid	Unknown	Emerged broadleaves and grasses. Direct spray to row middles
Rimsulfuron	Acetolactate synthase inhibitors	Broadleaves and grasses
Sethoxydim	Acetyl CoA carboxylase inhibitors	Grasses
Trifloxysulfuron	Acetolactate synthase inhibitors	Broadleaves and nutsedges. Direct spray to the base of transplanted tomato plants

crops, including tomato. A typical example of this control strategy is the use of mulches for weed suppression and to improve crop performance, while using pre-emergence herbicides to reduce the likelihood of troublesome weed emerging in the row middles.

Harvesting and handling

Detailed information on this topic is provided in Chapter 10. Depending on the tomato type and market, the crop can be harvested at diverse ripening stages. For instance, ripe tomatoes for local consumption and markets are generally handpicked when at least one-third of the fruits are pink or reddish. This allows for fruit to remain fresh for at least a week. Harvesting should be performed ideally in the early hours of the morning, once the moisture has disappeared from the surface of the fruit, or as the last practice of the day in the afternoon. The reason for this is to avoid harvesting fruit under high temperatures, causing loss of firmness and susceptibility to fungi and bacteria during transport. Fruit should be collected with great care to prevent bruising and injury that lowers quality. The fruits should be placed in collection boxes or specialized plastic containers and should be positioned preventing adjacent calyxes or peduncles from touching the skin of neighbouring fruits. Fruits should not be exposed to direct sunlight or placed on the floor once harvested and must be transported to their destination in cool or cold storage, if possible. Other postharvest management recommendations depend on the needs of the market. However, in most cases, fruit must be stored at temperatures between 4°C and 10°C and with high relative humidity (> 80%).

For beefsteak determinate tomato, harvesting begins between 65–80 days after transplanting and it is continued once or twice a week, depending on the cultivar, management, market demands and environmental conditions. Fruits may be harvested with or without calyx, depending on the market requirements. Depending on the type and cultivar, it is harvested in whole clusters or individual fruits. To harvest clusters, previously disinfected shears or scissors should be used, cutting fruit trusses as close to the stem as possible. Individual fruits can be harvested by hand, applying a small turn to the stem before pulling, as long as fruit maturity allows. Maturity and ripeness classes (Fig. 5.7) for fresh market tomatoes are: immature, mature-green A, mature-green B, mature-green C, breaker, turning, pink, light red, red, full red (Cantwell and Kasmire, 2002). These maturity levels are based on surface morphology and structure, external colour, internal colour and structure (Reid, 2002).

In large field operations in the USA and Mexico that require tomato transport over long distances, fruits are harvested at the mature-green stage, which has fully formed seed and juicy locules. Tomato pickers use buckets to collect the fruits and place them in large bins that hold 400–600 kg of fruit. The tomatoes are then transported to packing houses, where they are dumped on a conveyor belt leading to chlorinated water (100–150 mg Cl per l). Afterwards, fruits are sorted as extra-large (more than 7 cm in diameter), large (7–6.4 cm in diameter), medium (6.3–5.7 cm in diameter) and cull (less than 5.7 cm in diameter, or fruits with visible blemishes or injuries). They are placed in cardboard boxes and stored at 13–16°C for several weeks. In many cases, the tomatoes are treated with ethylene at a concentration of at least 100 ml/m^3 until the terminal end of the fruit starts to turn pink (Csizinszky, 2005). In California, as well as other parts of the world, processing tomatoes are harvested mechanically in a single harvest when fruits are ripe. Thompson (2002) indicated that tomato for processing is machine harvested because injuries do not significantly affect the quality of processed product.

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9

Greenhouse Tomato Production

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GREENHOUSE INDUSTRY OVERVIEW

Although definitive numbers are not available, the greenhouse area used for the production of vegetables worldwide has been increasing over the years. In comparing different geographical areas, large differences are seen in climatic conditions (light intensity and temperature), greenhouse design and equipment, as well as technical expertise (Montero *et al.*, 2011). This results in yield differences between regions. While it might be expected that regions with higher light would have higher production, the level of greenhouse technology utilized is often a more important factor. For example, average tomato yields in Almeria, Spain, are far lower (28 kg/m^2) than in The Netherlands or Canada (60 kg/m^2) (see Chapter 1), even though the daily light sum or daily light integral (DLI) is on average 5 times higher in winter and 60% higher on an annual basis in Spain than in The Netherlands (Costa and Heuvelink, 2000). This difference is partly caused by a difference in cropping season. In Almeria, the temperature in summer is too high for balanced crop growth and good product quality of tomatoes, therefore greenhouses are left empty in summer. On the other hand, in arid or semi-arid regions such as Mexico and south-western USA, tomatoes are grown very successfully in summer in greenhouses equipped with advanced climate control technology. Differences in yield between regions are also due to cultivar choice. For example, pink-type cultivars grown widely in Japan have been shown to be less productive than modern Dutch cultivars. The average annual greenhouse tomato yield has reportedly doubled from 30 kg/m^2 in the 1980s to 60 kg/m^2 in 2005 in The Netherlands (Higashide and Heuvelink, 2009). More recently a yield level of 100 kg/m^2 was anecdotally reached in south-western USA in a high-tech, semi-closed greenhouse by combining crop scheduling and advanced environmental control. This remarkable achievement is due to both technology advancements in controlled environment and the use of special cultivars that are highly productive and especially bred for greenhouse cultivation (Higashide and Heuvelink, 2009).

More recently, along with increasing interest in urban farming, greenhouses have now been erected in metropolitan areas such as New York City. Rooftop greenhouses, once considered economically impractical, are now successfully operating and attracting consumers who appreciate locally produced food. Entrepreneurs and academia are examining the feasibility of tomato production inside warehouses equipped with modern artificial lighting (i.e. 'plant factories'), but the economics of such production systems need to be further investigated under various scenarios.

Special cultivars exist for tomato production in greenhouses. The large red multi-locular 'beefsteak' type tomato with an indeterminate growth habit is the industry standard in North America. However, this cultivar type is no longer popular in North American greenhouses, due to competition from open field production. Cluster tomatoes or on-the-vine tomatoes (TOV) are now the common types produced in North American greenhouses (Fig. 9.1). Smaller two- to three-locule round fruits (47–65 mm) or cherry tomatoes (< 15 mm), nearly all cluster types, are the most common types grown in Europe. There is also a market for large fruit, which may replace smaller fruits as market trends change. Attributes for cluster types include uniformity of fruit size, harvesting the entire truss at once, simultaneous ripening of all the fruit on the truss and fruit remaining on the vine/truss after harvest.

Heirloom tomato varieties are grown in greenhouses for niche markets in North America, though scale of production is still small. They are open-pollinated and maintained for many generations among farmers in Europe and North America. They exhibit large variations in colour, flavour, shape and other traits. Many of these cultivars are not as productive as the widely used commercial hybrids but they are richer in flavour, creating specialized markets in retail and restaurants.

GREENHOUSE STRUCTURES AND SYSTEMS

Structure types and orientations

Greenhouses have been developed to protect the crop from 'hostile' weather conditions such as low temperatures, frost, wind, rain, hail or snow. Tomatoes can be grown in every type of greenhouse, provided it is high enough to train the plants vertically. Tunnels covered with plastic films are widely used as low-cost structures, typically for seasonal production. In many countries worldwide, the Dutch so-called Venlo type of glasshouse is now used (Atherton and Rudich, 1986; Teitel *et al.*, 2012). At the same time, many countries have their own greenhouse types with different dimensions, structures and coverings (Montero *et al.*, 2011). For example, a typical Chinese greenhouse (solar greenhouse) consists of short north–south oriented raised beds inside the east–west oriented lean-to structure with the north-side wall made of soil and/or bricks.

Fig. 9.1. Various types of tomatoes grown in greenhouse: (a) cluster type (TOV), medium size; (b) cluster type (TOV), mini Roma; (c) cherry type. (Photographs: Chieri Kubota.)

In southern Spain, flat-roof greenhouses covered by single-layer polyethylene film ('parral' greenhouses) are widely used. In Israel and Mexico, where temperatures are generally higher, screen houses are common and effectively used for tomato production.

The frames of modern Dutch-style greenhouses are generally made of aluminum or galvanized steel. More traditional and older or smaller greenhouses as well as high tunnels may be wood-framed. The shape of greenhouses used for tomato cultivation (Fig. 9.2) varies depending on several factors, such as: (i) the weight of the tomato plants on the wire; (ii) expected snow load; (iii) amount of natural ventilation required; (iv) place of vents (roof or side); (v) whether the bays are stand-alone or joined at the gutters; (vi) type of covering (glass or plastic); (vii) growing system; (viii) use of screens; and (ix) use of artificial lighting, etc.

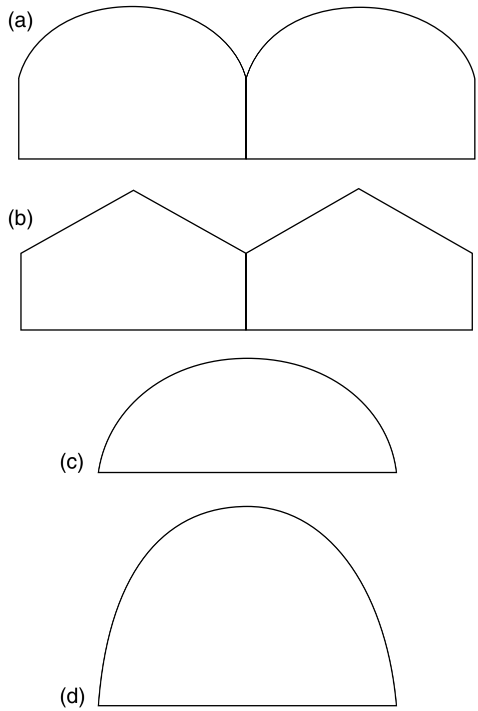

Fig. 9.2. Shapes of greenhouse frames: (a) gutter-connected straight sidewall with arch roof; (b) ridge-and-furrow style straight sidewall with gable roof; (c) hoop or quonset style; (d) Gothic-arch frame.

Sidewall and support column height has been increasing, and in modern Venlo-type greenhouses ranges from 3.5 m to 7 m. High sidewalls and support columns allow for a taller crop and for more climate control equipment to be installed above the crop, such as horizontal airflow fans, energy screens, shade screens, fogging, lighting and heating. High sidewalls also increase the effectiveness of natural ventilation in greenhouses with roof venting. However, increasing sidewall height may increase the heating costs in some cases, such as in single-span greenhouses, as it increases the surface/floor area ratio. Space near the sides can be used more efficiently with straight sidewalls than in so-called Quonset-style structures (Fig 9.2). Gothic-arch frame structures, which have a peak at the top but curving sides, provide adequate sidewall height without loss of strength and can be free-standing or part of a range of multi-span, gutter-connected units. The advantage of Gothic-arch structures in double-polyethylene greenhouses is better runoff of condensation from the inner layer of plastic. The runoff can be channelled outside the greenhouse, reducing greenhouse humidity.

Venlo-type greenhouses protect the crop against adverse weather conditions and at the same time provide high light transmittance, allowing good crop growth (Bakker *et al.*, 1995). Venlo greenhouses typically use glass covering but a variety of synthetic covering materials is also used.

The parral greenhouse, which is the standard in Almeria, Spain, consists of a supporting structure of vertical wooden, concrete or metal posts. The lower ends of the posts are set in the soil and their upper ends are connected by a flexible wire grid. The plastic film covering material is supported by this wire grid and is held in place by a second wire grid placed over the film. The roof pitch varies from flat to 13 degrees. Holes are made in the plastic film along the valleys to allow rainwater to pass through the cover and be collected using plastic or galvanized steel gutters. Since the cost of mechanical ventilation is unacceptably high for typical parral greenhouses, they use natural ventilation. This is the cheapest, most practical and commonly used method to ensure an acceptable greenhouse climate during both warm and cool periods (Brugger *et al.*, 2005) but does not allow production in mid-summer (high temperature season).

Greenhouse light transmission and spatial uniformity of light intensity inside the greenhouse are important. They are affected by the direct/diffuse ratio in the incident light as well as diffusion properties of covering materials (Hemming *et al.*, 2008, 2016), greenhouse design (roof angle and shape, number of spans, and amount of light intercepting structural elements), time of day, season and location (latitude). Computer-based simulations have been used to investigate influences of such factors on incident light intensity and distribution over the plant canopy (e.g. Gupta *et al.*, 2012; Castellano *et al.*, 2016). In general, seasonal changes in greenhouse light transmission are greater at higher latitudes. Greenhouses transmit more light (direct light) in winter when they are oriented east–west, while north–south oriented greenhouses have a more uniform light environment inside the greenhouse. In large, multi-span greenhouses, optimizing for light may be less critical than optimizing for wind direction if the greenhouse is to be naturally ventilated by roof vents. Ideally the greenhouse is placed perpendicular to the direction of the prevailing winds during the hottest times of the production cycle. An additional consideration is that the distribution of shaded and non-shaded areas should be uniform over the course of the day. Ideally, all areas of the greenhouse should receive the same amount of light over a 24 h period in order for plant growth to be uniform throughout the greenhouse.

Covering materials

There are three main types of greenhouse covering: glass; rigid plastics; polyethylene or other synthetic film. Film coverings can be either double or single. In cold climates, double layers are separated by an insulating layer of air, often continuously inflated with an air pump, usually about 10 cm thick, to conserve energy. Modern synthetic materials can have similar or better attributes than glass, but are often more expensive.

Traditionally, greenhouses have been made from glass; hence the use of the terms ‘glasshouse’. Glass maximizes photosynthetically active radiation

(PAR) transmission and only requires regular cleaning and sealing of the edges. Glass also has lower transmission of long-wave radiation (3000 nm or greater) and therefore glass provides more thermal protection than plastics. Glass can be used in large panels, reducing structural shading (Giacomelli and Roberts, 1993). However, glass is expensive compared with polyethylene plastic film and some rigid plastics.

Life expectancy of all plastics with a few exceptions (e.g. ethylene-co-tetrafluoroethylene (ETFE)) is reduced by extreme temperatures, dust, sand, air pollutants and ultraviolet (UV) radiation. Traditional polyethylene coverings would need to be replaced every 2–4 years to maintain acceptable light transmission. Innovation in plastic formulations and advanced extrusion technologies has led to high-quality greenhouse films with extended film life. Also different types of plastic layers are combined to modify thermal properties or to reduce condensate dripping. Near-infrared (NIR) blocking additive for plastic covering materials, however, typically reduces the transmission of PAR and needs evaluation under various conditions (e.g. Kempkes and Hemming, 2012). Some companies also offer wavelength-selective plastics said to reduce disease or insect pressure or to control plant height growth. At this time, because of their higher cost, wavelength-selective plastics are not widely used by growers.

Three types of coverings were compared in an experiment in Ontario, Canada: glass, rigid synthetics and polyethylene plastic film. Tomato and cucumber yields were reported to be similar under all three types of coverings (Papadopoulos and Hao, 1997a, b). Probably other factors compensated for the lower light level, such as more diffuse light and more favourable humidity in summer under plastic than under glass, leading to different plant balance. Presumably light was less limiting than in earlier studies done in The Netherlands, which compared single- and double-glass greenhouses. In these studies, 1% light loss resulted in about 1% loss of production (Van Winden *et al.*, 1984). In a more recent literature study, Marcelis *et al.* (2006) showed that 1% less light resulted in 0.7–1% less growth and production in most greenhouse vegetable crops (see also Chapter 4). Whatever the type of covering material used, optimal light transmission should be a priority for manufacturers of greenhouses and greenhouse covering materials.

Rigid plastics used for greenhouse covering include fibreglass reinforced polyester, polycarbonate and acrylic. Some are energy efficient, have good light transmission in the first year of usage and last at least 10 years. But rigid plastics are more expensive than polyethylene films (Giacomelli and Roberts, 1993). Acrylic and fibreglass panels deteriorate from dust more rapidly than glass and they are fire hazards. Like glass panes, rigid plastics are strong and can be installed as large panels to reduce shading from support structures. Insulated rigid double-walled plastic panels are sometimes used to conserve energy but they reduce the rate of snowmelt compared with glass or plastic film. This is a severe disadvantage in regions with snowfall, because snow accumulation not only blocks the natural light but also can lead to collapse of the greenhouse.

Double-layer polyethylene greenhouses are also energy efficient and in case of snowfall the double layers can be deflated, to increase melting rate. Thus, in Canada, Mexico and the USA, it is more common for new greenhouses to be covered with a double layer of plastic film than with glass or rigid plastic panels. In north-west Europe, use of glass is common. Double-poly houses often have rounded roofs (Quonset-style) (Fig. 9.2), which contribute to condensate dripping on to leaves. Designs with roof openings (Fig. 9.3) are available for plastic film greenhouses as well as acrylic (Giacomelli and Roberts, 1993).

Among recently developed covering materials, one that transforms direct light to diffuse light without substantially reducing the transmission of global light (diffuse plus direct light) has been evaluated. Converting direct light to diffuse light can increase plant productivity in greenhouses (see also Chapter 4), because diffuse light is non-directional and can reach deeper into the canopy and thus increase photosynthesis of the whole plant stand (Hemming *et al*., 2008, 2016; Li *et al*., 2014). For high-wire cucumber plants, Hemming *et al*. (2008) demonstrated that yield was increased by 8–10% by applying diffusing covering material made of ETFE in comparison with using glass. The effectiveness of diffusing covering materials can vary regionally and is minimum in areas with predominantly overcast skies, where the natural global light contains a high percentage of diffuse light. Diffusing covering materials are more effective

Fig. 9.3. Plastic greenhouse with roof opening for natural ventilation. (Photograph courtesy of Chieri Kubota.)

in promoting plant growth in regions with high levels of direct sunlight, such as arid and semi-arid regions. Other new technologies recently developed and evaluated are glass coatings that reduce reflection and hence increase the transmission of light (i.e. anti-reflection coatings), and low-emission coatings that reduce emissivity of long-wave radiation for energy saving (Hemming *et al.*, 2011). Most recent innovations of covering technology include a combination of glass and ETFE film to enhance thermal insulation while retaining high transmission.

Growing systems

In most commercial large-scale greenhouse complexes worldwide, greenhouse tomatoes are grown in 'soilless' cultures with drip irrigation. A large proportion of tomato cultivation uses growing media (also known as substrates or aggregates). The term 'hydroponic' can refer to substrate-based soilless culture or to water cultures such as nutrient film technique (NFT). NFT is used far less than substrate-based soilless culture for commercial tomato production.

For small-scale and for organic production, greenhouse tomatoes are produced in the soil (see Chapter 11). Growing in soil requires measures to reduce soilborne diseases and pests. Possible measures are crop rotation, grafting, pre-planting soil sterilization by steaming or fumigation, or companion planting and the use of biologicals that boost plant defence mechanisms.

Among various substrates available in crop production, stone wool has been the most common substrate for soilless culture. However, due to the higher price and (in some countries) issues associated with the disposal of stone wool, more growers are using coconut coir as an alternative. Both substrates are commonly used as substrate slabs, typically 15 cm wide, 7.5 cm high and 90 or 120 cm long, wrapped in white thin plastic film. In a typical setting, slabs are placed in gutters (or gullies) to collect the efflux solution typically called 'drain'. The drain is either discharged or recycled, typically called 'recirculated'. Recirculation involves pumping the drain back into the nutrient tank, ideally after sterilization. Then the nutrient solution must be adjusted to account for nutrients that were taken out by the plants. Many large-scale commercial growers use a raised, hanging gutter system so that harvestable fruits are at a convenient height for workers. Hanging gutter systems are also incorporated in the closed and semi-closed greenhouse system described elsewhere in this chapter.

Other substrates used in soilless culture of tomatoes include perlite, peat and to some extent pumice. Properties of various substrates are summarized in Schwarz *et al.* (2014). Locally available materials such as sand and volcanic rocks (e.g. 'tezontle' in Mexico) are used also. These loose substrates can be placed in buckets, pots or bags as shown in Fig. 9.4. Regarding differences between substrates in their effect on tomato plant growth and yield, a substrate

Fig. 9.4. Tomato plants grown in pots containing perlite in a greenhouse. (Photograph courtesy of Wim van Ieperen.)

trial at the University of Arizona (Jensen, 2002) showed that there were no significant differences in yield of greenhouse tomatoes between five different media (coconut coir, perlite, peat-lite, coir/perlite and stone wool) when irrigation was managed properly. A rule of thumb is to irrigate with a volume of 100 ml/plant (or 200 ml/plant with double shoots) per irrigation event, while timing of the irrigation events is controlled so that the efflux solution volume is about 30% of influx solution volume. Frequent irrigation with a small volume each time rather than infrequent irrigation with a large volume is advantageous in terms of maintaining the root zone conditions in an optimal range. (See also Chapter 6.)

Several types of substrate can be reused for a new planting, though some substrates suffer loss of structure and quality over time. Substrates are usually sterilized before a new crop is planted. Some growers reuse the substrates without sterilization, which was supported by studies conducted in The Netherlands (Sonneveld and Welles, 1984) as well as in the USA (Hochmuth *et al.*, 1991). However, it seems that single usage of substrate is more common among large greenhouse growers, presumably to minimize the risk of diseases.

CROPPING SCHEDULES AND MANAGEMENT UNIQUE TO GREENHOUSE

The tomato plant is a short-lived perennial and can be maintained for a period of a year or more in favourable environments. Most production schemes are based on an annual cycle, with allowance of about a month between crops for clean-up and pest and disease control. This means that typically, tomato plants are in the greenhouse for 11 months and in production for about 9 months per year. Year-round production can be achieved with interplanting (or intercropping, Fig. 9.5). The time chosen to be out of production is usually based on unfavourable prices or environmental conditions. In hot and humid climates, many growers seed in late summer or autumn and carry the crop until early summer of the next year. In this way they avoid summertime cooling, poor fruit set and quality, pest build-up and competition from field tomatoes. In many cases, large commercial greenhouses produce tomatoes almost or, in some cases, all year-round, in order to increase return on investments and to avoid buyers switching to alternative sources. In some cases interplanting is performed, which is essentially planting young seedlings among the finishing old plants, to ensure minimal interruption in supply.

December planting, with production from March until November, is common practice at northern latitudes in the northern hemisphere. In the southern hemisphere the same is done but with a 6 months offset. Supplemental lighting is used to support plant growth and enable tomato production in periods with low natural light. Some growers, especially in the south-eastern USA, and some in The Netherlands (approximately 100 ha), grow separate autumn and spring crops, leaving short production breaks both mid-winter and mid-summer.

There is considerable interest in organic greenhouse tomato production. With organic production, crop production must be certified by an authorized organization that regulates the types of material that can be used. In Europe, hydroponics and soilless cultural practices are not allowed under organic schemes. In the USA, the National Organic Program does not preclude soilless or hydroponic production, though synthetic substrates such as stone wool are not allowed. All guidelines relative to organic fertilizers and pesticides must be followed and the choice of cost-effective soluble organic fertilizers is still

Fig. 9.5. Willcox Arizona greenhouse showing new intercropped tomato plants with older plants on either side. Note severe leaf pruning on older plants in order to provide additional light to the new crop. The pipe-rail heating is visible in the middle. (Photograph courtesy of Chieri Kubota.)

limited. Use of high-analysis chemical fertilizers and most chemical pesticides is prohibited. Biological factors, such as soil condition and fertility and the use of beneficial insects, are the main factors used to assure a vital, healthy crop and good fruit quality. Total area of organic production in The Netherlands is 2% and that for tomato in the USA is 3% (USDA, 2013). (See Chapter 11 for organic cultivation.)

Transplant production

Transplant quality is defined by absence of pests and diseases, and the capacity to quickly grow and become established. Transplant production requires 3–6 weeks (or as long as 8 weeks for grafted two-headed plants), depending on temperature and light conditions and stage of transplanting: either as plugs or as mature plants with visible first truss. A good transplant is one that is as wide as it is tall and is not yet flowering. Supplemental light (50–100 μmol/m^2/s photosynthetic photon flux (PPF)) in situations with low natural light and carbon dioxide (CO_2) enrichment (800–1000 μmol/mol) during transplant production increase plant growth rates and plant quality, and have been a practice in specialized nurseries.

Seedlings for stone wool systems are generally started in a sterile inert medium, such as stone wool plugs at a density of around 600–1000 plants/m^2. Use of coated seeds is common, which is especially advantageous for machine seeding and for uniform germination. Germination is often conducted in a separate compartment with controlled temperature and humidity suitable for germination. In some cases, especially for grafted seedlings, a sorting by size may be conducted at the cotyledon stage to create a more uniform seedling stand.

Grafting, which involves combining a rootstock and a scion, was originally done for attaining resistance to soilborne diseases and pests. For soil-based production systems, rootstocks are selected mainly for resistance to diseases such as Fusarium, Verticillium wilt and bacterial wilt, as well as root-knot nematodes (*Meloidogyne* sp.). Increasingly, greenhouse growers use grafted plants even in soilless culture to achieve greater yields resulting from the stronger root system (Kubota *et al.*, 2008). Grafting is normally conducted when the second true leaf emerges and is followed by 5–7 days of healing. Grafted plants are sometimes pinched after healing to induce two lateral shoots per plant. The strong rootstock can support two heads and thus reduces the number of transplants needed for the production area without reducing the overall yields. Grafted plant performance depends on the combination of rootstock and scion. Rootstocks used for grafting tomato are either interspecific hybrids or intraspecific hybrids. The advantage of the former is generally greater vigour but they have the disadvantage of non-uniform germination. The standard method for grafting tomatoes is tube grafting, where scion and rootstock seedlings are cut at a sharp angle of 30–45 degrees and held together using an elastic plastic or silicon tube (Fig. 9.6). Machine grafting for tomato is available but hand grafting is still common in most nurseries.

During the transplanting stage growing in large cubes, plant density should be 20–22 plants/m^2. A rule of thumb is that transplant leaves should not touch. Sub-irrigation is always recommended, to minimize diseases on wet foliage. Transplanting into the final production greenhouse is performed just before the first flower opens (anthesis stage). This minimizes the time between transplanting and first harvest in the greenhouse and also ensures that the first flower grows out to a harvestable tomato fruit. Increasingly, growers purchase seedlings grown in commercial nurseries and in some cases the seedlings travel long distance over multiple days or are imported from another country. Long-distance transportation of seedlings requires appropriate temperature control during transportation, since unfavourable conditions could cause flower abortion or abnormal development (Kubota and Kroggel, 2006). Virus and viroid introduction associated with transporting plants is a serious issue and some countries have already banned importation of live tomato seedlings to minimize the risk of outbreak or

Fig. 9.6. Grafting tomato seedlings. Tube grafting is a common method for tomato that allows 100–400 grafts per hour per person. (Photograph courtesy of Chieri Kubota.)

introduction of new disease. In other cases, new certification programmes (e.g. Good Seed and Plant Practices (GSPP)) have been introduced to prevent the spread of diseases through vertically integrated hygiene practices. In Japan, use of isolated growth rooms with artificial lighting (300 μmol/m^2/s PPF for 16–18 h photoperiod) and CO_2 enrichment (about 1000 μmol/mol) is becoming standard for high-quality tomato transplant production partly because of increasing demand for transplants free from diseases and pests (Kozai, 2015) (Fig. 9.7).

Carefully designed watering schedules and nutrient programmes (especially with regard to nitrogen and potassium concentration) are necessary to control plant growth, as plants tend to be overly vegetative after transplanting. An example nutrient management programme is shown in Table 9.1.

Planting density/stem density

In a typical 3.2 m Venlo greenhouse span, there are four rows of tomato plants and two pathways. Generally, tomatoes are set out in double rows, normally around 0.5 m apart with 1.1 m access pathways between the double rows. Plant populations are adjusted at the start of the crop by altering the in-row spacing and later in the season by allowing extra heads (side shoots) to develop. Atherton and Rudich (1986) gave detailed information of the relation between plant density and yield per plant and consequences of spacing for mean fruit

Fig. 9.7. Indoor transplant production system using artificial lighting. More commercial nurseries in Japan are introducing this type of high-tech propagation facility to ensure hygiene and quality control. (Photograph courtesy of Chieri Kubota.)

weight and harvest costs. In general, under north European conditions a plant density of 2.5 plants/m^2 has been found to give the best financial margin. In more southerly latitudes, a higher plant density may be used because of higher light intensity. Similarly, the number of plant stems or side shoots allowed to develop should be based on light intensity. This ensures not only a high yield but also optimal quality, including taste. Uniformity of fruit size is also improved when the number of side shoots is matched to incident light (Ho, 2004). For example, Canada has about 2 times higher radiation in winter and 40% higher radiation in spring compared with The Netherlands. Anecdotally, optimal plant spacing in mid-December is 50 cm in Canada rather than 55 cm in The Netherlands. Extra stems are left starting in week 5 in Canada versus in week 9 in The Netherlands. For spring, recommended spacing in Canada is 40 cm in the row compared with 44 cm in The Netherlands. The use of strong rootstock allows the making of double-headed grafted seedlings. This allows growers to halve the number of plants while maintaining the shoot density (shoots/m^2). Increasingly, growers adopt the so-called 'V-system' with the two shoots per plant trained in opposite directions along the gutter to form double rows of plants.

Table 9.1. Final delivered nutrient solution concentration (ppm or mg/l) and electrical conductivity (EC) recommendations for tomatoes grown in Florida in stone wool, perlite or nutrient film technique (Hochmuth and Hochmuth, 1995). Numbers in bold denote changes from previous stage.

	Stage of growth				
Nutrient	Transplant to first truss	First truss to second truss	Second truss to third truss	Third truss to fifth truss	Fifth truss to termination
N	70	**80**	**100**	**120**	**150**
P	50	50	50	50	50
K	120	120	**150**	150	**200**
Ca[a]	150	150	150	150	150
Mg	40	40	40	**50**	50
S[a]	50	50	50	**60**	60
Fe	2.8	2.8	2.8	2.8	2.8
Cu	0.2	0.2	0.2	0.2	0.2
Mn	0.8	0.8	0.8	0.8	0.8
Zn	0.3	0.3	0.3	0.3	0.3
B	0.7	0.7	0.7	0.7	0.7
Mo	0.05	0.05	0.05	0.05	0.05
EC (dS/m)	0.7	**0.9**	**1.3**	**1.5**	**1.8**

[a]Calcium (Ca) and sulfur (S) concentrations may need to consider Ca and Mg concentrations in well water and amount of sulfuric acid used for acidification.

Training system

In the 1970s and early 1980s, plants were trained up to the wire, then allowed to drape down the other side (the up-and-down system). The negative effect on yield of plant heads hanging down in the shade of the canopy was the reason to move to the 'plant lowering' method, which later developed into the high-wire system. The most widespread training system in The Netherlands and North America at the present time is the high-wire system, which allows one crop to be carried over several seasons. In this system, the growing tip remains at the top of the canopy, but the stem is lowered and trails along the base of the plants (Fig. 9.8). This system combines the yield-improving advantages of maximum light interception by young leaves with increased labour efficiency resulting from easier removal of leaves and fruit. However, it requires a high enough greenhouse structure to accommodate the high horizontal wires used in training the plants. The greenhouse needs to be extra high if any screening materials are to be utilized for shading or energy conservation.

The high-wire system requires early training of the main stem. As soon after transplanting as possible, plant stems should be secured to plastic twine hung from horizontal wires (Fig. 9.9). The end of the twine is attached to the base of the stem with a non-slip loop. The twine is then wound around the

Fig. 9.8. Greenhouse worker standing on an electrical lift to sucker and train plants. The lift moves down the row on tracks, which are also the hot-water pipes (pipe-rail system). (Photograph courtesy of Arie de Gelder.)

stem in two or three spirals for each truss. The length of the supporting twine should allow an extra 10–15 m to unwind. Usually this extra twine is held in a winding hook placed near the wire. As an alternative to winding string around the stem, the stem can be clipped to the string every 20–30 cm. Clips can be sterilized and reused, but string should be discarded after each crop. If the stems are to be 'leaned and lowered' (see below), it is useful to wind twine around the lower stem, as the twine provides a better support than clipping.

The objective of 'leaning and lowering' is to keep the head of the plant upright for light interception and still have the harvestable fruits at a convenient

Fig. 9.9. Stone wool cubes containing two seedlings each are placed on stone wool slabs in a Dutch greenhouse. Vine twine is wound around the stem. The tube on the right in this high-wire system is a grow pipe with hot water for promoting fruit development. Plastic tubing for CO_2 enrichment is placed along with the grow pipe between the rows of plants. (Photograph courtesy of Arie de Gelder.)

height for workers even when crops are in the greenhouse for long periods. When plant tops near (or approach) the overhead wire, the strings are unwound from the twine hook hangers and string and plant are both moved sideways down the horizontal wire. This 'lowering' process is a delicate operation in order to avoid breaking the stems. At the same time the hooks are moved over, say, about 30 cm so that the plants start to 'lean' (all in the same direction, of course). The fourth flower truss is a good developmental stage for this operation, as the stem is relatively vigorous and should resist breakage. Leaning and lowering should be performed every 7–10 days. In some greenhouses, especially those using upright bags and hanging gutters, the stems rest on special holders designed to give additional support. At the end of the double row, the stems are wound around a corner and back down the next row. Upright rods or wire supports are placed or posts are used at the corners to turn the stems and this is another spot where stems frequently break. Black plastic drain tube or other types of 'bumpers' are sometimes placed on the rods to protect the stem as it is turned. Stems breaks can sometimes be successfully mended with duct tape.

Interplanting (Fig. 9.5) is a variation of the high-wire system that minimizes down time between crops and allows high annual yield and year-round production. Each pair of adjacent rows of old plants is aligned to one side of the trough, in order to create space on the other side of the trough for the new crop. Young plants are then transplanted into the open space, next to the row of old plants that are in their final stage. In the remaining old plants, the shoot tips are pinched to terminate the shoot growth and leaves are removed progressively during the transition time, except for the top four leaves. The height of the leafy part of the old canopy is adjusted to increase light penetration to the new plants. Disadvantages of intercropping are that greenhouse clean-up is more difficult. Also, diseases may be carried over from old plants to new transplants.

Pollination management

Before the early 1990s, each flower cluster had to be vibrated with an electric pollinator at least 3 times per week to release pollen and thus promote pollination. Poor pollination results in flower abortion and/or small, puffy or misshapen fruits. It is particularly important to get good fruit set on the first three trusses to establish an early pattern of generative growth. Pollination should take place around midday when humidity conditions are most favourable (50–70%). If humidity is too high in winter, temperatures can be raised by 2°C at midday to reduce humidity. Too-high mean daily temperatures will reduce pollen development and release (Sato *et al*., 2000), and too-low temperatures at night (below 16°C) have the same effect (Portree, 1996).

Nowadays bumblebees are used for pollination. They are shipped as hives containing varying numbers of bees, depending on price. Generally, one worker bee can service 40–75 m^2 of greenhouse area (Portree, 1996) and therefore about 5–7.5 hives/ha are required. As well as saving labour, the use of bumblebees increases yield and quality compared with manual vibration (Portree, 1996). Hives are placed on stands 1.5 m above the ground along the centre aisle (Fig. 9.10) and protected from ants with sticky bands or water troughs. Hives should be shaded by foliage or covers, and marked distinctively above the hive and around the entrance so that bees can return to the correct hive (Portree, 1996). Bumblebees are docile, unless the hive is disturbed or an individual is squeezed, but it is still a good idea to maintain first-aid supplies on site. No broad-spectrum insecticides or those with residual action should be used once a hive is in place. All pesticides should be checked for effects on bumblebees, and, if compatible, application should be done at night with the hive closed and covered. Some pesticides may be used when the hives remain closed during 3 days after treatment. The health of the hive can be monitored by observing activity and looking for brown bruise marks on the anther cone as evidence of flower visitation. At least 75% of withered flowers should have evidence of bee visits. Most hives will need to be replaced within 2 months or less.

Fig. 9.10. Bee hives are placed on stands in a greenhouse. A slide on the top front of the box can be used to open or close access to the hive. (Photograph courtesy of Koppert Biological Control.)

Other crop managements

Side shoots will develop from every axil. These side shoots should be removed weekly, leaving only one main stem as a growing point. An exception is that one side shoot may be maintained when light intensity is high compared with fruit load. Often this is done on only one in four plants, and some weeks later again on one in four plants as a method to increase stem density gradually. In The Netherlands, management of side shoots is an important tool for optimizing fruit load of the crop and hence yield (De Koning, 1994). Leutscher *et al.* (1996) presented an economic evaluation of the optimal number of additional side shoots, based on a modelling approach.

In the UK (Ho, 2004), uniform fruit size is maintained by fruit pruning and thereby controlling the number of fruits left on the truss, as well as by letting an additional side shoot develop as light increases. To address this problem associated with the seasonal change in fruit size, Cockshull and Ho (1995) demonstrated that letting additional side shoots develop could help to minimize the undesirable extra-large fruit produced in summer. With the high-wire system, side shooting and other operations may be done standing on an electrical lift (Fig. 9.8).

When plants are trained, lower leaves are removed (de-leafing or leaf pruning). Typically, all leaves are removed below the bottom fruit truss. The number of leaves to remain on the stem is determined by the grower and is often between 12 and 18 fully expanded compound leaves. The optimal number of leaves can be approached theoretically based on light interception and microclimate inside the canopy (Sarlikioti *et al*., 2011). When a new crop is interplanted next to the old one, de-leafing may be more severe to improve the light conditions for the young seedlings (Fig. 9.5).

An important factor to be considered in de-leafing is the effect on diseases, pests and beneficials. Removing lower leaves from the greenhouse and then destroying them will also remove immature whiteflies that are developing on the lower leaves. However, if beneficials have been introduced, they will have parasitized the immatures. Destroying these leaves will also prevent the beneficials from emerging. Sometimes, if parasitized pupae are known to be present, leaves are removed but left piled in the greenhouse for a few days to allow emergence. In this case, de-leafing and leaf removal represent a trade-off between emergence of whiteflies, emergence of beneficials and spread of disease from the discarded leaves.

The purpose of fruit pruning is to increase fruit size and fruit quality, and to create the right balance between fruit load and leaf area. Fruit pruning can also be used to maintain uniform fruit size. Misshapen fruits and undersized fruits at the end of a cluster are always removed, as these will generally not grow to marketable size and are thought to reduce the size of other fruits on the truss. In some cases, all trusses are pruned to leave only the four fruits nearest to the plant (proximal fruits). Whether or not trusses are pruned depends on a number of factors such as the expected fruit size for that cultivar, how many fruits normally form on the truss, growing conditions and the size demanded by the market. A typical practice for cluster or TOV cultivars is to have four to six fruits per truss. With some cultivars such as 'Campari', producing fruit in size between cherry type and typical TOV type, up to eight to ten fruits can be allowed to develop on each truss.

When some greenhouse tomato cultivars are grown under relatively low light conditions, the peduncles of the inflorescences (the truss stems) are too weak to support the weight of the fruits they bear and are prone to bending (Horridge and Cockshull, 1998) or 'kinking'. Another reason sometimes given for kinking is too high a temperature during the vegetative phase, which causes the truss to become almost vertical ('stick trusses'). As fruits develop on these trusses, they may become kinked (Fig. 9.11). Truss hooks prevent the peduncle from bending sharply under the weight of the fruit and prevent heavy trusses from pulling off the stem. Another type of truss support is 'truss braces', which are applied to the truss before fruit development, when the stem is still flexible.

Five to eight weeks before the anticipated crop termination date, the growing point is removed. A week later, all remaining flowers are removed. An individual fruit requires 6–9 weeks from anthesis to harvest, so flowers or small fruits present after topping will not have enough time to develop to maturity. Some growers leave some shoots at the top (or do not top at all) to provide shade to top fruits and also increase the transpiration to reduce the

Fig. 9.11. Kinking of the peduncle of cluster tomato (TOV). (Photograph courtesy of Arie de Gelder.)

risks of fruits cracking and russeting. (For further discussion of factors contributing to cracking and russeting, see Chapter 5.)

Nutrient and irrigation management

In systems with drip irrigation, nutrients are usually injected into the irrigation water (fertigation) from concentrated solutions in stock tanks (Sonneveld and Voogt, 2009). The fertilizers must be separated into at least two tanks to avoid precipitation of calcium phosphate and calcium sulfate. Often a third tank is present with acids for pH correction, and some large commercial greenhouses are fitted with six or more separate tanks to give better control of the nutrient solution. Some greenhouses have duplicate sets of stock tanks, so that irrigation will not be interrupted while solutions are remade. Greenhouses where interplanting is applied need to be equipped with two irrigation lines for each row to cater for the distinct difference in nutrient requirement between young seedlings and old plants.

In an open (non-recirculating) system, approximately 30% excess irrigation is applied. The main problem with open systems is that in areas of intensive production there may be significant discharge of nutrients into the environment. There has been increasing interest over the past decade in nutrient-recirculation systems with provision for disinfection of the water and/or replenishment of nutrients before reuse. Recirculation can decrease fertilizer costs by 30–40% and water usage by 50–60% (Portree, 1996). Provisions for reuse or at least recapture of greenhouse runoff should be designed into new greenhouses, as already required in many countries, as recapture systems are not easy to retrofit. In modern systems, the solution is monitored for salts and water, and specific nutrients may be replenished.

The main problem with recirculation systems is the risk of pathogens spreading through the system. To prevent this, the return water (or drain) is disinfected before it is pumped back into the nutrient tank. Ikeda *et al.* (2001) listed the following types of water disinfection: (i) physical methods including heat treatment, UV radiation and filtration; (ii) chemical methods including ozone, chlorination, iodination, hydrogen peroxide and metal ions; and (iii) biological methods including rhizosphere bacteria and antagonistic fungi, mycoparasistic fungi and biosurfactants. The most widely used water disinfection method is a combination of filtration and UV radiation. When coconut coir is used as substrate, UV radiation and heat treatment may be the preferred method, because coir particles tend to reduce the UV light transmission.

Large amounts of high-quality water are needed for plant transpiration, which serves both to cool the leaves and to trigger transport of nutrients from roots to leaves and fruits. Water consumption of 0.9 m^3/m^2/year is estimated for greenhouses in The Netherlands (Anonymous, 1995) and 0.8 m^3/m^2/year for British Columbia, Canada (Portree, 1996). Before building a greenhouse, it is important to ensure adequate water availability and quality. It is also important to consider whether evaporative cooling will be used (pad-and-fan cooling, fogging, misting, roof sprinklers, etc.) and to estimate the amount of water needed for that in addition to irrigation. In some regions, evaporative cooling water use largely exceeds the irrigation water use in the greenhouse. In a study reported by Sabeh *et al.* (2011), tomato production in a semi-arid greenhouse (located in Arizona) consumed 0.40–0.78 m^3/m^2 over a 7-month period (early March to early October) while additional water consumption for pad-and-fan cooling was 1.45 m^3/m^2 during this period. Water used for irrigation in soilless culture should have an electrical conductivity of below 0.5 dS/m, pH from 5.4 to 6.3

and alkalinity below 2 meq/l. Water treatment to lower alkalinity and adjust pH is usually possible. Lowering electrical conductivity (EC) by reverse osmosis is often not economically feasible.

The optimal frequency of irrigation varies with substrate rooting volume and water-holding capacity. In stone wool slabs, rooting volume is very restricted and slabs may be watered up to 6 times per hour in peak radiation, and up to 30 times per day under summer conditions. Daily timing of irrigation cycles also varies with water demand. In stone wool systems, fertigation should begin a few hours after sunrise and end 1–2 h before sunset to decrease diseases as well as summertime russetting and fruit cracking. Night watering may be needed, however, during the winter when night-time heating decreases relative humidity and in summer if conditions are hot and dry (OMAFRA, 2001). In The Netherlands an irrigation model has been developed for recirculating systems, based on leaf area, air temperature and season (De Graaf, 1988). Some growers use accurate weighing units (lysimeters), so that moisture content of the slab as well as plant transpiration can be monitored every hour to avoid stress to plants. The water content of the slab can also be directly measured by sensors (Balendonck *et al.*, 2005).

Steering plant growth (reproductive versus vegetative)

The balance between vegetative and reproductive (or generative) growth in the tomato crop needs to be controlled. A well balanced plant (OMAFRA, 2001) has a stem approximately 1 cm thick at about 15 cm below the shoot tip, dark green leaves and closely spaced, large flower clusters that set well. Thicker stems indicate excessive vegetative growth and are usually associated with poor fruit set and low productivity. Thinner stems usually indicate carbohydrate starvation, slow growth and ultimately low overall productivity. There are a number of ways to control this balance, including the environmental controls summarized in Table 9.2. EC, water supply and the ratio of nitrogen to potassium in the feed also influence plant balance. EC influences plant growth through its effect on plant/water relations (Heuvelink *et al.*, 2003). High salinity in the root environment, infrequent irrigation and low volume of irrigation water reduce water availability to plant roots, therefore decreasing water uptake and overall growth rate, steering the plant towards generative growth. High temperature and low relative humidity also have a generative effect because they make water less available, resulting in ‘hard’ plants and slow growth. Lowering nitrogen or maintaining a high potassium-to-nitrogen ratio in the fertilizer feed is another technique to reduce the rate of growth and steer plants towards generative development (OMAFRA, 2001).

Temperatures can also be used to steer the plant towards a particular growth pattern (Table 9.2). In the long cropping cycles typical of greenhouse production, tomatoes tend to cycle between being overly vegetative at the

Table 9.2. Regulating plant growth by adjusting environment and nutrition. Adapted from Reading the Plant (Portree, 1996).

Plant part	Observation	Recommendation
Plant head	Thick head	Too vegetative. Increase day temperature 1–2°C, especially during peak light period; increase the spread between day/night temperature settings by 1–8°C. The bigger the difference, the stronger the 'generative' signal to the plant
	Thin head	Too generative. Bring day/night temperatures closer together. Reduce the 24 h average in low radiation situation, e.g. early spring, late autumn. Target a 10–12 mm diameter head measured approximately 15 cm from growing tip or at the first fully expanded leaf before the flowering truss
	Head is 'tight' – leaves do not unfold until late in the day	Vegetative imbalance. Increase the 24 h average by increasing the temperature between midnight and sunrise. Curl should be out between 11am and 4pm. Target a slightly higher temperature in afternoon (+1 to 2°C)
	Heads are purple	Vegetative imbalance. Slight purpling is acceptable. Increase night temperature
	Grey heads	High tissue temperatures in combination with high CO_2 levels. Can be in early spring when venting is limited. Reduce CO_2 levels and shut off earlier in the day
	Chlorosis in head	Chlorosis in the head can occur if the media water/air ratio is not in balance. If the slab is dry, increase the EC. If the slab is wet, increase only the micronutrients 10%. Maintain a temperature differential between head temperatures and root temperatures of more than 5°C
Flower/truss	Flower colour pale yellow, especially in morning	Colour should be bright 'egg-yolk' yellow. If the climate is humid, low vapour pressure deficit (< 2 g/m^3) will often occur in the morning. Recommend increase VPD, especially early in the morning ,from 3 to 7 g/m^3 VPD, by opening vents for short period of time to let the moisture out, followed by heating. Flowering rate should be 0.8–1.0 truss/week
	Long, straight flower trusses (kink trusses)	Aggravated by low light and high temperature. Decrease 24 h temperature by decreasing day temperature. Promote an active climate 3 g/m^3 VPD. Avoid increasing plant density too early in the season when light levels are low (< 6 MJ/m^2/day)

Table 9.2. Continued.

Plant part	Observation	Recommendation
Flowers	'Sticky' flowers in which the sepal does not roll back	Caused by too humid a climate, especially in morning. Activate the plant in the morning with a minimum pipe and crack the vent. If left unchecked these flowers result in poor-quality fruit. Higher day temperature = higher VPD = less sticky
	Flowers too close to the head, < 10 cm below growing tip	Plant is too generative. Go down with day, up with night, i.e. bring day/night temperatures closer together. Late April/early May this is desirable in order to get enough fruit on the plant for summer fruit loads
Leaves	Short leaves in the head, e.g. < 35 cm in length	This condition occurs in late spring. The plant is in a vegetative imbalance. Fruit load is low (< 85 fruit/m^2). Increase differential between day and night

beginning of cropping (too much growth and too little fruiting) and later being overly generative (too little growth and excessive fruit loads). Where uniform production is desired, it is helpful to be able to moderate these swings in plant balance. Temperature is considered to be the most important factor affecting plant development and therefore an important tool to control flowering and fruit ripening. Quantitative data on the effects of temperature on flowering, fruit set, fruit growth and yield were used to develop yield prediction models for The Netherlands (De Koning, 1994).

GREENHOUSE ENVIRONMENTAL CONTROL AND SYSTEMS

Because of their increasing sophistication, ease of use and affordability, computers are used to optimize temperature, relative humidity, CO_2 concentration and light intensity. They are also very useful in providing a history of the crop environment over time and alerting operators to malfunctions. Computers can control many mechanical devices within a greenhouse, such as vents, heaters, fans, evaporative pads, fogging, roof sprinklers, CO_2 enrichment systems, irrigation valves, fertilizer injectors, shade-cloths and energy-saving curtains. Control is based on pre-set criteria, such as irradiance, outside temperature and other weather conditions as well as required inside conditions (temperature, humidity and CO_2 levels). More importantly, computers can integrate the readings of different sensors and process all the data and then steer the devices to achieve a desired result, such as a particular temperature or humidity regime. Computerized greenhouses would assist growers in selecting conditions that would help to balance plant growth using environmental controls. (Computer control of irrigation and fertilization regimes based on environmental conditions is discussed in Chapter 6.)

Heating and cooling

Optimal day and night temperatures for different crop developmental stages are shown in Table 9.3. Within the so-called linear range, there is a linear relationship between temperature and growth and development. Therefore, within the linear range, 24 h mean temperature has a significant influence on plant development rate. Therefore, theoretically, if daytime temperatures are warm, night-time temperatures can be allowed to fall to the level that does not cause any adverse physiological effect, to conserve energy.

A particular strategy was developed in The Netherlands, called ‘temperature integration’, which aims at maintaining a particular average daily temperature level rather than maintaining specific day and night temperatures (De Koning, 1990). The background is that this strategy can be much more energy efficient, because it allows maintaining a high temperature when heating is cheap, for instance when a screen is closed or when the sun is out. Energy savings of 10–15% have been realized by using temperature integration compared with using regimes of high day and low night temperatures regardless of the 24 h mean. Work summarized in Papadopoulos *et al.* (1997) suggested that plants tolerate some variation around the optimal temperature over periods ranging from 24 h to several days. For example, the temperature in a tomato crop can safely deviate 3°C below standard for 6 days, provided that the following 6 days are 3°C above standard, as long as the average temperature over the 12-day period stays the same. Even a deviation as high as 6°C can be tolerated if the 7-day temperature average is unaffected (Portree, 1996).

Energy conservation measures are widely used in northern latitude greenhouses to reduce heating costs. Pulling thermal curtains of porous polyester or an aluminum foil fabric over the plants at night reduces the radiative heat

Table 9.3. Growing recommendations for tomato cropping (adapted from OMAFRA, 2001).

	Germination	Plant raising	Transplanting	Fruit production
Temperature (°C)				
Day	25	19–21	24	20–22
Night	25	19–21	24	17–19
EC (dS/m)	0.0–0.1	2.5–3.0	2.5–3.0	2.7–4.0
pH	5.8	5.8	5.8	5.8
Volume of feed (l/day)	–	0.2–0.3	0.2–0.3	0.5–2.5

transfer and decreases the overall heat loss by as much as 70% while closed, or 20% over an entire year (Dieleman and Kempkes, 2006). Dual-purpose lightweight retractable curtains are sometimes used for energy conservation at night and for shade in daytime. Combining screens with forced ventilation and temperature integration is a next practice in energy conservation (De Gelder *et al.*, 2012b; De Gelder and Dieleman, 2012).

Typical greenhouse heating for small-scale operations is based on overhead air heating systems. Large greenhouse operations usually use a natural gas-fired central boiler with water and sometimes with steam for heat transfer. The heating pipes are often located a short distance (e.g. 10 cm) above the ground, between crop rows. They serve as a radiative heating system as well as a track for trolleys for internal transport and for working in the crop. The pipes are known as pipe-rail heating system (Figs 9.5 and 9.8). Additional heating pipes are sometimes used inside the canopy (Fig. 9.9). These pipes are called 'grow pipes' and are attached with flexible hoses at both ends so that they can be moved vertically to deliver heat to a selected zone of canopy to control plant growth and/or fruit ripening.

Extremely high temperatures are considered to limit greenhouse tomato production in summer in lower latitudes. This is especially true where evaporative cooling is less effective because of high humidity (high wet-bulb temperature). There are three common methods of greenhouse cooling: natural ventilation; mechanical cooling; and evaporative cooling. In The Netherlands, a new system called 'closed greenhouse' has been developed (De Gelder *et al.*, 2012a), which utilizes heat storage, heat pumps, heat exchangers and cooling plates to control greenhouse temperatures, as described later in this chapter.

Natural ventilation is popular in areas with relatively few days with high ambient temperatures. For natural ventilation, some part of the greenhouse, usually at the peak or ridge, is opened and air movement is created by wind pressures or by gradients in air temperatures. Single-span greenhouses are often ventilated via sidewalls that can be opened. Air exchanges driven by temperature gradients in greenhouses are generally enhanced with increasing difference in height between roof vent and side vent (so-called chimney effect). Multi-span greenhouses are vented through the roof vents.

Any influence of side vents on the natural ventilation rate would be small, especially for multi-span greenhouses. With any type of natural ventilation system, however, insect netting in the ventilation opening to prevent pest entry and escape of pollinators or beneficials reduces ventilation capacity by approximately 20%. Computer simulations based on computational fluid dynamics (CFD) effectively demonstrate how structural design, wind direction relative to vent opening, and/or insect netting affect the air exchange and temperature distribution inside the greenhouse (e.g. Boulard, 2011).

In arid and semi-arid climates, fan-and-pad evaporative cooling can be used effectively. While natural ventilation is used in some warm-climate greenhouses, hot conditions outside and lack of wind reduce its effectiveness.

In south-eastern USA, pest pressures, high humidity and high temperatures force most growers to invest in active mechanical cooling, usually with a combination of fans and pads. With mechanical cooling, low-pressure propeller-blade fans are placed opposite the air intake, which is covered by cellulose evaporative cooling pads. Louvres are placed outside the cooling pads and are closed when the greenhouse is not venting. Ventilation fans are normally sized to allow one air exchange per minute, though researchers in North Carolina and Israel have documented increased cooling at higher rates, especially when combined with evaporative cooling (Willits, 2000), while in arid climates increasing ventilation may not necessarily increase the cooling efficiency as the intake air is often higher in temperature and drier. Cooling system design needs better engineering approaches and there has been much effort in this area, along with the effort to make greenhouse tomato production year-round in mild climates and in arid/semi-arid regions, where traditionally greenhouses were not used in summer due to the unfavourable thermal conditions for plant growth. High-pressure fogging systems have been introduced as greenhouse cooling systems to a limited extent. The control logics and fogging nozzle distribution need to be further optimized to reduce large temperature fluctuations as well as wetting the plants, especially when the fogging system is combined with natural ventilation.

Presumably, the temperature-averaging method described above as a way to reduce heating costs can also be applied to warm conditions where cooling costs are a concern. This question has not been addressed directly, but Peet *et al.* (1997) reported that over the range 25–29°C, the actual day and night temperatures and the day/night differential were less important than the daily average in accounting for declines in fruit set, yield, fruit number and seediness, all of which are typical results of thermal stress. This suggests that in areas where summertime night temperatures are low, day temperatures can be allowed to exceed the normal maximal levels. In areas where temperatures are excessive, lowering night-time temperatures by a heat-pump (air conditioning) may be useful. Although the applicability of temperature averaging to above-optimal conditions for plants setting fruit has not been tested directly, data collected in south-eastern USA in an experiment with night-time air conditioning (Willits and Peet, 1989) suggested the potential feasibility of such an approach. With the closed glasshouse system in The Netherlands, temperatures can be kept below 26°C (De Gelder *et al.*, 2005).

Shade screen and supplemental lighting

Daily averaged light transmission of greenhouse is generally 70–80%, depending on the covering and the greenhouse structure and orientation. Therefore in almost all regions, CO_2 and irradiance (light intensity) are the most limiting factors for maximizing yield. Supplemental lighting is a common approach to promote growth in areas with short days in winter. One problem is that the bulb assembly, reflector, transformer and starter, when placed overhead, reduce the interception of natural light by the crop. Artificial lighting has been used successfully in The Netherlands (Marcelis *et al.*, 2002; Heuvelink *et al.*, 2006) and use of supplemental lighting for tomato crop is increasing nowadays (see also Chapter 4). New technologies such as light-emitting diodes (LEDs) have been tested as intra-canopy lighting to deliver photosynthetically efficient light to lower leaves of a canopy (Dueck *et al.*, 2012; Hao *et al.*, 2012; Gómez and Mitchell, 2016). However, the economics of supplemental LED lighting has not been proved yet and more on-farm evaluations are necessary. An empirical approach suggested that profitability of supplemental lighting for tomato crop was dependent on the lamp photon efficiency, electricity price, lamp photon utilization efficiency and heating fuel cost offset by the heat generated by the lamps (Kubota *et al.*, 2016). Therefore, efficient photon delivery methods such as intra-canopy lighting are expected to reduce the lighting cost and to increase profitability.

Shade-cloths and screens are used in southern production areas to protect fruit at the top of the canopy from sunscald, russetting and cracking caused by high temperatures and to reduce greenhouse temperatures. See Chapter 5 for additional discussion of the causes of these disorders. However, reduction of PAR decreases plant growth and yield.

CO_2 enrichment

Carbon dioxide can be added to the greenhouse in several ways. Natural gas or propane burners hooked up to sensors can be used to generate CO_2. Different fuel sources provide different amounts of CO_2. Burning 1 m^3 of natural gas, 1 l of kerosene or 1 l of propane provides 1.8 kg, 2.4 kg and 5.2 kg of CO_2, respectively (Portree, 1996). Flue gases from a hot-water boiler burning natural gas can be captured and recirculated. All these sources of CO_2 will add heat and water vapour to the greenhouse, as well as potential pollutants. Water vapour can be condensed before injecting to the greenhouse. Low NO_x (nitrous oxide and nitrogen dioxide) burners are available to minimize risks of pollutants reducing yield. Ethylene is a typical pollutant causing physiological disorders for tomato plants even at a very low concentration (less than 0.1 μmol/mol) when exhaust gas of an inefficient combustion heater is introduced into the greenhouse for the purpose of CO_2 enrichment. The most expensive but safest option is compressed or liquid CO_2. Some areas of The Netherlands have distribution nets for liquid CO_2 (Organic CO_2 for Assimilation by Plants, OCAP) coming from waste CO_2 from industrial sources. Sensors for CO_2 concentration should be calibrated periodically and located near the top of the plant. CO_2 distribution within the greenhouse should also be as uniform as possible to avoid yield differences and for efficient utilization.

CO_2 enrichment to 750–800 µmol/mol increases yields compared with standard outside conditions (about 390 µmol/mol) (see also Chapter 4). A standard approach to enrichment (Nederhoff, 1994) is to inject CO_2, as a by-product of combustion of natural gas, at a level of 800 µmol/mol during heating. At low ventilation rates (< 10% opening), this level is reduced to 500 µmol/mol when CO_2 cost is significant. With further vent opening, the goal is to maintain a base level of 450 µmol/mol, but this is not always possible. Currently, in The Netherlands during May to September, CO_2 is maintained at ambient levels. In Japan, a CO_2 controller was developed and made commercially available to maintain CO_2 concentration precisely at the same as outside level (i.e. null balance CO_2 enrichment) (Kozai *et al.*, 2015).

Computer control models (Aikman *et al.*, 1996; Stanghellini *et al.*, 2012) have been developed to convert the increase in photosynthesis by CO_2 enrichment into an anticipated financial return from fruit sales by linking biological processes with a fruit price model. Since fruit prices are very difficult to predict, use of such models is limited. Bailey (2002) considered strategies for CO_2 enrichment both with liquid CO_2 and with CO_2 from greenhouse heaters or combined heat and power (CHP) units. The heat produced by burning gas in a boiler or CHP can be stored in a heat storage tank to be used during nighttime, while CO_2 is used during daytime. The exhaust gas of a CHP needs to be 'cleaned' before it can be used in the greenhouse. On the basis of the financial margin between crop value and the combined costs of CO_2 and natural gas, Bailey (2002) showed that the most economic CO_2 control point with liquid CO_2 depended on its price. With exhaust gas CO_2 and CHP units, financial margins depended on whether there was heat storage.

In southern latitudes, greenhouses are vented at a higher rate and so CO_2 enrichment is not practical. In Raleigh, North Carolina, tomatoes could only be CO_2 enriched for 2–3 h daily for most of the growing season (Willits and Peet, 1989). In any case, when temperatures are above 25°C, in North American conditions CO_2 enrichment may not be cost effective (Portree, 1996) and may increase stomatal resistance, which reduces transpiration and increases leaf temperature. In the 'closed' greenhouse and 'semi-closed' greenhouse, high levels of CO_2 (> 1000 µmol/mol) will be used year-round.

Management of relative humidity or vapour pressure deficit

Relative humidity in a greenhouse is a result of the balance between transpiration of the crop, soil evapotranspiration, condensation on the greenhouse cover and vapour loss during ventilation, as well as air temperature inside the greenhouse. Therefore relative humidity inside the greenhouse exhibits diurnal and seasonal changes. Energy conservation features, such as the use of double layers of polyethylene films, have increased relative humidity (Hand, 1988).

Although computer control programs can be very sophisticated, there are limitations on the effectiveness of humidity control. For example, as vents are opened and closed to control temperature, relative humidity and CO_2 levels also change. If relative humidity levels become too high, while temperatures remain in an acceptable range, some combination of heating and ventilation may be necessary to maintain acceptable relative humidity and temperature. When the outside air temperature is lower than the set point, the heater should be turned on to maintain air temperature and the vents opened or the fans operated. With the 'closed' greenhouse, humidity control may be achieved without influencing other climatic factors, because cooling is provided by heat exchange with aquifer rather than ventilation.

Use of vapour pressure deficit (VPD) of the air, i.e. the difference between the saturated vapour pressure at that temperature and the current vapour pressure, is more encouraged to understand the plant response (i.e. transpiration and leaf temperature), as it shows the capacity of the air to hold vapour and directly influences evapotranspiration rates. In The Netherlands, growers use VPD expressed as weight of vapour (g) per volume (m^3) of moist air (also called 'humidity deficit') but the rest of the world uses pascal (Pa or kPa), an SI unit for partial pressure. Conversion from the Dutch unit to kPa needs temperature-specific coefficients, as mass and volume relationship of vapour differs at different temperatures. The University of Arizona developed a web-based application (https://cals.arizona.edu/vpdcalc/) to compute VPD and humidity deficit from air temperature and relative humidity. In a study in The Netherlands (Bakker, 1990), high relative humidity (low VPD) reduced leaf area because of calcium deficiency, and also increased stomatal conductance, reduced final yield and reduced mean fruit weight. This study was conducted over a fairly limited range of VPDs, however: 0.35–1.0 kPa in daytime and 0.21–0.71 kPa at night. It is unclear to what extent low humidity (high VPD) is deleterious to the plant if adequate water is available, but in general VPDs > 1.0 kPa are considered potentially stressful. In northern Europe, VPDs > 1 kPa are rarely seen, but in semi-arid parts of North America, VPDs will exceed this range without properly humidifying the air inside the greenhouse. A greenhouse temperature of 26°C and relative humidity of 60% would result in a VPD of 1.35 kPa or 9.7 g/m^3, for example. If plants transpire more water than can be supplied through the roots, fruit may develop blossom-end rot (BER) and stomata may close, resulting in poor growth.

The most important reason for reducing humidity and keeping leaf surfaces dry is disease prevention. Diseases spread rapidly when VPD is 0.2 kPa or less for prolonged periods of time and germination of fungal pathogen spores increases on wet leaf surfaces. This is most likely when warm, sunny days increase leaf transpiration and evaporation but moisture is held as water vapour until leaf or glazing internal surfaces cool to the dewpoint at night. Water vapour then condenses on to these cool surfaces and drips may fall on to the leaves. Wetting agents, either sprayed on the inside film or incorporated into the plastic, prevent condensation from dripping because moisture remains as a film, which slides off in a sheeting action, rather than dripping off on to the leaves. The problem of condensate dripping on leaves is most severe in Quonset-style double-poly greenhouses, because the rounded arch makes it hard to collect and remove condensation. Condensation over leaves may be reduced by increasing air movement to 1 m/s in the greenhouse. This is because increase in air movement would decrease boundary layer resistance over leaves to heat transfer. Enhanced heat transfer would reduce the differences in temperature between the leaf surface and the air and therefore prevent leaf surfaces from cooling below the dewpoint. Air movement can be increased by either running the fans on hot air furnaces or by horizontal airflow (HAF) fans. These small fans are placed above the canopy along the sides of the house to push air in one direction on one side of the greenhouse and in the opposite direction on the other side, and operate continuously, except when the exhaust fans are turned on for ventilation. HAF fans create a slow horizontal air movement, which also makes temperatures more uniform. In The Netherlands a condensation model has been developed (Rijsdijk, 1999) that enables growers to modify the heating regime during sunrise (the typical period when condensation is formed) as a function of measured fruit temperature rather than by use of ventilation fans. Condensation can form on the fruit because at sunrise air heats up (much faster than fruits) and plant transpiration increases the dewpoint of the air inside the greenhouse, while the fruit surface remains cold. Fruits heat more slowly than leaves and so if no condensation forms on the fruits, it should also not occur on the leaves. Therefore fruit temperature should be carefully monitored in the greenhouse during the time when condensation may occur.

Computerized climate control and monitoring system

The level of computerized climate control must be in accordance with the technical equipment. A high-tech greenhouse with sensors can be controlled by high-end climate computers to achieve optimal climate conditions. High-level computation such as neural network has been investigated and demonstrated for possible integration into greenhouse control systems (e.g. Fitz-Rodriguez *et al.*, 2012). For high tunnels and greenhouses with lower technology, less expensive temperature control equipment is adequate to regulate temperatures. In high-end instrumentation, sensors can give information on plant temperature, photosynthesis and transpiration, which can be incorporated into control algorithms. The sensor information can be integrated into 'speaking plant' control (Steppe, 2012), in which plant information is reflected in the decisions toward optimization of climate conditions (Van Straten *et al.*, 2011). Regardless of the level of technology, calibrating sensors periodically is critical to maintain the high accuracy of instrumentation. For example, air temperature and humidity sensors must be placed inside well aspirated housing with radiation shield. Measurement and reporting guidelines have been developed for disseminating more appropriate methods suitable in greenhouse climate measurements (Both *et al.*, 2015).

New environmental control technology

The 'closed' greenhouse is a recent innovation in The Netherlands to save energy as well as increase yields. The basic components of 'closed' greenhouses are mechanical cooling, heat-pump technology to produce warm water for heating and cold water for cooling, and seasonal storage of energy (cold and heat) in aquifers (De Gelder *et al.*, 2012a). Hanging gutter systems are used along with plastic tube air distribution systems (Fig. 9.12). This concept has the potential to fully control climate for optimal growth, because temperature, humidity and CO_2 levels can be controlled independently, but at present the system is not economically beneficial. In south-western USA, France and The Netherlands, 'semi-closed' greenhouse designs with limited ventilation have been successfully used commercially, because the limited ventilation allows CO_2 enrichment and also because more light transmission is expected from reduced numbers of roof vents that are often covered with insect screens and/or add more structural

Fig. 9.12. A semi-closed greenhouse with hanging gutter system and inflated air-distributing plastic tubes. (Photograph courtesy of Chieri Kubota.)

shade. In 'semi-closed' or 'closed' greenhouses, typically a large vertical temperature gradient exists due to a cooling system in the lower part of the greenhouse. However, this gradient (> 5°C during the day in summer) does not negatively affect tomato plant growth, development and yield (Qian *et al.*, 2015).

CONCLUSION: GREENHOUSE AS A SYSTEM TO CONTROL PLANT PRODUCTIVITY INTERACTIVELY

Although worldwide protected cultivation of tomato is small compared with open field production, it is an important industry sector in many countries. The cultivation system in the greenhouse offers the possibility of training the plants in an optimal way and optimizing climate conditions for maximal crop growth and production. It is a challenge to the grower to integrate information about plant development and climate into decisions about how to grow the plants to achieve maximal production of perfect quality with minimal costs of energy, labour and other production factors.

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10

Postharvest Biology and Handling of Tomatoes

Mikal E. Saltveit

INTRODUCTION

Originally, the tomato (*Solanum lycopersicum* L.) had an indeterminate growth habit; continuously producing flowers and fruit during the entire growing season. Cultivars (i.e. *culti*vated *var*ieties) with this indeterminate growth habit are grown where multiple harvests are economically justified, e.g. in the greenhouse, or in the field when plants are supported by a pole or trellis. These indeterminate fresh market cultivars have fruit at various stages of maturity on the plant, and hand harvesting is often necessary to maximize yield through multiple harvests. While this labour-intensive practice is economically justified for some fresh market field and glasshouse-grown fruit, processing tomatoes must be culturally managed to produce maximum yields with once-over mechanical harvests.

Breeding and cultural practices have modified the growth habit of processing cultivars and most field-grown fresh market tomatoes into a determinate growth habit in which the bush-like form produces side shoots that terminate in inflorescences. Determinate cultivars lend themselves to once-over mechanical harvesting and are therefore well suited for processing, since the plants are destroyed during harvest and the mechanical injuries the fruit sustain during harvest precludes their marketing as fresh fruit. Foliar sprays of ethylene-releasing chemicals are used before harvest to hasten the maturity of processing tomatoes and reduce the variability among the ripening fruit.

Processing tomatoes are harvested when fully ripe and quickly transported from the field to processing plants. The minimal postharvest care they receive during the hours between harvest and processing is often limited to shading the fruit from the sun or removing field heat by pre-cooling. In contrast, fresh market tomatoes are often harvested when mature, but less than fully ripe, and their ripening is carefully managed before, during and after transport to distant markets.

The protected greenhouse environment allows manipulation of many environmental parameters to maximize yield and quality. Greenhouse tomatoes can be grown in soil, peat moss, sawdust, stone wool, or other soilless and/or inert media, or by using a nutrient film technique or other hydroponic system. Fruit growth and development can be controlled by regulating pollination, adjusting nutrient availability to the stage of development (minerals as well as carbon dioxide), and modulating temperature and humidity.

Incorporation of naturally occurring ripening mutations into commercial lines has produced cultivars with significantly longer shelf-life but may have reduced the levels of valued indicators of quality. Traditional breeding with the incorporation of genes that convey chilling tolerance from related species has proved difficult and only resulted in cultivars with slightly increased chilling tolerance.

HARVESTING

Methods employed to harvest fresh market and processing tomatoes differ because their respective markets emphasize different measures of quality. Field-grown fresh market tomatoes are harvested at the mature-green through ripe stages of maturity, depending on the market demand and location. Mature-green fruit are firm enough and have sufficient shelf-life to survive the stress of being shipped considerable distances and still arrive at the market with enough shelf-life to be marketed at the wholesale and retail level. More mature fruit can be harvested for regional markets where transportation to the retail market is faster and less damaging to the softer, riper fruit. Ethylene is used to ripen mature-green fruit to red-ripe while in transit or at regional markets. Consumers value fruit with deep red and uniform colour, a lack of external blemishes and injuries, and characteristic aroma and flavour.

Processing tomatoes are once-over mechanically harvested when fully ripe and immediately transported to a processing plant. Pre-harvest applications of ethylene-releasing chemicals hasten and concentrate ripening and so a larger percentage of the fruit are at the proper stage of ripeness. Uniform deep red colour and characteristic aroma and flavour are also valued in processing tomatoes but surface blemishes and injuries are unimportant since the fruit are processed within a few hours of harvest and processing reduces the fruit to a peeled product or homogeneous paste.

Greenhouse tomatoes are usually harvested riper than fresh market field-grown fruit and are therefore more prone to mechanical injuries, because they are softer and have a shorter shelf-life than fruit harvested at the mature-green stage. Greenhouse-grown fruit are picked twice or three times a week as they reach the proper stage of development. More harvests are needed during warmer weather when the fruit develop faster. Fruit picked before fully developed (i.e. before reaching the mature-green stage) are more prone to mechanical injuries because of poorly developed epidermis and waxy cuticle than are more developed fruit. The thin skins and locular walls of greenhouse cultivars are valued by consumers, but these traits make them more susceptible to mechanical injury. Some greenhouse-grown fruit are harvested 'on the vine' at the breaker stage and marketed in clusters of three to six fruit.

Fruit that quickly reach the pink stage soon after harvest are called 'vine-ripened' tomatoes. Fewer mature-green and more vine-ripe fruit will probably be marketed in the future because of the development of non-toxic, gaseous inhibitors of ethylene action (e.g. 1-methylcyclopropene (1-MCP)) (Sisler *et al.*, 1996) and the incorporation of mutant genes that delay ripening through traditional plant breeding or genetic engineering to retain firmness during the later stages of ripening. However, maturity at harvest will still need to be uniform to minimize handling during grading and packing, and to ensure the consistent response of the fruit to the application of postharvest treatments.

Processing tomatoes are machine harvested in most industrialized countries (Fig. 10.1). The once-over, destructive harvest starts when at least 90% of the fruit are ripe. This determination is done by periodic visual inspections of fruit in the field. Application of chemicals to stimulate ripening (e.g. ethylene-releasing compounds) can be done a few weeks prior to harvest to maximize the percentage of coloured fruit. Harvest begins with the entire plant being lifted from the soil on to a mobile platform where the fruit are mechanically removed from the plant, which is discarded back into the field. The separated fruit are passed across an inspection platform where people manually discard defective (e.g. unripe, misshapen, diseased, etc.) fruit. The selected

Fig. 10.1. Fully ripe processing tomato fruit being mechanically harvested.

fruit are dumped into large gondolas or field bins being pulled alongside the harvester.

The mechanical injuries inflicted on the fruit as a result of machine harvesting are not severe enough, given the rapidity of processing, to significantly reduce the quality of the processed product. In comparison, the often extended duration of time between harvest and the consumption of fresh market tomatoes would allow excessive water loss and pathogenic infections to occur as a consequence of injuries to the surface of the fruit during mechanical harvesting, and thereby render the fruit unacceptable for the fresh market. Also, small bruises and skin injuries that do not affect internal quality are visually apparent and render the fruit unacceptable to many consumers. Processed tomato products include peeled whole, quartered and diced fruit, and concentrates for ketchup, juice, pulp, paste and soup. The market for tomato concentrate is larger than that for fresh fruit.

While appearance, texture and flavour are the important quality attributes of fresh market tomato, soluble solids, pH, titratible acidity, viscosity and colour are the major quality components of processing tomatoes. Other quality characteristics of processing tomatoes include uniform colour throughout the fruit (i.e. high lycopene and low chlorophyll content), freedom from decay, and firmness. Soluble solids include sugars, organic acids and other dry matter constituents such as pectic fragments that remain in solution. The production of tomato concentrates and paste requires removal of water, which is an energy-intensive process. Therefore it is less expensive to produce concentrates from fruit with high soluble solids and dry matter content. Viscosity or consistency is a complex physical property that is influenced by the amount and suspension of tomato solids, the size and linkage of pectins, and the solution of salts, proteins, sugars and organic acids in the processed product. Colour in tomato products is predominantly the result of the two lipid soluble pigments: lycopene and β-carotene.

Over the years, a great deal of work has been done by traditional plant breeding and recently by genetic engineering to increase the soluble solids, viscosity and colour of processing tomato cultivars. Significantly improving soluble solids content is difficult, because the trait is polygenetic, the environment exerts a large effect on expression of the trait, and there is a negative relation between soluble solids and other desirable characteristics such as high yield and concentrated ripening. For these reasons, the product of yield times soluble solids content (often expressed as °Brix) is more useful in estimating the productivity of processing tomatoes. In general, yields have been significantly increased, but the soluble solids content of the fruit has at most increased only slightly (Stevens, 1994; Zamir *et al*., 1999). The main reason for this relatively slow progress in increasing solids is the lack of genetic variability for high soluble solids in processing tomato germplasm. Attempts to increase soluble solids by introducing high solids potential from wild species of *Lycopersicon chmielewskii* and *L. cheesemanii* have been unsuccessful. This is because most of the high solids potential has disappeared by the time the appropriate genes have been introgressed into an acceptable horticultural type (Stevens, 1994). Modern genetic engineering may be able to introduce specific genes or clusters of genes that will increase soluble solids, but it will require a better understanding of the relationship among the partitioning of photosynthate within the plant, the mechanisms of soluble solids accumulation within the fruit, and the various other components of yield.

For the fresh market, multiple hand harvests are often needed to produce an economic yield when indeterminate cultivars are grown. Development of determinate, bush-type plants almost eliminated the growth of field-grown indeterminate tomatoes that were grown on plants supported by a pole or trellis. Plant decline is rapid after harvest of the major set from determinate plants, and few additional harvests are economical. However, the demand for high quality and heirloom fruit in regional up-scale markets has made cultural practices that favour multiple hand harvesting of riper fruit economically justifiable in some instances.

Harvested greenhouse-grown fruit are commonly hand carried to packing stations within the greenhouse and then moved on small carts to collection centres. In some large greenhouse operations, transportation systems are incorporated into the structures to protect the fruit from damage and reduce labour costs during harvest. Canals have been built under main greenhouse floors to float the fruit from areas of production to the packing area. The pickers empty their baskets into the troughs. The flowing water carries the fruit to a roller elevator where the fruit passes over a dryer prior to grading and packing. This can be an efficient system but cracking of the skin can be a problem and shelf-life can be decreased if the fruit remain in the water for more than 6 hours.

A number of rail systems have been developed that are economical and efficient for use within a greenhouse. The rail system can be temporarily installed during harvest or permanently installed so it can be utilized during the rest of the production cycle. A cart or trolley can run on heating pipes or specially made rails positioned among the rows. In the over-the-crop system, the heating pipes or rails are suspended between the rows and the suspended trolley runs along them. In both rail systems, the baskets of harvested fruit are transferred to a motorized cart that is hauled to the packing area. In the suspended system, the rails often continue to the packing area. Overhead wire cables can also be installed in a continuous loop to bring picking carts or baskets from the picking areas to the packing area. Some systems use self-propelled 'robotic' carts that follow sensing strips embedded in the greenhouse floor and can be programmed to go from one specific location within the greenhouse to another without manual intervention. When properly designed, such systems reduce labour and fruit damage and shorten the time from harvest to packing.

Most field-grown fresh market tomatoes are hand harvested at the mature-green stage of maturity. Fruit grown on indeterminate plants, supported by a pole or trellis in the field or greenhouse, are usually harvested at more advanced stages of maturity. Mechanical harvesters are being developed to assist in harvesting fresh market fruit, but none are yet commonly used in a commercial setting. Pickers put the fruit into buckets, bins or on to conveyer belts that move the fruit into large field bins or gondola trailers (Fig. 10.2). Fruit should be shielded from the sun during transport to the packing facility to prevent excessive heating of exposed fruit. Without the plant canopy to shade the fruit, direct sunlight can heat the surface of the fruit to a high enough temperature to damage the exposed tissue and adversely affect subsequent ripening.

In most commercial cultivars, ripe fruit abscise at the stem scar, but mature-green fruit may retain part of the pedicle in jointed fruit. Joints or knuckles form when a zone of abscission develops in the pedicle. Care should be taken with fruit harvested in clusters, or with a portion of the stem or pedicle attached, so that the protruding tissue does not puncture adjacent fruit during harvesting, handling, packing and transport. However, retention of the pedicle is seen as a positive quality aspect in some markets (Fig. 10.3). For example, in cluster tomatoes, the entire cluster is cut off at the main stem and kept together either by placing them in a single layer in boxes or by putting them into mesh bags.

Fig. 10.2. Fresh market mature-green tomato fruit being hand harvested into buckets and field bins.

Fig. 10.3. Breaker fruit in small plastic field trays with pedicle attached.

PACKAGING

Upon reaching the packing shed, the fruit can be flushed out of the trailer with chlorinated water. Smaller field bins or boxes can be unloaded by hand or gently dumped into water or on to a padded table. Although it would seem advantageous to start to remove field heat as soon as possible by using cold water in the dump tank, in reality the water should be a few degrees warmer than the fruit. Cold water will cool gases within the fruit and facilitate water intrusion through the stem scar. However, this cause of water intrusion is minor in comparison with that caused by hydrostatic pressure resulting from immersing the fruit too deeply in water. Water entering the fruit carries with it chemical and biological contaminants that accumulate in the dump water and for this reason water intrusion into harvested fruit should be avoided.

An inclined mesh conveyer belt is often used to remove the fruit from the dump tank and at the same time eliminate trash and undersized fruit. A spray of clean chlorinated water on the fruit as they are removed from the dump tank on the inclined conveyer belt facilitates removal of dirt and other field debris. The fruit are then dried with jets of air or sponge rollers. A visual inspection is often used to eliminate cull, over-mature, misshapen and otherwise unmarketable fruit. If regulations allow, the fruit may then be sprayed with an edible food-grade wax that may contain a fungicide. However, consumers in many markets object to the application of fungicide and waxes for purely cosmetic purposes.

After sorting for defects and colour differences, the fruit are segregated into several size categories by weight or diameter. Moving belts with increasing diameter holes (Fig. 10.4) or diverging rollers can sort by size, while spring-loaded pans can sort by weight. Other belts running underneath and perpendicular to the sorting belts convey the sized fruit to an area for one additional quality inspection where any remaining unmarketable fruit are eliminated (Fig. 10.5).

These sorting and grading steps can be injurious to softer fruit and they are therefore done primarily by hand when more mature fruit are being packed. In small packing sheds, breaker and turning fruit can be unloaded from small field bins onto sorting lines where the fruit are segregated into ripeness classes and hand packed (Fig. 10.6). The minimal handling that the fruit receive in this type of packing house permits riper fruit to be picked, packed and marketed.

The design of the many inspection lines in a packing shed requires serious attention by people trained in ergonomics. Inspection to remove unmarketable fruit from a mass of fruit rapidly moving past someone on a conveyer belt can quickly induce visual fatigue and degrade the inspector's effectiveness. The position of the inspector in relation to the conveyer belt, the speed of the belt, its width, the source, quality and direction of lighting, the arm movements necessary to remove the unwanted fruit and the placement of the receptacle

Fig. 10.4. A series of wide belts with holes of increasing diameter are used to sort fruit by size.

Fig. 10.5. Part of a packing line for mature-green tomato fruit where fruit are inspected for external quality defects.

for the eliminated fruit are all important factors that must be considered in the design of the inspection line. A simple movement to identify and remove a single fruit may be extremely easily and rapidly accomplished the first time but delayed and neglected the thousandth time. The incorporation of various analytical instruments can assist and/or eliminate human inspectors.

Mature-green fruit that appear the same externally can be at different stages of maturity as shown by differences in respiration, ethylene production, firmness of locular tissue and incipient lycopene synthesis in the columella tissue. Instruments that measure colour (e.g. the appearance of the red-coloured lycopene pigment) have been developed for the commercial sorting of fruit maturity by the development of external colour. However, these instruments are not effective at determining the ripening stage of mature-green fruit, because they cannot detect the lycopene that first accumulates in the inner columella tissue of mature-green fruit as they start to ripen. The highly variable content of chlorophyll in mature-green fruit makes it an inaccurate measure of their maturity. Internal development of lycopene, evolution of ethylene and liquefaction of the locular contents have all been shown to be reliable indicators of

Fig. 10.6. Small packing house where breaker and turning fruit are inspected for external quality defects and packed with minimal handling.

fruit maturity. Non-destructive methods to ascertain internal defects and the level of fruit maturity include acoustic, near-infra-red (NIR), NMR, MRI and many other technologies that use either penetrating radiation or resonance properties of the fruit to disclose the fruit's internal properties (Abbott *et al.*, 1997; Milczarek *et al.*, 2009). However, instruments employing these methods in a commercial packing shed have not yet been extensively adopted.

Mature-green and breaker fruit can be packed loose by weight or volume, while riper fruit should be placed on trays in cardboard boxes. Since mature-green fruit are usually treated with ethylene and ripened while in these boxes, the boxes must be resistant to moisture produced by the fruit during ripening. The boxes must also have a sufficient number of properly spaced holes to provide adequate ventilation. When placed on a pallet, the boxes must either be arranged with air channels between them, or aligned so that the holes in adjacent boxes line up to permit easy air movement through all the boxes on the pallet. Improper placing of the boxes can occlude holes in adjacent boxes and effectively isolate interior boxes from the movement of air providing ethylene and carrying away the products of respiration (i.e. heat, carbon dioxide, water vapour).

The enhanced production of heat, moisture and carbon dioxide by ripening fruit must be carried away from the fruit by fresh air containing the low concentration of ethylene used to stimulate the natural ripening of the fruit. Airflow and exchange should keep carbon dioxide levels in the boxes below 0.2%, because high levels of carbon dioxide can inhibit the ripening action of ethylene (Saltveit, 1997). Forced air is used for cooling after packing and for maintaining the proper temperature and atmosphere during the application of ethylene.

The superior appearance and flavour of greenhouse tomatoes can be used to justify a higher selling price that is needed to offset the higher production costs. This may be difficult in the USA, where supermarket mark-up is higher than in European countries and where greenhouse fruit must compete with low-cost field-grown tomatoes for much of the year, and with imported fruit during the off-season.

RIPENING

Tomatoes are climacteric fruit that are characterized by the onset of a climacteric rise in respiration and ethylene production coincident with ripening; i.e., softening and the first appearance of red colour (Fig. 10.7). Treatment with propylene, a biological analogue of ethylene, can also stimulate ripening.

The manifold changes accompanying ripening are presented in detail in Chapter 5. This section will concentrate on the postharvest initiation of ripening with ethylene and how this step fits into the general postharvest handling of tomato fruit.

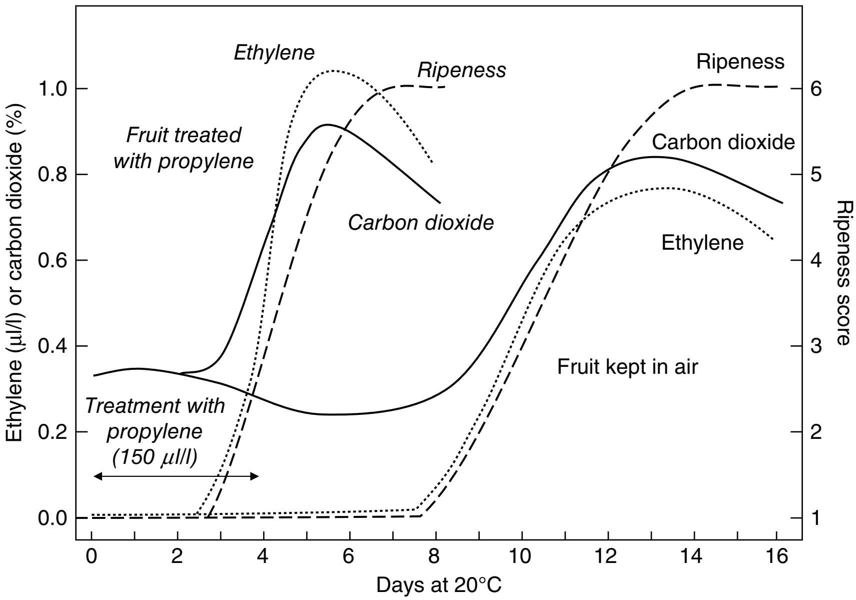

Fig. 10.7. Climacteric rise in respiration and ethylene production with the onset of ripening with or without treatment with 150 µl/l propylene in air for 4 days at 20°C.

Tomato fruit ripening requires the coordination of thousands of genes involved in softening, pigment destruction and/or synthesis, and accumulation of characteristic levels of sugars, acids and volatiles. In climacteric fruit like tomato, many of these changes are coordinated by ethylene. As the tomato fruit reaches physiological maturity, the response of fruit tissue to ethylene switches from a negative (System 1) to a positive (System 2) feedback. In immature tomato fruit tissue, ethylene inhibits its synthesis (System 1) thereby maintaining a low endogenous concentration that does not stimulate ripening. In physiologically mature tissue, ethylene promotes its own synthesis and this increasing endogenous level induces ripening-associated genes (Klee and Giovannoni, 2011).

Identification and characterization of ripening mutants has helped to explain how ethylene is critically involved in tomato fruit ripening. The ripening control mutants *ripening-inhibitor* (*rin*), *nonripening* (*nor*) and *Colorless nonripening* (*Cnr*) produce modified transcription factors that affect both ethylene and non-ethylene activities, while the *Never-ripe* (*Nr*) mutant affects the NR ethylene receptor (Klee and Giovannoni, 2011). Tomatoes incorporating the *rin* or *nor* gene are said to have the extended shelf-life (ESL) trait and ripen more slowly than conventional fruit. However, the quality of ESL fruit is severely affected if they are harvested at the MG2 stage of maturity (Cantwell, 2010). The minimum harvest maturity for ESL fruit should be at the Pink (USDA Color Stage 4) stage to maximize quality of the ripened fruit.

Tomato fruit ripening can be controlled by genetically altering (whether by incorporating naturally occurring mutant genes by traditional plant breeding, or by genetically engineering with sense or antisense technology) ethylene production, perception and/or action. However, these techniques can inadvertently compromise fruit quality traits. New techniques, such as RNAi silencing that can target specific genes, and has been used to reduce ACC synthesis activity in tomato, could be employed to alter ripening behaviour without unintentionally altering fruit quality (Gupta *et al*., 2013). Pigment synthesis and accumulation can be adversely affected in some slower ripening cultivars (Klee and Tieman, 2013). Surprisingly, increasing the level of anthocyanin pigments (natural antioxidants) in harvested tomatoes not only reduced disease susceptibility but also doubled the fruit's shelf-life (Zhang *et al*., 2013). Obviously, ripening is a complicated, multifaceted process with interrelated components that are still not fully integrated into an overarching system so that one aspect of fruit ripening can be manipulated without inadvertently altering other aspects of fruit quality.

The period of growth and development from fertilization to maturity lasts about 45–55 days, depending on cultivar and climate. Ripening is the final stage of maturation when fruit develop the characteristic colour, flavour, texture and aroma associated with optimum quality. Whether initiated naturally or induced by ethylene, ripening is accompanied by many changes, including an increased production of heat from respiration. Proper stacking of boxes on the pallet is important to ensure adequate air flow to remove this heat of respiration and the accompanying water vapour driven from the fruit because of their elevated temperature (Fig. 10.8).

Ventilation and refrigeration capacity of the storage facility must be matched with the increased production of heat during the climacteric rise in respiration. The rate of respiration can increase from 20% to 50% during ripening from a mature-green to red stage of development. The increased rate of evaporation of water from ripening fruit due to the elevated respiratory production of heat must be minimized by decreasing the vapour pressure deficit between the fruit and its surrounding atmosphere. This is done by maintaining high relative humidities (e.g. 85–95% RH) in the ripening rooms.

Fig. 10.8. Properly stacked boxes of mature-green tomato fruit on a pallet being moved by a fork-lift to a controlled temperature room for ethylene treatment.

Exposure to certain ppm (µl/l) levels of ethylene in air (e.g. 100 ppm) or to dips in ethylene-releasing compounds such as Ethrel promotes the ripening of mature-green tomato fruit to acceptable levels of quality (Table 10.1). Ethylene promotes the destruction of chlorophyll and the synthesis of lycopene in ripening fruit (see Chapter 5, this volume). However, ethylene can also induce ripening (i.e. lycopene synthesis and softening) in fruit that are so immature that good quality is never attained. In one fresh market cultivar, fruit that were about 90% of final size (i.e. more than 42 days after anthesis) were able to go through a normal climacteric and ripened to an acceptable level of quality after harvest (Saltveit, unpublished data). Fruit less mature did turn red and soften if given enough time, but the ripened fruit were of inferior quality. Fruit that were older than 31 days coloured without added ethylene, while younger fruit required added ethylene to colour and soften. Exposure of immature fruit as young as 17 days after anthesis to 1000 ppm ethylene in air induced a respiratory and ethylene climacteric, colour change and softening, but the fruit were of inferior quality.

Once ripening is initiated at the transition from the mature-green to the breaker stage, internal levels of ethylene rise and continue to support the auto-catalytic production of ethylene and further ripening. In vegetative and immature fruit tissue, ethylene suppresses its synthesis so that a low level of ethylene is maintained in the tissue. Once ripening is initiated in climacteric fruit like tomato, this negative feedback of ethylene on ethylene synthesis changes to a positive feedback, and the increasing levels of endogenous ethylene promote even more synthesis and accumulation of internal ethylene. The relatively impermeable cuticle of the mature tomato fruit minimizes ethylene loss to that which occurs through the stem scar and effectively isolates the internal tissue from depletion of ethylene resulting from diffusion to the external atmosphere.

Removal of ethylene from the external atmosphere does not significantly lower internal levels that are sustained by cellular synthesis, or alter the pattern of ripening. Introducing gaseous inhibitors of ethylene action (e.g. carbon dioxide, 1-MCP) into the environment of the fruit can slow ripening, as can lowering the temperature to near the chilling threshold. A few hours of

Table 10.1. Effect of ethylene treatment, either as 10 ppm (µl/l) ethylene in air or 2000 ppm (µg/g) ethephon dip, on the rate of ripening of mature-green tomato fruit.

Days of treatment	Days to breaker	Days from breaker to red-ripe	Days from red-ripe to discard	Total days
0	10.9	6.9	15.9	33.7
1	7.8	6.3	17.1	31.2
2	8.1	7.2	16.4	31.7
4	5.9	5.8	22.2	33.9
Continuous	4.4	6.3	15.2	25.9

exposure to ethanol vapour can inhibit ripening of tomato fruit harvested at various degrees of ripeness without affecting subsequent quality (Saltveit and Sharaf, 1992).

An ethylene concentration of 100–150 ppm is commonly used to initiate and hasten the ripening of harvested mature-green fruit at 18–22°C (Fig. 10.9). Advanced mature-green fruit treated in this way will reach the breaker stage in 24–36 h, depending on temperature. This atmosphere can be produced by a number of methods. The 'shot' method introduces a relatively large volume of ethylene into the ripening room by metering ethylene from compressed gas cylinders. Care should be taken when using this method, since ethylene in air mixtures between 3.1% and 32% (31,000–320,000 ppm) are explosive. While these explosive concentrations are more than 200-fold higher than the recommended levels for ripening, in rare instances they have been reached when metering equipment has malfunctioned. Using compressed gas mixtures containing around 3.1% ethylene in nitrogen (i.e. banana gas) to inject ethylene into a ripening room eliminates this problem. The concentration of ethylene in the ripening room is diluted as the gas is injected and the ethylene concentration can never reach the explosive limit. Catalytic converters are devices that employ heat and a metal catalyst to convert proprietary ethanol mixtures into ethylene. They deliver a continuous flow of a low concentration of ethylene gas into the storage room, thereby eliminating any possibility of reaching an explosive mixture of ethylene in air. This method also eliminates the storage and exchange of high pressure gas cylinders in favour of plastic bottles containing mixtures of denatured ethanol.

Sprays of aqueous solutions or applications of viscous paste containing ethylene-releasing compound are used to hasten the ripening of fruit still

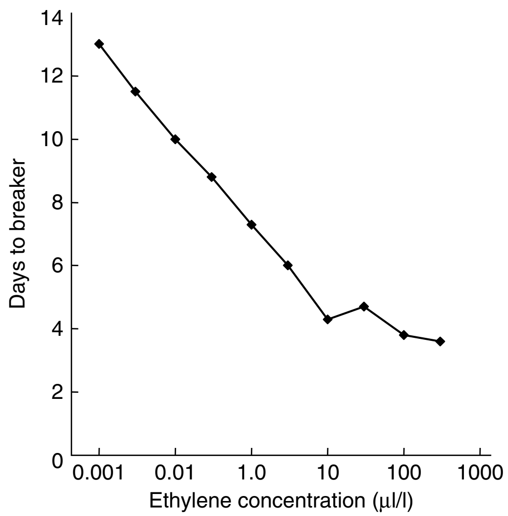

Fig. 10.9. Relation between externally applied ethylene concentration and the days it takes mature-green tomato fruit to reach the breaker stage of ripeness.

attached to the plants. Ethephon (2-chloroethylphosphonic acid, CAS Number: 82375-49-3; also known as Ethrel, Bromeflor, Arvest, CEPA) decomposes to innocuous products and ethylene in plant tissue. While stable at acidic pH, ethephon quickly breaks down to ethylene as the temperature and pH values increase (Table 10.2). Between 10°C and 20°C the Q_{10} is 6, while between 30°C and 40°C the Q_{10} is 5. Decomposition follows first-order kinetics; in any given unit of time the same fraction will decompose. The half-times for decomposition at different temperatures and pH values are given in Table 10.2. A spray or dip in a 2000 ppm aqueous Ethrel solution produced the same effect as gassing the fruit with 10 ppm (µl/l) ethylene in air (Saltveit, unpublished data). Although ethephon has very low toxicity and can be used on processing tomatoes and fresh market tomatoes before harvest, many governments strictly limit its residual levels in ripened fruit to such an extent that it is not approved for commercial use in the USA as a postharvest treatment.

Because multiple picks are usual for fresh market tomatoes, pre-harvest ethephon application is not practical since the ethylene produced can accelerate flower abscission, leaf drop and general senescence of the whole plant. Localized application of pastes containing ethylene-releasing compounds to a single truss (e.g. on greenhouse-grown indeterminate varieties) eliminates this problem.

Almost all fruit harvested at the mature-green stage are gassed with ethylene either at the packing warehouse or upon receipt at the terminal wholesale market. Fruit can be exposed to ethylene either before or after packing, but most are treated after packing. Treating before packing has some advantages. Fruit that develop decay while ripening in the warm, humid ripening room atmosphere can be eliminated before packing. After ripening, uniformly ripe fruit can be selected from those that are less ripe to produce a more uniform pack. Although ripening is accelerated and made more uniform by gassing with ethylene, the inherent variability in maturity of mature-green fruit can result in uneven ripening within a lot of fruit. Packed mature-green fruit may therefore produce a 'checkerboard' of fruit with different maturities after ripening. This may require removal of fruit from the boxes and their re-sorting and repacking to produce boxes containing fruit of uniform maturity that many

Table 10.2. Kinetics of Ethrel decomposition to ethylene at various pH values and temperatures.

	Half-time for ethephon decomposition to ethylene		
Temperature (°C)	pH6	pH7	pH8
10	70 days	14 days	7 days
20	10 days	2 days	1 day
30	30 hours	10 hours	5 hours
40	10 hours	2 hours	1 hour

Fig. 10.10. Variable ripening of mature-green tomato fruit produces a 'checkerboard' pattern of ripe and green fruit within a box.

retailers demand (Fig. 10.10). Repacking is both expensive and damaging to the riper, softer fruit and should be avoided.

TEMPERATURE MANAGEMENT

The optimum storage temperature that both slows ripening and also retains the quality of fruit ripened after storage varies with cultivar and stage of maturity. Mature-green fruit can be stored for 2 weeks at 12.5–15°C, ripening fruit at 10–12.5°C for a week, and fully ripe fruit at 7–10°C for 3–5 days. Mature-green fruit ripens best at 15–20°C, with ripening above 25°C producing soft, poorly coloured fruit.

Both the tomato plant and the harvested tomato fruit are chilling sensitive, incurring physiological damage if stored below about 12°C for durations dependent on the cultivar, stage of maturity and prior environmental exposure (Fig. 10.11). This sensitivity to non-freezing but chilling temperatures delineates the growing season and the geographical location where tomatoes can be grown under field conditions, and imposes a storage temperature well above the recommended 0°C for crops that are not chilling sensitive. The rate of many biological activities, e.g. ripening, can more than double for every 10°C rise in temperature, and this rise in metabolic activity can halve the storage life of chilling-sensitive crops in comparison with chilling-tolerant crops.

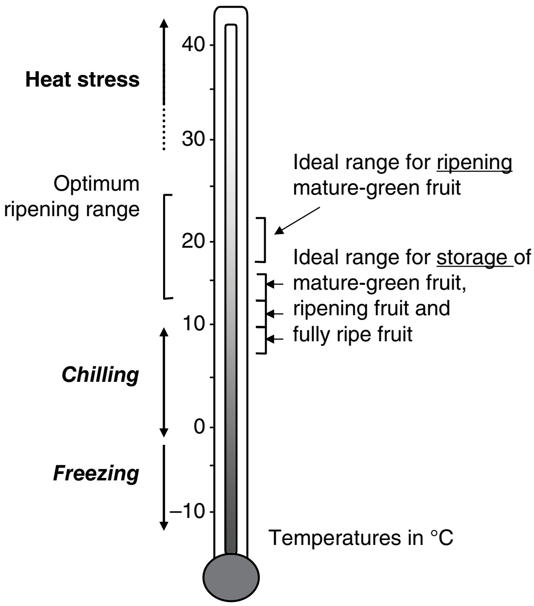

Fig. 10.11. Temperature limitations for the proper ripening and storage of fresh market tomato fruit.

Chilling injury is characterized by slow and abnormal ripening, increased disease susceptibility (often to specific decay organisms), reduced acidity, accelerated water loss and surface pitting. As with other crops, the degree of chilling sensitivity is related to the temperature the fruit were previously exposed to in the field. Chilling can occur in the field before harvest and contribute to the sensitivity of the fruit to chilling after harvest. Fruit grown in a hot climate are more sensitive to chilling than are fruit grown in a cooler climate. Also, the temperature at harvest appears to influence the level of chilling sensitivity. Fruit harvested hot are more chilling sensitive than fruit harvested when cool. Pre-conditioning at low, non-chilling or high heat-shock temperatures have been used to increase tolerance of mature-green fruit to subsequent chilling. Holding fruit at elevated temperatures may induce the heat-shock response that is protective against subsequent chilling injury. However, the shortened shelf-life resulting from holding the fruit at 35°C must be offset by a greater extension of storage life at what would have previously been chilling temperatures.

Attempts to increase chilling tolerance through conventional breeding have only slightly increased the chilling tolerance of commercial tomato cultivars (Venema *et al.*, 2005; Rugkong *et al.*, 2010). Chilling tolerance appears to be a multi-gene trait. The germplasm of resistant genotypes could be used to increase chilling tolerance of commercial cultivars. However, it has proved refractory to major improvements using traditional breeding to amplify existing genes related to chilling tolerance or to the introgression of genes from related species with greater chilling tolerance (e.g. *Solanum peruvianum*, *S. hirsutum* and *S. chilense*) (Venema *et al.*, 2005). Over the past few decades, the main focus of research has been on chilling-induced wilting and impaired photosynthetic efficiency of tomato plants; neither of which is of concern with harvested tomato fruit. Genetic engineering has produced lines with significantly increased chilling tolerance, but their lack of consumer acceptance makes their commercial adaptation doubtful.

Attempts to dissect the physiological basis of chilling sensitivity has eliminated some possibilities (e.g. membrane phase transition) but has failed to identify a critical physiological parameter that could be genetically manipulated to produce truly chilling-resistant tomato plants and fruit (Luengwilai *et al.*, 2012a). Chilling does not uniformly affect genes associated with tomato fruit ripening; some are promoted, some repressed, and some are unaffected (Rugkong *et al.*, 2011). Using postharvest treatments (e.g. heat-shock) that significantly alter the chilling sensitivity of genetically identical tissue could provide a means to dissect, through either genetic or metabolic profiling, the genetic and physiological events responsible for chilling sensitivity. When properly done (Saltveit, 2002), the rate of ion leakage from aged excised tomato pericarp tissue into an isotonic aqueous solution (e.g. 0.2 M mannitol) is an excellent indicator of chilling-induced changes in membrane permeability (Saltveit, 2005). Other indicators of chilling-induced injury (e.g. enhanced respiration and ethylene production upon removal to warm temperatures; abnormal colour development and softening; increased disease susceptibility) can be more variable and often require weeks to become obvious.

There is a diurnal variation in chilling tolerance that appears to be governed by carbohydrate levels in tomato plants and by temperature-induced alteration in membrane composition in fruit (Saltveit and Cabrera, 1987). Exposure to elevated or near-chilling temperatures, i.e. temperature conditioning, can increase the chilling tolerance of harvested mature-green tomatoes (Saltveit, 2005). Since the onset of chilling injury symptoms appears to occur only after a certain duration of chilling (the duration being dependent on the chilling temperature, the cultivar, the growing conditions and any prior exposure to environmental stress), interrupting long-term storage at chilling temperatures with periods of warm non-chilling temperatures, so that each chilling interval never exceeds that required to induce chilling injury, will ameliorate the development of chilling injury symptoms (Biswas *et al.*, 2012). This practice of intermittent warming suffers from two major drawbacks: increased energy costs to repeatedly warm and cool the fruit; and increased microbial growth if condensation occurs on the cool fruit as it is warmed by moist air.

Heat treatments can significantly increase chilling tolerance of harvested tomato fruit (Luengwilai *et al.*, 2012b) but, since the effect is localized to the heated tissue and not systemic throughout the whole fruit (Lu *et al.*, 2010), care must be taken to ensure that the heat treatment is uniformly applied to all fruit and to all portions of each fruit. While the exact optimal temperature and duration of exposure varies among cultivars and previous environmental

Table 10.3. Effect of temperature on quality and ripening of mature-green tomato fruit. (Kader, 1986.)

	Days to table ripeness (stage 5)					
Ripeness stage	12.5°C	15.0°C	17.5°C	20.0°C	22.5°C	25.0°C
Mature-green	18	15	12	10	8	7
Breaker	16	13	10	8	6	5
Turning	13	10	8	6	4	3
Pink	10	8	6	4	3	2

exposure, the heat-shock response is broad enough so that regional recommendations may be applicable to large production areas.

After ethylene treatment, the optimum transport and storage temperature is 12.5–15°C. Within these limits, ripening will be faster at higher temperatures (Table 10.3). Ripening of mature-green fruit will be uneven and abnormal below 10°C, while ripening may be too fast and decay more extensive above 25°C. Since chilling injury of mature-green fruit primarily affects subsequent ripening, riper pink fruit are almost by definition more chilling resistant than mature-green fruit and can be stored at lower, near-chilling temperatures. The shipping temperature for pink tomatoes will depend largely on the number of days in transit and the degree of ripeness which the person receiving the fruit may desire.

High temperatures (e.g. 25–30°C) inhibit ripening. Fruit stored at these high temperatures turn orange instead of red, because lycopene synthesis is inhibited above 30°C. Brief exposures of fruit on the plant to high temperatures, as during a hot day when shaded fruit may reach 30°C, are reversible with normal ripening proceeding at cool temperatures. A brief exposure to higher temperatures (e.g. exposure to the full sun) can cause permanent damage, cell death and sunscald. Exposure of harvested fruit to direct sunlight after harvest should be avoided by coverings and shading.

CONTROLLED ATMOSPHERE

Controlled and modified atmospheres are slightly beneficial during both transit and short-term storage but only as a supplement to proper temperature control. Atmospheres containing 3–5% oxygen and 0–3% carbon dioxide extend storage by a few days at 12°C (Leshuk and Saltveit, 1990; Saltveit, 1997). Reduced oxygen levels reduce the respiration rate, the production of and sensitivity to ethylene, and ripening. Injury occurs at oxygen levels below 2% and carbon dioxide concentrations above 5%. However, these values must be approached very cautiously. For example, elevated carbon dioxide atmospheres are problematic because atmospheres containing 2% or more carbon dioxide have caused increased softening and uneven ripening under some conditions. A controlled or modified atmosphere may be beneficial but only when used in conjunction with established postharvest management practices that have established and will maintain optimal temperature and humidity.

Unlike many other fruit in which gas diffusion takes places over the entire surface, gas diffusion in a tomato fruit occurs primarily through the stem scar. Any impediment to gas diffusion, such as the development of unusual amounts of corking on the stem scar or the application of excessive wax, can significantly alter the internal gas concentrations even in a constant ambient environment. The variable results reported for the storage of tomatoes under modified atmospheres may have resulted from the disparity between the imposed ambient atmosphere and the internal atmosphere that developed as the result of diffusion, respiration and metabolism.

Modified atmosphere packaging can be used to produce atmospheres low in oxygen and high in carbon dioxide which slow ripening and extend the market life of tomato fruit. A semi-permeable plastic film is usually incorporated into a more rigid package. The combination of gas diffusion through the film and the respiration of the tomato fruit produces and maintains a modified atmosphere. Changes in respiration resulting from ripening (i.e. the climacteric) or changes in temperature can significantly alter the atmosphere in the package. Selecting fruit of the desired maturity and maintaining the package at the proper temperature are crucial. The semi-permeable films also allow ethylene to accumulate and the relative humidity to reach saturation. Elevated levels of ethylene can stimulate ripening and nullify the inhibitory effects of the modified atmosphere on ripening. While the high relative humidity within the package lessens water loss from the fruit, it also fosters microbial growth since fluctuations in the storage temperature can result in water condensation on the fruit within the package. Packets of absorbents that reduce ethylene and water vapour in the package are useful but they add cost to an already costly package.

Although an atmosphere containing 5–10% carbon monoxide will reduce disease, its extreme toxicity has severely limited commercial application. Carbon monoxide has slight ethylene-like activity and will promote ripening of mature-green fruit held in air.

SUMMARY

Harvesting fruit at the proper maturity and careful handling during harvest, transport, packing and marketing will ensure that high-quality fruit reach the consumer. External colour changes are good indicators of the physiological maturity of the fruit of almost all tomato cultivars. Multiple picks by hand are common for fresh market fruit, while once-over mechanical harvesting is customary for processing tomatoes. Cultivars with genetically modified ripening

behaviour have extended shelf-life but additional care may be needed to ensure that other quality attributes are also maintained after harvest. Application of about 100 ppm (μl/l) levels of ethylene can hasten and coordinate the ripening of mature-green fruit. Once fruit have reached the breaker stage, internally synthesized ethylene promotes its own synthesis and sustains ripening even when external ethylene levels are reduced to zero. Temperature management is crucial since temperatures below about 12°C cause chilling injury and temperatures above about 25°C inhibit lycopene synthesis. Controlled atmospheres have some benefit in maintaining quality, but they are usually minimal compared with the benefits of avoiding physical injuries, and proper management of temperature, relative humidity and sanitation.

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11

Organic Tomato

Martine Dorais and Dietmar Schwarz

PRINCIPLES AND STANDARDS OF ORGANIC AGRICULTURE

Organic agriculture is defined by the International Federation of Organic Agriculture Movements (IFOAM) as a 'production system that sustains the health of soils, ecosystems and people'. Organic agriculture endeavours to minimize system inputs and adverse environmental impact through sustainable waste management, minimal use of energy, nutrient-balanced approaches, and mechanical and biological control of pests. It excludes genetically modified varieties, synthetic fertilizers and pesticides, sewage sludge, synthetic hormones or antibiotics. In most countries, plants have to be grown in soil.

For most of the European countries and for all member states of the European Union (EU), organic farming is strictly defined by the European Commission (EC). In 2009, the Commission revised guidelines from 1999 for the production of organic crops, including tomato (EC 834/2007; EC 889/2008). Certification of the product is compulsory, because it provides the consumer with further confidence. By law, all EU products labelled as organic must bear the country code, the name of the last operator who handled the product (such as the producer, the processor or the distributor), as well as the name or code number of their inspection body. A logo is used to supplement the labelling and increase the visibility of organic products for consumers. Some guidelines are country-specific or specific to grower associations; see, for example, Bioland (www.bioland.de) and Demeter International (www.demeter.net), both originating in Germany. If an EU tomato grower decides to become an organic grower, they need to work at least 2 years under the organic production guidelines before they can start providing certified organic tomatoes. In North America, no conversion period is required when production takes place in allowable demarcated soil beds or containers.

National governments and private organizations have developed their own interpretations and rules for tomato cultivation (e.g. Anonymous, 2013; Skal

Biocontrole, 2010). Some countries (USA, Mexico, Israel and Canada) allow tomato plants to be grown in demarcated soil beds or containers with organic growing media such as peat- and compost-based soils, where soil biological activity provides the major part of the nutrients to the tomato plants. Except in the USA hydroponics is prohibited. In Italy, heating is not allowed, whereas Demeter (Germany) allows heating up to 15°C. In Sweden, 80% of the energy used should be renewable. Although Spain does not allow carbon dioxide (CO_2) greenhouse enrichment for organic tomato crops, there is no restriction on its use in most EU countries or in North America. In Ireland, however, CO_2 should come from renewable sources, whereas in The Netherlands CO_2 should be a by-product of heating (e.g. natural gas). Artificial lighting can be used in North America, Sweden and Iceland or is limited to plant propagation (van der Lans *et al.*, 2011).

PRODUCTIVITY AND PROFITABILITY OF ORGANIC TOMATO

There are no complete statistical data available on organic tomato production. In 2010, however, tomato (6600 ha) was the fifth most important organic vegetable after potato (29,327 ha), lettuce (11,705 ha), carrot (8095 ha) and pea (8048 ha) (Granatstein *et al.*, 2010), while the total area of organic greenhouse crops was estimated at around 8302 ha in 2016, which is almost entirely used for lettuce and fruit vegetables such as tomato (Dorais and Cull, 2017). The ratio of organic yield to conventional yield ranges from 0.69 to 1.19 for protected tomato crops and from 0.60 to 1.75 for field-grown tomatoes (Table 11.1). Differences between organic and conventional tomato yield depend on climate conditions, growing systems and cultural management. For greenhouse tomato, yield decreases can reach as high as 40% owing to soilborne pathogens, root-knot nematodes and salinity, while yield of demarcated soil beds is similar to conventional systems (Dorais and Alsanius, 2015).

Organically grown tomato (Fig. 11.1) can be more profitable than a conventional crop when a premium price of 10–50%, or even more, depending on the market area, is added (Kaiser and Ernst, 2011). Nevertheless, there may be higher production costs for organic tomato due to additional expenses for weed and pest control, compost and cover crop management (Clark *et al.*, 1999; Kaiser and Ernst, 2011). For unheated polytunnel tomato crops grown in the UK, the labour cost (38.5%) has the highest impact on organic crop profitability (Schmutz *et al.*, 2011). Only 1.8% and 3% of costs were due to fertilizers and crop protection, respectively. Other variable costs were similar to those for conventional tomato crops: 20% for seedlings, 15% for commission, 7.2% for packaging, 5.6% for transport and 0.4% for irrigation. Protected organic crops can be very profitable, as shown for the UK (Schmutz *et al.*, 2011) and

Table 11.1. Yield of tomato grown in different production systems: ratio of organic (Org) to conventional (conv).

System	Org : conv.	References
Heated greenhouse		
Soil	0.69	Bakker, 2011 (Netherlands)
Container/coir	0.94–1.02	Gravel *et al.*, 2011a (Canada)
20 l pot	0.91–0.92	Papadopoulos *et al.*, 2011 (Canada)
Soil/soil and stone wool	0.85–1.19	Gravel *et al.*, 2010 (Netherlands)
Peat/compost (beefsteak), 15 l pot	0.98–1.00	Zhai *et al.*, 2009 (Canada)
Peat/compost (truss), 15 l pot	0.81	Zhai *et al.*, 2009 (Canada)
Peat/coir/bark, 19 l bag	1.15–1.18	Rippy *et al.*, 2004 (USA)
Unheated greenhouse		
Peat/coir/bark, 19 l bag	0.92–1.00	Reeve and Drost, 2012 (USA)
Field	0.70	De Pascale *et al.*, 2016 (Italy)
	0.84	de Bon *et al.*, 2012 (Vietnam)
	0.75–0.81	Chellemi *et al.*, 2012 (USA)
	1.00	Campanelli and Canali, 2012 (Italy)
	0.78	Aldrich *et al.*, 2010 (USA)
	0.62	Riahi *et al.*, 2009 (Tunisia)
	0.6–1.11	Pieper and Barrett, 2009 (USA)
	0.88–0.97	Tabaglio *et al.*, 2008 (Italy)
	0.95–1.75	Mitchell *et al.*, 2007 (USA, Canada)
	0.87–1.13	Barrett *et al.*, 2007 (USA, Canada)
	0.91–0.94	Poudel *et al.*, 2002 (USA, Canada)
	0.89–1.03	Colla *et al.*, 2002 (USA, Canada)
	0.83–0.87	Clark *et al.*, 1999 (USA, Canada)
	1.06	Eggert, 1983 (USA, Mexico)

for Canada (20–80% more profitable than conventional greenhouse tomato) (Dorais and Cull, 2017).

CULTIVAR SELECTION AND SEEDLING PRODUCTION

Cultivars and rootstocks

Depending on organic markets targeted, on production systems and on the growing environment, the type and diversity of the tomato varieties selected vary greatly to meet the demand for an optimal yield and quality. After having disappeared because of their sensitivity to diseases and pests, heirloom varieties are becoming more important and are returning to the market thanks to

Fig. 11.1. (A) Soil-grown organic tomato cultivated in a shade-house. (B) Soil-grown organic tomato cultivated in a glasshouse. (C) Demarcated raised beds newly amended with a mixed organic amendment on the top of the peat-based growing soil. (D) Organic tomato grown in soil under high plastic tunnels (Photographs Martine Dorais).

their taste and appearance. Although interest in these heirloom varieties is increasing, tolerance and resistance to major diseases and pests as well as to climatic conditions and production techniques cannot be ignored in the selection process. Besides flavour, consumers are paying increasing attention to characteristics related to human health (Dorais *et al.*, 2008; Dorais and Alsanius, 2015). A practical method of adapting heirloom and sensitive cultivars to counteract abiotic and biotic stresses is grafting these cultivars on to selected vigorous rootstocks. However, the rootstock–scion combination and scion performance must be adjusted to the organic growing conditions and consideration should be given to unfavourable conditions, including saline soils, soil pH (alkalinity) stress, nutrient deficiency, toxicity of heavy metals, thermal stress, drought and flooding, and persistent organic pollutants (Schwarz *et al.*, 2010). In recent years many new rootstocks have been bred and released to the market (Kleinhenz, 2015). Unfortunately, among the huge list of commercial rootstocks (see, for example, www.vegetablegrafting.org) only a few are bred and tested for organic farming. Recommended rootstocks for organic cultivation are 'Maxifort', 'Estamino', 'Fortamino', or cold-tolerant ones such as 'Armada', 'Arnold', 'Actimino', 'Top Bental' or 'Top Gun'.

Tomatoes are susceptible to varying degrees to diseases, and complete resistance does often not exist. The use of rootstocks is particularly successful against soilborne pathogens, such as Verticillium wilt (*Verticillium dahliae*), bacterial wilt (*Ralstonia solanacearum*), Fusarium wilt (*Fusarium oxysporum* f.sp. *lycopersici*), corky root rot (*Pyrenochaeta lycopersicum*), southern blight (*Sclerotium rolfsii*), *Rhizoctonia solani*, *Pythium* and *Phytophthora* species, and nematodes (see Chapter 7). Some rootstocks, such as 'Maxifort', 'Multifort', 'Survivor' and 'Brigeor', have been able to reduce root galling by 80%. Others are offered by the seed companies as completely resistant. Unfortunately, the results vary widely for different environmental conditions and scion cultivars. Moreover, tolerance disappears when the same rootstocks are used over a longer period and real resistance is absent.

Seeds

Some associations allow the use of any type of seed, whereas others do not allow all types of hybrids and prefer traditionally bred seeds (heirloom varieties). Biodynamic cultivation associations (Demeter) do not accept cytoplasmic male sterility in hybrids produced by cytoplast–protoplast fusion and recommend refraining completely from the use of hybrids. All seeds, seedlings and rootstocks must come from organic production and no genetically modified organisms are allowed. Some tools and breeders offer growers the selection of seeds suitable to their demands (see, for example, http://www.organicxseeds.com). Seeds should be treated only by methods approved for organic cultivation. Organic seed treatments against fungi and bacterial diseases by copper-based compounds, vinegar, plant and compost extracts, mustard seed extract, chitin, essential oils, lime or milk powder, combined or not with beneficial microorganisms, along with thermal treatment could be effective, depending on the concentrations used and the pathogens involved. However, the literature reports contradictory results, phytotoxicity effects and moderate or low efficacy. The use of acidified nitrite in citric acid (300 mmol/l $NaNO_2$ in citric acid buffer at pH 2; 20.7 g sodium nitrite in 1 l of 57.63 g citric acid buffer; applied for 10–20 min) to treat tomato seeds infected by *Didymella lycopersici* has been reported to be highly effective compared with biological control agents, acetic acid, mustard flour, milk powder and elicitors of plant resistance (Kasselaki *et al*., 2008). Moreover, for efficacy similar to that achieved with conventional fungicides used in tomato seed treatment, acidified nitrite treatment appears to be less phytotoxic. The release of gaseous compounds such as nitric oxide has antimicrobial activity through the seed coat, allowing pathogen removal from the seed's surface and inner tissues. Although nitrite is permitted as a food additive in organic food, the status of its agronomic use as a seed disinfection treatment is unclear. For *Clavibacter michiganensis*, the immersion of tomato seeds for 10 min in 300 mmol/l acidified nitrite has been found to result in 98% pathogen removal, whereas the use of copper hydroxide, certain strains of *Bacillus* spp. and compost water extracts (1:2; green waste, green waste mixed with manure, digestates from co-fermentation of cow and green waste or biowaste) resulted in 100% pathogen-free seeds without any phytotoxic effects (Kasselaki *et al.*, 2011).

Organic seedling production

The transplant medium usually consists of a mixture of peat and mature compost (20–30%; < 2 dS/m in 1.5 water extract v/v) amended with around 300 mg nitrogen (N) per litre of commercial organic fertilizers and 2 g limestone per litre to reach a pH value of 6.0–6.5 (Koller *et al.*, 2004). Coir, sawdust, composted bark, perlite, vermiculite, sand, clay, topsoil or other local materials may replace all or part of the peat substrate. Yucca or other certified ingredients may be used as wetting agents. Horn and feather meals are excellent animal-based nitrogen fertilizers, while plant-based products such as potato protein, malt sprouts and vinasse are interesting alternatives. However, plant-based products have to be mixed with the growing medium at least 2 weeks before sowing, to prevent any phytotoxic effects. The use of vermicompost or plant-based amendments (sesame or lucerne meal) in the transplant medium was found to increase transplant biomass and early yield in organic tomato compared with the use of thermogenic compost, made with the same feedstock as the vermicompost, and standardized industrial organic transplant media (Jack *et al.*, 2011). The level of phosphorus (P) (about 100–200 mg P_2O_5 per litre), potassium (K) (about 200 mg K_2O per litre), magnesium (Mg) and micronutrients provided by at least 20% of compost is normally sufficient to fulfil plant requirements. Additional fertilizers such as horn chips (2–3 g), rock phosphate and potassium sulfate may also be added to the growing medium. To ensure good seedling vigour during the final weeks in the nursery, liquid organic fertilizers are often provided as complements.

ORGANIC GROWING SYSTEMS

Soil

In contrast to conventional crops, soil biological activity is the main determinant of soil fertility for organic crops, given that nutrients are provided mainly by compost and manure. Organic soils have a larger pool of organic carbon (C) and microbial biomass, and higher enzyme activity compared with soils in conventional farming. To ensure high soil activity, several cultural aspects should be considered, as set out below.

Carbon:nitrogen ratio

A C:N ratio below 15 prevents microorganisms from immobilizing soil N and ensures a constant release of nutrients, whereas a C:N ratio between 20 and 30 is desired for timely decomposition of organic matter and subsequent N release. On the other hand, a C:N ratio of 25–40 promotes microorganism growth and humification. The cut-off point for the C:N ratio with respect to N mineralization and immobilization has therefore been defined at about 25–30. However, tomato fertilizer amendments used by growers generally have a C:N ratio ranging from 4 to 25.

Organic matter

A soil organic C content of 2% in the topsoil (0–30 cm), corresponding to 3.4% organic matter, is indicated by the European Commission as a threshold below which a reduction in soil chemical, biological and physical fertility (water-holding capacity, soil aggregation, air and water porosity, hydraulic conductivity, gas diffusivity, nutrients) as well as an increase in erosion can be observed. However, this threshold is soil- and site-dependent. To ensure a well balanced nutrient budget, the recommendations are 30 t compost supply per hectare per year for greenhouse tomato and 7–9 t/ha per year for field tomato. Too much or too little organic matter can result in nutrient imbalances (excess or deficiency) and nutrient leaching to groundwater.

Cover crops and green manure

Cover crops constitute a sustainable soil-building technique with significant advantages for the soil and nutrient supply (Table 11.2). For field tomato, incorporation of a cover crop at the time of planting was shown to result in the lowest weed and cover-crop regrowth compared with incorporation 17–19 days after planting or with no incorporation and, consequently, made N more available to the crop (Madden *et al*., 2004). The combination of rye and hairy vetch (*Vicia villosa*) in a no-tillage tomato system was found to provide the most weed suppression (Delate *et al*., 2012). Moreover, combining a summer cowpea cover crop and a resistant cultivar brought *Meloidogyne incognita* under control and improved the yield of tomato in rotation with pepper.

Crop rotation

For organic field tomato, rotation to non-solanaceous crops for 3 years is recommended to avoid pest problems. It was reported that the best tomato crops follow a crop of clover, sweet clover or lucerne in a 3- or 4-year rotation. For greenhouse tomato, the high infrastructure investment required means

Table 11.2. Nutrient contribution of cover crops used under northern or subtropical growing conditions.

Legumes	N (kg/ha)	K (kg/ha)	P (kg/ha)	Biomass (t/ha DW)	Non-legume species	N (kg/ha)	K (kg/ha)	P (kg ha)	Biomass (t/ha DW)
Hairy vetch *V. villosa*	111–224 (70–92%)[a]	149	20	2.58–5.60	Rye *S. cereale*	54–159 (78%)	121	19	3.36– 11.55
Pigeon bean *C. indicus*	171–236			6.40–8.29	Flax *L. usitatissimum*	141			10.16
Velvet bean *Mucuna* spp.	173–286	156–214	23–32	6.70–11.10	Sorghum sudangrass *S. bicolor*	30–110 (30–50%)	64–188	19–57	3.30–12.00
Cowpea *V. unguiculata*	75–243 (90%)	111–124	15	2.80–11.70	Ryegrass *L. multiflorum*	33			2.24–10.08
Sunn hemp *C. juncea*	190–561 (fast)	114–190	15–26	6.70–19.70	Wheat *T. aestivum*				3.36–8.96
Red clover *T. pratense*	78–168			2.24–5.60	Oat *A. sativa*				2.24–11.20
Crimson clover *T. incarnatum*	78–146 (89%)	160	18	3.92–6.16	Millet *Pennisetum* spp.	65			6.66
Subclover *T. subterraneum*	84–224			3.36–9.52	Buckwheat *F. esculentum*	48			2.24–4.48
Berseem clover *T. alexandrinum*	84–246			6.72–11.20	Mexican sunflower *T. diversifolia*	70–140			7.90–12.60
White clover *T. repens*	90–224			2.24–6.72	Radish *R. sativus*	56–224			4.48–7.84
Sweet clover *M. officinalis*	111–190			3.36–5.60	Rapeseed *B. rapa*	45–179			2.24–5.60
Austrian winter peas *P. sativum*	111–168 (77%)	178	21	4.48–5.60	Mustards *Brassica* spp.	33.6–134 43–224			3.36–10.37

DW, dry weight.

[a]Percentage of N release within the growing season. Note: These values vary depending on climate, soil, seeding rate and cropping system; potential rate of N decreases with plant maturity. (From Sullivan, 2003; Wang *et al*., 2009; Clark, 2007; Campiglia *et al*., 2011; Partey *et al*., 2011.)

that crop rotations are generally limited to a few species with similar pest susceptibility, such as aubergine, sweet pepper and cucumber.

Tillage

Organic tomato field producers use tillage to maximize the mineralization rate and nitrification processes of cover crops or other organic amendments and to control weeds. Nevertheless, from a sustainability perspective, minimum tillage or a combination of conventional and minimum tillage is preferable. Indeed, long-term tillage contributes to reduce both organic matter and microbial activity, worsen soil structure, increase greenhouse gas emissions and increase the potential for nitrate leaching to groundwater. For irrigated organic tomato, a no-tillage roller/crimper system using rye–hairy-vetch or wheat–winter-pea covers was found to provide results for plant growth, number of fruits and yield (40–63 t/ha) similar to those achieved in a tilled system (Delate *et al.*, 2012). Consequently, similar yields in no-tillage and tilled tomato crops are feasible when cover-crop regrowth does not occur and adequate irrigation is provided. However, the long-term effect of no-tillage on the weed seedbank has to be considered.

Containers, raised beds and organic hydroponics

This section relates to systems in non-EU countries. It describes greenhouse organic systems for container-grown plants in North America, Israel and some other countries, along with raised bed systems as found in Scandinavia and organic hydroponics as allowed in the USA.

Growing protected organic tomato in enclosed beds or containers can be profitable where soils are contaminated (e.g. by pests, salinity, pesticides, heavy metals) or have to be heated during a large portion of the year. Such systems also facilitate a grower's transition from conventional to organic farming when soil quality is the main limiting production factor. From an environmental perspective, enclosed soil also offers the possibility of collecting and recycling drained water and nutrients and, consequently, reducing pollutant emissions. To fulfil plant nutrient requirements, amendments are provided every 1–3 weeks (see Table 11.1 for yield performance).

Organic hydroponics using fish or digestates or other organic liquid fertilizers, although not allowed in most countries as main source of nutrients, provide beneficial compounds (e.g. amino acids, humic acids) and microorganisms that promote plant growth and productivity as well as protect the plant against pathogens. For instance, crude fish effluent was shown to reduce tomato root colonization by *Pythium ultimum* and *F. oxysporum*, and had a stimulating effect on tomato plant height, leaf area and root dry weight (Gravel *et al.*, 2014). Previous studies also showed the fertilizer value of digester effluents from the anaerobic digestion of animal manure.

ORGANIC FERTILIZATION AND AMENDMENT

Nutrient supply

Soil fertility management in organic tomato production systems is more complex than in conventional systems, because it is based on agroecological strategies, making good knowledge of the pedoclimatic situation and nutrient flow through the agroecosystem a requirement (Tittarelli *et al.*, 2016). Consequently, unlike in conventional tomato production systems, fertilization decisions in soil-based organic growing systems involve taking into account storage, nutrient-use efficiency, disease resistance and the well-being of beneficial soil microorganisms. Manure can be used for field tomato crops provided that a 120-day delay prior to harvest is respected. In addition, the regulations allow farmers to purchase mineral fertilizers of natural origin, such as rock phosphate, crude potassium salts, calcium and magnesium carbonate, magnesium sulfate, calcium chloride, calcium sulfate, lime, elemental sulfur and stone meals. Other amendments such as fish powder, algal products, feather meal, guano, blood meal (not permitted everywhere, owing to bovine spongiform encephalopathy) and vermicompost (see EC 889/2008, annex 1) are used. Liquid fertilizers are allowed as complements in most countries when the products are registered as permitted substances. In the EU, a maximum level of manure N is set (170 kg/ha per year) and fertilizers have to comply with the EU Nitrate Directive. The use of Chilean nitrate is prohibited in most countries and restricted to 20% of the crop N requirement in the USA (Hartz and Johnstone, 2006). Table 11.3 presents commercially used fertilization scenarios for organic field and greenhouse tomato crops.

The current strategy of nutrient supply is based on soil analyses to determine the plant-available minerals. Depending on plant demand application: (i) base dressing before planting (50–66%; in North America 56 kg N/ha) and (ii) side dressings when first fruits are about 2 cm in diameter (in North America 28–56 kg N/ha) (Schonbeck *et al.*, 2006) are provided. The recommended N fertilizer rate is up to 220 kg/ha (Zotarelli *et al.*, 2009). A mixture of animal meal by-products, rock phosphate and kelp meal is commonly given at transplanting as supplemental N (Diver *et al.*, 2012). Some farmers also provide biofertilizers during the first 6 weeks of production as biweekly foliar applications of fish extract (0.07 kg N/ha, 0.008 kg P/ha, 0.03 kg K/ha) combined or not with seaweed powder (0.007 kg N/ha, 0 kg P/ha, 0.02 kg K/ha). For soil with an excess of P or K, nitrogen is mainly provided by legume cover crops. Plant nutrient recommendations for fresh and processing organic tomatoes in the USA range from 56 to 146 kg N/ha, 112 to 280 kg P_2O_5/ha and 112 to 336 kg K_2O/ha (Diver *et al.*, 2012). For organic greenhouse tomato crops, top dressings are applied throughout the growing season to support the longer-term intensive production (up to 42 weeks of harvesting; yield of 500 t/ha) and higher crop demand compared with field tomato (40–100 t/ha), whereas additional

Table 11.3. Nutrient requirements for organic field and greenhouse tomato production.

Fertilizer inputs Field tomato[a]	Application rate (t DW/ha)[b]	N (kg/ha per year)	N (%)	C:N
Composted manure (composted chicken manure)	7–9	114–439	2.3–3.9	5.4–15.4
Cover crops (legumes or legume–grass mixtures)	–	81–148	1.8–3.9	10.2–21.7
Tomato plant uptake		105–189		
Fertilizer inputs Greenhouse tomato[b]	Application rate (t/ha/y)	N (kg/ha/y)	P (kg/ha/y)	K (kg/ha/y)
Base dressing	59[c]	776	191	636
Top dressing	17[d]	685	135	1751
Total	76	1461	326	2387

[a]Field tomato crops need up to 3.4 kg N per hectare per day (237 kg N/ha) and 6.75 kg K_2O per hectare per day (496 kg K_2O/ha), considering plant uptake of 100–150 kg N/ha, 20–40 kg P_2O_5/ha and 150–300 kg K_2O/ha and a yield of 40–50 t/ha (see Chapter 6). Source: Clark *et al.*, 1999.
[b]Fertilization scenario for greenhouse production based on the commercial practices of eight producers located in The Netherlands and according to a 3-year crop rotation (tomato, cucumber and sweet pepper). Source: Voogt *et al.*, 2011. DW is Dry Weight.
[c]In kg N:P:K content per tonne fresh weight: compost (7.1:1.4:4.8) 30 t/ha, manure (7.4:2.5:8.1) 19.3 t/ha, dried manure (22:7.85:12.45) 3.3 t/ha, mixed fertilizer (50:26.2:107.9) 1.4 t/ha, feather meal (130:0:0) 0.72 t/ha, bone meal (60.0:69.8:0) 0.28 t/ha, blood meal (130:0:0) 0.42 t/ha, seaweed (14.6:1.40:32.9) 0.96 t/ha, castor oil waste (50:8.5:12) 0.78 t/ha, hydrolysed plant material (41.3:9.1:27.7) 0.07 t/ha.
[d]In kg N:P:K content per tonne fresh weight: mixed fertilizer (50:26.2:107.9) 2.76 t/ha, blood meal (130:0:0) 0.82 t/ha, feather meal (130:0:0) 0.80 t/ha, brewery waste (50:13.08:41.50) 1.94 t/ha, vinasse (20:0:249) 3.50 t/ha, castor oil waste (44.7:5.8:70.0) 2.04 t/ha, dried chicken manure (22.0:7.8:12.4) 1.91 t/ha, hydrolysed plant material (30:4.2:26) 1.0 t/ha, bone meal (60.0:69.8:0) 0.10 t/ha, K_2SO_4 0.74 t/ha, $MgSO_4$ 1.04 t/ha.

fertilizers (e.g. K_2SO_4) are usually provided via the irrigation system. Many different waste products from the agricultural and fishery industries are used and may have fast or slow release rates, as shown in Table 11.4. Liming is sometimes needed to adjust the pH to about 6.5.

Nitrogen

For organic crops, N is the main limiting factor for productivity. The calculation of the fertilization is based on the N requirement and N release rate (Table 11.4). Owing to rapid N mineralization, the application of high-N waste products from agricultural and fishery industries is the most practical option for in-season N fertilization. In fact, 47–60% of the organic N in blood

Table 11.4. Nutrient values (g/kg fresh weight) of different types of amendments used for organic tomato production.

Sources	N	P_2O_5[a]	K_2O	CaO	MgO	OM[b]	C:N	N release (%) (< 1 year)
Bone meal	73–114	160	–	22	0.6	558–934	4.5	55–66
Blood meal	12–15	2–3	1	0.3	–			64–70
Feather meal	101	11.5	2.7	–	–	880	4.8	64–82
Meat meal	81	93	6	–	–	667	4.8	70
Guano	160	202	30	–	–	–	–	82–93
Cattle manure	5.7–8	4–9	5–14			210	16.6	20–45
Sheep manure	11.5	7	23			260		20
Pig manure	5–11	2–43	3–21				3.3	65
Poultry manure	14–28	23–28	7–25			479–577	14.5	50–70
Turkey manure	22–28	31–40	25–32			454–530		50
Fish meal	10–90	5–90	0–0.5	–	–			Fast
Crab waste	2	3.6	0.2	–	–			Slow
Shrimp waste	2.9	10	–	–	–			–
Algae products	6–7.5	1–1.2	11–15	3–4	1–2	200	–	Slow
Alfalfa meal	2.5	0.3	1.9	–	–			Medium
Cotton-seed meal	6–7	1.1–3	1.5	0.4	0.9			70
Peanut meal	7	1.5	1.2	0.4	0.3			–
Soybean meal	7	0.7–1.2	1.5–2.4	0.4	0.3			70
Castor cake	57	30	20	–	–	824	8.4	55– 66
Wheat bran	2.9	1.4	1.3	–	–			–
Green waste[a]	5–10	3–5	3–10	7–40	2–4	450	11–25	–
Molasses	4–5	0.3	6					40–50

[a]In general, 65–80% of P and 80–100% of K are available during the growing season.
[b]OM, organic matter. Note that heavy metal analyses (Cd, Cr, Cu, Hg, Ni, Pb, Se, Zn, As, Mo) should be performed to comply with organic production regulations.

meal, feather meal, guano and fish powder is mineralized within 2 weeks and up to 74% is mineralized after 8 weeks (Hartz and Johnstone, 2006). With *Tithonia diversifolia*, in contrast to leguminous species, no N immobilization occurs and an ideal N:P ratio of 10 is maintained during decomposition (Partey *et al.*, 2011). Under an organic farming management system in northern California, using a 4-year rotation (tomato, safflower, maize and bean with a biculture of oat–vetch mix), the potentially mineralizable N was 112% higher than under a 2-year conventional system (tomato, wheat) (Poudel *et al.*, 2002). Cost per unit of available N varies widely among fertilizers, with feather meal the cheapest and fish powder the most expensive. Nevertheless, too high levels of N and NH_4 were found to increase plant infection in organic tomato. For example, high concentrations of NH_4-N and total C and a low calcium (Ca) concentration were correlated with a higher level of corky root rot (Hasna *et al.*, 2007). Similarly, Fusarium crown rot of tomato was observed to be more severe under NH_4-N compared with NO_3-N, a higher total N concentration, low Ca fertilization and high NaCl levels (Woltz *et al.*, 1992). High NH_4-N form can be observed under cold conditions given the slower rate of nitrification at low temperatures.

Micronutrients

Organic amendments such as compost and manure provide micronutrients that are seldom applied by farmers, such as manganese (Mn), zinc (Zn) and other micronutrients. In addition to soil-incorporated fertilizers, growers use foliar feed with fish emulsion, seaweed and compost, weed teas, or other certificated substances. However, the effectiveness of foliar spraying varies, depending on: (i) the environmental conditions (e.g. temperature, light, relative humidity); (ii) leaf phenology (e.g. surface wetness); (iii) plant physiological status (e.g. open stomata, age, injury, nutrient status, osmotic potential, root temperature) at the time of treatment; (iv) the physical-chemical properties of the solution (e.g. pH, components, adjuvants, surface-active agents, surfactants); and (v) the amount and mode of application (e.g. droplet size, volume, concentration) (Fernández and Eichert, 2009).

Nutrient balance

The NPK composition of the organic fertilizers rarely matches plant demands, resulting in unbalanced soil nutrient content and soil salinization (e.g. SO_4, Na, Cl). Consequently, the production of organic tomato, particularly in greenhouses, is becoming more and more difficult with respect to European legal N and P standards and regulations, owing to the limited manure application permitted (170 kg N/ha per year), restrictions in the use of compost depending on its quality, and upcoming discussions on limiting the total nutrient supply amounts below a threshold of some 1250 kg N/ha per year (Skal Biocontrole, 2010; Tittarelli *et al*., 2016). Tremendous difficulties involved in synchronizing demand and supply were revealed in a study with ten organic greenhouse growers using a tomato, cucumber and sweet pepper crop rotation. The average N and P supply was more than double the crop demand (680 kg N and 150 kg P per hectare and year) resulting in a surplus of approximately 945 kg N and 250 kg P per hectare per year (Voogt *et al*., 2011). Moreover, unbalanced ratios between N, P and K were observed (42:11:47) compared with the crop demand (37:6:57). Therefore, to meet the annual N demand of a tomato crop of about 1110 kg/ha yielding 50 kg/m^2, the following fertilization was recommended: approximately 470 kg N/ha as compost, 170 kg N/ha as manure, 560 kg N/ha as additional fertilizers (Voogt *et al*., 2011). Models to calculate nutrient availability have also been developed to improve fertilization practices (Janssen, 2011).

When the nutrient application rate exceeds the crop uptake or when amendments contain high levels of residual salts such as Na, Cl and SO_4, soil salinization occurs, limiting plant growth, yield and fruit quality (Gravel *et al*., 2011b). To control soil salinity, growers over-irrigate their soil, causing N leaching and environmental pollution. Therefore, a dynamic nutrient balance approach is needed to simulate N availability, fine-tune manure applications to crop demand, and apply innovative approaches to the collection and treatment of crop effluents as part of sustainable fertilization management. Otherwise, nutrient losses (leaching, denitrification, fixation) not only harm the environment but may also affect the reputation of organic farming.

Compost

The most appropriate C:N ratio of composts used for organic tomato production ranges between 25:1 and 40:1, because these composts (1–3 months) have good biological activity and are a good source of humus.

Nutrient value

Increased yield of field tomato grown in a sandy loam was observed with 62 and 124 t of compost per hectare (Maynard, 1995). Although commercial composts usually have a low N content (< 1.5%), cow manure-based composts amended with grape marc (1:1) and wheat straw (2:1) with an organic matter content of 70–53%, a C:N ratio of 15.6–13.1 and a pH of 7.1–6.8 were found to provide 2.63–2.39% N in the form of NH_4-N (137–195 mg/kg) and NO_3-N (15–1644 mg/kg), 487–211 mg PO_4^{-3}-P/kg and 10,101–14,781 mg K/kg (Raviv *et al*., 2005). However, when the C:N ratio is higher (25–30) due to amendment with maize straw, wheat bran and cotton-seed cake, cow manure-based compost was found to have 1.21% N, 0.6% P and 1.58% K (Xie *et al*., 2011). Table 11.4 provides the nutrient values of composts/manures generally used for organic tomato production.

Disease suppressiveness

Organic matter inputs in amounts that provide significant levels of disease control (25–50% v:v) are commonly used in organic production systems (Giotis *et al.*, 2009). Depending on the type of compost, multiple studies have reported a major or moderate suppressive effect of compost/compost tea on tomato diseases such as bacterial spot (Xanthomonas *vesicatoria*), early blight (*Alternaria solani*), bacterial canker (*C. michiganensis*), corky root rot (*P. lycopersici*), grey mould (*Botrytis cinerea*), *F. oxysporum*, *P. infestans*, *P. ultimum*, *R. solani*, *V. dahliae* and nematodes (*M. javanica*) (Noble and Coventry, 2005). In addition, the number of fungal- and bacterial-feeding nematodes increased significantly. Compost may, however, have negative effects, such as stimulation of corky root rot by horse manure compost. The microbial content, nutrients and organic molecules (e.g. humic and phenolic compounds) in compost are the main factors in the inhibition or prevention of plant diseases. While microorganisms may act through pathogen antagonism, parasitism, antimicrobial production or induction of plant resistance, organic compounds in compost may induce systemic resistance as well or may have a direct toxic effect on the pathogen. The inoculation of compost with biofertilizers may increase its suppressive effect, though interactions with external factors and fine-tuning are extremely important for the effectiveness of this form of biological control (Van der Wurff *et al.*, 2016).

Biostimulants

The stimulation of crops with plant growth-promoting agents is a common strategy used by organic producers. *Pseudomonas*, *Bacillus* and *Streptomyces* spp. can control *B. cinerea* in organic greenhouse tomato. Suppression of corky root rot has been correlated with cellulolytic actinomycetes and total actinomycete populations (Workneh and van Bruggen, 1994). *P. fluorescens* strains have been shown to control bacterial pathogens effectively. However, abiotic factors such as water stress and nutrient deficiency and the indigenous microflora and microfauna affect the efficacy of these promoters. The beneficial response of tomato to a growth promoter may also be dependent on the tomato genotype. Biostimulants containing non-living substances, such as humic/fulvic acids, protein hydrolysates, chitosan compounds or silicon, may exhibit positive effects on leaf chlorophyll content, plant growth, yield and fruit components. Seaweed products are a rich source of Ca, K, P, Mg and Mn, iron (Fe), Zn, molybdenum (Mo), boron (B) and cobalt (Co). Their plant hormones (cytokinins, auxins, gibberellins), amino acids and vitamin content may promote the growth of beneficial soil microbes, while their rich content in polyphenols can have antifungal activity (Sangha *et al.*, 2014). Seaweed extracts (*Ecklonia maxima* or *Ascophyllum nodosum*) applied to soil reduced tomato root infestation by *M. incognita* or *M. javanica*, probably as a result of the betaine content in the extract (Chitwood, 2002). Despite the apparent benefits of some biostimulants, there is controversy about the real advantages for organic tomato crops.

WATER MANAGEMENT

Irrigation systems and monitoring tools used for organic tomato are similar to those for conventional crops (see Chapter 6), though special care is given to optimize microbial soil activity and mineralization rate (Dorais *et al.*, 2016). Good water management practices, however, by means of root zone or plant sensors as well as evapotranspiration and water-uptake models, still need to be adapted to organic crops. In fact, the fine-tuning of irrigation for organic crops is much more complex than in conventional crops, due to plant heterogeneity and ion imbalances as well as the optimization of soil biological activity, which is a main factor in plant productivity. Based on soil activity, physiological parameters, yield and fruit quality, matric potential thresholds of −40 mbar to −100 mbar were proposed for organic greenhouse tomato grown in a sandy loam soil, though similar mineralization rates were observed for organic soils at −35 mbar and −250 mbar (Pepin *et al.*, 2008). Maintaining a higher potential (< −30 mbar) resulted in a lower soil mineral content. For a greenhouse tomato crop grown in containers filled with a peat–sawdust-based growing medium, a matric potential ranging from −22 mbar to −15 mbar relative to a container capacity of −6 mbar was beneficial for improving plant photosynthetic activity and yield (Lemay *et al.*, 2012). Because tomato plants are sensitive to high soil moisture and poor soil oxygen content, excessive irrigation on poorly draining organic soils limits productivity by reducing or inhibiting root respiration, mineral uptake and water movement into the roots. This results in reduced stomatal conductance and photosynthesis and enhanced plant susceptibility to root diseases such as *Pythium* spp. For soil-grown organic greenhouse tomato, oxygen (O_2) enrichment of the irrigation water (16 μmol/l) or O_2 injection at a depth of 25 cm (25% soil O_2) increased soil biological activity, nutrient availability, leaf mineral content, plant photosynthetic capacity and, in some cases, fruit yield (Dorais and Pepin, 2011; Dorais *et al.*, 2011). For field-grown tomato in heavy clay, it was observed that subsurface irrigation with aerated water (12% air in water) stimulated above-ground growth, increased fruit yield by 21% and improved plant water use efficiency (Bhattarai *et al.*, 2006).

Water management of organic tomato should also take into consideration the build-up of salinity and ion imbalances when unbalanced amendment and fertilizers are used. Over-irrigation to control soil salinity should be limited, to prevent nutrient emission into the environment. This is particularly challenging for EU organic producers, as a zero emission target by 2027 has been established for greenhouse soil-grown crops to fulfil the water and nitrates framework directives (2000/60/EC; 91/676/EEC). Moreover, effluent recycling is difficult with soil-grown organic field or greenhouse tomato crops, because collecting drains have to be installed with minimal soil disturbance and contamination from other agricultural or industrial activities nearby. Organic tomato crop effluents can also be rich in organic matter, making the disinfection process to reuse this wastewater and limit pathogen propagation challenging. However, tailwater ponds designed to capture runoff and tailwater return systems installed to recycle runoff back to the field seem to be successful management practices for nutrient cycling on an organic tomato farm (Smukler *et al.*, 2012). Such systems significantly reduce total discharge, ammonium and dissolved organic matter compared with fallow fields. Artificial wetlands are also being used to treat greenhouse effluents from organic tomato grown in demarcated beds (Gruyer *et al.*, 2013a,b). Multiple technological approaches integrating biological processes (e.g. sand filter, wetland, passive bioreactor), membrane filtration, ozonation, ultraviolet (UV) treatment and thermal treatment may also be used to recycle effluents of organic tomato crops.

PLANT PROTECTION

The diseases and pests observed in conventional production (see Chapter 7) also occur in organic farming. Nevertheless, the improvements in soil fertility, organic matter and polysaccharide content as well as the higher and more diverse microbial biomass in organic tomato systems reduce the incidence of diseases. The main challenge in organic production is therefore to optimize the management system in such a way that makes either the tomato plants healthier or the conditions for the pests less favourable. Possible options for optimizing the equilibrium of the agroecosystem as a whole are: (i) the application of agronomic practices (e.g. rotation, cover crop, manure, compost, biostimulants, climate management and nutrition control in greenhouses); (ii) the introduction of natural pests and microbes; (iii) the use of banker plants; and (iv) the application of habitat management and sanitary control. Only when this approach of indirect crop protection is insufficient do control measures come into play. These measures depend on the incidence itself, the environmental conditions and the permissibility of supplying a potential protectant. In addition, the control measures should comply with the following rules: (i) must be necessary; (ii) should be organic; (iii) should not be harmful to the environment or human health; (iv) should have no negative effect on produce quality; and (v) should not meet resistance or opposition from consumers (Speiser *et al.*, 2006).

Control of fungal, bacterial and nematode pathogens

Soil steaming and soil solarization are the most widely used techniques to reduce soilborne diseases, though both techniques negatively affect microflora and fauna. Soil anaerobic disinfection and soil biofumigation in organic tomato crops by means of *Brassica* green manure can be an efficient alternative tool (Gamliel and van Bruggen, 2016) as well as using *Brassica* break crops (e.g. kohlrabi, white mustard) (Giotis *et al.*, 2009). The addition of fresh *Brassica* tissue or chitin, or both, to organic greenhouse soils infected by *P. lycopersici*, *V. albo-atrum*, or both, was found to increase fruit yield by reducing disease severity in tomato roots owing to the release of isothiocyanates or the induced resistance effects of chitin (Noble and Coventry, 2005). Intercropping organic tomato with companion species reduced diseases, but with different effects on yield and quality. For example, marigold (*Tagetes erecta*) was shown to reduce the incidence of early blight, *A. solani* (Gómez-Rodriguez *et al.*, 2003) and intercropping with turfgrass as living mulch reduced leaf blight incidence (Xu *et al.*, 2012). Essential oils and individual monoterpenoids from lavender (*Lavandula stoechas*), oregano (*Origanum vulgare* ssp. *hirtum*), sage (*Salvia fruticosa*) and spearmint (*Mentha spicata*) were found to prevent mycelial growth and conidial production of *Aspergillus terreus*, *F. oxysporum*, *Penicillium expansum* and *V. dahliae* (Woltz *et al.*, 1992; Kadoglidou *et al.*, 2011). A similar approach to reduce disease incidence is the application of suppressive soils or plant growth media (Castaño *et al.*, 2011). Copper-based products are largely used in organic tomato production to control anthracnose, bacterial speck, bacterial spot, early and late blight, grey leaf mould, powdery mildew and septoria leaf spot, and elemental sulfur (S) is used to control powdery mildew and mites (Diver *et al.*, 2012). However, growers should rely on alternative control methods, because copper is toxic for non-targeted microorganisms and fauna (e.g. earthworms) and leads to soil and water contamination, increasing the risk of plant toxicity. Potassium bicarbonate is often used to prevent anthracnose, early blight, leaf blight, leaf spot and powdery mildew.

Pest control

Pest control is based mainly on frequent releases of natural predators and the use of natural repellents (e.g. garlic, onion, zinnia, marigold, nasturtium). Trap crops are effective by attracting different pests. For example, maize invites tomato fruitworm, whereas marigold is attractive to the fruit borer. Biopesticides such as pyrethrum are commonly used when pest populations reach economically damaging levels (Diver *et al.*, 2012). Microbial insecticides such as *B. thuringiensis* may control several pests at their immature stage. Also potentially helpful are biocontrol agents, including the nuclear polyhedrosis virus, used against the fruit borer, and *Trichoderma* strains successfully used as an egg parasitoid against several *Lepidoptera* pests.

Weed control

Weed control, particularly within crop rows, is one of the main problems in organic tomato cultivation. Few efficient weed control products are available for use. Thus, exclusively physical control methods provide ‘instantaneous’ weed control (Raffaelli *et al.*, 2011). However, physical control requires on average 50% higher labour inputs than conventional weed control. Other strategies are cover crops and mulches, such as natural materials, paper and plastic, but they may increase soil moisture and N content, delaying tomato fruit ripening. Soil solarization in greenhouse tomato acts as a prophylactic method. It may kill most of the weed seeds and induce dormancy in the remaining seeds, resulting in higher crop yields.

HEALTH AND NUTRITIVE COMPOUNDS

The increasing consumer demand for organic products is mainly due to health and safety values over pesticide residues, antibiotics and genetically modified organisms in foods. Beyond the fact that the health benefits of organic foods are controversial, organic tomatoes have lower pesticide residues than conventional fruit (Dorais and Alsanius, 2015). Consequently, the pesticide risks posed by organically grown tomatoes are substantially smaller than those posed by tomatoes from other sources. However, risk assessments of natural pesticides remain to be scientifically attested. Furthermore, there is no evidence that organic tomatoes are more susceptible to microbiological contamination than conventional horticultural products if appropriate farm management practices are used.

Besides consumer perception, direct evidence showing that organic tomatoes have a higher nutritional value than conventionally grown fruits still has to be scientifically attested. Nonetheless, organic tomatoes have lower nitrate content and higher levels of vitamin C and phenolic compounds as compared with conventionally grown ones (Table 11.5). When tomato plants are grown under an organic regime, their natural defence mechanisms may be more active than those of conventional plants, resulting in higher defence-related secondary metabolites, which can often be considered beneficial for human health (Vallverdú-Queralt *et al.*, 2012; Orsini *et al.*, 2016). In fact, it has been shown that organic tomato experiencing oxidative stress has higher vitamin C and phenolic compounds (Oliveira *et al.*, 2013; López-Martinez *et al.*, 2016). Stresses related to reduced foliage development and higher fruit exposure to ambient light may also increase the vitamin C content of organically grown tomatoes. Because the flavonoid content in tomatoes is inversely related to available N, reduced N supply for organic tomato may enhance the flavonoid level compared with conventional tomato (Mitchell *et al.*, 2007). For carotenoids including β-carotene and lycopene, no consensus between differences of organic and conventional grown tomatoes has be made (Dorais *et al.*, 2008; Dorais and Alsanius, 2015). Given that Cu and Zn are common constituents of animal feed supplements, manure applications as an important source of nutrients provide additional Cu and Zn, resulting in a higher plant uptake. The use of

Table 11.5. Differences in the nutritional value of tomatoes grown under organic and conventional practices and their ratio (organic to conventional). Only compound values that were significantly different were reported.

Compounds (units[a])	Cultivars	Units	Organic[b]	Org:conv	Sampling period[c]	References
Soluble solids	Mean of 10 cvs	°Brix	4.4	0.956	2 y	Aldrich *et al.*, 2010
Soluble solids	Firenze, HyPeel 108	°Brix	4.58–5.27	0.867–1.114	1 y	Riahi *et al.*, 2009
Soluble solids	AB2	°Brix	5.05–6.66	1.002–1.273	2 y	Pieper and Barrett, 2009
Soluble solids	HM 830, Bos 315	°Brix	5.39–5.96	1.072–1.157	1 y	Barrett *et al.*, 2007
Soluble solids	Ropreco, Burbank	°Brix	5.4–5.5±0.5	1.102–1.170	3 y	Chassy *et al.*, 2006
Soluble solids	Sahel	°Brix	5.5±0.5	1.28	1 y	Lopez-Martinez *et al.*, 2016
Soluble solids	Licata, CH2000, Sansone	°Brix	5.2±0.3	0.97	3 exp	De Pascale *et al.*, 2016
Total solids	AB2	%	5.65–6.93	1.014–1.262	2 y	Pieper and Barrett, 2009
Titratable acidity	Firenze, Perfectpeel	%	0.33–0.46	1.138–1.278	1 y	Riahi *et al.*, 2009
Titratable acidity	HM 830, Bos 315	%	0.25–0.37	1.190–1.193	1 y	Barrett *et al.*, 2007
Titratable acidity	Licata, CH2000, Sansone	% DM	6.3	0.75	3 exp	De Pascale *et al.*, 2016
Crude protein	PS1296	g/kg FW	14.9	1.419	1 y	Rossi *et al.*, 2008
Glutamic acid	AB2	g/kg DW	14.3–36.7	0.709–0.991	2 y	Pieper and Barrett, 2009
Glutamine	AB2	g/kg DW	10.4–49.8	0.336–0.922	2 y	Pieper and Barrett, 2009
Tyrosine	AB2	g/kg DW	0.22–0.54	0.426–0.96	2 y	Pieper and Barrett, 2009
Ash	PS1296	g/kg FW	0.65	1.326	1 y	Rossi *et al.*, 2008
Nitrate	Plato de Egipto	% increase	–	0.667	5 y	Herencia *et al.*, 2011
NH_4-N	AB2	mg/kg DW	104–126	0.456–0.727	2 y	Pieper and Barrett, 2009
Total N	AB2	% DW	2.30	0.830	2 y	Pieper and Barrett, 2009
Total N	Brigade	g/kg DW	~20.0	0.717	2 y	Colla *et al.*, 2002
P	Brigade	g/kg DW	~3.88–4.02	1.165–1.589	2 y	Colla *et al.*, 2002
P	AB2	% DW	0.45	1.184	2 y	Pieper and Barret, 2009

K	AB2	% DW	4.00	1.149	2 y	Pieper and Barrett, 2009
Na	Brigade	g/kg DW	~347	0.529	2 y	Colla *et al.*, 2002
Ca	Many varieties	mg/kg DW	2126	2.070	2 y	Kelly and Bateman, 2010
Ca	AB2	% DW	0.13	0.929	2 y	Pieper and Barrett, 2009
Zn	Many varieties	mg/kg DW	~32	1.231	2 y	Kelly and Bateman, 2010
Mn	Many varieties	mg/kg DW	11	0.611	2 y	Kelly and Bateman, 2010
Mn	AB2	mg/kg DW	11–15	0.647–0.652	2 y	Pieper and Barrett, 2009
Cu	Many varieties	mg/kg DW	~8	1.333	2 y	Kelly and Bateman, 2010
B	AB2	mg/kg DW	17	0.895	2 y	Pieper and Barrett, 2009
Cd	PS1296	µg/kg FW	33.0	16.50	1 y	Rossi *et al.*, 2008
Pb	PS1296	µg/kg FW	37.8	11.12	1 y	Rossi *et al.*, 2008
Ascorbic acid	Mean of 10 cvs	g/kg FW	0.198	1.070	2 y	Aldrich *et al.*, 2010
Ascorbic acid	PS1296	mg/kg FW	118.2	0.552	1 y	Rossi *et al.*, 2008
Ascorbic acid	Burbank	mg/100 g FW	22.1	1.263	3 y	Chassy *et al.*, 2006
Ascorbic acid	Izabella, Félicia	mg/100 g FW	16.2–17.5	1.326–1.687	1 y	Caris-Veyrat *et al.*, 2004
Dehydro-ascorbic ac.	Rogers 1570, HM 830, Bos 315, HyPeel 45	µg/g FW	2448–1587	0.74–1.394	1 y	Barrett *et al.*, 2007
Salicylic acid	PS1296	mg/kg FW	0.745	1.613	1 y	Rossi *et al.*, 2008
Caffeic acid	Daniella	µg/g FW	41.70	1.822	2 y	Vallverdú-Queralt *et al.*, 2012
Chlorogenic acid	Daniella	µg/g FW	56.99	1.546	2 y	Vallverdú-Queralt *et al.*, 2012
Ferulic acid	Daniella	µg/g FW	35.11	1.619	2 y	Vallverdú-Queralt *et al.*, 2012
p-coumaric acid	Daniella	µg/g FW	34.25	1.663	2 y	Vallverdú-Queralt *et al.*, 2012
Chlorogenic acid	Félicia, Paola	mg/100 g FW	0.55–0.65	0.859–1.354	1 y	Caris-Veyrat *et al.*, 2004
Apigenin-*7-O*-glucoside	Daniella	µg/g FW	31.63	1.118	2 y	Vallverdú-Queralt *et al.*, 2012
Naringenin	Izabella, Félicia	mg/100 g FW	0.09–0.10	1.125–1.250	1 y	Caris-Veyrat *et al.*, 2004
Naringenin	Halley 3155	mg/g DW	39.6	1.311	10 y	Mitchell *et al.*, 2007
Naringenin	Daniella	µg/g FW	87.38	2.397	2 y	Vallverdú-Queralt *et al.*, 2012

Continued

Table 11.5. Continued.

Compounds (units[a])	Cultivars	Units	Organic[b]	Org:conv	Sampling period[c]	References
Naringenin-*7-O* glucoside	Daniella	µg/g FW	13.91	1.811	2 y	Vallverdú-Queralt *et al.*, 2012
Kaempferol	Halley 3155	mg/g DW	63.3	1.974	10 y	Mitchell *et al.*, 2007
Kaempferol	Ropreco, Burbank	mg/100 g FW	1.47–1.58	1.17–1.195	3 y	Chassy *et al.*, 2006
Kaempferol-*3-O*-rutinoside	Daniella	µg/g FW	12.70	2.106	2 y	Vallverdú-Queralt *et al.*, 2012
Quercetin	Daniella	µg/g FW	11.42	2.007	2 y	Vallverdú-Queralt *et al.*, 2012
Quercetin	Halley 3155	mg/g DW	115.5	1.788	10 y	Mitchell *et al.*, 2007
Quercetin	Burbank	mg/100 g FW	3.42	1.295	3 y	Chassy *et al.*, 2006
Rutin	Daniella	µg/g FW	272.75	2.276	2 y	Vallverdú-Queralt *et al.*, 2012
Rutin	Paola, Félicia, Izabella	mg/100 g FW	0.14–0.22	1.20–5.50	1 y	Caris-Veyrat *et al.*, 2004
Total phenolics	Sahel	mg/100 g FW	14.45	1.086	1 y	Lopez-Martinez *et al.*, 2016
Carotenoids	Licata, CH2000, Sansone	mg/100 g FW	7.85	1.16	2 exp	De Pascale *et al.*, 2016
Lycopene	PS1296	mg/100 g FW	3.6–4.2	0.760–1.125	1 y	Rossi *et al.*, 2008
Lycopene	Paola, Félicia, Izabella	mg/100 g FW		1.105	1 y	Caris-Veyrat *et al.*, 2004
Lycopene	Sahel	mg/100 g FW	4.37±0.30	0.897	1 y	Lopez-Martinez *et al.*, 2016
β-carotene	Paola, Félicia, Izabella	mg/100 g FW	1.03–1.35	1.198–1.626	1 y	Caris-Veyrat *et al.*, 2004
Antioxidant activity	Mean of 10 cvs	µmol TEAC/kg FW	1870	1.078	2 y	Aldrich *et al.*, 2010
Antioxidant activity	Sahel	µmol TEAC/kg FW	1437	1.026	1 y	Lopez-Martinez *et al.*, 2016

[a]FW, fresh weight; DW, dry weight; TEAC, Trolox equivalent antioxidant capacity.
[b]~: values estimated from published figures.
[c]exp: experiments; y: year(s).

Bordeaux mixture (copper sulfate and lime solution) to help to control fungal and bacterial diseases also provides additional Cu.

ENVIRONMENTAL IMPACT OF ORGANIC FARMING

The environmental benefits of organic farming have been documented most clearly for biodiversity and for water and soil conservation. In general, organic systems enhance floral and faunal diversity and overall microbial activity, and have a high erosion control potential via soil fertility management. Organic farming can also reduce soil CO_2 emissions and increase C sequestration compared with conventional cropping systems. C sequestration content of soil organic C in an organic system generally exceeds that in a conventional system through successive cropping years. In Europe, organic farming practices have positive impacts on the environment as shown by higher soil organic matter content, lower nutrient losses (N leaching, N_2O and NH_3 emissions) per unit of cultivated area compared with conventional farming, and lower energy requirements. However, due to a lower yield per cultivated area, higher N losses have been observed per product unit (Tuomisto *et al.*, 2012). On the other hand, the CO_2 footprint of an organic greenhouse tomato grown in The Netherlands was 10% higher than that of a conventional crop (1950 kg CO_2 eq/t versus 1775 kg CO_2 eq/t tomatoes), assuming that the gas requirement to heat the greenhouse is responsible for 85% of the total CO_2 emissions (Vermeulen and van der Lans, 2011). However, when combined heat and power is used (cogeneration), the organic and conventional tomato crops have an equal CO_2 footprint of 890 kg CO_2 eq/t tomatoes. Under the growing conditions in Quebec, Canada, climate control was responsible for 81–96% of the total CO_2 footprint per tonne of tomatoes (Dorais *et al.*, 2014). Organic greenhouse tomato systems use more energy than conventional production systems in northern regions (e.g. 33 GJ/t versus 26.3 GJ/t in northern central Europe; cumulative energy demand of 97 GJ/t versus 80 GJ/t in Quebec) owing to a lower yield per unit of cultivated area as well as to strict climate control to prevent disease (Vermeulen and van der Lans, 2011; Dorais *et al.*, 2014).

CONCLUSIONS

High production risks related to limited effective tools for pest, disease and weed management still constitute a major barrier for the expansion of organic farming. Therefore, new biological control agents, biopesticides and cultural practices increasing plant resilience to biotic and abiotic stresses as well as a balanced nutrient supply following plant demands without any leaching or emission into the environment have to be used and optimized to make organic tomato production competitive. More research is needed on the nutritive value of organic tomato, though lower pesticide residues compared with conventionally grown tomatoes is clear. The enhancement of microbial, floral and faunal diversity as a result of the use of different types of organic amendments also needs further verification and publication.

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